

Programming for Biomedical Informatics

Week 2 - FTP Bulk Download & APIs

This week we're going to practice using bulk download and API in a practical example. This week we are going to be looking at a resource called GenCC - The Gene Curation Collection - <https://thegencc.org>

GenCC is primarily focussed on providing a unified source of gene-disease relationships that is harmonised across the various international efforts that operate in this space. In the process of doing so they are establishing standards for curation and annotation to maximise the consistency and useability of their data.

Our efforts today will focus on a full FTP download of the GenCC data, some parsing, and use of the API at Bioportal to gather more information about some of the elements of the GenCC data. This is a **slightly** contrived example, but hopefully it is both interesting and useful to give you an idea of how these two approaches can be used together.

```
In [2]: # Step One - Bulk Download of GenCC data

'''The url to the latest version of the data is here - https://search.thege

#pseudocode 1 - download the data using the urllib libraryd
'''###YOUR CODE HERE###'''
import urllib.request

url = "https://search.thegencc.org/download/action/submissions-export-tsv"
urllib.request.urlretrieve(url, "gencc_data.tsv")

#pseudocode 2 - read the data into a pandas dataframe *you can see from the
'''###YOUR CODE HERE###'''
import pandas as pd
df = pd.read_csv("gencc_data.tsv", sep="\t")

#pseudocode 3 - look ay the first few rows of the dataframe
'''###YOUR CODE HERE###'''
df.head()
```

Out [2]:

	uuid	gene_curie	gene_symbol	disease_curie	disease_title
0	GENCC_000101- HGNC_10896- OMIM_182212- HP_0000006...	HGNC:10896	SKI	MONDO:0008426	Shprintzen- Goldberg syndrome
1	GENCC_000101- HGNC_16636- OMIM_171300- HP_0000006...	HGNC:16636	KIF1B	MONDO:0008233	pheochromocytoma
2	GENCC_000101- HGNC_16636- OMIM_118210- HP_0000006...	HGNC:16636	KIF1B	MONDO:0007308	Charcot-Marie- Tooth disease type 2A1
3	GENCC_000101- HGNC_17939- OMIM_617532- HP_0000007...	HGNC:17939	SLC45A1	MONDO:0044322	intellectual developmental disorder with neuro...
4	GENCC_000101- HGNC_11071- OMIM_616291- HP_0000007...	HGNC:11071	SLC9A1	MONDO:0014572	Lichtenstein-Knorr syndrome

5 rows × 30 columns

In [7]:

```
# Step Two – Gene Counts by Disease

'''Now we want a summary of how many genes have been annotated to each disease'''

#pseudocode 4 – group the data by disease and count the number of genes for each disease
'''###YOUR CODE HERE###'''
grouped = df.groupby('disease_title').agg({'gene_symbol': 'nunique'}).reset_index()

#pseudocode 5 – sort the data in descending order
'''###YOUR CODE HERE###'''
grouped = grouped.sort_values(by='gene_symbol', ascending=False)

#pseudocode 6 – look at the first few rows of the dataframe
'''###YOUR CODE HERE###'''
grouped.head()

'''There's a nice package called PrettyTable that can be used to print the contents of a dataframe'''

from prettytable import PrettyTable

#make some simple two-column data
data = [['disease1', 1000], ['disease2', 2000], ['disease3', 3000], ['disease4', 4000]]

# create a table
table = PrettyTable()

# add the columns
table.field_names = ['Disease Name', 'Number of Genes']
```

```

# add the rows
for i in range(len(data)):
    table.add_row(data[i])

# print the table
print(table)

# pseudocode 7 – create a table using the PrettyTable package
'''###YOUR CODE HERE###'''
table = PrettyTable()
table.field_names = ["Disease Name", "Number of Genes"]
for index, row in grouped.iterrows():
    table.add_row([row['disease_title'], row['gene_symbol']])
print(table)

'''There is a very nice plotting package called seaborn that can be used to
Here's an example of how to create a horizontal bar chart using seaborn'''

#plot a horizontal bar chart of the data
import seaborn as sns
import pandas as pd

# set the style
sns.set_style('darkgrid')

# plot the data
sns.barplot(x='Number of Genes', y='Disease Name', data=pd.DataFrame(data, c

#pseudocode 8 – plot the 10 diseases with the most genes
'''###YOUR CODE HERE###'''
top10 = grouped.head(10)
sns.barplot(x='gene_symbol', y='disease_title', data=top10)

```

Disease Name	Number of Genes
disease1	1000
disease2	2000
disease3	3000
disease4	4000
disease5	5000

Disease Name	Number of Genes	Disease
mental disorder	255	complex neurodevelop
al disorder	145	neurodevelopment
disease	127	mitochondrial
drome	115	Leigh syn
gmentosa	102	retinitis pi
enia	88	schizophr
c hearing loss	86	nonsyndromic geneti
ndrome	76	Tourette sy
omal recessive	73	hearing loss, autos
ual disability	68	syndromic intellect
isability	60	intellectual d
omyopathy	58	dilated cardi
pileptic encephalopathy		genetic developmental and e

54		
		male infertility with azoospermia or oligo
		zoospermia due to single gene mutation
53		
		primary ciliary
		dyskinesia
53		
		familial isolated dila
		ted cardiomyopathy
45		
		autosomal recessive non-syndrom
		ic intellectual disability
44		
		autosomal dominant nonsy
		ndromic hearing loss
43		
		amyotrophic later
		al sclerosis
43		
		hypertrophic car
		diomyopathy
41		
		autism spectru
		m disorder
41		
		undetermined early-onset ep
		ileptic encephalopathy
41		
		ciliopa
		thy
39		
		congenital hea
		rt disease
34		
		autosomal dominant non-syndrom
		ic intellectual disability
34		
		non-syndromic X-linked in
		tellectual disability
33		
		cone-rod dy
		strophy
30		
		systemic lupus e
		rythematosus
30		
		epilep
		sy
30		
		hereditary breas
		t carcinoma
29		
		mitochondrial compl
		ex I deficiency
29		
		familial idiopathic steroid-re

28	Leigh syndrome with leukodystrophy	28	Leigh syndrome with leukodystrophy
26	familial thoracic aortic aneurysm and aortic dissection	26	familial thoracic aortic aneurysm and aortic dissection
25	autosomal recessive primary microcephaly	25	autosomal recessive primary microcephaly
25	Leber congenital amaurosis	25	Leber congenital amaurosis
25	multiple congenital anomalies/dysmorphic syndrome	25	multiple congenital anomalies/dysmorphic syndrome
25	Bardet-Biedl syndrome	25	Bardet-Biedl syndrome
24	inherited retinal dystrophy	24	inherited retinal dystrophy
24	pulmonary arterial hypertension	24	pulmonary arterial hypertension
23	Joubert syndrome	23	Joubert syndrome
23	Kallmann syndrome	23	Kallmann syndrome
22	X-linked complex neurodevelopmental disorder	22	X-linked complex neurodevelopmental disorder
22	arrhythmogenic right ventricular cardiomyopathy	22	arrhythmogenic right ventricular cardiomyopathy
22	Fanconi anemia	22	Fanconi anemia
21	Diamond-Blackfan anemia	21	Diamond-Blackfan anemia
21	breast cancer	21	breast cancer
21	Brugada syndrome	21	Brugada syndrome
19	familial ovarian cancer	19	familial ovarian cancer
19	cystic fibrosis	19	cystic fibrosis

|
 drome
 19 |
 |
 e to sperm motility disorder
 19 |
 |
 oth disease
 18 |
 |
 hypogonadism
 18 |
 |
 ear cataract
 18 |
 |
 fibrillation
 17 |
 |
 ndrome
 16 |
 |
 esis disorder
 16 |
 |
 spasms
 16 |
 |
 toma-paraganglioma
 16 |
 |
 drome
 16 |
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 efficiency disease
 15 |
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 et cataract
 15 |
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 glycanopathy, type A
 14 |
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 ancer
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 cancer
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 sis colon cancer
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 um disorders

Noonan syn
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 peroxisome biogen
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 infantile
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 hereditary pheochromocy
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 Meckel syn
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 cytochrome-c oxidase d
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 total early-ons
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 muscular dystrophy-dystro
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 prostate c
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 common variable im
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 colorectal
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 hereditary nonpolypo
 |
 Zellweger spectr
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13		
		maturity-onset diabe
tes of the young		
13		diabetes mellitus, non
insulin-dependent		autosomal recessive limb-gi
13		
		X-linked syndromic inte
rdle muscular dystrophy		
13		congenital stationary
llectual disability		congenital m
13		
		permanent neonatal d
night blindness		
12		Brugada syn
yopathy		monogenic d
12		
		acute myeloid
iabetes mellitus		
12		Jeune syn
drome 1		postsynaptic congenital
12		
		early-onset lamel
iabetes		
12		catecholaminergic polymorphic
leukemia		hereditary spasti
12		
		Lynch syn
drome		
12		Senior-Loken
myasthenic syndrome		tooth age
12		
		dyskeratosis
lar cataract		
11		
ventricular tachycardia		
11		
c paraplegia		
11		
drome		
11		
syndrome		
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nesis		
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congenita		
11		
		Coffin-Siris
syndrome		
11		
		mitochondrial comple
x III deficiency		
11		
		osteogenesis impe
rfecta type 3		
11		
		Leber hereditary op
tic neuropathy		
11		
		congenital heart defec
ts, multiple types		
11		
		neonatal diabet
es mellitus		
11		
		heart conducti
on disease		
10		
		osteogenesis impe
rfecta type 4		
10		
		presynaptic congenital
myasthenic syndrome		
10		
		catara
ct		
10		
		holoprosenc
ephaly		
10		
		autosomal dominant mac
rothrombocytopenia		
10		
		Costello sy
ndrome		
9		
		Parkinson d
isease		
9		
		46 XX gonadal
dysgenesis		
9		
		Cowden di
sease		
9		
		cardiofaciocutane
ous syndrome		
9		
		short QT sy
ndrome		
9		

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 cataract
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 left ventricular
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 congenital anomaly of kid
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 malignant migrating partia
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 46,XY partial gona
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 microphthalmia, isolat
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 Meier–Gorlin
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 autis
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 congenital fiber-type di
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 cataract – microco
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 autosomal agamma
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		situs inv
ersus		
8		renal agenesis,
unilateral		tetralogy of
8		
		malignant pancrea
fallot		
8		early-onset posterio
tic neoplasm		Omenn syn
8		
		muscle-eye-bra
r polar cataract		
8		hearing loss
drome		orofaciodigital sy
8		
		Seckel syn
in disease		
8		mitochondrial proton-transporting A
disorder		atypical Rett
8		
		nephronopht
ndrome type 6		
8		Hirschsprung
drome		epidermodysplasia
8		
		Dravet syn
TP synthase complex deficiency		
7		X-linked intellect
syndrome		amelogenesis impe
7		
		trichothiody
hisis 2		
7		
disease		
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verruciformis		
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		hypotrichosis
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		MERRF syn
drome		
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		pheochromo
cytoma		
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		childhood-onset nem
aline myopathy		
7		
		autosomal dominant nocturna
l frontal lobe epilepsy		
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		46,XY complete gona
dal dysgenesis		
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		isolated anophthalmia-mic
rophthalmia syndrome		
7		
		septo optic d
ysplasia		
7		
		maternally-inherited
Leigh syndrome		
7		
		Lennox-Gastaut
syndrome		
7		
		young-onset Parki
nson disease		
7		
		Fuchs' endotheli
al dystrophy		
7		
		achromato
psia		
7		
		autosomal dominant polyc
ystic kidney disease		
7		
		nephronopht
hisis 1		
7		
		xeroderma pig
mentosum		
7		
		myoclonic-astat
ic epilepsy		
7		
		combined pituitary hormone de
ficiencies, genetic form		
7		

6		
		MELAS syn
drome		
6		
		pontocerebellar hyp
oplasia type 1		
6		
		connective tiss
ue disorder		
6		
		Berardinelli-Seip conge
nital lipodystrophy		
6		
		bilateral striopallido
dentate calcinosis		
6		
		sarcom
a		
6		
		Stickler sy
ndrome		
6		
		split hand-foot
malformation		
6		
		Perrault sy
ndrome		
6		
		Ehlers-Danlos
syndrome		
6		
		erythrokeratoderm
ia variabilis		
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		congenital g
laucoma		
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		postaxial polyda
ctyly type A		
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		progressive myocl
onus epilepsy		
6		
		Ellis-van Creve
ld syndrome		
6		
		cerebellar ataxia, intellectual d
isability, and dysequilibrium		
6		
		chronic granuloma
tous disease		
6		
		congenital muscular dystrophy w
ith cerebellar involvement		
6		
		osteogenesis impe

refectia type 2	
6	
ge syndrome	Cornelia de Lan
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refectia type 1	amelogenesis impe
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ancer	ovarian c
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pancreatitis	hereditary chronic
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s mellitus	type 2 diabete
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ystrophy	ovarioleukod
6	
smorphism	digitotalar dy
6	
ylcostal dysostosis	autosomal recessive spond
5	
otor neuropathy	distal hereditary m
5	
h motor neuron disease	frontotemporal dementia wit
5	
corneal dystrophy	posterior polymorphous
5	
febrile seizures plus	generalized epilepsy with
5	
splasia	skeletal dy
5	
nd syndrome	Shwachman-Diamo
5	
l dysplasia	cranioectoderma
5	
ly hair disorder	isolated familial woo
5	
ruption syndrome	pituitary stalk inter
5	
erekplexia	hereditary hyp
5	

5	active cardiomyopathy	5	familial isolated restri
5	uploidy syndrome	5	mosaic variegated ane
5	multiple lentigines	5	Noonan syndrome with m
5	us candidiasis	5	chronic mucocutaneo
5	lmia	5	nanophtha
5	with glycosylation defect	5	congenital myasthenic syndromes
5	rophy	5	cone dyst
5	malformations	5	conotruncal heart
5	nishing white matter 1	5	leukoencephalopathy with va
5	mperfecta	5	osteogenesis i
5	cy disease	5	immunodeficien
5	ith intellectual disability	5	congenital muscular dystrophy w
5	stromal tumor	5	gastrointestinal
5	omaly	5	Peters an
5	s involved in pituitary development or function	5	hypothyroidism due to deficient transcription factor
5	susceptibility to	5	neural tube defects,
5	ferative syndrome	5	autoimmune lymphoprolif
5	external ophthalmoplegia	5	autosomal dominant progressive
5	rygium syndrome	5	lethal multiple pte

5		
		colorectal
adenoma		
5		
		leukoencephalopathy with v
anishing white matter		
5		
		generalized junctional epidermoly
sis bullosa non-Herlitz type		
5		
		early-onset posterior s
ubcapsular cataract		
5		
		Sheldon-hall
syndrome		
5		
		autosomal recessive S
tickler syndrome		
5		
		osteosar
coma		
5		
		disseminated superficial
actinic porokeratosis		
5		
		autoimmune
disease		
5		
		spermatogenic
failure		
5		
		pontocerebellar hyp
oplasia type 2		
5		
		megacystis-microcolon-intestina
l hypoperistalsis syndrome		
5		
		short rib-polydactyly syndr
ome, Verma-Naumoff type		
5		
		severe congenital ne
maline myopathy		
5		
		junctional epidermolysis bu
llosa, non-Herlitz type		
5		
		Joubert syndrome wit
h ocular defect		
5		
		congenital fibrosis of
extraocular muscles		
5		
		immunodeficiency-centromeric instab
ility-facial anomalies syndrome		
5		
		hereditary sph

	primary Fancon
i syndrome	
4	
	Charcot-Marie-Tooth
disease type 3	
4	
	focal epi
lepsy	
4	
	familial men
ingioma	
4	
	autosomal dominant severe
congenital neutropenia	
4	
	46,XX ovotesticular disord
er of sex development	
4	
	pachyonychia
congenita	
4	
	Leigh syndrome with
cardiomyopathy	
4	
	autosomal recessive cen
tronuclear myopathy	
4	
	hereditary hemophagocytic
lymphohistiocytosis	
4	
	hereditary hemorrhagi
c telangiectasia	
4	
	Loeys-Dietz
syndrome	
4	
	familial or sporadic h
emiplegic migraine	
4	
	hereditary breast ovari
an cancer syndrome	
4	
	mismatch repair can
cer syndrome 1	
4	
	inherited o
besity	
4	
	male infer
tility	
4	
	autosomal thrombocytopenia
with normal platelets	
4	
	autosomal dominant R
obinow syndrome	

4		
		early-onset anterior
polar cataract		
4		Beckwith-Wiedema
nn syndrome		Warburg micro
4		
		bilateral rena
syndrome		
4		atypical glycine e
l agenesis		Usher syndrom
4		
		amelogenesis imper
ncephalopathy		
4		Dowling-Degos
e type 2		autosomal dominant hypohidro
4		
		medullobla
fecta, type 3A		
4		early-onset autosomal domi
disease		Treacher-Collin
4		
		intermediate maple sy
tic ectodermal dysplasia		
4		hereditary persistence of fetal hemo
stoma		osteogenesis impe
4		
		lung can
nant Alzheimer disease		
4		overgrowth syndrome and/or cerebral malformation
s syndrome		congenital muscular dystrophy wit
4		
		athyreo
rup urine disease		
4		
globin-beta-thalassemia syndrome		
4		
rfecta type 1		
4		
cer		
4		
s due to abnormalities in MTOR pathway genes		
4		
hout intellectual disability		
4		

sis		
4		
		Muir-Torre s
syndrome		
4		
		complex hereditary sp
astic paraplegia		
4		
		isolated plag
iocephaly		
4		
		craniosyno
stosis		
4		
		familial congenital
mirror movements		
4		
		xeroderma pigmentosum-Cock
ayne syndrome complex		
4		
		Moyamoya d
isease		
4		
		cerulean ca
taract		
4		
		complex neurodevelopmental dis
order with motor features		
4		
		Ullrich congenital mu
scular dystrophy		
4		
		familial bicuspid
aortic valve		
4		
		Bethlem my
opathy		
4		
		familial glucocorti
coid deficiency		
4		
		autosomal recessive
optic atrophy		
4		
		AP-4 deficienc
y syndrome		
4		
		Waardenburg
syndrome		
4		
		autosomal recessive hypohidro
tic ectodermal dysplasia		
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		MHC class II d
eficiency		
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 s of genetic origin
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 lity and spastic paraplegia
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 ifuss muscular dystrophy
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 nomic neuropathy type 1
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 e type 3
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 line myopathy

renal tubular dysgenesi
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 AP4-related intellectual disabi
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 autosomal dominant Emery-Dre
 |
 autosomal recessive
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 monileth
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 MHC class I d
 |
 hereditary late onset
 |
 hereditary macul
 |
 hereditary sensory and auto
 |
 homozygous familial hy
 |
 22q11.2 deletio
 |
 hereditary sensory and auto
 |
 Usher syndrom
 |
 isolated brac
 |
 juvenile myoclon
 |
 mitochondrial comple
 |
 juvenile amyotrophic
 |
 diabetes me
 |
 intermediate nema
 |

4		hereditary persistence of fetal hemogl
		obin-sickle cell disease syndrome
4		
		atrial fibri
		llation
4		
		autosomal systemic lupus
		erythematosus type 16
4		
		autosomal dominan
		t cutis laxa
3		
		classic maple syrup
		urine disease
3		
		lissencephaly spec
		trum disorders
3		
		familial sleep-related
		hypermotor epilepsy
3		
		Cockayne syndr
		ome type 2
3		
		autoinflammator
		y syndrome
3		
		Stargardt d
		isease
3		
		3-M synd
		rome
3		
		neuromuscular
		disease
3		
		Ehlers-Danlos syndro
		me, classic type
3		
		hypomyelination-hypogonadotropic hy
		pogonadism-hypodontia syndrome
3		
		familial throm
		bocytosis
3		
		Noonan syndrome-like disorde
		r with loose anagen hair
3		
		neurometabolic disorder du
		e to serine deficiency
3		
		familial vesicour
		eteral reflux
3		
		hypomyelination neuropathy-

arthrogryposis syndrome	
3	
	familial chilb
lain lupus	
3	
	cirrhosis, f
amilial	
3	
	infantile convulsions a
nd choreoathetosis	
3	
	junctional epidermolysis
bullosa Herlitz type	
3	
	Joubert syndrome wi
th renal defect	
3	
	familial hyper
cholanemia	
3	
	central areolar chor
oidal dystrophy	
3	
	juvenile polypos
is syndrome	
3	
	congenital vertebral-cardiac-
renal anomalies syndrome	
3	
	epidermolysis bul
losa simplex	
3	
	familial medullary t
hyroid carcinoma	
3	
	hemolytic uremic syndrome, atyp
ical, susceptibility to, 1	
3	
	self-limited epilepsy with
centrotemporal spikes	
3	
	familial hypofib
rinogenemia	
3	
	Desbuquois d
ysplasia	
3	
	familial infantile bilate
ral striatal necrosis	
3	
	familial long Q
T syndrome	
3	
	muscular dystrophy, limb-gir
dle, autosomal dominant	
3	

	Zimmermann-Laba
nd syndrome	
3	
	familial isolated hyp
erparathyroidism	
3	
	self-healing col
lodion baby	
3	
	familial focal epilepsy
with variable foci	
3	
	3MC synd
rome	
3	
	early-onset non-syn
dromic cataract	
3	
	familial dysfib
rinogenemia	
3	
	autosomal dominant epilepsy
with auditory features	
3	
	X-linked intellectual disabili
ty with marfanoid habitus	
3	
	junctional epiderm
olysis bullosa	
3	
	Sotos syn
drome	
3	
	LADD synd
rome	
3	
	intermittent maple sy
rup urine disease	
3	
	Sengers sy
ndrome	
3	
	rigid spine
syndrome	
3	
	hereditary diffuse gast
ric adenocarcinoma	
3	
	male infertility due t
o globozoospermia	
3	
	inclusion body myopathy with Paget diseas
e of bone and frontotemporal dementia	
3	
	Duane retractio
n syndrome	

3		
		neuronopathy, distal here
ditary motor, type 5A		
3		
		dysosteoscl
erosis		
3		
		lymphatic mal
formation		
3		
		Doyne honeycomb ret
inal dystrophy		
3		
		hemoglobin M
disease		
3		
		familial cerebral cave
rnous malformations		
3		
		anirid
ia		
3		
		mitochondrial complex II def
iciency, nuclear type 1		
3		
		amyotrophic lateral
sclerosis type 1		
3		
		isolated Klippel-
Feil syndrome		
3		
		patterned macula
r dystrophy		
3		
		hereditary thrombocytopenia and hemato
ogic cancer predisposition syndrome		
3		
		autosomal recessive
osteopetrosis 6		
3		
		short rib-polydactyly syn
drome, Majewski type		
3		
		congenital fibrino
gen deficiency		
3		
		thyroid hyp
oplasia		
3		
		glycine encep
halopathy		
3		
		Bartter syndro
me type 4		
3		
		cholesta

sis		
3		
		brachydactyly
type A1		
3		
		Bernard–Soulie
r syndrome		
3		
		Rh deficiency
syndrome		
3		
		Bethlem myop
athy 1A		
3		
		brachydactyly
type A2		
3		
		Ritscher–Schinz
el syndrome		
3		
		high bone mass osteog
enesis imperfecta		
3		
		movement di
sorder		
3		
		thromboph
ilia		
3		
		Hermansky–Pudlak syndrome wi
thout pulmonary fibrosis		
3		
		sodium channelopathy-related
small fiber neuropathy		
3		
		cerebral
palsy		
3		
		megalencephaly-polymicrogyria-postaxial
polydactyly-hydrocephalus syndrome		
3		
		autosomal recessive polyc
ystic kidney disease		
3		
		multiple synosto
ses syndrome		
3		
		thiamine-responsive maple
syrup urine disease		
3		
		primary pigmented nodular
adrenocortical disease		
3		
		branchio-oto-ren
al syndrome		
3		

	pseudohypoparathyroidism type 1B	pseudohypoparathyroidism type 1B
3		
	deficiency	C1Q deficiency
3		
	congenitally primary vitreous	persistent hyperplastic primary vitreous
3		
	atrial heart block	progressive familial atrial heart block
3		
	erythromatous hemorrhagic, type 1	telangiectasia, hereditary hemorrhagic
3		
	hypertrophic	myopathies
3		
	transient myoglobinuria	hereditary recurrent hemoglobinuria
3		
	anemia	Heinz body anemia
3		
	congenitally dispermic mole	complete hydatidiform mole
3		
	neurodegenerative disease	maple syrup urine disease
3		
	epidemic	Weaver syndrome
3		
	Werner's syndrome	activated PI3K-dependent
3		
	deaf hearing loss	X-linked nonsyndromic deafness
3		
	osteopetrotic, familial, 1	tumoral calcinosis, hyperphosphatemic
3		
	ectodermal dysplasia	ectodermal dysplasia, hypohidrotic
3		
	Werner's syndrome	Weill-Marchesani syndrome
3		
	congenitally associated dysgyria	tubulinopathy-associated
3		
	epidemic	Alport syndrome
3		
	congenital myopathy	tubular aggregation

3		
		multiple epiphyseal dysplasia
due to collagen 9 anomaly		
3		
		pyridoxine-depend
ent epilepsy		
3		
		UV-sensitive
syndrome		
3		
		triple-A sy
ndrome		
3		
		atypical juvenile
parkinsonism		
3		
		Mobius syn
drome		
3		
		Unverricht-Lundb
org syndrome		
3		
		multiple congenital anomalies/dysmorphi
c syndrome-intellectual disability		
3		
		hereditary periphe
ral neuropathy		
3		
		Hennekam sy
ndrome		
3		
		Axenfeld-Riege
r syndrome		
3		
		pyruvate dehydrogenase
E1-alpha deficiency		
3		
		Ullrich congenital mus
cular dystrophy 1A		
3		
		microphth
almia		
3		
		multiple acyl-CoA dehyd
rogenase deficiency		
3		
		complex neurodevelopmental disorder wi
th or without congenital anomalies		
3		
		auriculocondyla
r syndrome		
3		
		linear skin defects with mult
iple congenital anomalies		
3		
		visceral het

erotaxy	
3	
	Pendred sy
ndrome	
3	
	nervous system
disorder	
3	
	cold-induced swea
ting syndrome	
3	
	spastic quadriplegic
cerebral palsy	
3	
	IL10-related early-onset inf
lammatory bowel disease	
3	
	childhood absen
ce epilepsy	
3	
	mitochondrial ence
phalomyopathy	
3	
	Liddle syn
drome	
3	
	inherited neurodegen
erative disorder	
3	
	hereditary gingiva
l fibromatosis	
3	
	autosomal dominant seco
ndary polycythemia	
3	
	advanced sleep ph
ase syndrome	
3	
	hereditary isolated
aplastic anemia	
3	
	T-B+ severe combined immunodeficiency
due to CD3delta/CD3epsilon/CD3zeta	
3	
	cap myop
athy	
3	
	nevroid basal cell ca
rcinoma syndrome	
3	
	epidermolysis bullosa si
mplex 1C, localized	
3	
	Leber plus
disease	
3	

	polycystic kidn
ey disease	
3	
	delta-beta-th
alassemia	
3	
	orofacial
cleft	
3	
	antenatal Bartt
er syndrome	
3	
	striate palmoplant
ar keratoderma	
3	
	Carney-Strataki
s syndrome	
3	
	agammaglobu
linemia	
3	
	isolated an
iridia	
3	
	late-onset nephr
onophthisis	
3	
	autosomal recessive cutis la
xa type 2, classic type	
3	
	female infertility due to
zona pellucida defect	
3	
	microvillus inclu
sion disease	
3	
	ectodermal dyspla
sia syndrome	
3	
	Haddad syn
drome	
3	
	mitochondrial neurogastrointe
stinal encephalomyopathy	
3	
	congenital afib
rinogenemia	
3	
	Fraser syn
drome	
3	
	hypertrophic card
iomyopathy 1	
3	
	collagen 6-relat
ed myopathy	

3		
		distal arthrogryp
osis type 2B1		
3		
		mutilating palmoplantar keratoderma wi
th periorificial keratotic plaques		
3		
		pseudohypoaldosteronism, type
IB1, autosomal recessive		
3		
		microcephaly and cho
rioretinopathy 1		
3		
		autosomal recessive distal
renal tubular acidosis		
3		
		dehydrated hereditar
y stomatocytosis		
3		
		neonatal glycine e
ncephalopathy		
3		
		epidermolysis bullosa simple
x 1A, generalized severe		
2		
		otospondylomegaepiphyseal dysp
lasia, autosomal recessive		
2		
		Imerslund-Grasbeck
syndrome type 1		
2		
		enhanced S-con
e syndrome		
2		
		Imerslund-Grasbe
ck syndrome		
2		
		X-linked progressive
cerebellar ataxia		
2		
		cutaneous po
rphyria		
2		
		ectodermal dysplasia an
d immune deficiency		
2		
		neurofibromatosis-N
oonan syndrome		
2		
		Hb Bart's hydro
ps fetalis		
2		
		IMAGe syn
drome		
2		
		intestinal hypom

agnesemia 1	
2	
	seizures, benign fami
lial neonatal, 2	
2	
	choanal atresia-hearing loss-cardiac defe
cts-craniofacial dysmorphism syndrome	
2	
	neurodevelopmental disorder with progressive spa
sticity and brain white matter abnormalities	
2	
	IgE responsiven
ess, atopic	
2	
	encephalopathy due to mitochondrial
and peroxisomal fission defect	
2	
	intracranial ber
ry aneurysm	
2	
	arthrogryposis-renal dysfunct
ion-cholestasis syndrome	
2	
	Hermansky-Pudlak syndrome w
ith pulmonary fibrosis	
2	
	Jervell and Lange-N
ielsens syndrome	
2	
	intrinsic card
iomypopathy	
2	
	semantic de
mentia	
2	
	sensorineural hearin
g loss disorder	
2	
	otospondylomegaepiph
yseal dysplasia	
2	
	cholesterol-ester transfe
r protein deficiency	
2	
	severe dermatitis-multiple allergi
es-metabolic wasting syndrome	
2	
	Jackson-Weiss
syndrome	
2	
	Rieger an
omaly	
2	
	severe neonatal lactic acidosis due
to NFS1-ISD11 complex deficiency	
2	

		Huntington
disease		
2		
		severe combined immunodeficiency, autosomal recessive,
T cell-negative, B cell-negative, NK cell-positive		
2		
		short stature due to isolated growth hormone def
iciency with X-linked hypogammaglobulinemia		
2		
		congenital amegakaryocyt
ic thrombocytopenia		
2		
		Hartnup di
sease		
2		
		choreatic d
isease		
2		
		craniometaphyseal
l dysplasia		
2		
		short stature with nonspecific
skeletal abnormalities 1		
2		
		Hutchinson-Gilford p
rogeria syndrome		
2		
		cryptorch
idism		
2		
		enlarged vestibular a
queduct syndrome		
2		
		demyelinating hereditary moto
r and sensory neuropathy		
2		
		scleroste
osis		
2		
		megalencephalic leukoencephalopa
thy with subcortical cysts 1		
2		
		acromesomelic d
ysplasia 2A		
2		
		Meniere di
sease		
2		
		progressive familial intrahe
patic cholestasis type 1		
2		
		acromeg
aly		
2		
		qualitative plat
elet defect		

2		
		radio-ulnar synostosis-amegakaryoc
ytic thrombocytopenia syndrome		
2		
		combined oxidative phosphor
ylation defect type 20		
2		
		acrofacial dysostos
is, Weyers type		
2		
		recessive X-linke
d ichthyosis		
2		
		Meesmann cornea
l dystrophy		
2		
		acro dysostosis with multip
le hormone resistance		
2		
		acro dysos
tosis		
2		
		recurrent infections associated with rar
e immunoglobulin isotypes deficiency		
2		
		megalencephalic leukoencephalop
athy with subcortical cysts		
2		
		Matthew-Wood
syndrome		
2		
		amelogenesis i
mperfecta		
2		
		combined immunodeficiency
due to STK4 deficiency		
2		
		Marinesco-Sjogr
en syndrome		
2		
		Marie Unna hereditar
y hypotrichosis		
2		
		acrocallosal
syndrome		
2		
		progressive encephalopathy with leuko
dystrophy due to DECR deficiency		
2		
		progressive cerebello
-cerebral atrophy		
2		
		megalencephaly-polymicrogyria-polyd
actly-hydrocephalus syndrome 1		
2		
		acral peeling sk

in syndrome	
2	
nd polymicrogyria	occipital pachygyria a
2	
nesemia 2	renal hypomag
2	
ma	melano
2	
tor V and factor VIII	combined deficiency of fac
2	
ability-young-onset diabetes syndrome	primary microcephaly-mild intellectual dis
2	
nent neonatal diabetes syndrome	primary microcephaly-epilepsy-perma
2	
ysplasia 2B	acromesomelic d
2	
rotic syndrome	amelocerebrohypohid
2	
splasia	anauxetic dy
2	
juvenile neurologic onset	Niemann-Pick disease type C,
2	
disease	adult Refsum
2	
drome	Ochoa syn
2	
d lipofuscinosis	adult neuronal ceroid
2	
halangism	proximal symp
2	
syndrome	Norman-Roberts
2	
thalmoplegia with mitochondrial myopathy	adult-onset chronic progressive external oph
2	
cial cleft	commissural fa
2	
nflammatory syndrome 3	proteasome-associated autoi
2	

	adenosine monophosphate
deaminase deficiency	
2	
	psoriasis 14,
pustular	
2	
	age related macular
degeneration 1	
2	
	Niemann–Pick disease type
C, severe perinatal form	
2	
	Niemann–Pick disease type C, severe
early infantile neurologic onset	
2	
	Niemann–Pick disease type C, lat
early infantile neurologic onset	
2	
	Niemann–Pick disease type C,
adult neurologic onset	
2	
	developmental and epilept
ic encephalopathy 93	
2	
	propionic a
acidemia	
2	
	ptosis, hereditary
congenital, 1	
2	
	Neu–Laxova sy
syndrome 1	
2	
	Neu–Laxova s
syndrome	
2	
	optic atrophy
ophy	
2	
	PEHO synd
retinopathy	
2	
	intellectual disability,
autosomal dominant 40	
2	
	open-angle glaucoma
glaucoma	
2	
	macrothrombocytopenia
nia, isolated	
2	
	alpha thalassemia
ia spectrum	
2	
	macrocephaly–autism
ism syndrome	

2		
		punctate palmoplantar
keratoderma type 1		
2		
		pustulosis palmari
s et plantaris		
2		
		alternating hemipleg
ia of childhood		
2		
		renal-hepatic-pancr
eatic dysplasia		
2		
		intellectual developmental disorder with dysmo
rphic facies and behavioral abnormalities		
2		
		schwannoma
tosis		
2		
		Keppen-Lubinsk
y syndrome		
2		
		dominant hypophosphatemia with ne
phrolithiasis or osteoporosis		
2		
		apolipoprotein A-
I deficiency		
2		
		retinitis pig
mentosa 7		
2		
		LCAT defic
iency		
2		
		Polymerase proofreading-relat
ed adenomatous polyposis		
2		
		LADD syndr
ome 1		
2		
		methylmalonic aciduria
and homocystinuria		
2		
		methylmalonic aciduria and h
omocystinuria type cblC		
2		
		osteopetr
osis		
2		
		neuronopathy, distal hereditary
motor, autosomal recessive 5		
2		
		cystinu
ria		
2		
		arterial calcificat

ion of infancy	
2	
	RA Sapat
hy	
2	
	retinitis puncta
ta albescens	
2	
	cone-rod synaptic disorder, c
ongenital nonprogressive	
2	
	infantile liver fai
lure syndrome 2	
2	
	cystic leukoencephalopathy
without megalencephaly	
2	
	microcephalic primordial dwarfism
-insulin resistance syndrome	
2	
	arthrogryposis mult
iplex congenita	
2	
	immunodeficiency, co
mmon variable, 1	
2	
	Kabuki syn
drome	
2	
	KID synd
rome	
2	
	early myoclonic e
ncephalopathy	
2	
	Joubert syndrome with Jeune asph
xyiating thoracic dystrophy	
2	
	early-onset generalized
limb-onset dystonia	
2	
	arthrogryposis
syndrome	
2	
	rolandic epilepsy-speech
dyspraxia syndrome	
2	
	Yunis-Varon
syndrome	
2	
	early-onset progressive diffuse brain atrophy-micro
cephaly-muscle weakness-optic atrophy syndrome	
2	
	schneckenbecken
dysplasia	
2	

		Lafora di
sease		
2		
		Larsen syn
drome		
2		
		MEDNIK syn
drome		
2		
		Lewy body d
ementia		
2		
		nonpapillary renal
cell carcinoma		
2		
		retinal dege
neration		
2		
		Lynch synd
rome 1		
2		
		colobo
ma		
2		
		metabolic d
isease		
2		
		non-severe combined immunodeficiency d
ue to polymerase delta deficiency		
2		
		aniridi
a 1		
2		
		inherited oocyte ma
turation defect		
2		
		dense deposit
disease		
2		
		aniridia-cerebellar ataxia-intel
lectual disability syndrome		
2		
		metabolic s
yndrome		
2		
		Li-Fraumeni
syndrome		
2		
		ankylosing sp
ondylitis		
2		
		annular epidermoly
tic ichthyosis		
2		
		anosmi
a		

2		
		methemoglobinemi
a, alpha type		
2		
		anterior segment
dysgenesis		
2		
		Leigh syndrome with n
ephrotic syndrome		
2		
		anterior segment
dysgenesis 1		
2		
		primary familial polycythemia d
ue to EPO receptor mutation		
2		
		anterior segment
dysgenesis 7		
2		
		metaphyseal an
adysplasia		
2		
		delayed puberty,
self-limited		
2		
		Pfeiffer sy
ndrome		
2		
		Pfeiffer syndr
ome type 1		
2		
		primary congenit
al glaucoma		
2		
		Pick dis
ease		
2		
		aortic dis
order		
2		
		infantile myofi
bromatosis		
2		
		distal hereditary motor
neuropathy type 7		
2		
		epidermolysis bullosa simplex 1
B, generalized intermediate		
2		
		cataract-intellectual disabil
ity-hypogonadism syndrome		
2		
		epidermolysis bullosa simplex 1D, generalized,
intermediate or severe, autosomal recessive		
2		
		temporal lobe

epilepsy		
2		
		neonatal diabetes mellitus with
congenital hypothyroidism		
2		
		CHARGE syn
drome		
2		
		mitral valve
prolapse		
2		
		geleophysic d
ysplasia		
2		
		neonatal inflammatory sk
in and bowel disease		
2		
		mitochondrial trifunction
al protein deficiency		
2		
		neonatal severe primary
hyperparathyroidism		
2		
		autosomal recessive c
utis laxa type 1		
2		
		nephrogenic diabe
tes insipidus		
2		
		mitochondrial non-syndromic s
ensorineural hearing loss		
2		
		nephronoph
thisis		
2		
		brittle cornea
syndrome		
2		
		hypercalciuria, a
bsorptive, 2		
2		
		hypercholanemia,
familial 1		
2		
		autosomal recessive cong
enital ichthyosis 11		
2		
		parkinsonian-pyram
idal syndrome		
2		
		syndactyly
type 4		
2		
		late-onset Parki
nson disease		
2		

		Caroli di
sease		
2		
		hydrolethalus
syndrome		
2		
		generalized juvenile polyposi
s/juvenile polyposis coli		
2		
		Wiskott-Aldric
h syndrome		
2		
		branchiootic
syndrome		
2		
		brachydactyl
y type E		
2		
		piebald
ism		
2		
		brachydactyly-elbow wris
t dysplasia syndrome		
2		
		global developmental delay with or witho
ut impaired intellectual development		
2		
		paroxysmal familial vent
ricular fibrillation		
2		
		bradyop
sia		
2		
		Wolfram sy
ndrome		
2		
		Bruck syn
drome		
2		
		nemaline my
opathy		
2		
		glaucoma 1, ope
n angle, E		
2		
		thrombocyth
emia 1		
2		
		glauco
ma		
2		
		pituitary gi
gantism		
2		
		autosomal recessive inherited
pseudoxanthoma elasticum		

2		
		autosomal recessive infa
ntile hypercalcemia		
2		
		geroderma osteo
dysplastica		
2		
		tetraamelia-multiple mal
formations syndrome		
2		
		autosomal recessive hypo
phosphatemic rickets		
2		
		C3 glomerulon
ephritis		
2		
		Carpenter s
yndrome		
2		
		hypercholesterolemi
a, familial, 1		
2		
		Cenani-Lenz syndac
tyly syndrome		
2		
		hypercoagulability syndrome due to glyc
osylphosphatidylinositol deficiency		
2		
		Charcot-Marie-Tooth
disease type 2		
2		
		Charcot-Marie-Tooth
disease type 2B2		
2		
		congenital neutropenia-myelofib
rosis-nephromegaly syndrome		
2		
		spondylometaphyse
al dysplasia		
2		
		autosomal dominant pri
mary microcephaly		
2		
		mitochondrial complex IV def
iciency, nuclear type 1		
2		
		keratosis follicularis
spinulosa decalvans		
2		
		Stormorken s
yndrome		
2		
		autosomal dominant optic
atrophy, classic form		
2		
		fibrochondro

genesis		
2		
		neural tube
defect		
2		
		hypertrichotic osteochondr
odysplasia Cantu type		
2		
		hypertriglyc
eridemia		
2		
		hypertriglyce
ridemia 1		
2		
		keratolytic wint
er erythema		
2		
		congenital radioul
nar synostosis		
2		
		polycystic lipomembranous osteodysplas
ia with sclerosing leukoencephaly		
2		
		congenital reticular ichth
yosiform erythroderma		
2		
		Charlevoix-Saguenay
spastic ataxia		
2		
		Griscelli syndr
ome type 3		
2		
		lipodystr
ophy		
2		
		kidney Wilm
s tumor		
2		
		parietal fo
ramina		
2		
		autosomal recessive cerebellar atax
ia-saccadic intrusion syndrome		
2		
		X-linked Emery-Dreifuss
muscular dystrophy		
2		
		hyperglyci
nuria		
2		
		platelet-type bleed
ing disorder 16		
2		
		TH-deficient dopa-res
ponsive dystonia		
2		

		congenital dyserythropo	
ietic anemia type 1			
2			
		frontometaphyseal	
l dysplasia			
2			
		platelet-type bleed	
ing disorder 19			
2			
		Charcot-Marie-Tooth dis	
ease axonal type 2T			
2			
		congenital myopathy with	
myasthenic-like onset			
2			
		autosomal recessive	
Robinow syndrome			
2			
		capillary infantile	
le hemangioma			
2			
		congenital myopathy 4A,	
autosomal dominant			
2			
		congenital dyserythr	
opoietic anemia			
2			
		subcortical band	
heterotopia			
2			
		capillary malformation-arteriovenous	
enous malformation syndrome			
2			
		autosomal recessive	
Alport syndrome			
2			
		kidney dis	
order			
2			
		paroxysmal nonkinesigenic	
enic dyskinesia 1			
2			
		leiomyosarcoma	
rcoma			
2			
		spondyloepimetaphyseal dysplasia	
lasia with joint laxity			
2			
		hereditary mucopolysaccharidosis	
leukokeratosis			
2			
		VACTERL association, X-linked, with or without hydrocephalus	
ith or without hydrocephalus			
2			
		hereditary neurologic disease	
gical disease			

2		
		ventricular tachyca
rdia, familial		
2		
		hereditary palmoplan
tar keratoderma		
2		
		hereditary palmoplantar keratod
erma, Gamborg-Nielsen type		
2		
		periodic paralysis with later-on
set distal motor neuropathy		
2		
		hereditary pediatric B
ehçet-like disease		
2		
		van der Woude
syndrome		
2		
		leukodyst
rophy		
2		
		van Maldergem
syndrome		
2		
		hereditary pulmonary al
veolar proteinosis		
2		
		hereditary renal
hypouricemia		
2		
		hereditary sensory and a
utonomic neuropathy		
2		
		Acromicric d
ysplasia		
2		
		uncombable hai
r syndrome		
2		
		head and neck squamou
s cell carcinoma		
2		
		Aicardi sy
ndrome		
2		
		myoclonus-dyston
ia syndrome		
2		
		behavioral variant of fro
ntotemporal dementia		
2		
		hereditary multiple
osteochondromas		
2		
		hereditary motor and sens

ory neuropathy type 6	
2	
	thyroid c
ancer	
2	
	hereditary mixed pol
yposis syndrome	
2	
	muscular dystrophy-dystroglycanopathy (congenit
al with brain and eye anomalies), type A13	
2	
	periventricular
leukomalacia	
2	
	congenital he
mangioma	
2	
	3-methylcrotonyl-CoA car
boxylase deficiency	
2	
	3-methylglutaconic
aciduria type 3	
2	
	hemoglobin H
disease	
2	
	hereditary diffuse leukoencephalopathy wit
h axonal spheroids and pigmented glia	
2	
	hemochromatos
is type 5	
2	
	permanent neonatal diabetes mellitus-pancr
eatric and cerebellar agenesis syndrome	
2	
	46,XY disorder of sex development due
to isolated 17,20-lyase deficiency	
2	
	46,XY disorder of sex development due to
testicular 17,20-desmolase deficiency	
2	
	von Hippel-Lind
au disease	
2	
	autosomal recessive si
deroblastic anemia	
2	
	hemochromatos
is type 2	
2	
	hereditary hemo
chromatosis	
2	
	hematuria, benign
familial, 1	
2	

	vitamin K-dependent clotting factor
s, combined deficiency of, type 1	
2	
	vitamin D-dependent
rickets, type 1	
2	
	autosomal recessive progressiv
e external ophthalmoplegia	
2	
	hereditary skeletal
muscle disorder	
2	
	Usher syndrom
e type 2C	
2	
	benign familial neonatal
-infantile seizures	
2	
	hereditary spastic
paraplegia 12	
2	
	persistent Mulleria
n duct syndrome	
2	
	Bardet-Biedl s
ndrome 1	
2	
	glycogen storage disease due to liver
phosphorylase kinase deficiency	
2	
	myopathy, lactic acidosis, a
nd sideroblastic anemia	
2	
	autosomal recessive nonsyn
dromic hearing loss 1A	
2	
	blue cone mon
ochromacy	
2	
	hereditary spastic
paraplegia 9A	
2	
	Townes-Brocks
syndrome	
2	
	autosomal recessive nonsyn
dromic hearing loss 12	
2	
	Barrett eso
phagus	
2	
	lethal congenital cont
racture syndrome 3	
2	
	Bartter diseas
e type 4B	

2		
		congenital factor X
III deficiency		
2		
		Wilson-Turner
syndrome		
2		
		autosomal recessive multip
le pterygium syndrome		
2		
		congenital myasthen
ic syndrome 1A		
2		
		phosphoenolpyruvate carb
oxykinase deficiency		
2		
		lethal acantholytic epi
dermolysis bullosa		
2		
		thyroid cancer, no
nmedullary, 2		
2		
		blepharocheilodon
tic syndrome		
2		
		Baraitser-Winter cerebrof
rontofacial syndrome		
2		
		trichohepatoente
ric syndrome		
2		
		Andersen-Tawil
syndrome		
2		
		benign neonata
l seizures		
2		
		lethal restrictiv
e dermopathy		
2		
		tubulinop
athy		
2		
		tuberous sclero
sis complex		
2		
		Alzheimer d
isease		
2		
		multiple epiphyse
al dysplasia		
2		
		tuberous sc
lerosis		
2		
		Alzheimer dise

ase type 1	
2	
	triphalangeal thumb-poly
syndactyly syndrome	
2	
	multicentric osteolysis-nodul
osis-arthropathy spectrum	
2	
	peeling skin syn
drome type A	
2	
	Antley-Bixler
syndrome	
2	
	multiple endocri
ne neoplasia	
2	
	Axenfeld a
nomaly	
2	
	autosomal recessive nonsyn
dromic hearing loss 4	
2	
	trichothiodystrophy
1, photosensitive	
2	
	BH4-deficient hyperph
enylalaninemia A	
2	
	multiple benign circumferenti
al skin creases on limbs	
2	
	panhypopitu
itarism	
2	
	hyperostosis cortica
lis generalisata	
2	
	Ehlers-Danlos/osteogenesi
s imperfecta syndrome	
2	
	esophageal
cancer	
2	
	cerebellar
ataxia	
2	
	autosomal dominant
Alport syndrome	
2	
	localized junctional epidermolys
is bullosa, non-Herlitz type	
2	
	exfoliative i
chthyosis	
2	

	autoinflammatory syndrome with pyogenic bacterial infection and amylopectinosis	
2		
		corpus callosum,
	agenesis of	
2		
		congenital central hypov
	entilation syndrome	
2		
		Saethre–Chotze
	n syndrome	
2		
		specific granule
	deficiency	
2		
		hypoplastic tibiae–postaxia
	polydactyly syndrome	
2		
		cerebellar ataxia–hypo
	gonadism syndrome	
2		
		hypotonia, infantile, with psychomotor r
	etardation and characteristic facies	
2		
		spastic paraplegia, optic
	atrophy, and neuropathy	
2		
		erythrocytosis,
	familial, 7	
2		
		STAT3–related early-onset mult
	isystem autoimmune disease	
2		
		neurodevelopmental disorder with mic
	rocephaly and dysmorphic facies	
2		
		pachydermoper
	riostosis	
2		
		cortisone reducta
	se deficiency	
2		
		centronuclear
	myopathy	
2		
		central nervous syst
	em malformation	
2		
		congenital diarrhea 7 with
	exudative enteropathy	
2		
		autosomal dominant cent
	ronuclear myopathy	
2		
		familial avascular necro
	sis of femoral head	

2		Ehlers-Danlos syndrome,
arthrochalasia type		
2		pontocerebellar hyp
oplasia type 7		
2		Ehlers-Danlos syndrom
e, classic type, 1		
2		hypomyelinating le
ukodystrophy 6		
2		cataract-glauco
ma syndrome		
2		spermatogenic
failure 5		
2		Ehlers-Danlos syndrome, mu
sculocontractural type		
2		Ehlers-Danlos syndrome,
periodontal type 1		
2		congenital diaphra
gmatic hernia		
2		Ehlers-Danlos syndrome,
periodontitis type		
2		corneal dystroph
y, Meesmann, 1		
2		pontocerebellar hypoplasia, hypotonia, and respir
atory insufficiency syndrome, neonatal lethal		
2		isolated scap
hocephaly		
2		Coats plus s
yndrome		
2		mitochondrial DNA de
pletion syndrome		
2		pontocerebellar hypo
plasia, type 13		
2		facioscapulohumeral m
uscular dystrophy		
2		autoimmune lymphoprolifer
ative syndrome type 1		
2		porokeratosis

of Mibelli		
2		
		episodic kinesigeni
c dyskinesia 1		
2		
		Gitelman sy
ndrome		
2		
		epidermolytic
ichthyosis		
2		
		immunodefici
ency 104		
2		
		Rubinstein-Tayb
i syndrome		
2		
		atypical hemolytic-
uremic syndrome		
2		
		Rotor syn
drome		
2		
		craniofacial m
icrosomia		
2		
		childhood kidney
Wilms tumor		
2		
		childhood onset GLUT1 de
ficiency syndrome 2		
2		
		Glanzmann thro
mbasthenia		
2		
		Fontaine progero
id syndrome		
2		
		epidermolysis bullosa simplex
5C, with pyloric atresia		
2		
		Glanzmann's thr
ombasthenia		
2		
		congenital bilateral abs
ence of vas deferens		
2		
		premature ovaria
n failure 1		
2		
		atrioventricu
lar block		
2		
		premature ovaria
n failure 10		
2		

	epidermolysis bullosa simplex 2
F, with mottled pigmentation	
2	
	atrial stan
dstill	
2	
	short-rib thoracic dysplasia 9
with or without polydactyly	
2	
	isolated congeni
tal anosmia	
2	
	microlissen
cephaly	
2	
	long QT syn
drome 2	
2	
	hypotricho
sis 4	
2	
	SMARCB1-related sc
hwannomatosis	
2	
	Fraser synd
rome 1	
2	
	isolated growth hormone
deficiency type II	
2	
	isolated growth hormone
deficiency type IB	
2	
	Freeman-Sheldo
n syndrome	
2	
	hypotrichosis simpl
ex of the scalp	
2	
	isolated growth hormone
deficiency type IA	
2	
	SIN3A-related intellectual disability
syndrome due to a point mutation	
2	
	isolated focal non-epidermolyti
c palmoplantar keratoderma	
2	
	posterior polymorphous
corneal dystrophy 1	
2	
	ichthyosis, annular
epidermolytic 1	
2	
	isolated ectop
ia lentis	

2			
			sitosterol
emia 1			
2			sitostero
lemia			
2			microphthalmia,
Lenz type			
2			idiopathic a
chalasia			
2			isolated trigo
nocephaly			
2			Oguchi di
sease			
2			Crouzon sy
ndrome			
2			mitochondrial complex I def
iciency, nuclear type 1			
2			congenital vert
ical talus			
2			lissencephaly due t
o LIS1 mutation			
2			spinocerebellar ataxia, autosomal rec
essive, with axonal neuropathy 2			
2			X-linked hypohidrotic e
ctodermal dysplasia			
2			familial infantile my
oclonic epilepsy			
2			familial isolated con
genital asplenia			
2			familial rhabd
oid tumor			
2			juvenile myelomono
cytic leukemia			
2			familial pore
ncephaly			
2			familial isolated hypoparathyroidis
m due to impaired PTH secretion			
2			D00RS syn

drome		
2		
		Singleton-Merte
n dysplasia		
2		
		DEND synd
rome		
2		
		familial multipl
e meningioma		
2		
		lipoyl transferase
1 deficiency		
2		
		familial prostat
e carcinoma		
2		
		D-2-hydroxygluta
ric aciduria		
2		
		spinocerebellar a
taxia type 40		
2		
		juvenile primary la
teral sclerosis		
2		
		familial pancreat
ic carcinoma		
2		
		congenital sodi
um diarrhea		
2		
		pancreatic hypoplasia-diabetes-con
genital heart disease syndrome		
2		
		familial colore
ctal cancer		
2		
		familial primary localized
cutaneous amyloidosis		
2		
		familial primary hypomagnesemia with
normocalciuria and normocalcemia		
2		
		familial patent a
rterial duct		
2		
		spinocerebellar a
taxia type 8		
2		
		Shprintzen-Golddb
erg syndrome		
2		
		Smith-McCort
dysplasia		
2		

		pancreat
itis		
2		
		familial sick si
nus syndrome		
2		
		Senior-Boichis
syndrome		
2		
		familial dilated c
ardiomyopathy		
2		
		polydactyly of a tri
phalangeal thumb		
2		
		congenital short b
owel syndrome		
2		
		hypokalemic perio
dic paralysis		
2		
		spondylocarpotarsal sy
nostosis syndrome		
2		
		Dubowitz sy
ndrome		
2		
		junctional epidermolysis bull
osa with pyloric atresia		
2		
		Cold-induced sweat
ing syndrome 1		
2		
		Cole-Carpenter
syndrome		
2		
		hypertrophic card
iomypathy 3		
2		
		pancreatic a
genesis		
2		
		familial colorectal
cancer type X		
2		
		mitochondrial complex I de
ficiency, nuclear type		
2		
		autosomal dominant
hypocalcemia		
2		
		leukoencephalopathy with brain stem and spina
l cord involvement-high lactate syndrome		
1		
		lipoid prot
einosis		

1		
		lymphatic malfo
rmation 11		
1		
		macular degeneratio
n, age-related, 3		
1		
		macular degeneratio
n, early-onset		
1		
		macrothrombocytopenia, isolat
ed, 1, autosomal dominant		
1		
		linear skin defects with multi
ple congenital anomalies 3		
1		
		long QT syn
drome 8		
1		
		lipase deficienc
y, combined		
1		
		lymphatic malf
ormation 7		
1		
		long QT synd
rome 16		
1		
		long QT syn
drome 9		
1		
		lymphatic malfo
rmation 10		
1		
		macrothrombocytopenia and granulocyte inclusions with
or without nephritis or sensorineural hearing loss		
1		
		macrocephaly/megalecephaly syn
drome, autosomal recessive		
1		
		macrocephaly-intellectual disability-neurodev
elopmental disorder-small thorax syndrome		
1		
		leukodystrophy, hypomyelinating, 7, with or without
oligodontia and/or hypogonadotropic hypogonadism		
1		
		long QT syn
drome 5		
1		
		leukodystrophy, hypomyelinating, 23, with ataxia, deafn
ess, liver dysfunction, and dilated cardiomyopathy		
1		
		lymphatic malf
ormation 6		
1		
		lipodystrophy, partial, acqu

ired, susceptibility to	
1	
ormation 3	lymphatic malf
1	
myelinating, 25	leukodystrophy, hypo
1	
g, 26, with chondrodysplasia	leukodystrophy, hypomyelinatin
1	
ormation 4	lymphatic malf
1	
ormation 8	lymphatic malf
1	
merulopathy	lipoprotein glo
1	
l partial, type 9	lipodystrophy, familia
1	
rmation 14	lymphatic malfo
1	
rmation 13	lymphatic malfo
1	
drome 3	long QT syn
1	
dystrophy	macular corneal
1	
rmation 12	lymphatic malfo
1	
l partial, type 8	lipodystrophy, familia
1	
myelinating, 27	leukodystrophy, hypo
1	
myelinating, 24	leukodystrophy, hypo
1	
ormation 9	lymphatic malf
1	
lay-facial dysmorphism-camptodactyly syndrome	macrothrombocytopenia-lymphedema-developmental de
1	
drome 6	long QT syn
1	

	macular degeneration,
X-linked atrophic	
1	
	lissencephaly spectrum disorder with
complex brainstem malformation	
1	
	lissenceph
aly 10	
1	
	macular dystrophy with cen
tral cone involvement	
1	
	leukoencephalopathy, diffuse he
reditary, with spheroids 1	
1	
	lysosomal stora
ge disease	
1	
	liver dis
order	
1	
	lysosomal acid phosph
atase deficiency	
1	
	liver disease, sev
ere congenital	
1	
	lysosomal acid lip
ase deficiency	
1	
	lymphedema-posterior choa
nal atresia syndrome	
1	
	leukoencephalopathy, motor delay, spa
sticity, and dysarthria syndrome	
1	
	loricrin ker
atoderma	
1	
	lissencephaly with cer
ebellar hypoplasia	
1	
	lymphoma, non-Hodg
kin, familial	
1	
	lissencephaly due to
TUBA1A mutation	
1	
	leukoencephalopathy, progressive, infant
ile-onset, with or without deafness	
1	
	macrocephaly-developmen
tal delay syndrome	
1	
	lysinuric protein
intolerance	

1		
		leukoencephalopathy, progress
ive, with ovarian failure		
1		
		lymphoproliferat
ive syndrome		
1		
		leukoencephalopathy-thalamus and brains
tem anomalies-high lactate syndrome		
1		
		lymphoproliferati
ve syndrome 2		
1		
		lower motor neuron syndrome
with late-adult onset		
1		
		linear nevus sebac
eous syndrome		
1		
		limb-girdle muscular dystroph
y due to POMK deficiency		
1		
		lymphoproliferati
ve syndrome 1		
1		
		lissencephaly type 1 due to d
oublecortin gene mutation		
1		
		limb-mammary
syndrome		
1		
		lower urinary tract obst
ruction, congenital		
1		
		lissencephaly 9 with complex
brainstem malformation		
1		
		lissenceph
aly 8		
1		
		leukoencephalopathy, developmental delay, and
episodic neurologic regression syndrome		
1		
		leukoencephalopathy, acute reversible, with
increased urinary alpha-ketoglutarate		
1		
		lymphatic malf
ormation 1		
1		
		long chain 3-hydroxyacyl-CoA
dehydrogenase deficiency		
1		
		macrocephaly, neurodevelopmental delay, lymphoid
hyperplasia, and persistent fetal hemoglobin		
1		
		leukoencephalopathy with ca

lcifications and cysts		
1		
		leukoencephalopathy with mild cerebel
lar ataxia and white matter edema		
1		
		long QT synd
rome 15		
1		
		lissenceph
aly 4		
1		
		long chain acyl-CoA dehy
drogenase deficiency		
1		
		macrocephaly, dysmorphic facies,
and psychomotor retardation		
1		
		lymphangioloio
myomatosis		
1		
		long QT synd
rome 14		
1		
		lymphedema-distich
iasis syndrome		
1		
		macrocephaly, acquired, with impa
ired intellectual development		
1		
		long QT synd
rome 13		
1		
		lung lymphangiole
iomymatosis		
1		
		linear skin defects with multi
ple congenital anomalies 2		
1		
		leukoencephalopathy with va
nishing white matter 4		
1		
		long QT synd
rome 12		
1		
		lissencephaly 6 wit
h microcephaly		
1		
		long QT synd
rome 11		
1		
		leukoencephalopathy with va
nishing white matter 5		
1		
		long QT synd
rome 10		
1		

		long QT syn
drome 1		
1		
		linear skin defects with multi
ple congenital anomalies 1		
1		
		lissencephaly 7 with ce
rebellar hypoplasia		
1		
		lipoic acid synthet
ase deficiency		
1		
		2-aminoadipic 2-oxo
adipic aciduria		
1		
		macular dystrophy with or w
ithout cone dysfunction		
1		
		mitochondrial complex I defi
ciency, nuclear type 39		
1		
		mitochondrial complex III de
ficiency nuclear type 3		
1		
		mitochondrial complex III de
ficiency nuclear type 2		
1		
		mitochondrial complex III de
ficiency nuclear type 1		
1		
		mitochondrial complex I def
iciency, nuclear type 9		
1		
		mitochondrial complex I def
iciency, nuclear type 8		
1		
		mitochondrial complex I def
iciency, nuclear type 7		
1		
		mitochondrial complex I def
iciency, nuclear type 6		
1		
		mitochondrial complex I def
iciency, nuclear type 5		
1		
		mitochondrial complex I def
iciency, nuclear type 4		
1		
		mitochondrial complex I defi
ciency, nuclear type 37		
1		
		mitochondrial complex I defi
ciency, nuclear type 25		
1		
		mitochondrial complex I defi
ciency, nuclear type 36		

iciency, nuclear type 20		
1		
		mitochondrial complex IV def
iciency, nuclear type 19		
1		
		mitochondrial complex IV def
iciency, nuclear type 18		
1		
		mitochondrial complex IV def
iciency, nuclear type 17		
1		
		mitochondrial complex IV def
iciency, nuclear type 16		
1		
		mitochondrial complex IV def
iciency, nuclear type 15		
1		
		mitochondrial complex IV def
iciency, nuclear type 14		
1		
		mitochondrial complex IV def
iciency, nuclear type 12		
1		
		mitochondrial complex IV def
iciency, nuclear type 11		
1		
		mitochondrial complex IV def
iciency, nuclear type 10		
1		
		mitochondrial complex III def
iciency, nuclear type 11		
1		
		mitochondrial complex III def
iciency, nuclear type 10		
1		
		mitochondrial complex III de
ficiency nuclear type 8		
1		
		mitochondrial complex I defi
ciency, nuclear type 26		
1		
		mitochondrial complex I defi
ciency, nuclear type 24		
1		
		mirror move
ments 2		
1		
		mitochondrial DNA depletion syndrome 1
4 (cardioencephalomyopathic type)		
1		
		mitochondrial DNA depletion synd
rome 6 (hepatocerebral type)		
1		
		mitochondrial DNA depl
etion syndrome 4a		
1		

1	mitochondrial DNA depletion synd
rome 3 (hepatocerebral type)	
1	
1	mitochondrial DNA depletion
syndrome 20 (mngie type)	
1	
1	mitochondrial DNA depl
etion syndrome 19	
1	
1	mitochondrial DNA depl
etion syndrome 18	
1	
1	mitochondrial DNA depl
etion syndrome 17	
1	
1	mitochondrial DNA depletion s
yndrome 16 (hepatic type)	
1	
1	mitochondrial DNA depletion syndr
ome 15 (hepatocerebral type)	
1	
1	mitochondrial DNA depl
etion syndrome 13	
1	
1	mitochondrial complex I defi
ciency, nuclear type 23	
1	
1	mitochondrial DNA depletion syndrome 12B (ca
rdiomyopathic type), autosomal recessive	
1	
1	mitochondrial DNA depletion syndrome 12A (ca
rdiomyopathic type), autosomal dominant	
1	
1	mitochondrial DNA depl
etion syndrome 11	
1	
1	mitochondrial DNA dep
letion syndrome 1	
1	
1	mitochondrial DNA deletion syndro
me with progressive myopathy	
1	
1	mismatch repair can
cer syndrome 4	
1	
1	mismatch repair can
cer syndrome 3	
1	
1	mismatch repair can
cer syndrome 2	
1	
1	mirror move
ments 4	
1	
1	mitochondrial DNA depletion synd
rome 7 (hepatocerebral type)	

1		
		mitochondrial DNA depl
etion syndrome 8a		
1		
		mitochondrial DNA dep
letion syndrome 9		
1		
		mitochondrial DNA depletion syndrome, encephalo
myopathic form with methylmalonic aciduria		
1		
		mitochondrial complex I defi
ciency, nuclear type 22		
1		
		mitochondrial complex I defi
ciency, nuclear type 21		
1		
		mitochondrial complex I def
iciency, nuclear type 2		
1		
		mitochondrial complex I defi
ciency, nuclear type 19		
1		
		mitochondrial complex I defi
ciency, nuclear type 18		
1		
		mitochondrial complex I defi
ciency, nuclear type 17		
1		
		mitochondrial complex I defi
ciency, nuclear type 16		
1		
		mitochondrial complex I defi
ciency, nuclear type 15		
1		
		mitochondrial complex I defi
ciency, nuclear type 14		
1		
		mitochondrial complex I defi
ciency, nuclear type 13		
1		
		mitochondrial complex I defi
ciency, nuclear type 12		
1		
		mitochondrial complex I defi
ciency, nuclear type 11		
1		
		mitochondrial complex I defi
ciency, nuclear type 10		
1		
		mitochondrial complex 2 def
iciency, nuclear type 4		
1		
		mitochondrial complex 2 def
iciency, nuclear type 3		
1		
		mitochondrial complex 2 def

iciency, nuclear type 2	
1	
ciency, nuclear type 35	mitochondrial complex 1 defi
1	
syndrome, myopathic form	mitochondrial DNA depletion
1	
ome, hepatocerebrorenal form	mitochondrial DNA depletion syndr
1	
se) deficiency, nuclear type 1	mitochondrial complex V (ATP syntha
1	
se) deficiency, nuclear type 2	mitochondrial complex V (ATP syntha
1	
se) deficiency, nuclear type 3	mitochondrial complex V (ATP syntha
1	
al venous malformations	multiple cutaneous and mucos
1	
lasia, Beighton type	multiple epiphyseal dysp
1	
asia, Al-Gazali type	multiple epiphyseal dyspl
1	
dysplasia type 5	multiple epiphyseal
1	
dysplasia type 4	multiple epiphyseal
1	
dysplasia type 1	multiple epiphyseal
1	
eoplasia type 4	multiple endocrine n
1	
eoplasia type 2B	multiple endocrine n
1	
eoplasia type 2A	multiple endocrine n
1	
eoplasia type 1	multiple endocrine n
1	
velopmental syndrome, X-linked	multiple congenital anomalies-neurod
1	
se) deficiency, nuclear type 4A	mitochondrial complex V (ATP syntha
1	

potonia-seizures syndrome 3	multiple congenital anomalies-hy
1	
potonia-seizures syndrome 2	multiple congenital anomalies-hy
1	
potonia-seizures syndrome 1	multiple congenital anomalies-hy
1	
ypotonia-seizures syndrome	multiple congenital anomalies-h
1	
l skin creases on limbs 1	multiple benign circumferentia
1	
multinucleated neurons-anhydramnios-renal dysplasia	
-cerebellar hypoplasia-hydranencephaly syndrome	
1	
lating fundus flavimaculatus	multifocal pattern dystrophy simu
1	
ulosis, and arthropathy	multicentric osteolysis, nod
1	
is with or without nephropathy	multicentric carpo-tarsal osteolys
1	
nal atresia	multiple intesti
1	
sfunctions syndrome 1	multiple mitochondrial dy
1	
functions syndrome 10	multiple mitochondrial dys
1	
sfunctions syndrome 2	multiple mitochondrial dy
1	
e, autosomal recessive 28	muscular dystrophy, limb-girdl
1	
e, autosomal recessive 27	muscular dystrophy, limb-girdl
1	
e, autosomal recessive 26	muscular dystrophy, limb-girdl
1	
e, autosomal recessive 23	muscular dystrophy, limb-girdl
1	
le, autosomal dominant 4	muscular dystrophy, limb-gird
1	
l, with or without seizures	muscular dystrophy, congenita

1		
		muscular dystrophy, congenital hearing los
s, and ovarian insufficiency syndrome		
1		
		multisystemic smooth muscl
e dysfunction syndrome		
1		
		multiple syste
m atrophy		
1		
		multiple synostos
es syndrome 4		
1		
		multiple synostos
es syndrome 3		
1		
		multiple synostos
es syndrome 1		
1		
		multiple symmetri
c lipomatosis		
1		
		multiple self-healing s
quamous epithelioma		
1		
		multiple mitochondrial dy
sfunctions syndrome 7		
1		
		multiple mitochondrial dy
sfunctions syndrome 6		
1		
		multiple mitochondrial dy
sfunctions syndrome 5		
1		
		multiple mitochondrial dy
sfunctions syndrome 4		
1		
		multiple mitochondrial dy
sfunctions syndrome 3		
1		
		mullerian aplasia and
hyperandrogenism		
1		
		mulibrey n
anism		
1		
		mucosulfat
idosis		
1		
		motor neuron
disorder		
1		
		mosaic variegated aneu
ploidy syndrome 4		
1		
		mosaic variegated aneu

ploidy syndrome 3	
1	
	mosaic variegated aneu
ploidy syndrome 2	
1	
	mosaic variegated aneu
ploidy syndrome 1	
1	
	monosomy 7 myelodysplasia a
nd leukemia syndrome 2	
1	
	monocytopenia with suscept
ibility to infections	
1	
	monilethr
ix-1	
1	
	moderately severe
hemophilia B	
1	
	moderately severe
hemophilia A	
1	
	mitral valve prolaps
e, myxomatous 2	
1	
	mitochondrial short-chain Enoyl
-CoA hydratase 1 deficiency	
1	
	mitochondrial pyruvate
carrier deficiency	
1	
	mitochondrial myopathy-lactic
acidosis-deafness syndrome	
1	
	mitochondrial myopathy-cerebellar atax
ia-pigmentary retinopathy syndrome	
1	
	mitochondrial myopathy, episodic, with optic a
trophy and reversible leukoencephalopathy	
1	
	mitochondrial myopathy with reversibl
e cytochrome C oxidase deficiency	
1	
	mitochondrial hypertrophic cardiomyopathy wit
h lactic acidosis due to MT01 deficiency	
1	
	mitochondrial complex V (ATP syntha
se) deficiency, nuclear type 5	
1	
	mitochondrial complex V (ATP syntha
se) deficiency, nuclear type 4B	
1	
	mosaic variegated aneuploidy syndrome 7 wit
h inflammation and tumor predisposition	
1	

	moyamoya di
sease 7	
1	mucopolysaccharidosi
s-plus syndrome	mucocutaneous ulcer
1	
	mucopolysaccharid
ation, chronic	
1	mucopolysacchari
osis, type 10	mucopolysacchari
1	
	mucopolysacchari
dosis type 9	
1	mucopolysacchari
dosis type 7	mucopolysacchari
1	
	mucopolysaccharid
dosis type 6	
1	mucopolysaccharid
osis type 4B	mucopolysaccharid
1	
	mucopolysaccharid
osis type 4A	
1	mucopolysaccharid
osis type 3D	mucopolysaccharid
1	
	mucopolysaccharid
osis type 3C	
1	mucopolysaccharid
osis type 3B	mucopolysaccharid
1	
	mucopolysacchari
osis type 3A	
1	mucopolysacchari
dosis type 3	mucopolysaccharidosis t
1	
	mucopolysaccharidosis typ
ype 2, severe form	
1	mucopolysacchari
e 2, attenuated form	mucopolysacchari
1	
	mucopolysacchari
dosis type 2	
1	
	mucopolysacchari
dosis type 1	
1	mucopolysacchari
type IV	mucopolysacchari

1		
		mucopolipidosis type
III, alpha/beta		
1		mucopolipidosis
type II		
1		mucopolipid
osis		
1		mirror move
ments 3		
1		mirror movements 1 and/or agene
sis of the corpus callosum		
1		macular dystroph
y, retinal, 4		
1		melanoma, cutaneous malignan
t, susceptibility to, 8		
1		metabolic encephalomyopathic crises, recurrent, with rhabd
		omyolysis, cardiac arrhythmias, and neurodegeneration
1		
		metabolic crises, recurrent, with variable enceph
lomyopathic features and neurologic regression		
1		mesoaxial synostotic syndactyly
with phalangeal reduction		
1		menstrual cycle-depend
ent periodic fever		
1		meningi
oma		
1		melorheostosis with
osteopoikilosis		
1		melanoma-pancreatic
cancer syndrome		
1		melanoma, uveal, susc
eptibility to, 1		
1		melanoma, cutaneous malignan
t, susceptibility to, 9		
1		melanoma, cutaneous malignan
t, susceptibility to, 3		
1		megacystis-microcolon-intestinal
hypoperistalsis syndrome 4		
1		melanoma, cutaneous malignan

t, susceptibility to, 2	
1	
	melanoma and neural sys
tem tumor syndrome	
1	
	megalencephaly-severe kyphosco
lios-osis-overgrowth syndrome	
1	
	megalencephaly-polymicrogyria-polyd
actyly-hydrocephalus syndrome 3	
1	
	megalencephaly-polymicrogyria-polyd
actyly-hydrocephalus syndrome 2	
1	
	megalencephaly-capillary malforma
tion-polymicrogyria syndrome	
1	
	megalencephalic leukoencephalopathy wi
th subcortical cysts 4, remitting	
1	
	megalencephalic leukoencephalopa
thy with subcortical cysts 3	
1	
	megalencephalic leukoencephalopathy with subcortical cysts
2B, remitting, with or without intellectual disability	
1	
	metabolic myopathy due to la
ctate transporter defect	
1	
	metachondro
matosis	
1	
	metachromatic le
ukodystrophy	
1	
	metachromatic leukodystrophy d
ue to saposin B deficiency	
1	
	methylmalonic aciduria and h
omocystinuria type cblD	
1	
	methylmalonic acidemia with h
omocystinuria, type cblX	
1	
	methylmalonic acidemia with h
omocystinuria, type cblJ	
1	
	methylmalonic acidemia due to tr
anscobalamin receptor defect	
1	
	methylmalonic acidemia due to methyl
malonyl-CoA epimerase deficiency	
1	
	methylmalonate semialdehyde
dehydrogenase deficiency	
1	

ciency type cblG	1	
ciency type cblE	1	
sferase deficiency	1	
mia type 4	1	
cy of methemoglobin reductase	1	
inemia	1	
ysplasia	1	
poplasia-brachydacty syndrome	1	
tinitis pigmentosa syndrome	1	
plasia, Spahr type	1	
lasia, Jansen type	1	
dysplasia 2	1	
ophy, juvenile form	1	
hy with subcortical cysts 2A	1	
hypoperistalsis syndrome 3	1	
ments 1	1	
th type B lipodystrophy	1	
lketonuria	1	
osis		

methylocobalamin defi
|
methylocobalamin defi
|
methionine adenosyltran
|
methemoglobine
|
methemoglobinemia due to deficien
|
methemoglob
|
metatropic d
|
metaphyseal dysplasia-maxillary hy
|
metaphyseal chondrodysplasia-re
|
metaphyseal chondrodys
|
metaphyseal chondrodysp
|
metaphyseal ana
|
metachromatic leukodysr
|
megalencephalic leukoencephalopat
|
megacystis-microcolon-intestinal
|
mirror move
|
mandibuloacral dysplasia wi
|
maternal pheny
|
mastocyt
|

1		
		mast synd
rome		
1		
		maple syrup urine dise
ase, mild variant		
1		
		maple syrup urine
disease type 2		
1		
		maple syrup urine d
isease type 1B		
1		
		maple syrup urine d
isease type 1A		
1		
		mandibulofacial dysostosis
-microcephaly syndrome		
1		
		mandibulofacial dysost
osis with alopecia		
1		
		mandibuloacral dysplasia wi
th type A lipodystrophy		
1		
		megacystis-microcolon-intestinal
hypoperistalsis syndrome 2		
1		
		mandibuloacral dysplasia
progeroid syndrome		
1		
		mandibuloacral
dysplasia		
1		
		mandibular hypoplasia-deafn
ess-progeroid syndrome		
1		
		malonic ac
iduria		
1		
		malignant hyperthermia,
susceptibility to, 5		
1		
		malignant hyperthermia,
susceptibility to, 1		
1		
		malignant hypertherm
ia of anesthesia		
1		
		maleylacetoacetate iso
merase deficiency		
1		
		mal de Me
leda		
1		
		maternal riboflav

in deficiency	
1	
yopathy and hearing loss	maternally-inherited cardiom
1	
abetes and deafness	maternally-inherited di
1	
pastic paraplegia	maternally-inherited s
1	
hypoperistalsis syndrome 1	megacystis-microcolon-intestinal
1	
l muscular dystrophy	megaconial type congenita
1	
ongenital	megabladder, c
1	
r hypoplasia and cortical malformations	mega-corpus-callosum syndrome with cerebella
1	
ge kidney	medullary spon
1	
ydrogenase deficiency	medium chain acyl-CoA deh
1	
rome 14	meckel synd
1	
of the young type 9	maturity-onset diabetes
1	
of the young type 8	maturity-onset diabetes
1	
of the young type 7	maturity-onset diabetes
1	
of the young type 6	maturity-onset diabetes
1	
of the young type 4	maturity-onset diabetes
1	
of the young type 3	maturity-onset diabetes
1	
of the young type 2	maturity-onset diabetes
1	
of the young type 14	maturity-onset diabetes
1	

	maturity-onset diabetes
of the young type 13	
1	
	maturity-onset diabetes
of the young type 11	
1	
	maturity-onset diabetes
of the young type 10	
1	
	maturity-onset diabetes
of the young type 1	
1	
	methylmalonic aciduria and h
omocystinuria type cblF	
1	
	methylmalonic aciduria due to methy
lmalonyl-CoA mutase deficiency	
1	
	methylmalonic acidu
ria, cblA type	
1	
	microcephaly-thin corpus callosum-i
ntellectual disability syndrome	
1	
	microphthalmia, isolate
d, with coloboma 5	
1	
	microphthalmia, isolate
d, with coloboma 3	
1	
	microphthalmia, isolate
d, with coloboma 10	
1	
	microphthalmia with
limb anomalies	
1	
	microphthalmia with brain
and digit anomalies	
1	
	micrognathia-recurrent infections-behavioral abnor
malities-mild intellectual disability syndrome	
1	
	microcytic anemia with
liver iron overload	
1	
	microcornea-myopic cho
rioretinal atrophy	
1	
	microcornea, rod-cone dystrophy, cata
ract, and posterior staphyloma 1	
1	
	microcephaly-short stature-intellectual d
isability-facial dysmorphism syndrome	
1	
	methylmalonic acidu
ria, cblB type	

1		
		microcephaly-polymicrogyria-corpu
s callosum agenesis syndrome		
1		
		microcephaly-intellectual disability-sensorineural he
aring loss-epilepsy-abnormal muscle tone syndrome		
1		
		microcephaly-congenital cataract-ps
oriasiform dermatitis syndrome		
1		
		microcephaly-complex motor and sens
ory axonal neuropathy syndrome		
1		
		microcephaly-capillary m
alformation syndrome		
1		
		microcephaly, short stature,
and limb abnormalities		
1		
		microcephaly, short stature, and
impaired glucose metabolism 2		
1		
		microcephaly, short stature, and
impaired glucose metabolism 1		
1		
		microcephaly, seizures, an
d developmental delay		
1		
		microphthalmia, isolate
d, with coloboma 6		
1		
		microphthalmia, isolate
d, with coloboma 7		
1		
		microphthalmia, isolate
d, with coloboma 9		
1		
		microphthalmia,
syndromic 1		
1		
		minimal pigment oculocuta
neous albinism type 1		
1		
		mild spondyloepiphyseal dysplasia due to COL2A
1 mutation with early-onset osteoarthritis		
1		
		mild phosphoribosylpyrophosphat
e synthetase superactivity		
1		
		mild phenylk
etonuria		
1		
		mild hyperpheny
lalaninemia		
1		
		mild hemoph

ilia B	
1	
ilia A	mild hemoph
1	
disease	mild Canavan
1	
a, susceptibility to, 13	migraine, with or without aur
1	
hemiplegic, 3	migraine, familial
1	
hemiplegic, 2	migraine, familial
1	
hemiplegic, 1	migraine, familial
1	
elliptocytosis, and nephrocalcinosis	midface hypoplasia, hearing impairment,
1	
ia	microt
1	
lentis and with or without secondary glaucoma	microspherophakia and/or megalocornea, with ectopia
1	
coloboma 11	microphthalmia/
1	
syndromic 2	microphthalmia,
1	
syndromic 12	microphthalmia,
1	
syndromic 11	microphthalmia,
1	
creased sister chromatid exchange 2	microcephaly, growth restriction, and in
1	
izures, and brain malformations	microcephaly, growth deficiency, se
1	
nesis, and ambiguous genitalia syndrome	microcephaly, facial dysmorphism, renal age
1	
autosomal recessive	microcephaly 22, primary,
1	
autosomal recessive	microcephaly 20, primary,
1	

1	microcephaly 2, primary, autosomal recessive, with or without cortical malformations	1
1	autosomal recessive	microcephaly 19, primary,
1	autosomal recessive	microcephaly 17, primary,
1	autosomal recessive	microcephaly 16, primary,
1	autosomal recessive	microcephaly 15, primary,
1	autosomal recessive	microcephaly 14, primary,
1	autosomal recessive	microcephaly 13, primary,
1	autosomal recessive	microcephaly 12, primary,
1	autosomal recessive	microcephaly 11, primary,
1	autosomal recessive	microcephaly 1, primary,
1	warfism, Alazami type	microcephalic primordial dwarfism
1	primordial dwarfism due to ZNF335 deficiency	microcephalic primordial dwarfism
1	primordial dwarfism due to RTTN deficiency	microcephalic primordial dwarfism
1	primordial dwarfism type II	microcephalic osteodysplastic primordial dwarfism type II
1	primordial dwarfism type I	microcephalic osteodysplastic primordial dwarfism type I
1	osteopetrosis, Saul-Wilson type	microcephalic osteodysplastic dwarfism
1	microangiopathy, pontine, autosomal dominant	microangiopathy and leukoencephalopathy
1	mevalonic aciduria	mevalonic aciduria
1	autosomal recessive	microcephaly 21, primary,

1		
		microcephaly 23, primary,
autosomal recessive		
1		
		microcephaly, epilepsy, an
d diabetes syndrome 2		
1		
		microcephaly 26, primar
y, autosomal dominant		
1		
		microcephaly, epilepsy, an
d diabetes syndrome 1		
1		
		microcephaly, developmental dela
y, and brittle hair syndrome		
1		
		microcephaly, cataracts, impaired intellectual dev
elopment, and dystonia with abnormal striatum		
1		
		microcephaly with or wi
thout short stature		
1		
		microcephaly with or without chorioretinopath
y, lymphedema, or intellectual disability		
1		
		microcephaly with intel
lectual disability		
1		
		microcephaly and cho
rioretinopathy 3		
1		
		microcephaly and cho
rioretinopathy 2		
1		
		microcephaly 9, primary,
autosomal recessive		
1		
		microcephaly 7, primary,
autosomal recessive		
1		
		microcephaly 6, primary,
autosomal recessive		
1		
		microcephaly 6 with or w
ithout short stature		
1		
		microcephaly 5, primary,
autosomal recessive		
1		
		microcephaly 4, primary,
autosomal recessive		
1		
		microcephaly 30, primary,
autosomal recessive		
1		
		microcephaly 3, primary,

autosomal recessive	
1	
	microcephaly 29, primary,
autosomal recessive	
1	
	microcephaly 28, primary,
autosomal recessive	
1	
	microcephaly 27, primar
y, autosomal dominant	
1	
	microcephaly 8, primary,
autosomal recessive	
1	
	intellectual disability, a
utosomal recessive 14	
1	
	leukodystrophy, hypo
myelinating, 22	
1	
	hypogonadotropic hypogonadism
8 with or without anosmia	
1	
	hypomagnesemia, seizures, and
intellectual disability 1	
1	
	hypomagnesemia, seizures, and
intellectual disability	
1	
	hypomagnesemia 7, renal, with or w
ithout dilated cardiomyopathy	
1	
	hypokalemic tubulopa
thy and deafness	
1	
	hypokalemic periodic
paralysis, type 2	
1	
	hypokalemic periodic
paralysis, type 1	
1	
	hypokalemic alkalosis, familial, w
ith specific renal tubulopathy	
1	
	hypoinsulinemic hypoglycemia
and body hemihypertrophy	
1	
	hypohidrosis–enamel hypoplasia–palmoplantar ke
ratoderma–intellectual disability syndrome	
1	
	hypogonadotropic hypogonadism
9 with or without anosmia	
1	
	hypogonadotropic hypogonadism
7 with or without anosmia	
1	

1	hypomaturation-hypoplastic amelogenesis imperfecta with taurodontism	1	hypomaturation-hypoplastic amelogenesis imperfecta with taurodontism
1	1	1	1
6	6 with or without anosmia	1	hypogonadotropic hypogonadism
1	1	1	1
5	5 with or without anosmia	1	hypogonadotropic hypogonadism
1	1	1	1
4	4 with or without anosmia	1	hypogonadotropic hypogonadism
1	1	1	1
3	3 with or without anosmia	1	hypogonadotropic hypogonadism
1	1	1	1
27	27 without anosmia	1	hypogonadotropic hypogonadism
1	1	1	1
26	26 with or without anosmia	1	hypogonadotropic hypogonadism
1	1	1	1
25	25 with anosmia	1	hypogonadotropic hypogonadism
1	1	1	1
24	24 without anosmia	1	hypogonadotropic hypogonadism
1	1	1	1
23	23 with or without anosmia	1	hypogonadotropic hypogonadism
1	1	1	1
22	22 with or without anosmia	1	hypogonadotropic hypogonadism
1	1	1	1
2	2 intellectual disability	1	hypomagnesemia, seizures, and
1	1	1	1
10	10 k dystrophy	1	hypomyelinating leukodystrophy
1	1	1	1
20	20 with or without anosmia	1	hypogonadotropic hypogonadism
1	1	1	1
2	2 isolated milium cysts	1	hypoparathyroidism, familial
1	1	1	1
1	1 lenticular keratoderma syndrome	1	hypopigmentation-punctate palmoplantar keratoderma syndrome
1	1	1	1
1	1 delayed myelination and development	1	hypopigmentation, organomegaly, and delayed myelination and development
1	1	1	1
2	2 autosomal recessive	1	hypophosphatemic rickets, autosomal recessive
1	1	1	1
1	1 autosomal recessive	1	hypophosphatemic rickets, autosomal recessive
1	1	1	1

1		
		hypophosphatem
ic rickets		
1		
		hypophosphatemic nephrolit
hiasis/osteoporosis 2		
1		
		hypophosphatemic nephrolit
hiasis/osteoporosis 1		
1		
		hypophosph
atasia		
1		
		hypoparathyroidism-retardati
on-dysmorphism syndrome		
1		
		hypoparathyroidism-deafness
-renal disease syndrome		
1		
		hypoparathyroidism, fa
milial isolated 1		
1		
		hypomyelinating leu
kodystrophy 11		
1		
		hypomyelination-cerebellar atrophy-hypopl
asia of the corpus callosum syndrome		
1		
		hypomyelination with brain stem and spina
l cord involvement and leg spasticity		
1		
		hypomyelinating le
ukodystrophy 9		
1		
		hypomyelinating leukodystrophy 8 with or without ol
igodontia and-or hypogonadotropic hypogonadism		
1		
		hypomyelinating le
ukodystrophy 5		
1		
		hypomyelinating le
ukodystrophy 4		
1		
		hypomyelinating le
ukodystrophy 3		
1		
		hypomyelinating le
ukodystrophy 2		
1		
		hypomyelinating leu
kodystrophy 13		
1		
		hypomyelinating leu
kodystrophy 12		
1		
		hypogonadotropic hypogonadism

21 with or without anosmia	
1	
	hypogonadotropic hypogonadism
2 with or without anosmia	
1	
	hyperpigmentation with or without hypo
pigmentation, familial progressive	
1	
	hypertrophic card
iomyopathy 14	
1	
	hypertrophic card
iomyopathy 6	
1	
	hypertrophic card
iomyopathy 4	
1	
	hypertrophic card
iomyopathy 26	
1	
	hypertrophic card
iomyopathy 25	
1	
	hypertrophic card
iomyopathy 20	
1	
	hypertrophic card
iomyopathy 2	
1	
	hypertrophic card
iomyopathy 18	
1	
	hypertrophic card
iomyopathy 17	
1	
	hypertrophic card
iomyopathy 16	
1	
	hypertrophic card
iomyopathy 15	
1	
	hypertrophic card
iomyopathy 13	
1	
	hypertrophic card
iomyopathy 8	
1	
	hypertrophic card
iomyopathy 12	
1	
	hypertrophic card
iomyopathy 11	
1	
	hypertrophic card
iomyopathy 10	
1	

	hypertriglyce
ridemia 2	
1	
	hypertrichosis-acromegaloid f
acial appearance syndrome	
1	
	hyperthyroxinemia, fami
lial dysalbuminemic	
1	
	hypertelorism,
Teebi type	
1	
	hyperprolinem
ia type 2	
1	
	hyperprolinem
ia type 1	
1	
	hyperproinsu
linemia	
1	
	hypertrophic card
iomyopathy 7	
1	
	hypertrophic card
iomyopathy 9	
1	
	hypogonadotropic hypogonadism
19 with or without anosmia	
1	
	hypogona
dism	
1	
	hypogonadotropic hypogonadism
18 with or without anosmia	
1	
	hypogonadotropic hypogonadism
17 with or without anosmia	
1	
	hypogonadotropic hypogonadism
16 with or without anosmia	
1	
	hypogonadotropic hypogonadism
15 with or without anosmia	
1	
	hypogonadotropic hypogonadism
14 with or without anosmia	
1	
	hypogonadotropic hypogonadism
13 with or without anosmia	
1	
	hypogonadotropic hypogonadism
12 with or without anosmia	
1	
	hypogonadotropic hypogonadism
11 with or without anosmia	

1		
		hypogonadotropic hypogonadism
10	with or without anosmia	
1		hypogonadotropic hypogonadism
1	with or without anosmia	
1		hypoglycemia, leu
cine-induced		
1		hypertrophic osteoarthropathy,
primary, autosomal dominant		
1		hypocomplementemic urt
icarial vasculitis		hypochondro
1		
		hypochondro
plasia		
1		
		hypochondro
genesis		
1		hypoalphalipoprotein
emia, primary, 2		hypoalphalipoprotein
1		
		hypervalinemia and hyperl
emia, primary, 1		
1		hyperuricemic nephropathy, f
eucine-isoleucinemia		hyperuricemia-pulmonary hypertension-
1		
		hypertrophic osteoarthropathy, pr
amilial juvenile type 4		
1		hypertrophic osteoarthropathy, pr
renal failure-alkalosis syndrome		hypoplasmino
1		
		hypoplastic left h
imary, autosomal recessive, 2		
1		hypoplastic left he
imary, autosomal recessive, 1		
1		immunodeficiency 107, susceptibility to in
genemia		
1		immunodeficiency 123 with
heart syndrome		
1		
art syndrome 1		
1		
vasive staphylococcus aureus infection		
1		

HPV-related verrucosis	
1	
th autoinflammation	immunodeficiency 121 wi
1	
ency 120	immunodefici
1	
h atopic dermatitis	immunodeficiency 11b wit
1	
ency 119	immunodefici
1	
ency 118	immunodefici
1	
ency 117	immunodefici
1	
th autoinflammation	immunodeficiency 115 wi
1	
munity and autoinflammation	immunodeficiency 113 with autoim
1	
lymphoproliferation	immunodeficiency 109 with
1	
bility to viral infections	immunodeficiency 106, suscepti
1	
utosomal recessive	immunodeficiency 14b, a
1	
ency 102	immunodefici
1	
la zoster virus-specific)	immunodeficiency 101 (varicel
1	
disorder	immune system
1	
hy-enteropathy-X-linked syndrome	immune dysregulation-polyendocrinopat
1	
ritis-recurrent infections-lymphopenia syndrome	immune dysregulation-inflammatory bowel disease-arth
1	
nity, and autoinflammation	immune dysregulation, autoimmu
1	
nuria	iminoglyci
1	

	idiopathic steroid-sensiti
ve nephrotic syndrome	
1	
	idiopathic pulmon
ary fibrosis	
1	
	idiopathic juvenil
e osteoporosis	
1	
	immunodefici
ency 14	
1	
	immunodefici
ency 15a	
1	
	hypoplastic pancreas-intestinal atresi
a-hypoplastic gallbalder syndrome	
1	
	immunodefici
ency 39	
1	
	immunodeficiency 63 with lymphopr
oliferation and autoimmunity	
1	
	immunodefici
ency 62	
1	
	immunodefici
ency 61	
1	
	immunodefici
ency 60	
1	
	immunodefici
ency 57	
1	
	immunodefici
ency 53	
1	
	immunodefici
ency 51	
1	
	immunodefici
ency 49	
1	
	immunodefici
ency 47	
1	
	immunodefici
ency 45	
1	
	immunodefici
ency 37	
1	
	immunodefici
ency 18	

1		
		immunodeficiency 36 with
lymphoproliferation		
1		
		immunodefici
ency 35		
1		
		immunodefici
ency 33		
1		
		immunodefici
ency 32B		
1		
		immunodefici
ency 31B		
1		
		immunodefici
ency 28		
1		
		immunodefici
ency 27A		
1		
		immunodefici
ency 25		
1		
		immunodefici
ency 23		
1		
		immunodefici
ency 19		
1		
		idiopathic inherited
hypercalciuria		
1		
		idiopathic apla
stic anemia		
1		
		ichthyotic keratoderma, spasticity, hypomyel
ination, and dysmorphic facial features		
1		
		hypothyroidism, congeni
tal, nongoitrous, 9		
1		
		hypotricho
sis 11		
1		
		hypotricho
sis 1		
1		
		hypotonia-cystinu
ria syndrome		
1		
		hypotonia, infantile, with psychomotor re
tardation and characteristic facies 3		
1		
		hypotonia, infantile, with psychomotor re

tardation and characteristic facies 2	
1	
	hypotonia, infantile, with psychomotor re
tardation and characteristic facies 1	
1	
	hypotonia, hypoventilation, impaired intellectual develo
pment, dysautonomia, epilepsy, and eye abnormalities	
1	
	hypotonia, ataxia, developmental dela
y, and tooth enamel defect syndrome	
1	
	hypotonia, ataxia, and delay
ed development syndrome	
1	
	hypotonia with lactic acide
mia and hyperammonemia	
1	
	hypothyroidism, congeni
tal, nongoitrous, 8	
1	
	ichthyosis, lamellar,
autosomal dominant	
1	
	hypothyroidism, congeni
tal, nongoitrous, 7	
1	
	hypothyroidism, congeni
tal, nongoitrous, 5	
1	
	hypothyroidism, congeni
tal, nongoitrous, 2	
1	
	hypothyroidism due to TS
H receptor mutations	
1	
	hypothyro
idism	
1	
	hypotaurinemic retinal degene
ration and cardiomyopathy	
1	
	hypospadias 2,
X-linked	
1	
	hypospad
ias	
1	
	hyposmia-nasal and ocular hypoplasia-hyp
ogonadotropic hypogonadism syndrome	
1	
	hypoproteinemia, h
ypercatabolic	
1	
	hypotricho
sis 12	
1	

	hypotricho
sis 13	
1	
	hypotricho
sis 14	
1	
	hypotricho
sis 15	
1	
	ichthyosis, hystrix-lik
e, with hearing loss	
1	
	ichthyosis, congenital, a
utosomal recessive 14	
1	
	ichthyosis, congenital, a
utosomal recessive 13	
1	
	ichthyosis, congenital, a
utosomal recessive 12	
1	
	ichthyosis, annular
epidermolytic, 2	
1	
	ichthyosis prematu
rity syndrome	
1	
	ichthyosis hystrix o
f Curth-Macklin	
1	
	ichthyo
sis	
1	
	ichthyosiform erythroderma, corneal
involvement, and hearing loss	
1	
	hypoxanthine guanine phosphoribosyl
transferase partial deficiency	
1	
	hypouricemia,
renal, 2	
1	
	hypouricemia,
renal 1	
1	
	hypotrichosis-lymphedema-telangie
ctasia-renal defect syndrome	
1	
	hypotrichosis-lymphedema-telangi
ectasia syndrome (grouping)	
1	
	hypotrichosis-lymphedema-t
elangiectasia syndrome	
1	
	hypotricho
sis 8	

1		
		hypotricho
sis 7		
1		hypotricho
sis 6		hypotricho
1		
		hypotricho
sis 5		
1		hypotricho
sis 3		hypotricho
1		
		hypotricho
sis 2		
1		hyperplastic polyp
osis syndrome		hyperphosphatasia with intellec
1		
		immunodeficiency 65, susceptib
tual disability syndrome 6		
1		hereditary spastic
ility to viral infections		hereditary spastic
1		
		hereditary spastic
paraplegia 45		
1		hereditary spastic
paraplegia 56		hereditary spastic
1		
		hereditary spastic
paraplegia 54		
1		hereditary spastic
paraplegia 53		hereditary spastic
1		
		hereditary spastic
paraplegia 52		
1		hereditary spastic
paraplegia 51		hereditary spastic
1		
		hereditary spastic
paraplegia 50		
1		hereditary spastic
paraplegia 49		hereditary spastic
1		
		hereditary spastic
paraplegia 48		
1		hereditary spastic
paraplegia 47		hereditary spastic
1		
		hereditary spastic

paraplegia 46	
1	
	hereditary spastic
paraplegia 44	
1	
	hereditary spastic
paraplegia 5A	
1	
	hereditary spastic
paraplegia 43	
1	
	hereditary spastic
paraplegia 42	
1	
	hereditary spastic
paraplegia 4	
1	
	hereditary spastic
paraplegia 3A	
1	
	hereditary spastic
paraplegia 39	
1	
	hereditary spastic
paraplegia 35	
1	
	hereditary spastic
paraplegia 33	
1	
	hereditary spastic
paraplegia 31	
1	
	hereditary spastic
paraplegia 30	
1	
	hereditary spastic
paraplegia 28	
1	
	hereditary spastic
paraplegia 57	
1	
	hereditary spastic
paraplegia 6	
1	
	hereditary spastic
paraplegia 23	
1	
	hereditary spheroc
ytosis type 2	
1	
	hernia, anterior
diaphragmatic	
1	
	hereditary von Will
ebrand disease	
1	

	hereditary thrombophilia due to co
ngenital protein S deficiency	
1	
	hereditary thrombophilia due to co
ngenital protein C deficiency	
1	
	hereditary thrombophilia due to congenital hist
idine-rich (poly-L) glycoprotein deficiency	
1	
	hereditary thrombocytosis wit
h transverse limb defect	
1	
	hereditary thrombocytopenia and hematological cance
r predisposition syndrome associated with RUNX1	
1	
	hereditary spheroc
ytosis type 5	
1	
	hereditary spheroc
ytosis type 4	
1	
	hereditary spheroc
ytosis type 3	
1	
	hereditary spheroc
ytosis type 1	
1	
	hereditary spastic
paraplegia 61	
1	
	hereditary spastic
paraplegia 8	
1	
	hereditary spastic
paraplegia 77	
1	
	hereditary spastic
paraplegia 75	
1	
	hereditary spastic
paraplegia 74	
1	
	hereditary spastic
paraplegia 73	
1	
	hereditary spastic
paraplegia 72	
1	
	hereditary spastic
paraplegia 7	
1	
	hereditary spastic
paraplegia 64	
1	
	hereditary spastic
paraplegia 63	

1		
		hereditary spastic
paraplegia 62		
1		hereditary spastic
paraplegia 26		
1		hereditary spastic
paraplegia 2		
1		
		hyperphosphatasia with intellec
tual disability syndrome 5		
1		hereditary cryohydrocytosis
with reduced stomatin		
1		hereditary leiomyomatosis
and renal cell cancer		
1		hereditary intrinsic
factor deficiency		
1		hereditary inclusion body myopathy-joint
contractures-ophthalmoplegia syndrome		
1		hereditary hypotrichosis with
recurrent skin vesicles		
1		hereditary hypophosphatemic ri
ckets with hypercalciuria		
1		hereditary hyperferritinemia
with congenital cataracts		
1		hereditary hypercarotenemia
and vitamin A deficiency		
1		hereditary glaucoma, pr
imary closed-angle		
1		hereditary fructos
e intolerance		
1		hereditary folate
malabsorption		
1		hereditary copr
oporphyria		
1		hereditary motor and sensory
neuropathy, Okinawa type		
1		hereditary arterial and articular m
ultiple calcification syndrome		
1		hereditary antithro

mbin deficiency	1	hereditary angioedema wi
th C1Inh deficiency	1	hereditary angio
edema type 3	1	hereditary angio
edema type 2	1	hereditary angio
edema type 1	1	hereditary North American In
dian childhood cirrhosis	1	hereditary ATTR
amyloidosis	1	hepatorenocardiac dege
nerative fibrosis	1	hepatoerythropoie
tic porphyria	1	hereditary methe
moglobinemia	1	hereditary mucoepith
elial dysplasia	1	hereditary spastic
paraplegia 18	1	hereditary sensory and auto
nomie neuropathy type 4	1	hereditary spastic
paraplegia 17	1	hereditary spastic
paraplegia 15	1	hereditary spastic
paraplegia 13	1	hereditary spastic
paraplegia 11	1	hereditary spastic
paraplegia 10	1	hereditary sensory neuropathy-
deafness-dementia syndrome	1	

1	hereditary sensory and autonomic neuropathy with spastic paraplegia	1
1	hereditary sensory and autonomic neuropathy type 7	1
1	hereditary sensory and autonomic neuropathy type 6	1
1	hereditary sensory and autonomic neuropathy type 5	1
1	hereditary sensorimotor neuropathy with hyperelastic skin	1
1	hereditary myopathy with lactic acidosis due to ISCU deficiency	1
1	hereditary sclerosing poikiloderma with tendon and pulmonary involvement	1
1	hereditary retinoblastoma	1
1	hereditary progressive mucinosis	1
1	hereditary progressive chorea without dementia	1
1	hereditary persistence of fetal alpha-fetoprotein	1
1	hereditary periodic fever syndrome	1
1	hereditary papillary renal cell carcinoma	1
1	hereditary neutrophilia	1
1	hereditary neuropathy with liability to pressure palsies	1
1	hereditary neuroendocrine tumor of small intestine	1
1	herpes simplex encephalitis, susceptibility to, 1	1
1	herpes simplex encephalitis, susceptibility to, 4	1
1	heterotaxy, viscera	1

1		
		hyperekple
xia 2		
1		hyperinsulinism due t
o INSR deficiency		
1		hyperinsulinism due to
HNF4A deficiency		
1		hyperinsulinism due to
HNF1A deficiency		
1		hyperinsulinemic hypogl
ycemia, familial, 8		
1		hyperinsulinemic hypogl
ycemia, familial, 4		
1		hyperinsulinemic hypogl
ycemia, familial, 2		
1		hyperinsulinemic hypogl
ycemia, familial, 1		
1		hyperimmunoglobulinemia D
with periodic fever		
1		hyperekple
xia 4		
1		hyperekple
xia 3		
1		hyperekple
xia 1		
1		hyperinsulinism due to gl
ucokinase deficiency		
1		hypercholesterolemi
a, familial, 4		
1		hypercholesterolemia, auto
somal dominant, type B		
1		hypercholesterolemia, au
tosomal dominant, 3		
1		hypercholesterolemia due to cholester
ol 7alpha-hydroxylase deficiency		
1		hypercholanemia,
familial, 2		
1		hypercalcemia, i

nfantile, 2	
1	
	hypercalcemia, i
nfantile, 1	
1	
	hyperbiliriver
dinemia	
1	
	hyperammonemic encephalopathy due to
carbonic anhydrase VA deficiency	
1	
	hyperammonemia due to N-acetylgl
utamate synthase deficiency	
1	
	hyperinsulinism due t
o UCP2 deficiency	
1	
	hyperinsulinism due to short chain 3-hydr
oxylacyl-CoA dehydrogenase deficiency	
1	
	heterotaxy, visceral, 11, autos
omal, with male infertility	
1	
	hyperop
ia	
1	
	hyperphosphatasia with intellec
tual disability syndrome 4	
1	
	hyperphosphatasia with intellec
tual disability syndrome 3	
1	
	hyperphosphatasia with intellec
tual disability syndrome 2	
1	
	hyperphosphatasia with intellec
tual disability syndrome 1	
1	
	hyperphenylalaninemia due
to DNAJC12 deficiency	
1	
	hyperparathyroidism, t
ransient neonatal	
1	
	hyperparathyr
oidism 4	
1	
	hyperparathyroidism 2
with jaw tumors	
1	
	hyperparathyr
oidism 1	
1	
	hyperostosis cran
ialis interna	
1	

	hypermethioninemia with deficiency of
S-adenosylhomocysteine hydrolase	
1	
	hyperinsulinism-hypera
mmonemia syndrome	
1	
	hypermetabolism due to uncoupled mitoch
ondrial oxidative phosphorylation 2	
1	
	hypermanganesemia w
ith dystonia 2	
1	
	hyperlysi
nemia	
1	
	hyperlipoproteine
mia, type 1D	
1	
	hyperlipoprotein
emia type V	
1	
	hyperlipoprotein
emia type 3	
1	
	hyperlipidemia, familial
combined, LPL related	
1	
	hyperlipidemia,
combined, 1	
1	
	hyperlipidemia due to hepatic tr
iglyceride lipase deficiency	
1	
	hyperkalemic perio
dic paralysis	
1	
	hyperaldosteronism, f
amilial, type IV	
1	
	hyper-IgM syndr
ome type 5	
1	
	hyper-IgM syndr
ome type 3	
1	
	holocarboxylase synth
etase deficiency	
1	
	holoprosencephaly-hypokinesia-con
genital contractures syndrome	
1	
	holoprosence
phaly 9	
1	
	holoprosence
phaly 7	

1		
		holoprosence
phaly 5		
1		
		holoprosence
phaly 4		
1		
		holoprosence
phaly 3		
1		
		holoprosence
phaly 2		
1		
		holoprosence
phaly 14		
1		
		holoprosencephaly 12 with or w
ithout pancreatic agenesis		
1		
		holoprosence
phaly 11		
1		
		histidin
emia		
1		
		hyper-IgM syndr
ome type 2		
1		
		hip dysplasia,
Beukes type		
1		
		high myopia-sensorineura
l deafness syndrome		
1		
		heterotopia, periventricul
ar, X-linked dominant		
1		
		heterotaxy, visceral, 9, autoso
mal, with male infertility		
1		
		heterotaxy, viscera
l, 8, autosomal		
1		
		heterotaxy, viscera
l, 7, autosomal		
1		
		heterotaxy, viscera
l, 6, autosomal		
1		
		heterotaxy, viscera
l, 5, autosomal		
1		
		heterotaxy, viscera
l, 4, autosomal		
1		
		heterotaxy, viscera

1, 2, autosomal	
1	
	homocystinuria due to methylene tetra
hydrofolate reductase deficiency	
1	
	hot water refle
x epilepsy	
1	
	human H0XA1 s
yndromes	
1	
	hyaline body
myopathy	
1	
	hyper-IgM syndr
ome type 1	
1	
	hyper-IgE recurrent infection syn
drome 5, autosomal recessive	
1	
	hyper-IgE recurrent infection syn
drome 4A, autosomal dominant	
1	
	hyper-IgE recurrent infection syn
drome 4, autosomal recessive	
1	
	hyper-IgE recurrent infection syn
drome 3, autosomal recessive	
1	
	hyper-IgE recurrent infection sy
ndrome 1, autosomal dominant	
1	
	hydroxyprol
inemia	
1	
	hydrops-lactic acidosis-sideroblastic an
emia-multisystemic failure syndrome	
1	
	hydrops fe
talis	
1	
	hydrolethalus
syndrome 2	
1	
	hydrolethalus
syndrome 1	
1	
	hydrocephalus, nonsyndromi
c, autosomal recessive 2	
1	
	hydrocephalus, nonsyndromi
c, autosomal recessive 1	
1	
	hydrocephalus, congenital,
5, susceptibility to	
1	

	hydrocephalus, congenit
al communicating, 1	
1	hydranence
phaly	hydatidiform mole,
1	
	hydatidiform mole,
recurrent, 4	
1	hydatidiform mole,
recurrent, 3	hydatidiform mole,
1	
	hydatidiform mole,
recurrent, 2	
1	hydatidiform mole,
recurrent, 1	hyaline fibromato
1	
sis syndrome	immunodefici
1	
ency 64	immunodefici
1	
	leukodystrophy, hypo
ency 66	
1	intellectual disability, a
myelinating, 21	intellectual disability, short stature, fa
1	
	intellectual disability,
utosomal recessive 58	
1	intellectual disability, a
cial anomalies, and joint dislocations	intellectual disability, a
1	
	intellectual disability, a
autosomal recessive 7	
1	intellectual disability, a
utosomal recessive 66	intellectual disability, a
1	
	intellectual disability, a
utosomal recessive 65	
1	intellectual disability, a
utosomal recessive 64	intellectual disability, a
1	
	intellectual disability, a
utosomal recessive 63	
1	intellectual disability, a
utosomal recessive 61	intellectual disability, a
1	
	intellectual disability, a
utosomal recessive 60	

1		
		intellectual disability,
autosomal recessive 6		
1		intellectual disability, a
utosomal recessive 59		
1		intellectual disability, a
utosomal recessive 57		
1		intellectual disability-expressive ap
hasia-facial dysmorphism syndrome		
1		intellectual disability, a
utosomal recessive 56		
1		intellectual disability, a
utosomal recessive 54		
1		intellectual disability, a
utosomal recessive 53		
1		intellectual disability, a
utosomal recessive 52		
1		intellectual disability, a
utosomal recessive 51		
1		intellectual disability, a
utosomal recessive 50		
1		intellectual disability,
autosomal recessive 5		
1		intellectual disability, a
utosomal recessive 47		
1		intellectual disability, a
utosomal recessive 46		
1		intellectual disability, a
utosomal recessive 45		
1		intellectual disability-epilep
sy-extrapramidal syndrome		
1		intellectual disability-facial dysmorphism s
yndrome due to SETD5 haploinsufficiency		
1		intellectual disability, a
utosomal recessive 43		
1		
		interstitial lun
g disease 2		
1		
		isobutyryl-CoA dehydro

genase deficiency		
1		
		ischio-vertebra
l syndrome		
1		
	intrauterine growth retardation, metaphyseal dysplasia, adrenal h	
ypoplasia congenita, genital anomalies, and immunodeficiency		
1		
	intrauterine growth restriction-short statu	
re-early adult-onset diabetes syndrome		
1		
		intracerebral
hemorrhage		
1		
	intestinal pseudoobstruction, neuron	
al, chronic idiopathic, X-linked		
1		
	intestinal obstruction in the newborn du	
e to guanylate cyclase 2C deficiency		
1		
		intestinal dysmoti
lity syndrome		
1		
		intestinal
cancer		
1		
	interstitial lung disease d	
ue to ABCA3 deficiency		
1		
		interstitial lu
ng disease		
1		
	intellectual disability-feeding difficulties-	
developmental delay-microcephaly syndrome		
1		
		intermediate severe
Salla disease		
1		
		intermediate DE
ND syndrome		
1		
		intellectual disability-
strabismus syndrome		
1		
	intellectual disability-sparse	
hair-brachydactyly syndrome		
1		
	intellectual disability-severe speech	
delay-mild dysmorphism syndrome		
1		
	intellectual disability-obesity-brain malf	
ormations-facial dysmorphism syndrome		
1		
	intellectual disability-microcephaly-strabi	
smus-behavioral abnormalities syndrome		
1		

	intellectual disability-hypotonic
facies syndrome, X-linked, 1	
1	intellectual disability-hypotonia-sp
asticity-sleep disorder syndrome	intellectual disability-hyperkinetic
1	
	intellectual disability, a
movement-truncal ataxia syndrome	
1	intellectual disability, a
utosomal recessive 44	intellectual disability, X
1	
	intellectual disability,
utosomal recessive 42	
1	intellectual disability,
-linked, syndromic, 35	intellectual disability,
1	
	intellectual disability,
autosomal dominant 22	
1	intellectual disability,
autosomal dominant 42	intellectual disability,
1	
	intellectual disability,
autosomal dominant 41	
1	intellectual disability,
autosomal dominant 4	intellectual disability,
1	
	intellectual disability,
autosomal dominant 39	
1	intellectual disability,
autosomal dominant 34	intellectual disability,
1	
	intellectual disability,
autosomal dominant 33	
1	intellectual disability,
autosomal dominant 30	intellectual disability,
1	
	intellectual disability,
autosomal dominant 3	
1	intellectual disability,
autosomal dominant 29	intellectual disability,
1	
	intellectual disability,
autosomal dominant 24	
1	intellectual disability,
autosomal dominant 16	intellectual disability,
1	
	intellectual disability,
autosomal dominant 45	

1		
		intellectual disability,
autosomal dominant 15		
1		intellectual disability,
autosomal dominant 14		
1		intellectual disability,
autosomal dominant 13		
1		intellectual disability,
autosomal dominant 11		
1		intellectual disability,
autosomal dominant 10		
1		intellectual disability,
autosomal dominant 1		
1		intellectual disability, anterior max
illary protrusion, and strabismus		
1		intellectual disability, X-link
ed, with panhypopituitarism		
1		intellectual disability, X-linked, wit
h or without seizures, ARX-related		
1		intellectual disability, X-lin
ked, syndromic, Houge type		
1		intellectual disability,
autosomal dominant 43		
1		intellectual disability,
autosomal dominant 46		
1		intellectual disability, a
utosomal recessive 34		
1		intellectual disability,
autosomal dominant 6		
1		intellectual disability,
autosomal recessive 3		
1		intellectual disability, a
utosomal recessive 27		
1		intellectual disability,
autosomal recessive 2		
1		intellectual disability, a
utosomal recessive 18		
1		
		muscular dystrophy–dystroglycanopathy (congenit

al with brain and eye anomalies), type A, 4	
1	
	intellectual disability, a
utosomal recessive 13	
1	
	intellectual disability, a
utosomal recessive 12	
1	
	intellectual disability,
autosomal recessive 1	
1	
	intellectual disability,
autosomal recessive	
1	
	intellectual disability,
autosomal dominant 9	
1	
	intellectual disability,
autosomal dominant 58	
1	
	intellectual disability,
autosomal dominant 47	
1	
	intellectual disability,
autosomal dominant 57	
1	
	intellectual disability,
autosomal dominant 56	
1	
	intellectual disability, autosoma
l dominant 55, with seizures	
1	
	intellectual disability,
autosomal dominant 54	
1	
	intellectual disability,
autosomal dominant 53	
1	
	intellectual disability,
autosomal dominant 52	
1	
	intellectual disability,
autosomal dominant 51	
1	
	intellectual disability,
autosomal dominant 50	
1	
	intellectual disability,
autosomal dominant 5	
1	
	intellectual disability,
autosomal dominant 48	
1	
	isolated Pierre-R
obin syndrome	
1	

	isolated anhidrosis with
normal sweat glands	
1	
	isolated asymptomatic elevation
of creatine phosphokinase	
1	
	lathostero
losis	
1	
	lethal congenital cont
racture syndrome 1	
1	
	lethal arthrogryposis-anterior
horn cell disease syndrome	
1	
	lethal arteriopathy syndrome d
ue to fibulin-4 deficiency	
1	
	lessel-kubisch
syndrome	
1	
	left ventricular n
oncompaction 8	
1	
	left ventricular n
oncompaction 7	
1	
	left ventricular no
ncompaction 10	
1	
	left ventricular n
oncompaction 1	
1	
	laurin-Sandrow
syndrome	
1	
	lattice corneal dy
strophy type I	
1	
	laterality defects, a
utosomal dominant	
1	
	lethal congenital cont
racture syndrome 2	
1	
	lateral scl
erosis	
1	
	lateral meningoc
ele syndrome	
1	
	late-onset retinal
degeneration	
1	
	late-onset junctional ep
idermolysis bullosa	

1		
		late-adult onset reti
nititis pigmentosa		
1		
		laryngo-onycho-cuta
neous syndrome		
1		
		language delay and attention deficit-hyperactivity disorder/
cognitive impairment with or without cardiac arrhythmia		
1		
		lacrimoauriculodentod
igital syndrome 3		
1		
		lacrimoauriculodentod
igital syndrome 2		
1		
		kyphosis-lateral tongue atrophy-m
yofibrillar myopathy syndrome		
1		
		lethal congenital contr
acture syndrome 11		
1		
		lethal congenital cont
racture syndrome 4		
1		
		isolated autosomal dominant hypo
magneemia, Glaudemans type		
1		
		leukocyte adhesion
deficiency 1		
1		
		leukodystrophy, hypo
myelinating, 20		
1		
		leukodystrophy, hypomyelinatin
g, 19, transient infantile		
1		
		leukodystrophy, hypo
myelinating, 18		
1		
		leukodystrophy, hypo
myelinating, 17		
1		
		leukodystrophy, hypo
myelinating, 16		
1		
		leukodystrophy, hypo
myelinating, 15		
1		
		leukodystrophy, hypo
myelinating, 14		
1		
		leukodystrophy and acquired microce
phaly with or without dystonia;		
1		
		leukocyte adhesion de

deficiency type II	
1	
	leukocyte adhesion
deficiency 3	
1	
	leukemia, acute lymphocytic
c, susceptibility to, 1	
1	
	lethal congenital contracture
fracture syndrome 6	
1	
	leukemia, acute lymphoblastic
c, susceptibility to, 3	
1	
	lethal polymalformative syndrome
syndrome, Boissel type	
1	
	lethal osteosclerotic
c bone dysplasia	
1	
	lethal occipital encephalocele-
skeletal dysplasia syndrome	
1	
	lethal left ventricular non-compaction-seizures-hypotonia-cataract-developmental delay syndrome
1	
	lethal fetal cerebrorenogenitourinary
renal agenesis/hypoplasia syndrome	
1	
	lethal congenital glycogen storage
storage disease of heart	
1	
	lethal congenital contracture
fracture syndrome 9	
1	
	lethal congenital contracture
fracture syndrome 8	
1	
	lethal congenital contracture
fracture syndrome 7	
1	
	kyphoscoliosis-lateral tongue atrophy-hereditary
hereditary spastic paraplegia syndrome	
1	
	kyphomelic dysplasia
ysplasia	
1	
	ketoacidosis due to monocarboxylate transporter-1 deficiency
ate transporter-1 deficiency	
1	
	isolated micropthalmia
phthalmia 4	
1	
	isolated sulfite oxidase
dase deficiency	
1	

	isolated sedoheptulok
inase deficiency	
1	isolated oxy
cephaly	isolated osteo
1	
	isolated optic ner
poikilosis	
1	isolated neonatal scle
ve hypoplasia	isolated microp
1	
	isolated microp
rosing cholangitis	
1	isolated microp
hthalmia 8	isolated microp
1	
	isolated microp
hthalmia 7	
1	isolated microp
hthalmia 6	isolated microp
1	
	isolated microp
hthalmia 5	
1	isolated microp
hthalmia 3	keratosis pilaris
1	
	isolated microp
atrophicans	
1	isolated hyperch
hthalmia 2	isolated growth hormone
1	
	isolated growth hormone
lorhidrosis	
1	isolated growth hormone
deficiency, type 5	isolated dy
1	
	isolated congenita
deficiency, type 4	
1	isolated congenital
deficiency type III	
1	
stonia	
1	
l megalocornea	
1	
digital clubbing	

1		
		isolated congenital
adermatoglyphia		
1		
		isolated cerebellar hy
poplasia/agenesis		
1		
		isolated thyroid-stimulati
ng hormone deficiency		
1		
		isovaleric a
cidemia		
1		
		joint contractures, osteochond
romas, and B-cell lymphoma		
1		
		joint laxity, short st
ature, and myopia		
1		
		keratosis palmoplan
taris striata 2		
1		
		keratosis linearis-ichthyosis congenit
a-sclerosing keratoderma syndrome		
1		
		keratosis follicularis spinu
losa decalvans, X-linked		
1		
		keratoderma-ichthyosis-deafness s
yndrome, autosomal recessive		
1		
		keratoderma heredit
arium mutilans		
1		
		keratocon
us 9		
1		
		keratocon
us 1		
1		
		keratoco
nus		
1		
		keratitis fugax
hereditaria		
1		
		karyomegalic interst
itial nephritis		
1		
		juvenile-onset diabetes mellitus-central an
d peripheral neurodegeneration syndrome		
1		
		juvenile sialid
osis type 2		
1		
		juvenile polyposis/hereditary hemor

rhagic telangiectasia syndrome	1		
1			
gle glaucoma	1		juvenile open an
1			
nson disease 19A	1		juvenile onset Parki
1			
hic cystinosis	1		juvenile nephropat
1			
fibromatosis	1		juvenile hyaline
1			
renal glucosuria syndrome	1		juvenile cataract-microcornea-
1			
to defect in LACC1	1		juvenile arthritis due
1			
t disease	1		juvenile Page
1			
ton disease	1		juvenile Hunting
1			
ked, syndromic, Bain type	1		intellectual disability, X-lin
1			
-linked, syndromic 33	1		intellectual disability, X
1			
ency 67	1		immunodefici
1			
athy due to MRPL44 deficiency	1		infantile hypertrophic cardiomyop
1			
editary spastic paralysis	1		infantile-onset ascending her
1			
pinal muscular atrophy	1		infantile-onset X-linked s
1			
c hyalinosis	1		infantile systemi
1			
e brain atrophy-basal ganglia disease syndrome	1		infantile spams-psychomotor retardation-progressiv
1			
h neuroaxonal dysplasia	1		infantile osteopetrosis wit
1			
endocrine-pancreatic disease	1		infantile multisystem neurologic-
1			

	infantile liver fai
lure syndrome 3	
1	
	infantile liver fai
lure syndrome 1	
1	
	infantile hypotonia-oculomotor anomalies-hyperki
netic movements-developmental delay syndrome	
1	
	infantile hypop
hosphatasia	
1	
	infantile epileptic-dyski
netic encephalopathy	
1	
	infantile-onset generalized dyski
sia with orofacial involvement	
1	
	infantile cerebral and cerebellar atrophy w
ith postnatal progressive microcephaly	
1	
	infantile cerebellar-re
tinal degeneration	
1	
	infantile cataract, skin abnormalities, glutamate
excess, and impaired intellectual development	
1	
	indifference to pain, congen
ital, autosomal dominant	
1	
	incontinentia
pigmenti	
1	
	inclusion body
myositis	
1	
	inclusion body myopathy with early-onset Paget dis
ease with or without frontotemporal dementia 3	
1	
	inclusion body myopathy with early-onset Paget dis
ease with or without frontotemporal dementia 2	
1	
	inclusion body myopathy with Paget disease of
bone and frontotemporal dementia type 1	
1	
	inclusion body myopathy and brai
n white matter abnormalities	
1	
	infantile-onse
t epilepsy	
1	
	infantile-onset mesial temporal lobe epil
epsy with severe cognitive regression	
1	
	inborn glycerol kin
ase deficiency	

1		
		inherited Creutzfeld
t-Jakob disease		
1		insulin-resistance
syndrome type A		
1		inosine triphosphat
ase deficiency		
1		inherited rippling
muscle disease		
1		inherited prekallik
rein deficiency		
1		inherited isolated adrenal insufficienc
y due to partial CYP11A1 deficiency		
1		inherited glutathione sy
nthetase deficiency		
1		inherited fatty acid m
etabolism disorder		
1		inherited epidermo
lysis bullosa		
1		inherited d
ystonia		
1		inherited bleeding diso
rder, platelet-type		
1		inflammatory skin and bowe
l disease, neonatal, 2		
1		infertility
disorder		
1		inflammatory skin and bowe
l disease, neonatal, 1		
1		inflammatory poikiloderma with hair a
bnormalities and acral keratoses		
1		inflammatory
disease		
1		inflammatory bowel disease, immuno
deficiency, and encephalopathy		
1		inflammatory bowe
l disease 30		
1		inflammatory bowe

l disease 28	
1	
	inflammatory bowe
l disease 25	
1	
	inflammatory bow
el disease 1	
1	
	inflammatory bowel disease (infantile ulcer
ative colitis) 31, autosomal recessive	
1	
	inflammatory bo
wel disease	
1	
	inborn organic
aciduria	
1	
	inborn disorder of lysine and
hydroxylysine metabolism	
1	
	intellectual disability, X-lin
ked syndromic, Turner type	
1	
	immunodeficiency 80 with or witho
ut congenital cardiomyopathy	
1	
	immunodeficiency 91 and
hyperinflammation	
1	
	immunodeficiency 89
and autoimmunity	
1	
	immunodefici
ency 88	
1	
	immunodeficiency 87
and autoimmunity	
1	
	immunodefici
ency 86	
1	
	immunodeficiency 85
and autoimmunity	
1	
	immunodefici
ency 84	
1	
	immunodeficiency 83, susceptib
ility to viral infections	
1	
	immunodeficiency 82 with
systemic inflammation	
1	
	immunodefici
ency 81	
1	

1		immunodeficiency 79
1		
1		immunodeficiency 93 and hypotrophic cardiomyopathy
1		
1		immunodeficiency 78 with autoimmunity and developmental delay
1		
1		immunodeficiency 77
1		
1		immunodeficiency 76
1		
1		immunodeficiency 75
1		
1		immunodeficiency 74, COVID-19-related, X-linked
1		
1		immunodeficiency 73c with defective neutrophil chemotaxis and hypogammaglobulinemia
1		
1		immunodeficiency 73b with defective neutrophil chemotaxis and lymphopenia
1		
1		immunodeficiency 72 with autoinflammation
1		
1		immunodeficiency 70
1		
1		immunodeficiency 69
1		
1		immunodeficiency 92
1		
1		immunodeficiency 95
1		
1		inborn disorder of cobalamin metabolism and transport
1		
1		immunodeficiency, common variable, 4
1		
1		immunodeficiency with osteosclerotic dysplasia and developmental abnormalities
1		
1		immunoglobulin-mediated membranoproliferative glomerulonephritis
1		
1		immunodeficiency-centromeric instability-facial anomalies syndrome 4

1		
		immunodeficiency-centromeric instabi
lity-facial anomalies syndrome 3		
1		
		immunodeficiency-centromeric instabi
lity-facial anomalies syndrome 2		
1		
		immunodeficiency-centromeric instabi
lity-facial anomalies syndrome 1		
1		
		immunodeficiency, developmental d
elay, and hypohomocysteinemia		
1		
		immunodeficiency, co
mmon variable, 7		
1		
		immunodeficiency, co
mmon variable, 6		
1		
		immunodeficiency, co
mmon variable, 5		
1		
		immunodeficiency, co
mmon variable, 3		
1		
		immunodefici
ency 96		
1		
		immunodeficiency, co
mmon variable, 2		
1		
		immunodeficiency, com
mon variable, 14		
1		
		immunodeficiency, com
mon variable, 12		
1		
		immunodeficiency, com
mon variable, 10		
1		
		immunodeficiency due to
ficolin3 deficiency		
1		
		immunodeficiency due to
MASP-2 deficiency		
1		
		immunodeficiency due t
o CD25 deficiency		
1		
		immunodeficiency 99 with hypogammaglob
ulinemia and autoimmune cytopenias		
1		
		immunodeficiency 98 with aut
oinflammation, X-linked		
1		
		immunodeficiency 97 wit

h autoinflammation	
1	
	intellectual developm
ental disorder 59	
1	
	intellectual developmental d
isorder 60 with seizures	
1	
	intellectual developm
ental disorder 61	
1	
	intellectual developmental disorder, autosomal recessive 75,
with neuropsychiatric features and variant lissencephaly	
1	
	intellectual disabili
ty, X-linked 101	
1	
	intellectual disabili
ty, X-linked 100	
1	
	intellectual disabil
ity, X-linked 1	
1	
	intellectual disabil
ity, FRA12A type	
1	
	intellectual disability a
nd myopathy syndrome	
1	
	intellectual developmental disorder
der, autosomal recessive 81	
1	
	intellectual developmental disorder, autosomal
recessive 80, with variant lissencephaly	
1	
	intellectual developmental disorder
der, autosomal recessive 79	
1	
	intellectual developmental disorder
der, autosomal recessive 78	
1	
	intellectual developmental disorder
der, autosomal recessive 76	
1	
	intellectual developmental disorder
der, autosomal recessive 74	
1	
	intellectual disabili
ty, X-linked 103	
1	
	intellectual developmental disorder
der, autosomal recessive 73	
1	
	intellectual developmental disorder
der, autosomal recessive 72	
1	

der, autosomal recessive 71	intellectual developmental disorder
1	
der, autosomal recessive 70	intellectual developmental disorder
1	
der, autosomal recessive 69	intellectual developmental disorder
1	
der, autosomal recessive 68	intellectual developmental disorder
1	
der, autosomal recessive 67	intellectual developmental disorder
1	
der, autosomal dominant 75	intellectual developmental disorder
1	
der, autosomal dominant 73	intellectual developmental disorder
1	
der, autosomal dominant 72	intellectual developmental disorder
1	
ity, X-linked 102	intellectual disability
1	
ity, X-linked 104	intellectual disability
1	
ental disorder 62	intellectual developmental disorder
1	
ity, X-linked 63	intellectual disability
1	
9, syndromic, female-restricted	intellectual disability, X-linked 9
1	
ity, X-linked 99	intellectual disability
1	
ity, X-linked 97	intellectual disability
1	
ity, X-linked 96	intellectual disability
1	
ity, X-linked 95	intellectual disability
1	
ity, X-linked 93	intellectual disability
1	
ity, X-linked 91	intellectual disability

1		
		intellectual disabil
ity, X-linked 90		
1		intellectual disabil
ity, X-linked 9		
1		intellectual disabil
ity, X-linked 72		
1		intellectual disabil
ity, X-linked 61		
1		intellectual disabili
ty, X-linked 105		
1		intellectual disabil
ity, X-linked 58		
1		intellectual disabil
ity, X-linked 49		
1		intellectual disabil
ity, X-linked 46		
1		intellectual disabil
ity, X-linked 45		
1		intellectual disabil
ity, X-linked 41		
1		intellectual disabil
ity, X-linked 30		
1		intellectual disabil
ity, X-linked 21		
1		intellectual disabil
ity, X-linked 19		
1		intellectual disabili
ty, X-linked 107		
1		intellectual disabili
ty, X-linked 106		
1		
		intellectual developmental disorder, autosomal
dominant 71, with behavioral abnormalities		
1		
		intellectual developmental diso
rder, autosomal dominant 70		
1		
		intellectual developmental diso
rder, autosomal dominant 69		
1		
		intellectual developmental disorder with h

ypotonia and behavioral abnormalities	
1	
intellectual developmental disorder with or	
without epilepsy or cerebellar ataxia	
1	
intellectual developmental disorder with ocular	
anomalies and distinctive facial features	
1	
intellectual developmental disorder	
with neuropsychiatric features	
1	
intellectual developmental disorder with nasal speech,	
dysmorphic facies, and variable skeletal anomalies	
1	
intellectual developmental disorder with muscle t	
one abnormalities and distal skeletal defects	
1	
intellectual developmental disorder with microcephaly and with or	
without ocular malformations or hypogonadotropic hypogonadism	
1	
intellectual developmental disorder with ma	
crocephaly, seizures, and speech delay	
1	
intellectual developmental disorder with language impairmen	
t and early-onset DOPA-responsive dystonia-parkinsonism	
1	
intellectual developmental disorder with i	
mpaired language and dysmorphic facies	
1	
intellectual developmental disorder with hypoto	
nia, impaired speech, and dysmorphic facies	
1	
intellectual developmental disorder with	
hypertelorism and distinctive facies	
1	
intellectual developmental diso	
rder, autosomal dominant 68	
1	
intellectual developmental disorder with gastroin	
testinal difficulties and high pain threshold	
1	
intellectual developmental disorder with dysmorphi	
c facies, seizures, and distal limb anomalies	
1	
intellectual developmental disorder w	
ith dysmorphic facies and ptosis	
1	
intellectual developmental disorder with	
cardiac defects and dysmorphic facies	
1	
intellectual developmental disorder with behavioral abnormalit	
ies and craniofacial dysmorphism with or without seizures	
1	
intellectual developmental disorder with autistic feat	
ures and language delay, with or without seizures	
1	

	intellectual developmental disorder with autism and macrocephaly	
1		
	intellectual developmental disorder with autism and dysmorphic facies	
1		
	intellectual developmental disorder with abnormal behavior, microcephaly, and short stature	
1		
	intellectual developmental disorder and retinitis pigmentosa; IDRP	
1		
	intellectual developmental disorder with or without peripheral neuropathy	
1		
	intellectual developmental disorder with paroxysmal dyskinesia or seizures	
1		
	intellectual developmental disorder with polymicrogyria and seizures	
1		
	intellectual developmental disorder with growth and with or without seizures or ataxia	
1		
	intellectual developmental disorder, autosomal dominant 67	
1		
	intellectual developmental disorder, autosomal dominant 66	
1		
	intellectual developmental disorder, autosomal dominant 65	
1		
	intellectual developmental disorder, autosomal dominant 64	
1		
	intellectual developmental disorder, autosomal dominant 63, with macrocephaly	
1		
	intellectual developmental disorder, X-linked, syndromic, with pigmentary mosaicism and coarse facies	
1		
	intellectual developmental disorder, X-linked, syndromic, Pilorge type	
1		
	intellectual developmental disorder, X-linked, syndromic, Hackmann-Di Donato type	
1		
	intellectual developmental disorder, X-linked, syndromic 37	
1		
	intellectual developmental disorder, X-linked 114	
1		
	intellectual developmental disorder, X-linked 113	

1			
			intellectual developmental
disorder, X-linked 112			
1			
			intellectual developmental
disorder, X-linked 111			
1			
			intellectual developmental
disorder, X-linked 108			
1			
		intellectual developmental disorder with speech del	
ay, dysmorphic facies, and t-cell abnormalities			
1			
		intellectual developmental disorder with spe	
ech delay, autism, and dysmorphic facies			
1			
		intellectual developmental disorder with spee	
ch delay and axonal peripheral neuropathy			
1			
		intellectual developmental disorder with shor	
t stature and variable skeletal anomalies			
1			
		intellectual developmental disorder with sho	
rt stature and behavioral abnormalities			
1			
		intellectual developmental disorder with	
severe speech and ambulation defects			
1			
		intellectual developmental disorder	
with seizures and language delay			
1			
		muscular dystrophy, limb-girdl	
e, autosomal recessive 29			
1			
		neurodevelopmental disorder with epile	
psy, spasticity, and brain atrophy			
1			
		muscular dystrophy-dystroglycanopathy (congenit	
al with brain and eye anomalies), type A, 7			
1			
		severe combined immunodeficien	
cy due to CD70 deficiency			
1			
		severe early-onset axonal neurop	
athy due to MFN2 deficiency			
1			
		severe congenital	
neutropenia			
1			
		severe congenital hypochromic ane	
mia with ringed sideroblasts			
1			
		severe combined immunodeficiency, autosomal recessive, T cell-negative, B	
cell-negative, NK cell-negative, due to adenosine deaminase deficiency			
1			
		severe combined immunodeficie	

1	severe combined immunodeficiency due to LCK deficiency	1
1	severe combined immunodeficiency due to LAT deficiency	1
1	severe combined immunodeficiency due to IKK2 deficiency	1
1	severe combined immunodeficiency due to DNA-PKcs deficiency	1
1	severe combined immunodeficiency due to DCLRE1C deficiency	1
1	severe combined immunodeficiency due to CTPS1 deficiency	1
1	severe combined immunodeficiency due to CORO1A deficiency	1
1	severe combined immunodeficiency due to CARMIL2 deficiency	1
1	seborrhea-like dermatitis with psoriasiform elements	1
1	severe combined immunodeficiency due to CARD11 deficiency	1
1	severe achondroplasia-developmental delay-acanthosis nigricans syndrome	1
1	severe X-linked mitochondrial encephalomyopathy	1
1	severe Canavan disease	1
1	sessile serrated polyposis cancer syndrome	1
1	sensory ataxic neuropathy, dysarthria, and ophthalmoparesis	1
1	seizures-scoliosis-macrocephaly syndrome	1
1	seizures, early-onset, with neurodegeneration and brain calcifications	1
1	seizures, benign familial neonatal, 1	1
1	seizures, benign familial infantile, 5	1

	seizures, benign fami
lial infantile, 3	
1	
	severe early-onset obesity-insulin resist
ance syndrome due to SH2B1 deficiency	
1	
	severe early-onset pulmonary alveolar
proteinosis due to MARS deficiency	
1	
	severe feeding difficulties-failure to thrive-mi
crocephaly due to ASXL3 deficiency syndrome	
1	
	severe growth deficiency-strabismus-extensive dermal
melanocytosis-intellectual disability syndrome	
1	
	short stature due to par
tial GHR deficiency	
1	
	short stature due to growth ho
rmone qualitative anomaly	
1	
	short stature due to
GHSR deficiency	
1	
	short stature and microcephal
y with genital anomalies	
1	
	short stature and advanced bone age, with or without early
-onset osteoarthritis and/or osteochondritis dissecans	
1	
	short chain acyl-CoA dehy
drogenase deficiency	
1	
	short QT syndr
ome type 3	
1	
	short QT syndr
ome type 2	
1	
	short QT syndr
ome type 1	
1	
	short QT syn
drome 7	
1	
	severe primary tri
methyaminuria	
1	
	severe phosphoribosylpyrophospha
te synthetase superactivity	
1	
	severe neurodegenerative syn
drome with lipodystrophy	
1	
	severe neonatal-onset enceph
lopathy with microcephaly	

1		
		severe motor and intellectual disabilities-sensorineural deafness-dystonia syndrome
1		
		severe intellectual disability-short stature-behavioral abnormalities-facial dysmorphism syndrome
1		
		severe intellectual disability-progressive spastic diplegia syndrome
1		
		severe intellectual disability-progressive postnatal microcephaly-midline stereotypic hand movements syndrome
1		
		severe intellectual disability-poor language-strabismus-grimacing face-long fingers syndrome
1		
		severe intellectual disability-corpus callosum agenesis-facial dysmorphism-cerebellar ataxia syndrome
1		
		severe hypotonia-psychomotor developmental delay-strabismus-cardiac septal defect syndrome
1		
		severe hemophilia B
1		
		severe hemophilia A
1		
		seizures, benign familial infantile, 2
1		
		sebocystomatosis
1		
		retinitis pigmentosa 74
1		
		retinitis pigmentosa 89
1		
		retinoblastoma
1		
		retinitis pigmentosa-juvenile cataract-short stature-intellectual disability syndrome
1		
		retinitis pigmentosa-hearing loss-premature aging-short stature-facial dysmorphism syndrome
1		
		retinitis pigmentosa with or without situs inversus
1		
		retinitis pigmentosa and erythrocytic microcytosis
1		
		retinitis pigmentosa

entosa 97		
1		
		retinitis pigm
entosa 96		
1		retinitis pigm
entosa 95		retinitis pigm
1		
		retinitis pigm
entosa 93		
1		retinitis pigm
entosa 90		retinitis pig
1		
		retinitis pigm
mentosa 9		
1		retinitis pigm
entosa 88		sea-blue histioc
1		
		retinitis pigmentosa 87 wit
yte syndrome		
1		retinitis pigm
h choroidal involvement		retinitis pigm
1		
		retinitis pigm
entosa 86		
1		retinitis pigm
entosa 85		retinitis pigm
1		
		retinitis pigm
entosa 84		
1		retinitis pigm
entosa 83		retinitis pigm
1		
		retinitis pigm
entosa 81		
1		retinitis pigm
entosa 80		retinitis pigm
1		
		retinitis pigm
entosa 79		
1		retinitis pigm
entosa 78		retinitis pigm
1		
		retinitis pigm
entosa 77		
1		retinitis pigm
entosa 76		
1		

		retinosch
isis		
1		
		rhabdoid tumor predis
osition syndrome 1		
1		
		rhabdoid tumor predis
osition syndrome 2		
1		
		rhabdomyolysis, susc
eptibility to, 1		
1		
		sclerosteo
sis 2		
1		
		sclerosteo
sis 1		
1		
		schizophre
nia 6		
1		
		schizophre
nia 15		
1		
		schizence
phaly		
1		
		scapuloperoneal spinal muscular
atrophy, autosomal dominant		
1		
		scalp-ear-nippl
e syndrome		
1		
		sarcosin
emia		
1		
		sandestig-stefan
ova syndrome		
1		
		sacral agenesis-abnormal ossification of the vertebr
al bodies-persistent notochordal canal syndrome		
1		
		rolandic epilepsy, intellectual disabil
ity, and speech dyspraxia, X-linked		
1		
		robinow syndrome, auto
somal recessive 2		
1		
		rippling muscle
disease 2		
1		
		ring dermoid
of cornea		
1		
		rigid spine muscul
ar dystrophy 1		

1		
		right atrial
isomerism		
1		
		ribose-5-P isomera
se deficiency		
1		
		rhizomelic limb shortening w
ith dysmorphic features		
1		
		rhizomelic chondrodyspla
sia punctata type 5		
1		
		rhizomelic chondrodyspla
sia punctata type 3		
1		
		rhizomelic chondrodyspla
sia punctata type 2		
1		
		rhizomelic chondrodyspla
sia punctata type 1		
1		
		rheumatoid a
rthritis		
1		
		short stature due to primary aci
d-labile subunit deficiency		
1		
		short stature, amelogenesis imperfecta, a
nd skeletal dysplasia with scoliosis		
1		
		short stature, facial dysmorphism, and skeletal a
nomalies with or without cardiac anomalies 1		
1		
		spastic paraplegia 87,
autosomal recessive		
1		
		spasticity-ataxia-gait
anomalies syndrome		
1		
		spastic tetraplegia-thin corpus callosum-pro
gressive postnatal microcephaly syndrome		
1		
		spastic tetraplegia and axia
l hypotonia, progressive		
1		
		spastic paraplegia-severe developm
ental delay-epilepsy syndrome		
1		
		spastic paraplegia-Paget di
sease of bone syndrome		
1		
		spastic paraplegia, intellectual dis
ability, nystagmus, and obesity		
1		
		spastic paraplegia 92,

autosomal recessive	
1	
	spastic paraplegia 90B,
autosomal recessive	
1	
	spastic paraplegia 90A,
autosomal dominant	
1	
	spastic paraplegia 89,
autosomal recessive	
1	
	spastic paraplegia 88,
autosomal dominant	
1	
	spastic paraplegia 86,
autosomal recessive	
1	
	short stature, facial dysmorphism, and skeletal a
nomalies with or without cardiac anomalies 2	
1	
	spastic paraplegia 85,
autosomal recessive	
1	
	spastic paraplegia 82,
autosomal recessive	
1	
	spastic paraplegia 81,
autosomal recessive	
1	
	spastic paraplegia 80,
autosomal dominant	
1	
	spastic paraplegia 79A, autoso
mal dominant, with ataxia	
1	
	spastic paraplegia 70,
autosomal recessive	
1	
	spastic paraparesis-cataract
s-speech delay syndrome	
1	
	spastic ataxia 9, aut
osomal recessive	
1	
	spastic ataxia 8, autosomal recessive,
with hypomyelinating leukodystrophy	
1	
	spastic at
axia 5	
1	
	spastic at
axia 4	
1	
	specific granule
deficiency 1	
1	

1		
		spermatogenic
failure 16		
1		spermatogenic
failure 15		
1		spermatogenic
failure 14		
1		spermatogenic
failure 13		
1		spermatogenic
failure 12		
1		spermatogenic
failure 11		
1		spermatogenic
failure 10		
1		
		spastic at
axia 3		
1		spastic at
axia 2		
1		spastic at
axia 1		
1		sialidosis
type 1		
1		short-rib thoracic dysplasia 8
with or without polydactyly		
1		short-rib thoracic dysplasia 7
with or without polydactyly		
1		short-rib thoracic dysplasia 6
with or without polydactyly		
1		short-rib thoracic dysplas
ia 20 with polydactyly		
1		short-rib thoracic dysplasia 19
with or without polydactyly		
1		short-rib thoracic dysplas
ia 18 with polydactyly		
1		short-rib thoracic dysplasia 17
with or without polydactyly		
1		short-rib thoracic dysplasia 16

with or without polydactyly	
1	
	short-rib thoracic dysplas
ia 15 with polydactyly	
1	
	short-rib thoracic dysplas
ia 14 with polydactyly	
1	
	short-rib thoracic dysplasia 13
with or without polydactyly	
1	
	short-rib thoracic dysplasia 11
with or without polydactyly	
1	
	short-rib thoracic dysplasia 10
with or without polydactyly	
1	
	short stature-pituitary and cerebellar d
effects-small sella turcica syndrome	
1	
	short stature-optic atrophy-Pe
lger-Huët anomaly syndrome	
1	
	short stature-onychodysplasia-facial d
ysmorphism-hypotrichosis syndrome	
1	
	short stature-delayed bone age due to th
teroid hormone metabolism deficiency	
1	
	short stature-brachydactyly-obesity-gl
obal developmental delay syndrome	
1	
	short stature-auditory canal atresia-mandibul
ar hypoplasia-skeletal anomalies syndrome	
1	
	short stature-advanced bone age-earl
y-onset osteoarthritis syndrome	
1	
	short stature, rhizomelic, with microcephal
y, micrognathia, and developmental delay	
1	
	short stature, microcephaly,
and endocrine dysfunction	
1	
	short stature, impaired intellectual development,
microcephaly, hypotonia, and ocular anomalies	
1	
	sialido
sis	
1	
	sialidosis
type 2	
1	
	southeast Asian
ovalocytosis	
1	

		sialur
ia		
1		
		solitary median maxillary c
entral incisor syndrome		
1		
		snowflake vitreoreti
nal degeneration		
1		
		snijders blok-fis
her syndrome		
1		
		skin fragility-woolly hair-palmo
plantar keratoderma syndrome		
1		
		skin creases, congenital symm
etric circumferential, 2		
1		
		skeletal syste
m disorder		
1		
		skeletal overgrowth-craniofacial dysmorphism-hyp
erelastic skin-white matter lesions syndrome		
1		
		skeletal dysplasia, mild, with join
t laxity and advanced bone age		
1		
		skeletal defects, genital hypoplasia
a, and intellectual disability		
1		
		sitosterol
emia 2		
1		
		sinoatrial node dysfun
ction and deafness		
1		
		silver-russell
syndrome 4		
1		
		sideroblastic
anemia 3		
1		
		sideroblastic
anemia 2		
1		
		sickle cell-hemoglobin
d disease syndrome		
1		
		sickle cell-hemoglobin
c disease syndrome		
1		
		sickle cell-hemoglobin
E disease syndrome		
1		
		sickle cell-beta-thalasse
mia disease syndrome		

1		
		sickle cell disease an
d related diseases		
1		
		sickle cell
disease		
1		
		sick sinus sy
ndrome 4		
1		
		sick sinus syndrome 2,
autosomal dominant		
1		
		sick sinus sy
ndrome 1		
1		
		retinitis pigm
entosa 75		
1		
		retinitis pigm
entosa 73		
1		
		spermatogenic
failure 35		
1		
		pulmonary fibrosis and/or bone marrow fa
ilure syndrome, telomere-related, 7		
1		
		pulmonary hypertens
ion, primary, 4		
1		
		pulmonary hypertens
ion, primary, 3		
1		
		pulmonary hypertens
ion, primary, 2		
1		
		pulmonary hypertens
ion, primary, 1		
1		
		pulmonary fibrosis and/or bone marr
ow failure, telomere-related, 6		
1		
		pulmonary fibrosis and/or bone marr
ow failure, telomere-related, 5		
1		
		pulmonary fibrosis and/or bone marr
ow failure, Telomere-related, 4		
1		
		pulmonary fibrosis and/or bone marr
ow failure, Telomere-related, 3		
1		
		pulmonary fibrosis and/or bone marr
ow failure, Telomere-related, 1		
1		
		pulmonary fibrosis and/or bone marrow fa

ilure syndrome, telomere-related, 9	
1	
	pulmonary fibrosis and/or bone marrow fa
ilure syndrome, telomere-related, 8	
1	
	pulmonary f
ibrosis	
1	
	pseudohypoaldosteronism, type
IB2, autosomal recessive	
1	
	pulmonary arterioven
ous malformation	
1	
	pulmonary alveolar proteinosis
with hypogammaglobulinemia	
1	
	pulmonary alveolar
microlithiasis	
1	
	pterin-4 alpha-carbinolamine
dehydratase 1 deficiency	
1	
	psychomotor retardation, epilepsy,
and craniofacial dysmorphism	
1	
	psychomotor regression-oculomotor apraxia-
movement disorder-nephropathy syndrome	
1	
	psoriasi
s 2	
1	
	psoriasis 15, pustular,
susceptibility to	
1	
	pseudoxanthoma elasticum-like skin manif
estations with retinitis pigmentosa	
1	
	pseudopseudohypop
arathyroidism	
1	
	pseudohypoparathyr
oidism type 1C	
1	
	pulmonary veno-occlusive disease and/or
pulmonary capillary haemangiomatosis	
1	
	pulmonary venooccl
usive disease	
1	
	pulmonary venoocclu
sive disease 2	
1	
	punctate palmoplant
ar keratoderma	
1	

	rapadilino s
yndrome	
1	
	radioulnar synostosis, nonsyn
dromic, susceptibility to	
1	
	radioulnar synostosis with amegak
aryocytic thrombocytopenia 2	
1	
	radioulnar synostosis with amegak
aryocytic thrombocytopenia 1	
1	
	question mark ea
rs, isolated	
1	
	pyruvate kinase h
yperactivity	
1	
	pyruvate kinase defici
ency of red cells	
1	
	pyruvate dehydrogenase ph
osphatase deficiency	
1	
	pyruvate dehydrogenase E3-bi
nding protein deficiency	
1	
	pyruvate dehydrogena
se E3 deficiency	
1	
	pyruvate dehydrogena
se E2 deficiency	
1	
	pyruvate dehydrogenase
E1-beta deficiency	
1	
	pyruvate carboxylase deficien
cy, severe neonatal type	
1	
	pyruvate carboxylase defic
iency, infantile form	
1	
	pyruvate carboxylase def
iciency, benign type	
1	
	pyruvate carboxylase d
eficiency disease	
1	
	pyropoikilocytosi
s, hereditary	
1	
	pyridoxine-dependent epilepsy
caused by ALDH7A1 mutant	
1	
	pyridoxal phosphate-re
sponsive seizures	

1		
		pyogenic bacterial infections
due to MyD88 deficiency		
1		
		pyogenic arthritis-pyoderma g
angrenosum-acne syndrome		
1		
		pyncnodysos
tosis		
1		
		purine nucleoside phosph
horylase deficiency		
1		
		pseudohypoparathyr
oidism type 1A		
1		
		pseudohypoaldoster
onism type 2E		
1		
		retinitis pigm
entosa 72		
1		
		progressive myoclonic
epilepsy type 3		
1		
		progressive spondyloepimetaphyseal dysplasia-short stature-s
hort fourth metatarsals-intellectual disability syndrome		
1		
		progressive sensorineural hearing loss-h
ypertrophic cardiomyopathy syndrome		
1		
		progressive scapulohumerope
roneal distal myopathy		
1		
		progressive retinal dystrophy due
to retinol transport defect		
1		
		progressive pseudorheumatoid
arthropathy of childhood		
1		
		progressive osseou
s heteroplasia		
1		
		progressive myoclonic ep
ilepsy with dystonia		
1		
		progressive myoclonic
epilepsy type 9		
1		
		progressive myoclonic
epilepsy type 8		
1		
		progressive myoclonic
epilepsy type 7		
1		
		progressive myoclonic

epilepsy type 6	
1	
	progressive muscular dystrophy
1	
	pseudohypoaldosteronism type 2D
1	
	progressive microcephaly-seizures-cortical blindness-developmental delay syndrome
1	
	progressive familial intrahepatic cholestasis type 3
1	
	progressive familial intrahepatic cholestasis type 2
1	
	progressive familial heart block, type 1A
1	
	progressive familial heart block type IB
1	
	progressive external ophthalmoplegia with mitochondrial dna deletions, autosomal recessive 6
1	
	progressive external ophthalmoplegia with mitochondrial DNA deletions, autosomal recessive 5
1	
	progressive external ophthalmoplegia with mitochondrial DNA deletions, autosomal recessive 4
1	
	progressive external ophthalmoplegia with mitochondrial DNA deletions, autosomal recessive 2
1	
	progressive external ophthalmoplegia with mitochondrial DNA deletions, autosomal dominant 5
1	
	progressive external ophthalmoplegia with mitochondrial DNA deletions, autosomal dominant 4
1	
	progressive supranuclear palsy-parkinsonism syndrome
1	
	prolidase deficiency
1	
	prolonged electroretinal response suppression 1
1	
	prolonged electroretinal response suppression 2
1	
	pseudohypoaldosteronism type 2C
1	

	pseudohypoaldoster
onism type 2B	
1	pseudohyperaldoste
onism type 2	pseudoachond
1	
	pseudo-TORCH s
roplasia	
1	pseudo-TORCH s
ndrome 3	pseudo-TORCH s
1	
	pseudo-TORCH
ndrome 2	
1	prune belly
ndrome 1	proximal myopathy with e
1	
	protoporphyrin, ery
syndrome	
1	protoporphyrin, ery
syndrome	proteasome-associated auto
1	
	proteinuria, chr
xtrapyramidal signs	
1	protein-losing
thropoietic, 2	protein Z de
1	
	proteasome-associated autoi
thropoietic, 1	
1	proteasome-associated autoi
inflammatory syndrome	proteasome-associated autoi
1	
	proteasome-associated autoi
onic benign	
1	proteasome-associated autoi
enteropathy	proteasome-associated autoi
1	
	proteasome-associated autoi
ficiency	
1	proteasome-associated autoi
nflammatory syndrome 5	proteasome-associated autoi
1	
	proteasome-associated autoi
nflammatory syndrome 4	
1	proteasome-associated autoi
nflammatory syndrome 2	proteasome-associated autoi
1	
	proteasome-associated autoi
nflammatory syndrome 1	

1		
		prostate cancer,
hereditary, 9		
1		prostate cancer,
hereditary, 1		
1		properdin deficie
ncy, X-linked		
1		recessive dystrophic ep
idermolysis bullosa		
1		recessive dystrophic epider
molysis bullosa inversa		
1		recessive dystrophic epidermols
is bullosa-generalized other		
1		retinitis pigm
entosa 27		retinitis pig
1		
mentosa 4		retinitis pigm
1		
entosa 39		retinitis pigm
1		
entosa 38		retinitis pigm
1		
entosa 37		retinitis pigm
1		
entosa 36		retinitis pigm
1		
entosa 35		retinitis pigm
1		
entosa 33		retinitis pigm
1		
entosa 31		retinitis pigm
1		
entosa 30		retinitis pig
1		
mentosa 3		retinitis pigm
1		
entosa 28		retinitis pigm
1		

entosa 26	
1	
	recessive intellectual disability-motor dysfun
ction-multiple joint contractures syndrome	
1	
	retinitis pigm
entosa 25	
1	
	retinitis pigm
entosa 23	
1	
	retinitis pigm
entosa 20	
1	
	retinitis pig
entosa 2	
1	
	retinitis pigm
entosa 19	
1	
	retinitis pigm
entosa 18	
1	
	retinitis pigm
entosa 17	
1	
	retinitis pigm
entosa 14	
1	
	retinitis pigm
entosa 13	
1	
	retinitis pigm
entosa 12	
1	
	retinitis pigm
entosa 11	
1	
	retinitis pigm
entosa 40	
1	
	retinitis pigm
entosa 41	
1	
	retinitis pigm
entosa 42	
1	
	retinitis pigm
entosa 43	
1	
	retinitis pigm
entosa 71	
1	
	retinitis pigm
entosa 70	
1	

1		
		retinitis pigm
entosa 45		
1		retinitis pigm
entosa 44		
1		retinitis pigm
entosa 10		
1		retinitis pig
mentosa 1		
1		retinal vasculopathy with cerebral leukoenc
ephalopathy and systemic manifestations		
1		renal hypomagnesemia 5 wi
th ocular involvement		
1		renal hypomag
nesemia 3		
1		renal hypodysplas
ia/aplasia 4		renal hypodysplas
1		
		renal hypodysplas
ia/aplasia 3		
1		renal hypodysplas
ia/aplasia 2		renal hypodysplas
1		
		renal hypodysplas
ia/aplasia 1		
1		renal dysplasia,
unilateral		renal dysplasia, cysti
1		
c, susceptibility to		renal dysplasi
1		
		renal dysp
a, bilateral		
1		renal cysts and dia
lasia		renal coloboma
1		
		renal carc
betes syndrome		
1		renal age
syndrome		
1		
inoma		
1		

nesis	
1	
	regressive spondylomet
aphyseal dysplasia	
1	
	reducing body
myopathy	
1	
	red-green color
blindness	
1	
	red color bl
indness	
1	
	recurrent metabolic encephalomyopathic crises-rhabdomyolys
is-cardiac arrhythmia-intellectual disability syndrome	
1	
	recurrent infections-inflammatory syndrome d
ue to zinc metabolism disorder syndrome	
1	
	recurrent Neisseria infections
due to factor D deficiency	
1	
	recombinase activating
gene 2 deficiency	
1	
	recombinase activating
gene 1 deficiency	
1	
	recessive mitochondria
l ataxia syndrome	
1	
	renal hypomag
nesemia 4	
1	
	renal hypomag
nesemia 6	
1	
	retinal macular dy
strophy type 2	
1	
	renal tubular
acidosis	
1	
	retinal dystrophy-ataxia-pituitary hormo
ne abnormality-hypogonadism syndrome	
1	
	retinal dystrophy, optic nerve edema, splenomegal
y, anhidrosis, and migraine headache syndrome	
1	
	retinal dystrophy wit
h leukodystrophy	
1	
	retinal dystrophy with inner retinal dys
function and ganglion cell anomalies	
1	

	retinal dystrophy
and obesity	
1	
	retinal dystrophy and microv
illus inclusion disease	
1	
	retinal cone d
ystrophy 4	
1	
	retinal cone dy
strophy 3A	
1	
	retinal arterial
tortuosity	
1	
	reticulate acropigment
ation of Kitamura	
1	
	reticular dystrophy of the re
tinal pigment epithelium	
1	
	reticular dy
sgenesis	
1	
	restrictive de
rmopathy 2	
1	
	restrictive de
rmopathy 1	
1	
	respiratory infections, recurrent, and fail
ure to thrive with or without diarrhea	
1	
	resistance to thyrotropin-re
leasing hormone syndrome	
1	
	resistance to thyroid hormone due to a mut
ation in thyroid hormone receptor beta	
1	
	renal-hepatic-pancre
atic dysplasia 2	
1	
	renal tubulopathy-encephalopat
hy-liver failure syndrome	
1	
	renal tubule
disorder	
1	
	renal tubular acidosis, dista
l, 4, with hemolytic anemia	
1	
	renal tubular acidosis, distal, 3, with o
r without sensorineural hearing loss	
1	
	renal tubular acidosis, distal, 2, with p
rogressive sensorineural hearing loss	

1		
		spermatogenic
failure 34		
1		
		spermatogenic
failure 36		
1		
		progressive external ophthalmoplegia with mitoch
ondrial DNA deletions, autosomal dominant 2		
1		
		thyrotoxic periodic paralysi
s, susceptibility to, 2		
1		
		torsion dys
tonia 2		
1		
		tooth and nail
syndrome		
1		
		tooth agenesis, selec
tive, X-linked, 1		
1		
		tooth agenesis,
selective, 8		
1		
		tooth agenesis,
selective, 7		
1		
		tooth agenesis,
selective, 4		
1		
		tooth agenesis,
selective, 3		
1		
		tooth agenesis,
selective, 1		
1		
		tibial muscular
dystrophy		
1		
		tibial hem
imelia		
1		
		tibial aplasia-ectro
dactyly syndrome		
1		
		thyroid hormone resistance, gene
ralized, autosomal recessive		
1		
		thrombophilia, X-linked, d
ue to factor 9 defect		
1		
		thyroid hormone resistance, gene
ralized, autosomal dominant		
1		
		thyroid hormone metab

olism, abnormal 1
 1 |
 |
 lary carcinoma
 1 |
 |
 carcinoma
 1 |
 |
 nogogenesis 6
 1 |
 |
 nogogenesis 5
 1 |
 |
 nogogenesis 4
 1 |
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 nogogenesis 3
 1 |
 |
 nogogenesis 2A
 1 |
 |
 nogogenesis 1
 1 |
 |
 ell carcinoma
 1 |
 |
 tonia 4
 1 |
 |
 tonia 6
 1 |
 |
 deficiency
 1 |
 |
 I deficiency
 1 |
 |
 nonphotosensitive
 1 |
 |
 nonphotosensitive
 1 |
 |
 nonphotosensitive
 1 |
 |
 3, photosensitive
 1 |
 |
 2, photosensitive
 1 |

|
 thyroid gland papil
 |
 thyroid gland
 |
 thyroid dyshormo
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 thyroid dyshormo
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 thyroid dyshormo
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 thyroid dyshormo
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 thyroid Hurthle c
 |
 torsion dys
 |
 torsion dys
 |
 transaldolase
 |
 transcobalamin I
 |
 trichothiodystrophy 6,
 |
 trichothiodystrophy 5,
 |
 trichothiodystrophy 4,
 |
 trichothiodystrophy
 |
 trichothiodystrophy
 |

	trichorhinophalangeal
syndrome, type III	
1	trichorhinophalangeal sy
ndrome type I or III	trichorhinophalangeal
1	
	trichomegaly-retina pigmentary de
syndrome type I	
1	trichome
generation-dwarfism syndrome	trichohepatoneurodevel
1	
	trichohepatoenter
galy	
1	trichohepatoenter
opmental syndrome	trichoepithelioma, mul
1	
	tricho-dento-osse
ic syndrome 2	
1	trichilemma
ic syndrome 1	tremor-ataxia-central hyp
1	
	tremor, hereditary
triple familial, 1	
1	tremor, hereditary
ous syndrome	transketolase
1	
	transient neonatal m
l cyst	
1	transient infantile hypertriglyc
omyelination syndrome	transient bullous dermol
1	
	thrombophilia, familial, due to decreased r
essential, 5	
1	release of tissue plasminogen activator
essential, 4	
1	
deficiency	
1	
asthenia gravis	
1	
eridemia and hepatosteatosi	
1	
ysis of the newborn	
1	
release of tissue plasminogen activator	

1		
		thrombophilia due to
thrombin defect		
1		
		syndromic X-linked intellectu
al disability Snyder type		
1		
		tarsal-carpal coal
ition syndrome		
1		
		tetraamelia s
ndrome 2		
1		
		tetraamelia s
ndrome 1		
1		
		testicular regres
sion syndrome		
1		
		testicular anomalies with or with
out congenital heart disease		
1		
		terminal osseous dysplasia-pi
gmentary defects syndrome		
1		
		tentamy sy
ndrome		
1		
		tentamy preaxial brach
ydactyly syndrome		
1		
		temperature-sensitive oculoc
utaneous albinism type 1		
1		
		telomere sy
ndrome		
1		
		telangiectasia, hereditar
y hemorrhagic, type 5		
1		
		telangiectasia, hereditar
y hemorrhagic, type 2		
1		
		tall stature-scoliosis-macrodacty
ly of the great toes syndrome		
1		
		thrombophilia due to protein S de
ficiency, autosomal recessive		
1		
		tall stature-intellectual disabil
ity-renal anomalies syndrome		
1		
		systemic primary carnitin
e deficiency disease		
1		
		systemic lupus ery

thematosis 17	
1	
	synpolydactyl
y type 2	
1	
	synpolydactyl
y type 1	
1	
	syndromic sensorineural deafness due to com
bined oxidative phosphorylation defect	
1	
	syndromic multisystem autoimmune d
isease due to ITCH deficiency	
1	
	syndromic microph
thalmia type 5	
1	
	syndromic micr
phthalmia	
1	
	syndromic constitutiona
l thrombocytopenia	
1	
	syndromic congenital
sodium diarrhea	
1	
	tetrahydrobiopterin-responsive hyper
phenylalaninemia/phenylketonuria	
1	
	thanatophoric dys
plasia type 1	
1	
	thanatophoric dys
plasia type 2	
1	
	thiamine-responsive
encephalopathy	
1	
	thrombophilia due to protein S de
ficiency, autosomal dominant	
1	
	thrombophilia due to protein C de
ficiency, autosomal recessive	
1	
	thrombophilia due to protein C de
ficiency, autosomal dominant	
1	
	thrombophilia due to activat
ed protein C resistance	
1	
	thrombomodulin-related
bleeding disorder	
1	
	thrombocytopenia-absen
t radius syndrome	
1	

	thrombocytopenia, anemi
a, and myelofibrosis	
1	
	thrombocytopenia, X-linked, with or
without dyserythropoietic anemia	
1	
	thrombocytopenia with congenita
l dyserythropoietic anemia	
1	
	thrombocyto
penia 7	
1	
	thrombocyto
penia 6	
1	
	thrombocyto
penia 5	
1	
	thrombocyto
penia 4	
1	
	thrombocyto
penia 3	
1	
	thrombocyto
penia 2	
1	
	thrombocytopenia 11 with multiple congen
ital anomalies and dysmorphic facies	
1	
	thrombocytop
enia 10	
1	
	thrombocyto
penia 1	
1	
	thrombocyth
emia 3	
1	
	thrombocyth
emia 2	
1	
	thoracic malf
ormation	
1	
	thoracic aorti
c aneurysm	
1	
	thiamine-responsive megalob
lastic anemia syndrome	
1	
	trichothiodystrophy 7,
nonphotosensitive	
1	
	trichothiodystrophy 8,
nonphotosensitive	

1		
		trichothiodystrophy 9,
nonphotosensitive		
1		
		visual impairment and prog
recessive phthisis bulbi		
1		
		vitelliform macula
r dystrophy 4		
1		
		vitelliform macula
r dystrophy 3		
1		
		vitelliform macula
r dystrophy 2		
1		
		vitamin K-dependent clotting factor
s, combined deficiency of, type 2		
1		
		vitamin D-dependent
rickets, type 3		
1		
		vitamin D-dependent
rickets, type 2A		
1		
		vitamin D-dependent
rickets, type 2		
1		
		vitamin D-dependent
rickets, type 1A		
1		
		vitamin D hydroxylation-def
icient rickets, type 1B		
1		
		vitamin B12-unresponsive methyl
malonic acidemia type mut0		
1		
		vitamin B12-unresponsive methyl
malonic acidemia type mut-		
1		
		visceral neuropathy, familia
l, 2, autosomal recessive		
1		
		trichotill
omania		
1		
		visceral neuropathy, familia
l, 1, autosomal recessive		
1		
		visceral myo
pathy 2		
1		
		visceral myo
pathy 1		
1		
		vesicoureteral

reflux 8		
1		
		vesicoureteral
reflux 3		
1		
		vesicoureteral
reflux 2		
1		
		very long chain acyl-CoA de
hydrogenase deficiency		
1		
		vertebral, cardiac, tracheoesopha
geal, renal, and limb defects		
1		
		vertebral, cardiac, renal, an
d limb defects syndrome 3		
1		
		vertebral, cardiac, renal, an
d limb defects syndrome 2		
1		
		vertebral, cardiac, renal, an
d limb defects syndrome 1		
1		
		vitelliform macula
r dystrophy 5		
1		
		vitreoretinopathy with phalan
geal epiphyseal dysplasia		
1		
		von Willebrand
disease 1		
1		
		von Willebrand
disease 2		
1		
		xerosis and growth failure with immune
and pulmonary dysfunction syndrome		
1		
		xeroderma pigmentos
um variant type		
1		
		xeroderma pigment
osum group G		
1		
		xeroderma pigment
osum group F		
1		
		xeroderma pigment
osum group E		
1		
		xeroderma pigment
osum group D		
1		
		xeroderma pigment
osum group C		
1		

	xeroderma pigment
osum group B	
1	
	xeroderma pigment
osum group A	
1	
	xanthinuria
type II	
1	
	xanthinuria
type I	
1	
	wrinkly skin
syndrome	
1	
	wooly hair-palmoplantar
keratoderma syndrome	
1	
	wooly hair, autosom
al recessive 3	
1	
	wooly hair-skin fra
gility syndrome	
1	
	white sponge
nevus 2	
1	
	white sponge
nevus 1	
1	
	warburg-cinott
i syndrome	
1	
	von Willebrand di
sease type 2N	
1	
	von Willebrand di
sease type 2M	
1	
	von Willebrand di
sease type 2B	
1	
	von Willebrand di
sease type 2A	
1	
	von Willebrand
disease 3	
1	
	vertebral hypersegmentation
and orofacial anomalies	
1	
	vertebral anomalies and variable en
docrine and T-cell dysfunction	
1	
	ventriculomegaly-cyst
ic kidney disease	

1		
		ulnar-mammary
syndrome		
1		tyrosinemia
type II		tyrosinemia
1		
		tyrosine hydroxyla
type I		
1		tyrosine hydroxyla
se deficiency		type II complement comp
1		
		type I complement comp
onent 8 deficiency		
1		type 1 diabetes
onent 8 deficiency		type 1 diabetes
1		
		type 1 diabetes
mellitus 20		
1		type 1 diabetes
mellitus 2		type 1 diabetes
1		
		type 1 diabetes
mellitus 10		
1		type 1 diabetes
mellitus 1		turnpenny-fry
1		
		tumoral calcinosis, hyperph
syndrome		
1		tumoral calcinosis, hyperph
osphatemic, familial, 3		tumor predispositi
1		
		tumor predispositi
osphatemic, familial, 2		
1		tubulointerstitial kidney dise
on syndrome 3		tuberous scl
1		
		tuberous scl
on syndrome 2		
1		tropical panc
ase, autosomal dominant, 2		
1		
erosis 2		
1		
erosis 1		
1		

reatitis	
1	
dactyly syndrome	trismus-pseudocampto
1	
erase deficiency	triosephosphate isom
1	
e deficiency syndrome	triokinase and FMN cyclas
1	
inuria	trimethylam
1	
haly 2	trigonocep
1	
type III	tyrosinemia
1	
syndrome 1	uncombable hair
1	
arthrogryposis	ventriculomegaly and
1	
syndrome 2	uncombable hair
1	
al defect 2	ventricular sept
1	
al defect 1	ventricular sept
1	
paroxysmal familial, 2	ventricular fibrillation,
1	
l syndrome	velocardiofacia
1	
DA2 deficiency	vasculitis due to A
1	
ormation	vascular malf
1	
rphyria	variegate po
1	
amyloidosis	variant ABeta2M
1	
syndrome 2	van der Woude
1	

	van der Woude
syndrome 1	
1	
	van den Ende-Gup
ta syndrome	
1	
	van Maldergem
syndrome 2	
1	
	van Maldergem
syndrome 1	
1	
	vacuolar Neur
omyopathy	
1	
	uveal coloboma-cleft lip and pal
ate-intellectual disability	
1	
	uterine corpu
s sarcoma	
1	
	uterine corpus
leiomyoma	
1	
	urofacial syndr
ome type 1	
1	
	urofacial sy
ndrome 2	
1	
	urocanic ac
iduria	
1	
	urinary bladde
r, atony of	
1	
	unilateral multicystic
dysplastic kidney	
1	
	uncombable hair
syndrome 3	
1	
	syndromic X-linked intellec
tual disability type 10	
1	
	syndromic X-linked intellectua
l disability Siderius type	
1	
	spermatogenic
failure 37	
1	
	spinocerebellar a
taxia type 1	
1	
	spinocerebellar a
taxia type 21	

1		
		spinocerebellar a
taxia type 2		
1		spinocerebellar ata
xia type 19/22		spinocerebellar a
1		
		spinocerebellar a
taxia type 18		
1		spinocerebellar a
taxia type 17		spinocerebellar ata
1		
		spinocerebellar a
xia type 15/16		
1		spinocerebellar a
taxia type 14		spinocerebellar a
1		
		spinocerebellar a
taxia type 13		
1		spinocerebellar a
taxia type 12		spinocerebellar a
1		
		spinocerebellar a
taxia type 11		
1		spinocerebellar a
taxia type 10		spinocerebella
1		
		spinal muscular atr
r ataxia 7		
1		spinocerebellar
ophy, type III		spinocerebellar
1		
		spinocerebellar
ataxia 50		
1		spinocerebellar
ataxia 49		spinocerebellar
1		
		spinocerebellar
ataxia 48		
1		spinocerebellar
ataxia 47		spinocerebellar
1		
		spinocerebellar
ataxia 46		
1		spinocerebellar
ataxia 45		spinocerebellar
1		
		spinocerebellar

ataxia 44		
1		
		spinocerebellar
ataxia 43		
1		
		spinocerebellar ataxia 42, early-onset, se
vere, with neurodevelopmental deficits		
1		
		spinocerebellar
ataxia 27A		
1		
		spinal muscular atrophy-progressiv
e myoclonic epilepsy syndrome		
1		
		spinocerebellar a
taxia type 23		
1		
		spinocerebellar a
taxia type 25		
1		
		spinocerebellar a
taxia type 26		
1		
		spinocerebellar a
taxia type 27		
1		
		spinocerebellar ataxia, a
utosomal recessive 29		
1		
		spinocerebellar ataxia, a
utosomal recessive 28		
1		
		spinocerebellar ataxia, a
utosomal recessive 27		
1		
		spinocerebellar ataxia, a
utosomal recessive 26		
1		
		spinocerebellar ataxia, a
utosomal recessive 25		
1		
		spinocerebellar ataxia, a
utosomal recessive 24		
1		
		spinocerebellar ataxia, a
utosomal recessive 23		
1		
		spinocerebellar ataxia, a
utosomal recessive 22		
1		
		spinocerebellar atax
ia with epilepsy		
1		
		spinocerebellar a
taxia type 7		
1		

	spinocerebellar a
taxia type 6	
1	spinocerebellar a
taxia type 5	spinocerebellar a
1	
	spinocerebellar a
taxia type 42	
1	spinocerebellar a
taxia type 41	spinocerebellar a
1	
	spinocerebellar a
taxia type 4	
1	spinocerebellar a
taxia type 38	spinocerebellar a
1	
	spinocerebellar a
taxia type 37	
1	spinocerebellar a
taxia type 36	spinocerebellar a
1	
	spinocerebellar a
taxia type 35	
1	spinocerebellar a
taxia type 34	spinocerebellar a
1	
	spinocerebellar a
taxia type 31	
1	spinocerebellar a
taxia type 29	spinocerebellar a
1	
	spinocerebellar a
taxia type 28	
1	spinal muscular at
rophy, type IV	spinal muscular at
1	
	syndromic X-linked intellectu
rophy, type II	
1	spermatogenic
al disability Shashi type	spermatogenic
1	
	spermatogenic
failure 52	
1	spermatogenic
failure 65	spermatogenic
1	
failure 64	

1		
		spermatogenic
failure 63		
1		spermatogenic
failure 62		
1		spermatogenic
failure 61		
1		spermatogenic
failure 6		
1		spermatogenic
failure 58		
1		spermatogenic
failure 57		
1		spermatogenic
failure 56		
1		spermatogenic
failure 55		
1		spermatogenic
failure 53		
1		spermatogenic
failure 50		
1		
		spinal muscular at
rophy, type 1		
1		spermatogenic
failure 48		
1		spermatogenic
failure 47		
1		spermatogenic
failure 46		
1		spermatogenic
failure 45		
1		spermatogenic
failure 44		
1		spermatogenic
failure 43		
1		spermatogenic
failure 42		
1		spermatogenic

failure 40	
1	
	spermatogenic
failure 4	
1	
	spermatogenic
failure 39	
1	
	spermatogenic
failure 38	
1	
	spermatogenic
failure 66	
1	
	spermatogenic
failure 69	
1	
	spermatogenic
failure 7	
1	
	spermatogenic
failure 70	
1	
	spinal muscular atrophy,
infantile, James type	
1	
	spinal muscular atrophy, dista
l, autosomal recessive, 6	
1	
	spinal muscular atrophy with r
espiratory distress type 2	
1	
	spinal muscular atrophy with c
ongenital bone fractures 2	
1	
	spinal muscular atrophy with c
ongenital bone fractures 1	
1	
	spinal muscula
r atrophy	
1	
	spheroid body
myopathy	
1	
	spermatogenic failu
re, X-linked, 9	
1	
	spermatogenic failu
re, X-linked, 7	
1	
	spermatogenic failu
re, X-linked, 6	
1	
	spermatogenic failu
re, X-linked, 5	
1	

	spermatogenic failu
re, X-linked, 3	
1	spermatogenic failu
re, X-linked, 2	spermatogenic
1	
failure 9	spermatogenic
1	
failure 83	spermatogenic
1	
failure 82	spermatogenic
1	
failure 81	spermatogenic
1	
failure 80	spermatogenic
1	
failure 8	spermatogenic
1	
failure 79	spermatogenic
1	
failure 77	spermatogenic
1	
failure 76	spermatogenic
1	
failure 74	spermatogenic
1	
utosomal recessive 30	spinocerebellar ataxia, a
1	
utosomal recessive 31	spinocerebellar ataxia, a
1	
utosomal recessive 32	spinocerebellar ataxia, a
1	
-like T-cell lymphoma	subcutaneous panniculitis
1	
tic stenosis	supravalvular aor
1	
progressive, 1	supranuclear palsy,
1	
lytic ichthyosis	superficial epidermo

1		
		sulfite oxidase deficiency due to moly
bdenum cofactor deficiency type C		
1		
		sulfite oxidase deficiency due to moly
bdenum cofactor deficiency type B		
1		
		sulfite oxidase deficiency due to moly
bdenum cofactor deficiency type A		
1		
		sulfide quinone oxidore
ductase deficiency		
1		
		sudden infant death-dysgenesi
s of the testes syndrome		
1		
		sudden cardiac fail
ure, infantile		
1		
		succinyl-CoA:3-ketoacid CoA
transferase deficiency		
1		
		succinic semialdehyde deh
ydrogenase deficiency		
1		
		structural heart defects and
renal anomalies syndrome		
1		
		spinocerebellar ataxia, autosomal rec
essive, with axonal neuropathy 1		
1		
		structural congenital heart dise
ase, multiple types - GATA4		
1		
		structural brain anomalies with impaired inte
llectual development and craniosynostosis		
1		
		stromal corneal
dystrophy		
1		
		striatonigral degenerati
on, childhood-onset		
1		
		striatal degeneration, a
utosomal dominant 2		
1		
		striatal degeneration,
autosomal dominant		
1		
		stiff skin s
yndrome		
1		
		sterol carrier prote
in 2 deficiency		
1		
		sterile multifocal osteomyelitis w

ith periostitis and pustulosis	
1	
	stapes ankylosis with br
oad thumbs and toes	
1	
	squalene synthas
e deficiency	
1	
	surfactant metabolism dysf
unction, pulmonary, 1	
1	
	surfactant metabolism dysf
unction, pulmonary, 2	
1	
	surfactant metabolism dysf
unction, pulmonary, 4	
1	
	surfactant metabolism dysf
unction, pulmonary, 5	
1	
	syndromic X-linked intellectua
l disability Raymond type	
1	
	syndromic X-linked intellectual
disability Nascimento type	
1	
	syndromic X-linked intellect
ual disability Najm type	
1	
	syndromic X-linked intellect
ual disability Lubs type	
1	
	syndromic X-linked intellectu
al disability Hedera type	
1	
	syndromic X-linked intellectual
disability Claes-Jensen type	
1	
	syndromic X-linked intell
ectual disability 94	
1	
	syndromic X-linked intel
lectual disability 5	
1	
	syndromic X-linked intell
ectual disability 34	
1	
	syndromic X-linked intell
ectual disability 14	
1	
	syndactyly-telecanthus-anogenital a
nd renal malformations syndrome	
1	
	syndactyly
type 8	
1	

1		syndactyly
type 5		
1		
1		syndactyly
type 3		
1		
1		symptomatic form of muscular dystrophy of
Duchenne and Becker in female carriers		
1		
1		symptomatic form of hemophil
ia B in female carriers		
1		
1		symptomatic form of hemophil
ia A in female carriers		
1		
1		symptomatic form of fragile X
syndrome in female carrier		
1		
1		symptomatic form of Coffin-Lowry
syndrome in female carriers		
1		
1		sympalangism, p
roximal, 1B		
1		
1		susceptibility to respiratory infections a
ssociated with CD8alpha chain mutation		
1		
1		susceptibility to mononeuropath
y of the median nerve, mild		
1		
1		susceptibility to localized
juvenile periodontitis		
1		
1		sporadic pheoch
romocytoma		
1		
1		spondyloperipher
al dysplasia		
1		
1		spondylometaphyseal dysplasia-c
one-rod dystrophy syndrome		
1		
1		spondyloepimetaphyseal dyspla
sia, Isidor-Toutain type		
1		
1		spondyloepimetaphyseal dys
plasia, Genevieve type		
1		
1		spondyloepimetaphyseal dys
plasia, Bieganski type		
1		
1		spondyloepimetaphyseal dysplasia
with multiple dislocations		
1		
1		spondyloepimetaphyseal dysplasi
a with joint laxity, type 3		

1		
		spondyloepimetaphyseal dysplasia with joint laxity, type 1, with or without fractures
1		
		spondyloepimetaphyseal dysplasia
1		
		spondylocostal dysostosis
6, autosomal recessive		
1		
		spondylocostal dysostosis 5
1		
		spondylocostal dysostosis
4, autosomal recessive		
1		
		spondylocostal dysostosis
3, autosomal recessive		
1		
		spondylocostal dysostosis
2, autosomal recessive		
1		
		spondylocostal dysostosis
1, autosomal recessive		
1		
		spondylocostal dysostosis
dysostosis		
1		
		spondylo-ocular syndrome
1		
		spondylo-megaepiphyseal-metaphyseal dysplasia
1		
		split-hand/foot malformation with long bone deficiency 1
1		
		split-foot malformation-mesodactylous polydactyly syndrome
1		
		split hand-foot malformation 6
1		
		split hand-foot malformation 4
1		
		split hand-foot malformation 3
1		
		split hand-foot malformation 1 with sensorineural hearing loss
1		
		split hand-foot malformation 1
1		
		spinocerebellar ataxia, autosomal recessive

essive, with axonal neuropathy 3	
1	
lasia, Guo-Campeau type	spondyloepimetaphyseal dysp
1	
splasia, Krakow type	spondyloepimetaphyseal dy
1	
sia, Sedaghatian type	spondylometaphyseal dyspla
1	
plasia, Missouri type	spondyloepimetaphyseal dys
1	
lasia, Schmidt type	spondylometaphyseal dysp
1	
asia, Kozłowski type	spondylometaphyseal dyspl
1	
a, 'corner fracture' type	spondylometaphyseal dysplasi
1	
a with corneal dystrophy	spondylometaphyseal dysplasi
1	
ed intellectual development, and leber congenital amaurosis	spondyloepiphyseal dysplasia, sensorineural hearing loss, impair
1	
asia, nishimura type	spondyloepiphyseal dyspl
1	
asia, kondo-fu type	spondyloepiphyseal dyspl
1	
asia, Stanescu type	spondyloepiphyseal dyspl
1	
asia, Kimberley type	spondyloepiphyseal dyspl
1	
ith metatarsal shortening	spondyloepiphyseal dysplasia w
1	
congenital joint dislocations	spondyloepiphyseal dysplasia with
1	
asia tarda, X-linked	spondyloepiphyseal dyspl
1	
dysplasia tarda	spondyloepiphyseal
1	
splasia congenita	spondyloepiphyseal dy
1	

		spondyloepiphyse
al dysplasia		
1		
		spondyloepimetaphyseal dysplasia-short l
imb-abnormal calcification syndrome		
1		
		spondyloepimetaphyseal dysplasia
-abnormal dentition syndrome		
1		
		spondyloepimetaphyseal dysp
lasia, sponastrime type		
1		
		spondyloepimetaphyseal dysp
lasia, matrilin-3 type		
1		
		spondyloepimetaphyseal dys
plasia, aggrecan type		
1		
		spondyloepimetaphyseal dys
plasia, Strudwick type		
1		
		spondyloepimetaphyseal dy
splasia, Shohat type		
1		
		spondyloepimetaphyseal dy
splasia, PAPSS2 type		
1		
		progressive external ophthalmoplegia with mitoch
ondrial DNA deletions, autosomal dominant 3		
1		
		progressive external ophthalmoplegia with mitoch
ondrial DNA deletions, autosomal dominant 1		
1		
		muscular dystrophy-dystroglycanopathy (congeni
tal with brain and eye anomalies), type A1		
1		
		neurodevelopmental disorder with movement abnorma
lities, abnormal gait, and autistic features		
1		
		neurodevelopmental disorder with or without var
iable movement or behavioral abnormalities		
1		
		neurodevelopmental disorder with or without
t variable brain abnormalities; NEDBA		
1		
		neurodevelopmental disorder with or with
out seizures and gait abnormalities		
1		
		neurodevelopmental disorder with or without hyperkin
etic movements and seizures, autosomal recessive		
1		
		neurodevelopmental disorder with or without hyperkin
etic movements and seizures, autosomal dominant		
1		
		neurodevelopmental disorder with or with
out early-onset generalized epilepsy		

1		
		neurodevelopmental disorder with or without autistic features and/or structural brain abnormalities
1		
		neurodevelopmental disorder with or without autism or seizures
1		
		neurodevelopmental disorder with or without anomalies of the brain, eye, or heart
1		
		neurodevelopmental disorder with nonspecific brain abnormalities and with or without seizures
1		
		neurodevelopmental disorder with neuromuscular and skeletal abnormalities
1		
		neurodevelopmental disorder with motor regression, progressive spastic paraplegia, and oromotor dysfunction
1		
		neurodevelopmental disorder with microcephaly, hypotonia, and variable brain anomalies
1		
		neurodevelopmental disorder with motor and speech delay and behavioral abnormalities
1		
		neurodevelopmental disorder with motor and language delay, ocular defects, and brain abnormalities
1		
		neurodevelopmental disorder with motor abnormalities, seizures, and facial dysmorphism
1		
		neurodevelopmental disorder with midbrain and hindbrain malformations
1		
		neurodevelopmental disorder with microcephaly, short stature, and speech delay
1		
		neurodevelopmental disorder with microcephaly, seizures, and neonatal cholestasis
1		
		neurodevelopmental disorder with microcephaly, seizures, and cortical atrophy
1		
		neurodevelopmental disorder with microcephaly, seizures, and brain atrophy
1		
		neurodevelopmental disorder with microcephaly, movement abnormalities, and seizures
1		
		neurodevelopmental disorder with microcephaly, impaired language, epilepsy, and gait abnormalities
1		
		neurodevelopmental disorder with microcephaly, impaired language, and gait abnormalities
1		
		neurodevelopmental disorder with poor g

rowth and behavioral abnormalities	
1	
	neurodevelopmental disorder with poor growth and skeletal anomalies
1	
	neurodevelopmental disorder with poor growth, large ears, and dysmorphic facies
1	
	neurodevelopmental disorder with poor growth, spastic tetraplegia, and hearing loss
1	
	neurodevelopmental disorder with speech delay and variable ocular anomalies
1	
	neurodevelopmental disorder with spasticity, seizures, and brain abnormalities
1	
	neurodevelopmental disorder with spasticity, cataracts, and cerebellar hypoplasia
1	
	neurodevelopmental disorder with spasticity and poor growth
1	
	neurodevelopmental disorder with spastic quadriplegia, optic atrophy, seizures, and structural brain anomalies
1	
	neurodevelopmental disorder with spastic quadriplegia and brain abnormalities with or without seizures
1	
	neurodevelopmental disorder with short stature, prominent forehead, and feeding difficulties
1	
	neurodevelopmental disorder with severe motor impairment, absent language, cerebral hypomyelination, and brain atrophy
1	
	neurodevelopmental disorder with severe motor impairment and absent language
1	
	neurodevelopmental disorder with seizures, spasticity, and complete or partial agenesis of the corpus callosum
1	
	neurodevelopmental disorder with seizures, microcephaly, and brain abnormalities
1	
	neurodevelopmental disorder with seizures, hypotonia, and brain imaging abnormalities
1	
	neurodevelopmental disorder with seizures and speech and walking impairment
1	
	neurodevelopmental disorder with seizures and nonepileptic hyperkinetic movements
1	
	neurodevelopmental disorder with seizures and gingival overgrowth
1	

	neurodevelopmental disorder with	
seizures and brain atrophy		
1		
	neurodevelopmental disorder with se	
izures and brain abnormalities		
1		
	neurodevelopmental disorder with relative macrocephaly	
and with or without cardiac or endocrine anomalies		
1		
	neurodevelopmental disorder with regression, abno	
rmal movements, loss of speech, and seizures		
1		
	neurodevelopmental disorder with progressi	
ve spasticity and brain abnormalities		
1		
	neurodevelopmental disorder with pro	
gressive movement abnormalities		
1		
	neurodevelopmental disorder with progressive mi	
crocephaly, spasticity, and brain anomalies		
1		
	neurodevelopmental disorder with poor	
language and loss of hand skills		
1		
	neurodevelopmental disorder with microcepha	
ly, hypotonia, nystagmus, and seizures		
1		
	neurodevelopmental disorder with microcep	
haly, hypotonia, and absent language		
1		
	neurodevelopmental disorder with dysmorphic facie	
s, sleep disturbance, and brain abnormalities		
1		
	neurodevelopmental disorder with hypotonia and c	
erebellar atrophy, with or without seizures		
1		
	neurodevelopmental disorder with hypotonia, imp	
aired speech, and behavioral abnormalities		
1		
	neurodevelopmental disorder with hypotonia, feeding diffi	
culties, facial dysmorphism, and brain abnormalities		
1		
	neurodevelopmental disorder with hypotonia, fa	
cial dysmorphism, and brain abnormalities		
1		
	neurodevelopmental disorder with hypotonia, d	
ysmorphic facies, and skin abnormalities		
1		
	neurodevelopmental disorder with hypotonia, dysmorphic fac	
ies, and skeletal anomalies, with or without seizures		
1		
	neurodevelopmental disorder with hypotonia, brain an	
omalies, distinctive facies, and absent language		
1		
	neurodevelopmental disorder with hypotonia and vari	
able intellectual and behavioral abnormalities		

1		
		neurodevelopmental disorder with hypotonia and speech delay, with or without seizures
1		
		neurodevelopmental disorder with hypotonia and gross motor and speech delay
1		
		neurodevelopmental disorder with hypotonia and dysmorphic facies
1		
		neurodevelopmental disorder with hypotonia and characteristic brain abnormalities
1		
		neurodevelopmental disorder with hypotonia and brain abnormalities
1		
		neurodevelopmental disorder with microcephaly, epilepsy, and hypomyelination
1		
		neurodevelopmental disorder with hypotonia and autistic features with or without hyperkinetic movements
1		
		neurodevelopmental disorder with hyperkinetic movements and dyskinesia
1		
		neurodevelopmental disorder with hearing loss and spasticity
1		
		neurodevelopmental disorder with growth retardation, dysmorphic facies, and corpus callosum abnormalities
1		
		neurodevelopmental disorder with gait disturbance, dysmorphic facies, and behavioral abnormalities, X-linked
1		
		neurodevelopmental disorder with facial dysmorphism, absent language, and pseudo-pelger-huet anomaly
1		
		neurodevelopmental disorder with eye movement abnormalities and ataxia
1		
		hepatitis, fulminant viral, susceptibility to
1		
		neurodevelopmental disorder with epilepsy, cataracts, feeding difficulties, and delayed brain myelination
1		
		neurodevelopmental disorder with epilepsy and hypoplasia of the corpus callosum
1		
		neurodevelopmental disorder with epilepsy and brain atrophy
1		
		neurodevelopmental disorder with hypotonia, language delay, and skeletal defects with or without seizures
1		
		neurodevelopmental disorder with hypo

tonia, microcephaly, and seizures	
1	
neurodevelopmental disorder with hypotonia, neonatal respiratory insufficiency, and thermoregulation	
1	
neurodevelopmental disorder with hypotonia, neuropathy, and deafness	
1	
neurodevelopmental disorder with microcephaly, epilepsy, and brain atrophy	
1	
neurodevelopmental disorder with microcephaly, cortical malformations, and spasticity	
1	
neurodevelopmental disorder with microcephaly, cerebral atrophy, and visual impairment	
1	
neurodevelopmental disorder with microcephaly, cataracts, and renal abnormalities	
1	
neurodevelopmental disorder with microcephaly, ataxia, and seizures	
1	
neurodevelopmental disorder with microcephaly, arthrogryposis, and structural brain anomalies	
1	
neurodevelopmental disorder with microcephaly and structural brain anomalies	
1	
neurodevelopmental disorder with microcephaly and speech delay, with or without brain abnormalities	
1	
neurodevelopmental disorder with microcephaly and movement abnormalities	
1	
neurodevelopmental disorder with language impairment, autism, and attention deficit-hyperactivity disorder	
1	
neurodevelopmental disorder with language impairment and behavioral abnormalities	
1	
neurodevelopmental disorder with language delay and variable cognitive abnormalities	
1	
neurodevelopmental disorder with language delay and seizures	
1	
neurodevelopmental disorder with language delay and behavioral abnormalities, with or without seizures	
1	
neurodevelopmental disorder with involuntary movements	
1	
neurodevelopmental disorder with intracranial hemorrhage, seizures, and spasticity	
1	

1	neurodevelopmental disorder with infantile epileptic spasms	1
1	neurodevelopmental disorder with impaired speech and hyperkinetic movements	1
1	neurodevelopmental disorder with impaired language, behavioral abnormalities, and dysmorphic facies	1
1	neurodevelopmental disorder with impaired language and ataxia and with or without seizures	1
1	neurodevelopmental disorder with impaired intellectual development, hypotonia, and ataxia	1
1	neurodevelopmental disorder with hypotonia, stereotypic hand movements, and impaired language	1
1	neurodevelopmental disorder with hypotonia, seizures, and absent language	1
1	neurodevelopmental disorder with speech impairment and dysmorphic facies	1
1	neurodevelopmental disorder with speech impairment and with or without seizures	1
1	neurodevelopmental disorder with speech or visual impairment and brain hypomyelination	1
1	nicotine dependence	1
1	nonsyndromic congenital nail disorder 3	1
1	nonsyndromic congenital nail disorder 1	1
1	nonepidermolytic palmoplantar keratoderma	1
1	non-syndromic intellectual disability	1
1	non-spherocytic hemolytic anemia due to hexokinase deficiency	1
1	non-small cell lung carcinoma	1
1	non-severe combined immunodeficiency due to CDPG1 deficiency	1
1	non-progressive predominantly posterior cavitating leukoencephalopathy with peripheral neuropathy	1

1		
		non-immune hydr
ops fetalis		
1		
		non-acquired combined pituitary hormone
deficiency with spine abnormalities		
1		
		night blindness, congenit
al stationary, type1i		
1		
		neutrophil immunodef
iciency syndrome		
1		
		neurodevelopmental disorder with structural
brain anomalies and dysmorphic facies		
1		
		neutropenia, severe congenita
l, 9, autosomal dominant		
1		
		neutropenia, severe congenita
l, 8, autosomal dominant		
1		
		neutropenia, severe congenita
l, 2, autosomal dominant		
1		
		neutropenia, severe congenita
l, 10, autosomal recessive		
1		
		neutrope
nia		
1		
		neutral lipid sto
rage myopathy		
1		
		neuropathy, hereditary
sensory, type 2C		
1		
		neuropathy, hereditary
sensory, type 1F		
1		
		neuropathy, hereditary
sensory, type 1D		
1		
		neuropathy, hereditary sensor
y and autonomic, type 2B		
1		
		neuropathy, hereditary sensor
y and autonomic, type 2A		
1		
		nonsyndromic congenita
l nail disorder 4		
1		
		noonan synd
rome 12		
1		
		normal pressure h

hydrocephalus		
1		
		normophosphatemic familia
l tumoral calcinosis		
1		
		oculocutaneous alb
inism type 1B		
1		
		oculocutaneous alb
inism type 1A		
1		
		oculocutaneous al
binism type 1		
1		
		oculocerebroren
al syndrome		
1		
		oculocerebrofacial synd
rome, Kaufman type		
1		
		oculocerebrodent
al syndrome		
1		
		oculoauricular
syndrome		
1		
		ocular motor aprax
ia, Cogan type		
1		
		ocular cyst
inosis		
1		
		ocular alb
inism		
1		
		occult macular
dystrophy		
1		
		occipital horn
syndrome		
1		
		obesity, hyperphagia, and
developmental delay		
1		
		obesity due to prohormone c
onvertase I deficiency		
1		
		obesity due to pro-opiome
lanocortin deficiency		
1		
		obesity due to melanocortin
4 receptor deficiency		
1		
		obesity due to leptin rec
eptor gene deficiency		
1		

	obesity due to congenita
1 leptin deficiency	
1	
	obesity due to SI
M1 deficiency	
1	
	obesity due to CEP
19 deficiency	
1	
	nystagmus 6, congen
ital, X-linked	
1	
	nystagmus 1, congen
ital, X-linked	
1	
	null synd
rome	
1	
	neuropathy, hereditary sensor
y and autonomic, type 1C	
1	
	neuropathy, hereditary sensor
y and autonomic, type 1A	
1	
	neuropathy, hereditary motor and sens
ory, type VIc, with optic atrophy	
1	
	neuronal ceroid li
pofuscinosis 7	
1	
	neuronal ceroid li
pofuscinosis 3	
1	
	neuronal ceroid li
pofuscinosis 2	
1	
	neuronal ceroid lip
ofuscinosis 13	
1	
	neuronal ceroid lip
ofuscinosis 11	
1	
	neuronal ceroid lip
ofuscinosis 10	
1	
	neuronal ceroid li
pofuscinosis 1	
1	
	neuromuscular disorder, congeni
tal, with dysmorphic facies	
1	
	neuromuscular disease caused by qualitativ
e or quantitative defects of dysferlin	
1	
	neuromuscular disease and ocular or audito
ry anomalies with or without seizures	

1		
		neurologic, endocrine, and pancreatic dis
ease, multisystem, infantile-onset 2		
1		
		neurologic, endocrine, and pancreatic dis
ease, multisystem, infantile-onset 1		
1		
		neurohypophyseal dia
betes insipidus		
1		
		neurogenic scapuloperoneal
syndrome, Kaeser type		
1		
		neurofibromato
sis type 1		
1		
		neurofibrom
atosis		
1		
		neuroferriti
nopathy		
1		
		neurofacioskeletal syndrome wit
h or without renal agenesis		
1		
		neurodevelopmental-craniofacial syndrome with
variable renal and cardiac abnormalities		
1		
		neurodevelopmental, jaw, ey
e, and digital syndrome		
1		
		neurodevelopmental disorder-craniofacial dysmor
phism-cardiac defect-hip dysplasia syndrome		
1		
		neurodevelopmental disorder, nonprogressive, w
ith spasticity and transient opisthotonus		
1		
		neurodevelopmental disorder, mitochondrial, with abnormal m
ovements and lactic acidosis, with or without seizures		
1		
		neurodevelopmental disorder with vis
ual defects and brain anomalies		
1		
		neuronal ceroid li
pofuscinosis 5		
1		
		neuronal ceroid li
pofuscinosis 8		
1		
		neuropathy, hereditary moto
r and sensory, type 6B		
1		
		neuronal ceroid lipofuscinosis
8 northern epilepsy variant		
1		
		neuropathy, hereditary moto

r and sensory, type 6A	
1	
hypomyelinating, 2	neuropathy, congenital
1	
ring impairment	neuropathy with hea
1	
l syndrome	neurooculorena
1	
ourinary syndrome	neurooculocardiogenit
1	
syndrome	neuroocular
1	
ditary motor, type 9	neuronopathy, distal here
1	
ditary motor, type 7B	neuronopathy, distal here
1	
ditary motor, type 7A	neuronopathy, distal here
1	
ditary motor, type 5C	neuronopathy, distal here
1	
ditary motor, type 5B	neuronopathy, distal here
1	
ditary motor, type 2D	neuronopathy, distal here
1	
ditary motor, type 2C	neuronopathy, distal here
1	
ditary motor, type 2B	neuronopathy, distal here
1	
ditary motor, type 2A	neuronopathy, distal here
1	
motor, autosomal recessive 8	neuronopathy, distal hereditary
1	
motor, autosomal recessive 7	neuronopathy, distal hereditary
1	
motor, autosomal recessive 4	neuronopathy, distal hereditary
1	
tosomal recessive 11, with spasticity	neuronopathy, distal hereditary motor, au
1	

|
 motor, autosomal dominant 8
 1 |
 |
 motor, autosomal dominant 10
 1 |
 |
 motor, autosomal dominant
 1 |
 |
 hereditary motor
 1 |
 |
 with dystonia and seizures
 1 |
 |
 neurodevelopmental disorder with dysmorphic
 facies, impaired speech, and hypotonia
 1 |
 |
 binism type 3
 1 |
 |
 sideroblastic anemia 2
 1 |
 |
 autosomal dominant
 1 |
 |
 a
 1 |
 |
 aggregate, 2
 1 |
 |
 aggregate, 1
 1 |
 |
 glasmic body
 1 |
 |
 linked, early-onset, severe
 1 |
 |
 with ophthalmoplegia
 1 |
 |
 autosomal recessive
 1 |
 |
 e, autosomal dominant
 1 |
 |
 early respiratory failure
 1 |
 |
 late-onset, with cardiomyopathy

neuronopathy, distal hereditary
 |
 neuronopathy, distal hereditary
 |
 neuronopathy, distal hereditary
 |
 neuronopathy, distal
 |
 neurodevelopmental disorder w
 |
 neurodevelopmental disorder with dysmorphic
 |
 oculocutaneous al
 |
 myopathy, lactic acidosis, an
 |
 myopia 21, autoso
 |
 myopi
 |
 myopathy, tubular
 |
 myopathy, tubular
 |
 myopathy, sarcop
 |
 myopathy, reducing body, X-li
 |
 myopathy, proximal, an
 |
 myopathy, myosin storage,
 |
 myopathy, myosin storag
 |
 myopathy, myofibrillar, 9, with
 |
 myopathy, myofibrillar, 12, infant
 |

1		
		myopathy, lactic acidosis, an
d sideroblastic anemia 1		
1		
		myopathy, centr
onuclear, 2		
1		
		myopathy, epilepsy, and prog
ressive cerebral atrophy		
1		
		myopathy, distal, 7, ad
ult-onset, X-linked		
1		
		myopathy, distal, 6, adult-o
nset, autosomal dominant		
1		
		myopathy, di
stal, 5		
1		
		myopathy, congenita
l, with tremor		
1		
		myopathy, congenital, with structure
d cores and z-line abnormalities		
1		
		myopathy, congenital, with respiratory
insufficiency and bone fractures		
1		
		myopathy, congenital, with diaphragmatic defects, r
espiratory insufficiency, and dysmorphic facies		
1		
		myopathy, congenital, progr
essive, with scoliosis		
1		
		myopathy, congenital proxima
l, with minicore lesions		
1		
		myopathy, centronuclear, 6, wit
h fiber-type disproportion		
1		
		myopia 23, autoso
mal recessive		
1		
		myopia 24, autoso
mal dominant		
1		
		myopia 25, autoso
mal dominant		
1		
		myopia 26, X-linked,
female-limited		
1		
		nemaline myo
pathy 2		
1		
		nemaline myo

pathy 10	
1	
	nasopharyngeal carcinoma,
susceptibility to, 3	
1	
	nasopalpebral lipoma-
coloboma syndrome	
1	
	narcolep
sy 7	
1	
	narcolep
sy 1	
1	
	nanophthal
mos 4	
1	
	nanophthal
mos 2	
1	
	nail-patella-like
renal disease	
1	
	nail-patella
syndrome	
1	
	nail and teeth abnormalities-marginal palmoplanta
r keratoderma-oral hyperpigmentation syndrome	
1	
	myotonic dystro
phy type 2	
1	
	myotonic dystro
phy type 1	
1	
	myotonia pe
rmanens	
1	
	myotonia fl
uctuans	
1	
	myotonia congenita, au
tosomal recessive	
1	
	myotonia congenita, a
utosomal dominant	
1	
	myostatin-related mu
scle hypertrophy	
1	
	myoscler
osis	
1	
	myopia, high, with cataract and
vitreoretinal degeneration	
1	

6		myopia
1		
		myopia 28, autosomal recessive
1		
		myopia
27		
1		
		myopathy, centronuclear, 5
1		
		myopathy with myalgia, increased serum creatine kinase, and with or without episodic rhabdomyolysis
1		
		neurodevelopmental disorder with dysmorphic facies and variable seizures
1		
		muscular dystrophy-dystroglycanopathy (congenital with intellectual disability), type B3
1		
		myoclonus, familial
1		
		myoclonic dystonia 26
1		
		myoclonic dystonia 11
1		
		myelodysplastic syndrome
1		
		myasthenic syndrome, congenital, 25, presynaptic
1		
		myasthenic syndrome, congenital, 24, presynaptic
1		
		myasthenic syndrome, congenital, 23, presynaptic
1		
		myasthenic syndrome, congenital, 22
1		
		myasthenic syndrome, congenital, 1B, fast-channel
1		
		muscular dystrophy-dystroglycanopathy type B6
1		
		muscular dystrophy-dystroglycanopathy type B5
1		
		muscular dystrophy-dystroglycanopathy (congenital with intellectual disability), type B2

1		
		myopathy with abnormal
lipid metabolism		
1		
		muscular dystrophy-dystroglycanopathy (congeni
tal with intellectual disability), type B1		
1		
		muscular dystrophy-dystroglycanopathy (congenit
al with brain and eye anomalies), type a, 8		
1		
		muscular dystrophy-dystroglycanopathy (congenita
l with brain and eye anomalies), type a, 12		
1		
		muscular dystrophy-dystroglycanopathy (congenita
l with brain and eye anomalies), type a, 11		
1		
		muscular dystrophy-dystroglycanopathy (congenita
l with brain and eye anomalies), type a, 10		
1		
		muscular dystrophy-dystroglycanopathy (congeni
tal with brain and eye anomalies), type A9		
1		
		muscular dystrophy-dystroglycanopathy (congeni
tal with brain and eye anomalies), type A6		
1		
		muscular dystrophy-dystroglycanopathy (congeni
tal with brain and eye anomalies), type A5		
1		
		muscular dystrophy-dystroglycanopathy (congeni
tal with brain and eye anomalies), type A3		
1		
		muscular dystrophy-dystroglycanopathy (congeni
tal with brain and eye anomalies), type A2		
1		
		muscular dystrophy-dystroglycanopathy (congenit
al with brain and eye anomalies), type A14		
1		
		myoclonus, fa
milial, 1		
1		
		myoclonus, fa
milial, 2		
1		
		myoclonus, intract
able, neonatal		
1		
		myofibrillar
myopathy		
1		
		myopathy due to myoadenylat
e deaminase deficiency		
1		
		myopathy due to calsequestrin a
nd SERCA1 protein overload		
1		
		myopathy caused by va

riation in POMT2	1		
riation in POMT1	1		myopathy caused by va
riation in POMGNT2	1		myopathy caused by var
riation in POMGNT1	1		myopathy caused by var
riation in GMPPB	1		myopathy caused by va
riation in FKTN	1		myopathy caused by v
riation in FKRP	1		myopathy caused by v
riation in CRPPA	1		myopathy caused by va
nt, autosomal recessive	1		myoglobinuria, acute recurre
infantile, 2	1		myofibromatosis,
infantile, 1	1		myofibromatosis,
yopathy 8	1		myofibrillar m
yopathy 7	1		myofibrillar m
yopathy 6	1		myofibrillar m
yopathy 5	1		myofibrillar m
yopathy 4	1		myofibrillar m
yopathy 3	1		myofibrillar m
yopathy 2	1		myofibrillar m
yopathy 11	1		myofibrillar m

	myofibrillar m
yopathy 10	
1	
	myofibrillar m
yopathy 1	
1	
	nemaline myo
pathy 5	
1	
	nemaline myopathy 5C,
autosomal dominant	
1	
	nemaline myo
pathy 6	
1	
	neurodegeneration with developmental delay, early respirato
ry failure, myoclonic seizures, and brain abnormalities	
1	
	neurodegenerati
ve disease	
1	
	neurodegeneration, infantile-onset, with
optic atrophy and brain abnormalities	
1	
	neurodegeneration, infantile-
onset, biotin-responsive	
1	
	neurodegeneration, early-onset, with choreoa
thetoid movements and microcytic anemia	
1	
	neurodegeneration, childhood-onset,
with progressive microcephaly	
1	
	neurodegeneration, childhood-onset, with multisyste
m involvement due to mitochondrial dysfunction	
1	
	neurodegeneration, childhood-onset, with hypotonia, respi
ratory insufficiency, and brain imaging abnormalities	
1	
	neurodegeneration, childhood-ons
et, with cerebellar atrophy	
1	
	neurodegeneration, childhood-onset, with c
erebellar ataxia and cognitive decline	
1	
	neurodegeneration, childhood-onset, with ataxia,
tremor, optic atrophy, and cognitive decline	
1	
	neurodegeneration, childhood-onset, stress-in
duced, with variable ataxia and seizures	
1	
	neurodegeneration with bra
in iron accumulation 9	
1	
	nemaline myo
pathy 7	

1		
		neurodegeneration with bra
in iron accumulation 8		
1		
		neurodegeneration with bra
in iron accumulation 7		
1		
		neurodegeneration with bra
in iron accumulation 6		
1		
		neurodegeneration with bra
in iron accumulation 5		
1		
		neurodegeneration with bra
in iron accumulation 4		
1		
		neurodegeneration with brai
n iron accumulation 2B		
1		
		neurodegeneration with brai
n iron accumulation 2A		
1		
		neurodegeneration with ataxia, dystoni
a, and gaze palsy, childhood-onset		
1		
		neurodegeneration with ataxia a
nd late-onset optic atrophy		
1		
		neurodegeneration and seizures du
e to copper transport defect		
1		
		neurocardiofaciodi
gital syndrome		
1		
		neurodegenerative syndrome due to cer
ebal folate transport deficiency		
1		
		neurodevelopmental disorder and structural brain an
omalies with or without seizures and spasticity		
1		
		neurodevelopmental disord
er plus optic atrophy		
1		
		neurodevelopmental disorder with abse
nt language and variable seizures		
1		
		neurodevelopmental disorder with dysmorp
hic facies and thin corpus callosum		
1		
		neurodevelopmental disorder with dysmorphic fa
cies and skeletal and brain abnormalities		
1		
		neurodevelopmental disorder with dysmorph
ic facies and ischiopubic hypoplasia		
1		
		neurodevelopmental disorder with dysmophi

c facies and distal skeletal anomalies	
1	
	neurodevelopmental disorder with dysmorp
hic facies and distal limb anomalies	
1	
	neurodevelopmental disorder with dysmorp
hic facies and cerebellar hypoplasia	
1	
	neurodevelopmental disorder with dysmorp
c facies and behavioral abnormalities	
1	
	neurodevelopmental disorder with craniofa
cial dysmorphism and skeletal defects	
1	
	neurodevelopmental disorder with coarse facie
s and mild distal skeletal abnormalities	
1	
	neurodevelopmental disorder with characteristic fac
ial and ectodermal features and tetraparesis 1	
1	
	neurodevelopmental disorder with cerebral a
trophy and variable facial dysmorphism	
1	
	neurodevelopmental disorder with cere
bellar hypoplasia and spasticity	
1	
	neurodevelopmental disorder with cerebella
r atrophy and with or without seizures	
1	
	neurodevelopmental disorder with cerebe
llar atrophy and motor dysfunction	
1	
	neurodevelopmental disorder with centr
al hypotonia and dysmorphic facies	
1	
	neurodevelopmental disorder with centra
l and peripheral motor dysfunction	
1	
	neurodevelopmental disorder with cataract
s, poor growth, and dysmorphic facies	
1	
	neurodevelopmental disorder with cardiomyopa
thy, spasticity, and brain abnormalities	
1	
	neurodevelopmental disorder with brain anomalies and
with or without vertebral or cardiac anomalies	
1	
	neurodevelopmental disorder with behavioral ab
normalities, absent speech, and hypotonia	
1	
	neurodevelopmental disorder with ataxic gait, abse
nt speech, and decreased cortical white matter	
1	
	neurodevelopmental disorder with al
opelia and brain abnormalities	
1	

	neurodevelopmental disorder with absent speech
and movement and behavioral abnormalities	
1	
	neuroblastoma, suscep
ptibility to, 3	
1	
	neuroblastoma, suscep
ptibility to, 2	
1	
	neuroblastoma, suscep
ptibility to, 1	
1	
	nephronophthisis-lik
e nephropathy 1	
1	
	nephronophth
hisis 7	
1	
	nephronophth
hisis 4	
1	
	nephronophth
hisis 3	
1	
	nephronophth
isis 20	
1	
	nephronophth
isis 19	
1	
	nephronophth
isis 18	
1	
	nephronophth
isis 16	
1	
	nephronophth
isis 15	
1	
	nephronophth
isis 14	
1	
	nephronophth
isis 13	
1	
	nephronophth
isis 12	
1	
	nephronophth
isis 11	
1	
	nephrolithiasis susceptibi
lity caused by SLC26A1	
1	
	nephrogenic syndrome of ina
ppropriate antidiuresis	

1		
		neonatal-onset encephalopathy
with rigidity and seizures		
1		neonatal-onset developmental an
d epileptic encephalopathy		
1		neonatal intrahepatic cholestas
is due to citrin deficiency		
1		neonatal ichthyosis-sclerosi
ng cholangitis syndrome		
1		neonatal encephalopathy with
non-epileptic myoclonus		
1		neonatal encephalomyopathy-cardiomyopa
thy-respiratory distress syndrome		
1		neonatal Marfa
n syndrome		
1		nemaline myo
pathy 9		
1		nemaline myo
pathy 8		
1		nephronopht
hisis 9		
1		nephronophthisis-lik
e nephropathy 2		
1		neuralgic am
yotrophy		
1		nephropathic c
ystinosis		
1		nephrotic syndr
ome, type 9		
1		nephrotic syndr
ome, type 8		
1		nephrotic syndr
ome, type 6		
1		nephrotic syndr
ome, type 3		
1		nephrotic syndro
me, type 24		
1		nephrotic syndro

me, type 23	
1	
	nephrotic syndro
me, type 22	
1	
	nephrotic syndro
me, type 21	
1	
	nephrotic syndro
me, type 20	
1	
	nephrotic syndr
ome, type 2	
1	
	nephrotic syndro
me, type 19	
1	
	nephrotic syndro
me, type 18	
1	
	nephrotic syndro
me, type 17	
1	
	nephrotic syndro
me, type 13	
1	
	nephrotic syndro
me, type 12	
1	
	nephrotic syndro
me, type 11	
1	
	nephrotic syndro
me, type 10	
1	
	nephrotic syndr
ome, IIa 26	
1	
	nephrotic syn
drome 16	
1	
	nephrotic syn
drome 15	
1	
	nephrotic syn
drome 14	
1	
	nephrotic s
yndrome	
1	
	nephropathic infant
ile cystinosis	
1	
	oculocutaneous al
binism type 2	
1	

	oculocutaneous al
binism type 4	
1	progressive external ophthalmoplegia
with mitochondrial DNA deletions	
1	polydactyly, posta
xial, type A8	
1	polymicrogyria with or without vascu
lar-type Ehlers-Danlos syndrome	
1	polymicrogyria with opt
ic nerve hypoplasia	
1	polyhydramnios, megalencephal
y, and symptomatic epilepsy	
1	polyglucosan body m
yopathy type 2	
1	polyglucosan body m
yopathy type 1	
1	polyglucosan body myopathy 1 with
or without immunodeficiency	
1	polyendocrine-polyneu
ropathy syndrome	
1	polydactyly-macroce
phaly syndrome	
1	polydactyly, posta
xial, type a7	
1	polydactyly, posta
xial, type a10	
1	polydactyly, posta
xial, type A9	
1	polydactyly, posta
xial, type A6	
1	polycystic kidne
y disease 4	
1	polydactyly, posta
xial, type A1	
1	polydactyly of a bip
halangeal thumb	
1	polydact
yly	

1		
		polycystic liver disease 4 wit
h or without kidney cysts		
1		
		polycystic liver disease 3 wit
h or without kidney cysts		
1		
		polycystic live
r disease 2		
1		
		polycystic lipomembranous osteodysplasia
with sclerosing leukoencephalopathy 2		
1		
		polycystic lipomembranous osteodysplasia
with sclerosing leukoencephalopathy 1		
1		
		polycystic kidne
y disease 8		
1		
		polycystic kidne
y disease 7		
1		
		polycystic kidney disease 6 with or
without polycystic liver disease		
1		
		polymicrogyria, bilateral
perisylvian, X-linked		
1		
		polymicrogyria, perisylvian, with cereb
ellar hypoplasia and arthrogryposis		
1		
		polyneuro
pathy		
1		
		polyposis syndrome, he
reditary mixed, 1		
1		
		pontocerebellar hypo
plasia, type 14		
1		
		pontocerebellar hypo
plasia, type 12		
1		
		pontocerebellar hypo
plasia, type 11		
1		
		pontocerebellar hyp
oplasia, IIA 17		
1		
		pontocerebellar hyp
oplasia type 9		
1		
		pontocerebellar hyp
oplasia type 8		
1		
		pontocerebellar hyp

oplasia type 6	
1	
oplasia type 5	pontocerebellar hyp
1	
oplasia type 4	pontocerebellar hyp
1	
oplasia type 3	pontocerebellar hyp
1	
oplasia type 2E	pontocerebellar hyp
1	
oplasia type 2D	pontocerebellar hyp
1	
oplasia type 2C	pontocerebellar hyp
1	
oplasia type 2B	pontocerebellar hyp
1	
oplasia type 2A	pontocerebellar hyp
1	
oplasia type 1B	pontocerebellar hyp
1	
oplasia type 1A	pontocerebellar hyp
1	
oplasia type 10	pontocerebellar hyp
1	
opathy with leukoencephalopathy	pontine autosomal dominant microangi
1	
isease syndrome	polyvalvular heart d
1	
tyly 4	polysyndac
1	
matory fibroid, gastrointestinal	polyps, multiple and recurrent inflam
1	
reditary mixed, 2	polyposis syndrome, he
1	
y disease 5	polycystic kidne
1	
without polycystic liver disease	polycystic kidney disease 3 with or
1	

is disorder 6B	peroxisome biogenes
1	
nglioma syndrome 4	pheochromocytoma/paraga
1	
ical disease, primary, 1	pigmented nodular adrenocort
1	
atoderma-skin carcinoma syndrome	pigmentation defects-palmoplantar ker
1	
ylase deficiency	phytanoyl-CoA hydrox
1	
synthetase superactivity	phosphoribosylpyrophosphate
1	
lysinuria	phosphohydroxy
1	
e deficiency, mitochondrial	phosphoenolpyruvate carboxykinas
1	
ase deficiency, cytosolic	phosphoenolpyruvate carboxykin
1	
inzel type	phocomelia, Sch
1	
nglioma syndrome 7	pheochromocytoma/paraga
1	
nglioma syndrome 6	pheochromocytoma/paraga
1	
nglioma syndrome 5	pheochromocytoma/paraga
1	
nglioma syndrome 3	pheochromocytoma/paraga
1	
y disease 2	polycystic kidne
1	
nglioma syndrome 2	pheochromocytoma/paraga
1	
nglioma syndrome 1	pheochromocytoma/paraga
1	
paraganglioma	pheochromocytoma-
1	
nuria	phenylketo

1		
s arteriosus		persistent truncu
1		
itreous, autosomal recessive		persistent hyperplastic primary v
1		
disorder type 3B		peroxisome biogenesis
1		
is disorder 9B		peroxisome biogenes
1		
is disorder 8B		peroxisome biogenes
1		
order 8A (Zellweger)		peroxisome biogenesis dis
1		
is disorder 7B		peroxisome biogenes
1		
ical disease, primary, 2		pigmented nodular adrenocort
1		
ical disease, primary, 3		pigmented nodular adrenocort
1		
ical disease, primary, 4		pigmented nodular adrenocort
1		
inochoroidal atrophy		pigmented paravenous ret
1		
y disease 1		polycystic kidne
1		
s nodosa		polyarteriti
1		
neutropenia		poikiloderma with
1		
y blastoma		pleuropulmonar
1		
ia, Torrance type		platyspondylic dysplas
1		
llebrand disease		platelet-type von Wi
1		
ing disorder 9		platelet-type bleed
1		
		platelet-type bleed

ing disorder 8	1	
ing disorder 20	1	platelet-type bleed
ing disorder 18	1	platelet-type bleed
ing disorder 17	1	platelet-type bleed
ing disorder 15	1	platelet-type bleed
ing disorder 12	1	platelet-type bleed
ing disorder 11	1	platelet-type bleed
ing disorder 10	1	platelet-type bleed
ool deficiency	1	platelet storage p
platelet abnormalities with eosinophilia an	1	
d immune-mediated inflammatory disease	1	
pituitary hormone defic	1	
iciency, combined, 6	1	
pituitary hormone defic	1	
iciency, combined, 2	1	
pituitary hormone defic	1	
iciency, combined, 1	1	
pituitary hormone deficiency,	1	
combined or isolated, 8	1	
pituitary adenoma, growth	1	
hormone-secreting, 2	1	
pili torti-developmental delay-neur	1	
ological abnormalities syndrome	1	
pontocerebellar hypo	1	
plasia, type 15	1	
pontocerebellar hypo	1	
plasia, type 16	1	
pontocerebellar hypo	1	
plasia, type 1C	1	

	primary ciliary d
yskinesia 22	
1	primary ciliary d
yskinesia 33	primary ciliary d
1	
	primary ciliary d
yskinesia 32	
1	primary ciliary d
yskinesia 30	primary ciliary
1	
	primary ciliary d
dyskinesia 3	
1	primary ciliary d
yskinesia 29	primary ciliary d
1	
	primary ciliary d
yskinesia 28	
1	primary ciliary d
yskinesia 27	primary ciliary d
1	
	primary ciliary d
yskinesia 26	
1	primary ciliary d
yskinesia 25	primary ciliary d
1	
	primary ciliary d
yskinesia 24	
1	primary ciliary d
yskinesia 23	primary ciliary d
1	
	primary ciliary d
yskinesia 21	
1	pontocerebellar hypo
plasia, type 1D	primary ciliary d
1	
	primary ciliary
yskinesia 20	
1	primary ciliary d
dyskinesia 2	primary ciliary d
1	
	primary ciliary d
yskinesia 19	
1	primary ciliary d
yskinesia 18	primary ciliary d
1	
	primary ciliary d
yskinesia 17	

1		
		primary ciliary d
yskinesia 16		
1		primary ciliary d
yskinesia 15		
1		primary ciliary d
yskinesia 14		
1		primary ciliary d
yskinesia 13		
1		primary ciliary d
yskinesia 12		
1		primary ciliary d
yskinesia 11		
1		primary ciliary d
yskinesia 34		
1		primary ciliary d
yskinesia 35		
1		primary ciliary
dyskinesia 5		primary ciliary
1		
dyskinesia 6		
1		
		progressive essential tremor-speech impairment-facial dysmor
phism-intellectual disability-abnormal behavior syndrome		
1		
		progressive demyelinating neuropathy
with bilateral striatal necrosis		
1		
		progressive bu
lbar palsy		
1		
		progeroid s
yndrome		
1		
		progeroid features-hepatocellular ca
rcinoma predisposition syndrome		
1		
		progeroid and marfanoid aspec
t-lipodystrophy syndrome		
1		
		primordial dwarfism-immunodefici
ency-lipodystrophy syndrome		
1		
		primary ovaria
n failure		
1		
		primary membranoproliferat

ive glomerulonephritis	
1	
	primary intraosseous v
enous malformation	
1	
	primary immunodeficiency with post-measles
-mumps-rubella vaccine viral infection	
1	
	primary immunodeficiency with natural-killer
cell deficiency and adrenal insufficiency	
1	
	primary immunodeficiency syndr
ome due to p14 deficiency	
1	
	primary hyperoxa
luria type 3	
1	
	primary hyperoxa
luria type 2	
1	
	primary hyperoxa
luria type 1	
1	
	primary failure of
tooth eruption	
1	
	primary eryth
ermalgia	
1	
	primary dysa
utonomia	
1	
	primary coenzyme Q1
0 deficiency 8	
1	
	primary ciliary dyskinesia-ret
initis pigmentosa syndrome	
1	
	primary ciliary
dyskinesia 9	
1	
	primary ciliary
dyskinesia 7	
1	
	primary ciliary d
yskinesia 10	
1	
	primary ciliary
dyskinesia 1	
1	
	primary angle-clo
sure glaucoma	
1	
	predisposition to invasive fungal d
isease due to CARD9 deficiency	
1	

	precocious
puberty	
1	
	potassium-aggrava
ted myotonia	
1	
	postural orthostatic ta
chycardia syndrome	
1	
	posterior uret
hral valve	
1	
	posterior polymorphous
corneal dystrophy 3	
1	
	posterior polymorphous
corneal dystrophy 2	
1	
	posterior column ataxia-retin
itis pigmentosa syndrome	
1	
	postaxial polydactyly-anterior pituitary
anomalies-facial dysmorphism syndrome	
1	
	postaxial acrofaci
al dysostosis	
1	
	portal hypertension,
noncirrhotic, 2	
1	
	portal hypertensio
n, noncirrhotic	
1	
	porphyria due to ALA deh
ydratase deficiency	
1	
	porokeratosis 9, m
ultiple types	
1	
	porokeratosis 8, disseminated
superficial actinic type	
1	
	porokeratosis 7, m
ultiple types	
1	
	porokeratosis 3, disseminated
superficial actinic type	
1	
	porokeratosis 1,
Mibelli type	
1	
	porencephaly-microcephaly-bilatera
l congenital cataract syndrome	
1	
	porenceph
aly 2	

1		
		popliteal pteryg
ium syndrome		
1		
		pontocerebellar hypo
plasia, type 2F		
1		
		pontocerebellar hypo
plasia, type 1F		
1		
		pontocerebellar hypo
plasia, type 1E		
1		
		precocious pubert
y, central, 2		
1		
		preeclampsia/e
clampsia 5		
1		
		primary CD59 d
eficiency		
1		
		pregnancy loss, recurrence
nt, susceptibility		
1		
		pretibial dystrophic ep
idermolysis bullosa		
1		
		prenatal-onset spinal muscular atroph
y with congenital bone fractures		
1		
		premature ovaria
n failure 9		
1		
		premature ovaria
n failure 8		
1		
		premature ovaria
n failure 7		
1		
		premature ovaria
n failure 6		
1		
		premature ovaria
n failure 5		
1		
		premature ovaria
n failure 3		
1		
		premature ovaria
n failure 2B		
1		
		premature ovaria
n failure 2A		
1		
		premature ovaria

n failure 21	
1	
n failure 20	premature ovaria
1	
n failure 19	premature ovaria
1	
n failure 18	premature ovaria
1	
n failure 17	premature ovaria
1	
n failure 16	premature ovaria
1	
n failure 14	premature ovaria
1	
n failure 13	premature ovaria
1	
n failure 12	premature ovaria
1	
n failure 11	premature ovaria
1	
onic lethality 2	preimplantation embryo
1	
onic lethality 1	preimplantation embryo
1	
susceptibility to, 3	pregnancy loss, recurrent,
1	
order 7A (Zellweger)	peroxisome biogenesis dis
1	
order 6A (Zellweger)	peroxisome biogenesis dis
1	
binism type 6	oculocutaneous al
1	
syndrome V	orofaciodigital
1	
dysplasia	osteofibrous
1	
tenosis	osteocranios
1	

and overlapping malformed digits

1	
ysplasia	
1	
dissecans	
1	
otension 2	
1	
otension 1	
1	
duria	
1	
ndrome type II	
1	
ndrome type 14	
1	
syndrome XV	
1	
syndrome IX	
1	
left 15	
1	
syndrome IV	
1	
syndrome III	
1	
syndrome I	
1	
syndrome 20	
1	
syndrome 19	
1	
syndrome 18	
1	
syndrome 17	
1	
syndrome 16	

[illegible]

1		
		orofaciodigita
l syndrome		
1		orofacial cleft 6, s
usceptibility to		
1		orofacial c
left 5		
1		osteogenesis imper
fecta type 10		osteogenesis imper
1		
		osteogenesis imper
fecta type 11		
1		osteogenesis imper
fecta type 12		osteogenesis imper
1		
		otitis m
fecta type 13		
1		osteosclerosis-developmental del
edia		osteoporosis-pseudo
1		
		osteoporosis, childhood- or juvenile
ay-craniosynostosis syndrome		
1		osteopor
glioma syndrome		osteopetrosis, autos
1		
		osteopathia striata wit
-onset, with developmental delay		
1		osteotohepatoent
osis		osteoglophonic
1		
		osteogenesis imper
omal dominant 3		
1		osteogenesis imper
h cranial sclerosis		osteogenesis imper
1		
		osteogenesis imper
eric syndrome		
1		
dysplasia		
1		
fecta, type 21		
1		
fecta, type 20		
1		

fecta, type 19	
1	
fecta, type 18	osteogenesis imper
1	
fecta, IIA 22	osteogenesis imper
1	
rfecta type 9	osteogenesis impe
1	
rfecta type 8	osteogenesis impe
1	
rfecta type 7	osteogenesis impe
1	
rfecta type 6	osteogenesis impe
1	
rfecta type 5	osteogenesis impe
1	
fecta type 17	osteogenesis imper
1	
fecta type 16	osteogenesis imper
1	
fecta type 15	osteogenesis imper
1	
fecta type 14	osteogenesis imper
1	
left 3	orofacial c
1	
left 11	orofacial c
1	
is disorder 5B	peroxisome biogenes
1	
l syndrome	oculotrichoana
1	
n defect 14	oocyte maturatio
1	
n defect 13	oocyte maturatio
1	
n defect 12	oocyte maturatio
1	

	onychodystrophy, osteodystrophy, impaired intel
lectual development, and seizures syndrome	
1	
	oligospe
rmia	
1	
	oligodontia-cancer pred
isposition syndrome	
1	
	oligoasthenotera
tozoospermia	
1	
	odontoleukod
ystrophy	
1	
	odontohypopho
sphatasia	
1	
	odontochondrodysplasia 2 with
hearing loss and diabetes	
1	
	odonto-onycho-der
mal dysplasia	
1	
	oculopharyngodist
al myopathy 4	
1	
	orofacial c
left 10	
1	
	oculopharyngodist
al myopathy 2	
1	
	oculopharyngodist
al myopathy 1	
1	
	oculopharyngodis
tal myopathy	
1	
	oculopharyngeal musc
ular dystrophy 2	
1	
	oculopharyngeal mus
cular dystrophy	
1	
	oculomotor-abduce
ns synkinesis	
1	
	oculogastrointestinal-neuro
developmental syndrome	
1	
	oculodentodigital dysplasi
a, autosomal recessive	
1	
	oculodentodigita
l dysplasia	

1		
		oculocutaneous al
binism type 8		
1		oculocutaneous al
binism type 7		
1		oocyte maturati
on defect 2		
1		oocyte maturati
on defect 3		
1		oocyte maturati
on defect 4		
1		oocyte maturati
on defect 5		
1		ornithine transloc
ase deficiency		
1		ornithine carbamoyltran
sferase deficiency		
1		ornithine aminotransf
erase deficiency		
1		
		optic atrophy-ataxia-peripheral neuropath
y-global developmental delay syndrome		
1		
		optic atrophy with or without deafness, ophthal
moplegia, myopathy, ataxia, and neuropathy		
1		
		optic atro
phy 9		
1		optic atro
phy 5		
1		optic atro
phy 3		
1		optic atro
phy 14		
1		optic atrophy 13 with retinal
and foveal abnormalities		
1		optic atro
phy 12		
1		optic atro
phy 11		
1		optic atrophy 10 with or without ataxia,

intellectual disability, and seizures	
1	
	opsismodys
plasia	
1	
	ophthalmoplegia, external, with
rib and vertebral anomalies	
1	
	oocyte/zygote/embryo ma
turation arrest 20	
1	
	oocyte/zygote/embryo ma
turation arrest 19	
1	
	oocyte/zygote/embryo ma
turation arrest 18	
1	
	oocyte/zygote/embryo ma
turation arrest 17	
1	
	oocyte maturati
on defect 9	
1	
	oocyte maturati
on defect 8	
1	
	oocyte maturati
on defect 7	
1	
	oocyte maturati
on defect 6	
1	
	otofacial neurodevelo
pmental syndrome	
1	
	otofaciocervical
syndrome 2	
1	
	otopalatodigital s
ndrome type 1	
1	
	patterned macular
dystrophy 2	
1	
	perinatal lethal hy
pophosphatasia	
1	
	pentosu
ria	
1	
	pelviscapular
dysplasia	
1	
	pellag
ra	
1	

1	peeling skin-leukonuchia-acral punctate ker
1	atoses-cheilitis-knuckle pads syndrome
1	peeling skin s
1	yndrome 6
1	peeling skin s
1	yndrome 5
1	peeling skin s
1	yndrome 4
1	peeling skin s
1	yndrome 1
1	pediatric l
1	ymphoma
1	patterned macular
1	dystrophy 3
1	patent ductus a
1	rteriosus 3
1	otopalatodigital s
1	yndrome type 2
1	partial lipodystrophy, congenital catara
1	cts, and neurodegeneration syndrome
1	partial deletion of the sho
1	rt arm of chromosome 3
1	partial corpus callosum agenesis-cerebellar vermis
1	hypoplasia with posterior fossa cysts syndrome
1	partial androgen insen
1	sitivity syndrome
1	paroxysmal nocturnal
1	hemoglobinuria
1	paroxysmal extreme
1	pain disorder
1	paroxysmal dy
1	skinesia
1	parkinsonism-dystoni
1	a, infantile, 2
1	parkinsonism-dysto
1	nia, infantile
1	parkinsonism with
1	polyneuropathy

1		
		parkinsonism due to A
TP13A2 deficiency		
1		
		periodic fever–infantile enterocol
itis–autoinflammatory syndrome		
1		
		periodontitis, a
aggressive 1		
1		
		peripheral ne
uropathy		
1		
		peripheral neuropathy, autosomal recessive, with
or without impaired intellectual development		
1		
		peroxisome biogenesis dis
order 5A (Zellweger)		
1		
		peroxisome biogenes
is disorder 4B		
1		
		peroxisome biogenesis dis
order 4A (Zellweger)		
1		
		peroxisome biogenesis dis
order 3A (Zellweger)		
1		
		peroxisome biogenes
is disorder 2B		
1		
		peroxisome biogenesis dis
order 2A (Zellweger)		
1		
		peroxisome biogenes
is disorder 1B		
1		
		peroxisome biogenesis dis
order 1A (Zellweger)		
1		
		peroxisome biogenes
is disorder 14B		
1		
		peroxisome biogenesis dis
order 13A (Zellweger)		
1		
		peroxisome biogenesis dis
order 12A (Zellweger)		
1		
		peroxisome biogenes
is disorder 11B		
1		
		peroxisome biogenesis dis
order 11A (Zellweger)		
1		
		peroxisome biogenes

is disorder 10B	
1	
order 10A (Zellweger)	peroxisome biogenesis dis
1	
xidase deficiency	peroxisomal acyl-CoA o
1	
abetes mellitus 1	permanent neonatal di
1	
ar heterotopia 9	periventricular nodul
1	
ar heterotopia 8	periventricular nodul
1	
ar heterotopia 7	periventricular nodul
1	
ar heterotopia 6	periventricular nodul
1	
crocephaly, autosomal recessive	periventricular heterotopia with mi
1	
rseness-hearing loss syndrome	peripheral neuropathy-myopathy-hoa
1	
eidocranial dysplasia	parietal foramina with cl
1	
amina 2	parietal for
1	
amina 1	parietal for
1	
ma, Nagashima type	palmoplantar keratoder
1	
triate, focal, or diffuse	palmoplantar keratoderma i, s
1	
dysmorphism-developmental delay syndrome	palatal anomalies-widely spaced teeth-facial
1	
ongenita 4	pachyonychia c
1	
ongenita 3	pachyonychia c
1	
ongenita 2	pachyonychia c
1	

		pachyonychia c
ongenita 1		
1		
	pachygyria, microcephaly, developmental delay, an	
d dysmorphic facies, with or without seizures		
1		
	oxoglutarica	
ciduria		
1		
	overhydrated heredita	
ry stomatocytosis		
1		
	overgrowth-macrocephaly-faci	
al dysmorphism syndrome		
1		
	overgrowth s	
yndrome		
1		
	ovarian hyperstimul	
ation syndrome		
1		
	ovarian dysg	
enesis 9		
1		
	ovarian dysg	
enesis 8		
1		
	ovarian dysg	
enesis 7		
1		
	ovarian dysg	
enesis 6		
1		
	ovarian dysg	
enesis 5		
1		
	ovarian dysg	
enesis 3		
1		
	ovarian dysg	
enesis 2		
1		
	ovarian dysge	
nesis 10		
1		
	ovarian dysg	
enesis 1		
1		
	otospondylomegaepiphyseal dysp	
lasia, autosomal dominant		
1		
	otoscleros	
is 11		
1		
	palmoplantar keratoder	
ma, Bothnian type		

1		
		palmoplantar keratoderma, no
nepidermolytic, focal 1		
1		
		parenti-mignot neurodev
elopmental syndrome		
1		
		palmoplantar keratoderma, nonepi
dermolytic, focal or diffuse		
1		
		parathyroid glan
d carcinoma		
1		
		parastreumatic
dwarfism		
1		
		paramyotonia congenita
of Von Eulenburg		
1		
		papillary renal c
ell carcinoma		
1		
		pantothenate kinase-associ
ated neurodegeneration		
1		
		panhypopituitari
sm, X-linked		
1		
		pancytopenia-developmen
tal delay syndrome		
1		
		pancytopenia due to
IKZF1 mutations		
1		
		pancreatic triacylglycer
ol lipase deficiency		
1		
		pancreatic insufficiency-anem
ia-hyperostosis syndrome		
1		
		pancreatic cancer, sus
ceptibility to, 5		
1		
		pancreatic cancer, sus
ceptibility to, 4		
1		
		pancreatic cancer, sus
ceptibility to, 3		
1		
		pancreatic cancer, sus
ceptibility to, 2		
1		
		pancreatic cancer, sus
ceptibility to, 1		
1		
		pancreatic ag

enesis 2	
1	
	pancreatic ag
enesis 1	
1	
	palmoplantar
keratosis	
1	
	palmoplantar keratoderma-s
clerodactyly syndrome	
1	
	palmoplantar keratoderma-esop
hageal carcinoma syndrome	
1	
	palmoplantar keratoderm
a-deafness syndrome	
1	
	palmoplantar keratoderma-XX sex reversal-predis
osition to squamous cell carcinoma syndrome	
1	
	palmoplantar keratoderm
a, punctate type 1A	
1	
	hepatoencephalopathy due to combined oxi
native phosphorylation defect type 1	
1	
	fetal akinesia deform
ation sequence 2	
1	
	hepatic veno-occlusive disease
-immunodeficiency syndrome	
1	
	Singleton-Merten
syndrome 2	
1	
	Spondyloenchondrodysplasia w
ith immune dysregulation	
1	
	Sotos synd
rome 1	
1	
	Sorsby fundus
dystrophy	
1	
	Snijders Blok-Cam
peau syndrome	
1	
	Sneddon sy
ndrome	
1	
	Smith-McCort d
ysplasia 2	
1	
	Smith-McCort d
ysplasia 1	
1	

		Smith-Magenis
syndrome		
1		
		Smith-Lemli-Opie
tz syndrome		
1		
		Skraban-Deardorff
ff syndrome		
1		
		Sjogren-Larsson
n syndrome		
1		
		Singleton-Merten
syndrome 1		
1		
	THOC6-related developmental delay-micro	
cephaly-facial dysmorphism syndrome		
1		
		Simpson-Golabi-Behmel
syndrome type 2		
1		
		Simpson-Golabi-Behmel
syndrome type 1		
1		
		Simpson-Golabi-Behmel
hmel syndrome		
1		
	Silverman-Handmaker type d	
yssegmental dysplasia		
1		
		Silver-Russell
syndrome 5		
1		
		Silver-Russell
syndrome 3		
1		
		Silver-Russell
syndrome		
1		
		Sifrim-Hitz-Wei
ss syndrome		
1		
		Siddiqi syndrome
ndrome		
1		
		Shwachman-Diamond
d syndrome 2		
1		
		Shwachman-Diamond
d syndrome 1		
1		
		Stankiewicz-Isidor
or syndrome		
1		
		Stargardt disease
sease 3		

1		
		Steel syn
drome		
1		
		Stickler syndr
ome type 1		
1		
		TEL02-related intellectual disabili
ty-neurodevelopmental disorder		
1		
		TEK-related prim
ary glaucoma		
1		
		TCR-alpha-beta-positive
T-cell deficiency		
1		
		TCF12-related cra
niosynostosis		
1		
		TARP synd
rome		
1		
		T-cell lymphopenia, infantile, with or with
out nail dystrophy, autosomal dominant		
1		
		T-cell immunodeficiency, congenita
l alopecia, and nail dystrophy		
1		
		T-cell immunodeficiency with epi
dermodysplasia verruciformis		
1		
		T-B+ severe combined immunodeficienc
y due to gamma chain deficiency		
1		
		T-B+ severe combined immunodefic
iency due to JAK3 deficiency		
1		
		T-B+ severe combined immunodeficien
cy due to IL-7Ralpha deficiency		
1		
		T-B+ severe combined immunodefic
iency due to CD45 deficiency		
1		
		Sweeney-Cox
syndrome		
1		
		Suleiman-El-Hatt
ab syndrome		
1		
		Stüve-Wiedemann
syndrome 1		
1		
		Stüve-Wiedeman
n syndrome		
1		
		Sturge-Weber

syndrome	
1	
	Stromme sy
ndrome	
1	
	Stickler syndrome, type
I, nonsyndromic ocular	
1	
	Stickler syndro
me, type 6	
1	
	Stickler syndro
me, type 5	
1	
	Stickler syndro
me, type 4	
1	
	Stickler syndr
ome type 2	
1	
	Shukla-Vernon
syndrome	
1	
	Short stature, Daub
er-Argente type	
1	
	Shashi-Pena
syndrome	
1	
	SYNGAP1-related developmental a
nd epileptic encephalopathy	
1	
	STT3B-congenital disord
er of glycosylation	
1	
	STT3A-congenital disord
er of glycosylation	
1	
	STING-associated vasculopath
y with onset in infancy	
1	
	SSR4-congenital disorde
r of glycosylation	
1	
	SRD5A3-congenital disord
er of glycosylation	
1	
	SPAST-related mo
tor disorder	
1	
	SOX3-related X-linked pituitary hormone deficiency wi
th or without intellectual developmental disorder	
1	
	SNRNP200-related domi
nant retinopathy	
1	

1		SMARCC1-associated developme	
1	ntal dysgenesis syndrome		
1			
1	orter deficiency syndrome	SLC6A3-related dopamine transp	
1			
1	CDG	SLC39A8-	
1			
1	der of glycosylation	SLC35A2-congenital disor	
1			
1	der of glycosylation	SLC35A1-congenital disor	
1			
1	l disability syndrome	SIN3A-related intellectua	
1			
1	ort stature	SHOX-related sh	
1			
1	drome	SHORT syn	
1			
1	itial lung disease	SFTPC- related interst	
1			
1	cial dysostosis	SF3B4-related acrofa	
1			
1	ithout or with macrocephaly/overgrowth	SETD2-related neurodevelopmental disorder w	
1			
1	ability-multiple congenital anomalies syndrome	SETD2-related microcephaly-severe intellectual dis	
1			
1	drome	SERKAL syn	
1			
1	d myopathy	SELENON-relate	
1			
1	tic liver disease	SEC61B-related polycys	
1			
1	elopmental disorder	SYNCRIP-related neurodev	
1			
1	ease	Salla dis	
1			
1	yndrome 9	Senior-Loken s	
1			
1	isease	Sandhoff d	

1		
		Senior-Loken s
yndrome 8		
1		Senior-Loken s
yndrome 7		
1		Senior-Loken s
yndrome 5		
1		Senior-Loken s
yndrome 4		
1		
		Seckel synd
rome 9		
1		Seckel synd
rome 8		
1		Seckel synd
rome 7		
1		Seckel synd
rome 6		
1		Seckel synd
rome 5		
1		Seckel synd
rome 2		
1		Seckel synd
rome 10		
1		Seckel synd
rome 1		
1		Scott syn
drome		
1		Schöpf-Schulz-Pass
arge syndrome		
1		Schwartz-Jampel sy
ndrome type 1		
1		Schwartz-Jampe
l syndrome		
1		Schuurs-Hoeijmak
ers syndrome		
1		Schnyder cornea
l dystrophy		
1		Schmid metaphyseal c

hondrodysplasia	
1	
	Schinzel-Giedio
n syndrome	
1	
	Schimke immuno-oss
eous dysplasia	
1	
	Scheie syn
drome	
1	
	Schaaf-Yang
syndrome	
1	
	TFRC-related combined
immunodeficiency	
1	
	TMEM165-congenital disor
der of glycosylation	
1	
	SCN4A-related myopathy,
autosomal recessive	
1	
	Visser-Bodmer
syndrome	
1	
	Waardenburg synd
rome type 4C	
1	
	Waardenburg synd
rome type 4B	
1	
	Waardenburg synd
rome type 4A	
1	
	Waardenburg synd
rome type 3	
1	
	Waardenburg synd
rome type 2E	
1	
	Waardenburg synd
rome type 2D	
1	
	Waardenburg synd
rome type 2A	
1	
	Waardenburg synd
rome type 1	
1	
	WHIM syndr
ome 2	
1	
	WHIM syndr
ome 1	
1	

		WHIM synd
rome		
1		
		Vici synd
rome		
1		
		TMEM199-
CDG		
1		
		Ververi-Brady
syndrome		
1		
		VISS synd
rome		
1		
		VACTERL with hy
drocephalus		
1		
		Usmani-Riazuddin syndrom
e, autosomal recessive		
1		
		Usmani-Riazuddin syndrom
e, autosomal dominant		
1		
		Usher syndrom
e, type 4		
1		
		Usher syndrom
e, type 1M		
1		
		Usher syndrom
e type 3B		
1		
		Usher syndrom
e type 3A		
1		
		Usher syndrom
e type 2D		
1		
		Usher syndrom
e type 2A		
1		
		Waardenburg synd
rome, IIa 2F		
1		
		Wagner di
sease		
1		
		Warburg micro
syndrome 1		
1		
		Warburg micro
syndrome 2		
1		
		Wolf-Hirschhor
n syndrome		

1		
		Wolcott-Ralliso
n syndrome		
1		Wiskott-Aldrich
syndrome 2		
1		Winchester s
yndrome		
1		Wilson di
sease		
1		Wilms tum
or 6		
1		Wilms tum
or 5		
1		Wilms tum
or 1		
1		Wiedemann-Stein
er syndrome		
1		Wiedemann-Rautenst
rauch syndrome		
1		Wieacker-Wolff syndrom
e, female-restricted		
1		Wieacker-Wolff
syndrome		
1		White-Kernohan
syndrome		
1		Werner syn
drome		
1		Weiss-Kruszka
syndrome		
1		Weill-Marchesani
syndrome 3		
1		Weill-Marchesani synd
rome 2, dominant		
1		Weill-Marchesani
syndrome 1		
1		Weill-Marchesani 4 sy
ndrome, recessive		
1		Webb-Dattani

syndrome		
1		
		Warsaw breakag
e syndrome		
1		
		Warburg micro
syndrome 4		
1		
		Warburg micro
syndrome 3		
1		
		Usher syndrom
e type 1J		
1		
		Usher syndrom
e type 1G		
1		
		Usher syndrom
e type 1F		
1		
		Tessadori-van Haaften neurod
developmental syndrome 1		
1		
		Tessadori-Van Haaften neurod
developmental syndrome 3		
1		
		Tenorio sy
ndrome		
1		
		Temple-Baraitse
r syndrome		
1		
		Teebi hypertelori
sm syndrome 2		
1		
		Tay-Sachs diseas
e AB variant		
1		
		Tay-Sachs d
isease		
1		
		Tatton-Brown-Rahman ov
ergrowth syndrome		
1		
		Tangier di
sease		
1		
		Tan-Almurshedi
syndrome		
1		
		TWIST1-related cra
niosynostosis		
1		
		TUBB4A-related neur
ologic disorder		
1		

	TUBB3-related tu
bulinopathy	
1	
	TTN-related
myopathy	
1	
	TSPAN12-related vit
reoretinopathy	
1	
	TRPV4-related bo
ne disorder	
1	
	TRPM1-related r
etinopathy	
1	
	TRIP11-related skel
etal dysplasia	
1	
	TRAF3 haploins
ufficiency	
1	
	TPM4-related plat
elet disorder	
1	
	TPM3-related
myopathy	
1	
	TPM2-related
myopathy	
1	
	TNF receptor 1-associated p
eriodic fever syndrome	
1	
	TMEM63B-related developmental and epi
leptic encephalopathy with anemia	
1	
	Tessadori-Van Haaften neurod
developmental syndrome 4	
1	
	Tessadori-van Haaften neurod
developmental syndrome 2	
1	
	Usher syndrom
e type 1D	
1	
	Tessier number 4
facial cleft	
1	
	Usher syndrom
e type 1C	
1	
	Usher syndrom
e type 1B	
1	
	Ullrich congenital mus
cular dystrophy 2	

1		
		Ullrich congenital mus
cular dystrophy 1C		
1		Ullrich congenital mus
cular dystrophy 1B		
1		UV-sensitive s
ndrome 3		
1		UV-sensitive s
ndrome 2		
1		UV-sensitive s
ndrome 1		
1		UROD-related inher
ited porphyria		
1		Turcot syndrome w
ith polyposis		
1		Troyer syn
drome		
1		Treacher Collins
syndrome 4		
1		Treacher Collins
syndrome 3		
1		Treacher Collins
syndrome 2		
1		Treacher Collins
syndrome 1		
1		Townes-Brocks
syndrome 2		
1		Townes-Brocks
syndrome 1		
1		Toriello-Carey
syndrome		
1		Tolchin-Le Caign
ec syndrome		
1		Timothy sy
ndrome		
1		Tietz syn
drome		
1		Thomsen and Bec

ker disease	
1	
Thiel-Behnke corn	
1	
SEC61A1 def	
1	
SATB2 associate	
1	
Melnick-Needle	
1	
Noonan synd	
1	
Oguchi dis	
1	
Oguchi dis	
1	
Ogden syn	
1	
OPA1-related optic atrophy with or	
1	
O'Donnell-Luria-R	
1	
Norum dis	
1	
North Carolina mac	
1	
Norrie di	
1	
Noonan syndrome-like disorder	
1	
Noonan syndrome-like disorder	
1	
Noonan syndrome and Noon	
1	
Noonan synd	
1	
PRPF31-related	
1	

	Noonan synd
rome 7	
1	
	Noonan synd
rome 6	
1	
	Noonan synd
rome 5	
1	
	Noonan synd
rome 4	
1	
	Noonan synd
rome 3	
1	
	Noonan synd
rome 2	
1	
	Noonan synd
rome 14	
1	
	Noonan synd
rome 13	
1	
	Noonan synd
rome 11	
1	
	Noonan synd
rome 10	
1	
	Noonan synd
rome 1	
1	
	Okur-Chung neurodevel
opmental syndrome	
1	
	Olmsted syn
drome 1	
1	
	Olmsted syn
drome 2	
1	
	Olmsted syndrom
e, X-linked	
1	
	PRKAR1B-related neurodegenerative dem
entia with intermediate filaments	
1	
	PRKAG2-related ca
rdiomyopathy	
1	
	PPFIA3-related neurodev
elopmental disorder	
1	
	PPARG-related familial p
artial lipodystrophy	

1		
		POLE-related polyposis and co
lorectal cancer syndrome		
1		
		POLD1-related polyposis and c
lorectal cancer syndrome		
1		
		PNPLA6-related spastic paraple
gia with or without ataxia		
1		
		PMM2-congenital disorde
r of glycosylation		
1		
		PLIN1-related familial p
artial lipodystrophy		
1		
		PLA2G6-associated ne
urodegeneration		
1		
		PIEZ01-related generalized lymphatic dysp
lasia with non-immune hydrops fetalis		
1		
		PHIP-related behavioral problems-intellectual dis
ability-obesity-dysmorphic features syndrome		
1		
		PHGDH defi
ciency		
1		
		PHARC syn
drome		
1		
		PGM1-congenital disorde
r of glycosylation		
1		
		PERCHING sy
ndrome		
1		
		PEHO-like s
ndrome		
1		
		PCWH synd
rome		
1		
		PCARE-related r
etinopathy		
1		
		PAX6-related ocul
ar dysgenesis		
1		
		PAX5-related B lymphopenia and
autism spectrum disorder		
1		
		PAICS defi
ciency		
1		
		P5CS defic

iciency		
1		
		Nizon-Isidor
syndrome		
1		
		Nijmegen breakage synd
rome-like disorder		
1		
		Nijmegen breaka
ge syndrome		
1		
		NARP synd
rome		
1		
		NACC1-related neurodevelopmental disorder with e
pilepsy, cataracts and episodic irritability		
1		
		NAA10-related
syndrome		
1		
		N-acetylaspartat
e deficiency		
1		
		Myhre syn
drome		
1		
		Mungan syn
drome		
1		
		Mullegama-Klein-Mar
tinez syndrome		
1		
		Muenke syn
drome		
1		
		Muckle-Wells
syndrome		
1		
		Moyamoya disease with ea
rly-onset achalasia		
1		
		Moyamoya di
sease 5		
1		
		Moyamoya di
sease 2		
1		
		Mowat-Wilson
syndrome		
1		
		Miyoshi my
opathy		
1		
		Miyoshi muscular
dystrophy 3		
1		

	Mitchell-Riley
syndrome	
1	
	Mitchell sy
ndrome	
1	
	Menkes di
sease	
1	
	Menke-Hennekam
syndrome 1	
1	
	Mendelian susceptibility to mycobacterial d
iseases due to partial STAT1 deficiency	
1	
	Mendelian susceptibility to mycobacterial d
iseases due to partial IRF8 deficiency	
1	
	Mendelian susceptibility to mycobacterial di
seases due to complete ISG15 deficiency	
1	
	Mendelian susceptibility to mycobacterial dis
eases due to complete IL12RB1 deficiency	
1	
	Mendelian susceptibility to mycobacterial di
seases due to complete IL12B deficiency	
1	
	NAD(P)HX dehydrata
se deficiency	
1	
	NDE1-related microh
ydranencephaly	
1	
	Niemann-Pick dise
ase, type C2	
1	
	NEK9-related lethal s
keletal dysplasia	
1	
	Niemann-Pick dise
ase, type C1	
1	
	Niemann-Pick dis
ease type B	
1	
	Niemann-Pick dis
ease type A	
1	
	Niemann-Pick
disease	
1	
	Newfoundland cone-
rod dystrophy	
1	
	Neu-Laxova sy
ndrome 2	

1		
		Netherton s
syndrome		
1		
		Nestor-Guillermo pr
ogeria syndrome		
1		
		Naxos dis
ease		
1		
		Nance-Horan
syndrome		
1		
		Nager acrofacial
dysostosis		
1		
		Naegeli-Franceschetti-
Jadassohn syndrome		
1		
		NYX-related re
tinopathy		
1		
		NTHL1-deficiency tumor pr
edisposition syndrome		
1		
		NR2F2 related multiple congenital
anomalies/dysmorphic syndrome		
1		
		NPHP3-related Mecke
l-like syndrome		
1		
		NOG-related symphalangis
m spectrum disorder		
1		
		NMNAT1-related
retinopathy		
1		
		NKX2.5-related congenital, conducti
on and myopathic heart disease		
1		
		NKX2-1 related choreoathetosis and congenital hypot
hyroidism with or without pulmonary dysfunction		
1		
		NIK defic
iency		
1		
		NGLY1-defi
ciency		
1		
		NF2-related schw
annomatosis		
1		
		PRPF19-related neurodev
elopmental disorder		
1		
		PRPH2-related r

etinopathy
 1 |
 |
 o EP300 haploinsufficiency
 1 |
 |
 etinopathy
 1 |
 |
 a syndrome
 1 |
 |
 ll syndrome
 1 |
 |
 syndrome
 1 |
 |
 myopathy
 1 |
 |
 etinopathy
 1 |
 |
 ive retinopathy
 1 |
 |
 ant retinopathy
 1 |
 |
 tinopathy
 1 |
 |
 ve retinopathy
 1 |
 |
 nt retinopathy
 1 |
 |
 um disorder
 1 |
 |
 rome
 1 |
 |
 y disorder
 1 |
 |
 drome
 1 |
 |
 r of glycosylation
 1 |
 |
 d syndrome
 1 |

Rubinstein-Taybi syndrome due t
 |
 RLBP1-related r
 |
 Radio-Tartagli
 |
 Rabson-Mendenha
 |
 Rabin-Pappas
 |
 RYR1-related
 |
 RPGR-related r
 |
 RPE65-related recess
 |
 RPE65-related domin
 |
 RP2-related re
 |
 RP1-related recessi
 |
 RP1-related domina
 |
 RNU4ATAC spectr
 |
 RIN2 synd
 |
 PRPS1 deficienc
 |
 RIDDLE syn
 |
 RFT1-congenital disorde
 |
 RECON progeroi
 |

	RDH5-related r
etinopathy	
1	
	RDH12-related recess
ive retinopathy	
1	
	RCBTB1-related
retinopathy	
1	
	RAB28-related r
etinopathy	
1	
	RAB23-related Carp
enter syndrome	
1	
	RAB18 defi
ciency	
1	
	Quebec platele
t disorder	
1	
	Pyle dis
ease	
1	
	Rafiq syn
drome	
1	
	Rahman syn
drome	
1	
	Rajab interstitial lung disease
with brain calcifications 1	
1	
	Rajab interstitial lung disease
with brain calcifications 2	
1	
	Rubinstein-Taybi syndrome d
ue to CREBBP mutations	
1	
	Rothmund-Thomson sy
ndrome, type 3	
1	
	Rothmund-Thomson s
yndrome type 2	
1	
	Rothmund-Thomson s
yndrome type 1	
1	
	Rothmund-Thomso
n syndrome	
1	
	Roifman sy
ndrome	
1	
	Roberts-SC phocom
elia syndrome	

1		
		Roberts sy
ndrome		
1		
		Ritscher-Schinze
l syndrome 4		
1		
		Ritscher-Schinze
l syndrome 3		
1		
		Ritscher-Schinze
l syndrome 2		
1		
		Ritscher-Schinze
l syndrome 1		
1		
		Riley-Day s
yndrome		
1		
		Rienhoff sy
ndrome		
1		
		Richieri Costa-Per
eira syndrome		
1		
		Revesz syn
drome		
1		
		Rett syndrome, cong
enital variant		
1		
		Rett synd
rome		
1		
		Renpenning s
yndrome		
1		
		Reis-Bucklers corn
eal dystrophy		
1		
		Rauch-Steindl
syndrome		
1		
		Rapp-Hodgkin
syndrome		
1		
		Ramon syn
drome		
1		
		Proteus-like
syndrome		
1		
		Proteus sy
ndrome		
1		
		Primrose sy

ndrome	
1	
sease, classic form	Pelizaeus-Merzbacher di
1	
anomaly	Pelger-Huet
1	
yndrome	Partington s
1	
dominant, susceptibility to	Parkinson disease 5, autosomal
1	
dominant, susceptibility to	Parkinson disease 26, autosomal
1	
dominant, susceptibility to	Parkinson disease 24, autosomal
1	
autosomal dominant	Parkinson disease 22,
1	
dominant, susceptibility to	Parkinson disease 18, autosomal
1	
sease 17	Parkinson di
1	
dominant, susceptibility to	Parkinson disease 11, autosomal
1	
syndrome	Parkes Weber
1	
re disease	Papillon-Lefev
1	
n syndrome	Pan-Chung-Belle
1	
syndrome	Pallister-Hall
1	
of bone 6	Paget disease
1	
of bone 3	Paget disease
1	
e 2, early-onset	Paget disease of bon
1	
o syndrome	Paganini-Miozz
1	

	PYCR1-related de B
arsy syndrome	
1	
	PURA-related severe neonatal hypotonia-seizures-en
cephalopathy syndrome due to a point mutation	
1	
	PTEN hamartoma tu
mor syndrome	
1	
	PSPH defic
iency	
1	
	PSAT defic
iency	
1	
	PRRT2-associated paroxysm
al movement disorder	
1	
	Pelizaeus-Merzbacher disea
se in female carriers	
1	
	Pelizaeus-Merzbacher dis
ease, connatal form	
1	
	Prieto syn
drome	
1	
	Pelizaeus-Merzbacher disea
se, transitional form	
1	
	Prader-Willi
syndrome	
1	
	PrP systemic a
myloidosis	
1	
	Potocki-Shaffe
r syndrome	
1	
	Potocki-Lupski
syndrome	
1	
	Poirier-Bienvenu neurode
velopmental syndrome	
1	
	Pitt-Hopkins-lik
e syndrome 2	
1	
	Pitt-Hopkins
syndrome	
1	
	Pilarowski-Bjorns
son syndrome	
1	
	Pierson sy
ndrome	

1		
		Pierpont sy
ndrome		
1		
		Phelan-McDermi
d syndrome		
1		
		Pfeiffer syndr
ome type 3		
1		
		Pfeiffer syndr
ome type 2		
1		
		Peutz-Jeghers
syndrome		
1		
		Peters plus
syndrome		
1		
		Perry syn
drome		
1		
		Perrault syn
drome 6		
1		
		Perrault syn
drome 4		
1		
		Perrault syn
drome 3		
1		
		Perrault syn
drome 2		
1		
		Perrault syn
drome 1		
1		
		Perlman sy
ndrome		
1		
		Pelizeaus-Merzbacher
spectrum disorder		
1		
		Wolff-Parkinson-W
hite syndrome		
1		
		Wolfram syn
drome 1		
1		
		Wolfram syn
drome 2		
1		
		arthrogryposis multiplex cong
enita 2, neurogenic type		
1		
		arthrogryposis, distal, with impa

ired proprioception and touch	
1	
stal, type 2B2	arthrogryposis, di
1	
stal, type 1C	arthrogryposis, di
1	
stal, type 1B	arthrogryposis, di
1	
stal, type 1A	arthrogryposis, di
1	
stal, type 12	arthrogryposis, di
1	
istal, IIa 11	arthrogryposis, d
1	
s, and impaired intellectual development	arthrogryposis, cleft palate, craniosynostosi
1	
plex congenita 5	arthrogryposis multi
1	
nic, with agenesis of the corpus callosum	arthrogryposis multiplex congenita 4, neuroge
1	
genita 3, myogenic type	arthrogryposis multiplex con
1	
1, neurogenic, with myelin defect	arthrogryposis multiplex congenita
1	
n, familial, 14	atrial fibrillatio
1	
posis	arthrogry
1	
ity syndrome	arterial tortuos
1	
sorder	arterial di
1	
eralized, of infancy, 2	arterial calcification, gen
1	
eralized, of infancy, 1	arterial calcification, gen
1	
ar dysplasia, familial, 14	arrhythmogenic right ventricul
1	

	arrhythmogenic right ven
tricular dysplasia 9	
1	arrhythmogenic right ven
tricular dysplasia 8	arrhythmogenic right ven
1	
	arrhythmogenic right ven
tricular dysplasia 5	
1	arrhythmogenic right ven
tricular dysplasia 2	arrhythmogenic right vent
1	
	arrhythmogenic right vent
ricular dysplasia 13	
1	arthrogryposis, renal dysfun
ction, and cholestasis 1	arthrogryposis, renal dysfun
1	
	arthrogryposis- oculomotor limitation-
ction, and cholestasis 2	
1	arthrogryposis-l
electroretinal anomalies syndrome	atrial fibrillatio
1	
	atrial fibrillatio
ike syndrome	
1	atrial conducti
n, familial, 12	atransferr
1	
	atelosteogenesi
n, familial, 11	
1	atelosteogenes
on disease	atelosteogenes
1	
	ataxia-telangiectasia
inemia	
1	ataxia-telangiectasia
s type III	ataxia-pancytopenia syndrome
1	
is type II	
1	
is type I	
1	
-like disorder 2	
1	
-like disorder 1	
1	
nia syndrome	

1		
		ataxia-hypogonadism-choroi
dal dystrophy syndrome		
1		
		ataxia, intention tremor, and hypot
onia syndrome, childhood-onset		
1		
		ataxia, early-onset, with oculomot
or apraxia and hypoalbuminemia		
1		
		ataxia telang
iectasia		
1		
		ataxia - oculomotor
apraxia type 4		
1		
		ataxia - intellectual disability - oculomot
or apraxia - cerebellar cysts syndrome		
1		
		asthma-related traits, s
usceptibility to, 5		
1		
		asthma, nasal polyps, and
aspirin intolerance		
1		
		asphyxiating thorac
ic dystrophy 5		
1		
		asphyxiating thorac
ic dystrophy 4		
1		
		asphyxiating thorac
ic dystrophy 3		
1		
		asphyxiating thorac
ic dystrophy 2		
1		
		aspartylgluco
saminuria		
1		
		arrhythmogenic right vent
ricular dysplasia 12		
1		
		arrhythmogenic right vent
ricular dysplasia 11		
1		
		arrhythmogenic right vent
ricular dysplasia 10		
1		
		anterior segment
dysgenesis 8		
1		
		anterior segment
dysgenesis 4		
1		
		anterior segment

dysgenesis 3	
1	
gium syndrome	antecubital ptery
1	
ophageal atresia syndrome	anophthalmia/microphthalmia-es
1	
ts-cleft lip/palate syndrome	ankyloblepharon-ectodermal defec
1	
a 3	aniridi
1	
a 2	aniridi
1	
iciency-osteopetrosis-lymphedema syndrome	anhidrotic ectodermal dysplasia-immunodefic
1	
editary, 7	angioedema, her
1	
editary, 6	angioedema, her
1	
editary, 5	angioedema, her
1	
editary, 4	angioedema, her
1	
ritis syndrome	aneurysm-osteoarth
1	
nial berry, 12	aneurysm, intracra
1	
ly 2	anencepha
1	
ly 1	anencepha
1	
blastic, 5	anemia, sidero
1	
ic, due to G6PD deficiency	anemia, nonspherocytic hemolyt
1	
thropoietic, type 1a	anemia, congenital dysery
1	
vity syndrome	androgen insensiti
1	

	anauxetic dys
plasia 2	
1	
	anaplastic as
trocytoma	
1	
	amyotrophic n
euralgia	
1	
	anterior segment
dysgenesis 6	
1	
	aortic aneurysm, fami
lial thoracic 10	
1	
	arrhythmogenic right ven
tricular dysplasia 1	
1	
	aortic aneurysm, familial thor
acic 11, susceptibility to	
1	
	arrhythmogenic cardiomyopathy wit
h wooly hair and keratoderma	
1	
	arrhythmogenic cardiomyopathy with v
ariable ectodermal abnormalities	
1	
	aromatic L-amino acid dec
arboxylase deficiency	
1	
	aromatase exces
s syndrome	
1	
	aromatase de
ficiency	
1	
	ariboflavi
nosis	
1	
	arhinia, choanal atresi
a, and microphthalmia	
1	
	argininosuccini
c aciduria	
1	
	arginase def
iciency	
1	
	apraxi
a	
1	
	apparent mineraloco
rticoid excess	
1	
	apolipoprotein A-I
I amyloidosis	

1		
		aplasia of lacrimal an
d salivary glands		
1		
		aplasia cutis-enamel d
ysplasia syndrome		
1		
		aortic valve
disease 3		
1		
		aortic valve
disease 2		
1		
		aortic valve
disease 1		
1		
		aortic aneurysm, fam
ilial thoracic 9		
1		
		aortic aneurysm, fam
ilial thoracic 8		
1		
		aortic aneurysm, fam
ilial thoracic 7		
1		
		aortic aneurysm, fam
ilial thoracic 6		
1		
		aortic aneurysm, fam
ilial thoracic 4		
1		
		aortic aneurysm, fami
lial thoracic 12		
1		
		atrial fibrillatio
n, familial, 13		
1		
		atrial fibrillatio
n, familial, 15		
1		
		Wolfram-like
syndrome		
1		
		autoinflammatory syndrome
with immunodeficiency		
1		
		autosomal dominant Opi
tz G/BBB syndrome		
1		
		autosomal dominant Ken
ny-Caffey syndrome		
1		
		autosomal dominant Ehlers-Danl
os syndrome, vascular type		
1		
		autosomal dominant Charcot-Ma

rie-Tooth disease type 2W	
1	
	autosomal dominant Charcot-Ma
rie-Tooth disease type 2M	
1	
	autosomal dominant Charcot-Ma
rie-Tooth disease type 2K	
1	
	autosomal dominant Charcot-Marie-Tooth
disease type 2 due to TFG mutation	
1	
	autosomal dominant Charcot-Marie-Tooth d
isease type 2 due to KIF5A mutation	
1	
	autoinflammatory-pancytopenia synd
rome due to DNASE2 deficiency	
1	
	autoinflammatory syndrome, famil
ial, X-linked, Behcet-like 2	
1	
	autoinflammatory syndrome,
familial, Behcet-like 1	
1	
	autoinflammatory disease, sy
stemic, with vasculitis	
1	
	atrial fibrillatio
n, familial, 18	
1	
	autoinflammatory disease, multisystem,
with immune dysregulation, X-linked	
1	
	autoinflammation-PLCG2-associated antib
ody deficiency-immune dysregulation	
1	
	autoinflammation, panniculitis, and derm
atosis syndrome, autosomal recessive	
1	
	autoinflammation, immune dysre
gulation, and eosinophilia	
1	
	autoinflammation with pulmonar
y and cutaneous vasculitis	
1	
	autoinflammation with episodic
fever and lymphadenopathy	
1	
	autoinflammation with episodic fe
ver and immune dysregulation	
1	
	autoinflammation with art
hritis and vasculitis	
1	
	autoinflammation with arth
ritis and dyskeratosis	
1	

	autoimmune polyendocri
ne syndrome type 1	
1	
	autoimmune lymphoproliferative syndrome,
type III caused by mutation in PRKCD	
1	
	autosomal dominant Pa
rkinson disease 1	
1	
	autosomal dominant Pa
rkinson disease 4	
1	
	autosomal dominant Pa
rkinson disease 8	
1	
	autosomal dominant Ro
binow syndrome 1	
1	
	autosomal domina
nt keratitis	
1	
	autosomal dominant intermediate Charcot-Mar
ie-Tooth disease with neuropathic pain	
1	
	autosomal dominant intellectual disability-cran
iofacial anomalies-cardiac defects syndrome	
1	
	autosomal dominant ic
hthyosis vulgaris	
1	
	autosomal dominant hypop
hosphatemic rickets	
1	
	autosomal dominant
hypocalcemia 2	
1	
	autosomal dominant
hypocalcemia 1	
1	
	autosomal dominant hyperinsulin
ism due to SUR1 deficiency	
1	
	autosomal dominant hyperinsulini
sm due to Kir6.2 deficiency	
1	
	autosomal dominant hereditary sen
sory and autonomic neuropathy	
1	
	autosomal dominant familial hematuria-retinal a
rteriolar tortuosity-contractures syndrome	
1	
	autosomal dominant distal
renal tubular acidosis	
1	
	autosomal dominant distal axonal motor neu
ropathy-myofibrillar myopathy syndrome	

1		
		autosomal dominant deafness -
onychodystrophy syndrome		
1		
		autosomal dominant complex sp
astic paraplegia type 9B		
1		
		autosomal dominant childhood-onset proximal sp
inal muscular atrophy without contractures		
1		
		autosomal dominant childhood-onset proximal s
pinal muscular atrophy with contractures		
1		
		autosomal dominant cerebellar ata
xia, deafness and narcolepsy		
1		
		autosomal dominan
t brachyolmia		
1		
		autosomal dominant aud
itory neuropathy 1		
1		
		autosomal dominant aplasi
a and myelodysplasia		
1		
		autosomal dominant Ro
binow syndrome 3		
1		
		autosomal dominant Ro
binow syndrome 2		
1		
		autoimmune lymphoprolifera
tive syndrome type 2B		
1		
		autoimmune lymphoprolifera
tive syndrome type 2A		
1		
		autoimmune lymphoproliferative syndrom
e due to CTLA4 haploinsufficiency		
1		
		atypical hemolytic-uremic synd
rome with MCP/CD46 anomaly		
1		
		atypical hemolytic-uremic synd
rome with DGKE deficiency		
1		
		atypical hemolytic-uremic s
yndrome with C3 anomaly		
1		
		atypical hemolytic-uremic synd
rome with B factor anomaly		
1		
		atypical dentin dysplasia d
ue to SMOC2 deficiency		
1		
		atypical Werne

1	attenuated Chédiak-H
1	attention deficit-hype
1	atrophoderma v
1	atrioventricular septal defe
1	atrioventricular s
1	atrachia with pap
1	atrial stand
1	atrial septal
1	atrial septal
1	atrial septal
1	atrial septal
1	atrial septal
1	atrial septal
1	atrial septal
1	atrial septal
1	atrial fibrillatio
1	atrial fibrillatio
1	atrial fibrillatio

	atypical hemolytic-uremic synd	
rome with I factor anomaly		
1		
	atypical hemolytic-uremic syndrom	
e with thrombomodulin anomaly		
1		
	autoimmune interstitial lung d	
isease-arthritis syndrome		
1		
	auditory neu	
ropathy		
1		
	autoimmune hemolytic anemia-autoimmune thromboc	
ytopenia-primary immunodeficiency syndrome		
1		
	autoimmune enteropathy and endocrinopathy - sus	
ceptibility to chronic infections syndrome		
1		
	autoimmune disease, su	
sceptibility to, 6		
1		
	autoimmune disease, multisys	
tem, infantile-onset, 3		
1		
	autism, susceptibilit	
y to, X-linked 5		
1		
	autism, susceptibilit	
y to, X-linked 4		
1		
	autism, susceptibilit	
y to, X-linked 2		
1		
	autism, susceptibilit	
y to, X-linked 1		
1		
	autism, susceptib	
ility to, 20		
1		
	autism, susceptib	
ility to, 19		
1		
	autism, susceptib	
ility to, 17		
1		
	autism, susceptib	
ility to, 16		
1		
	autism suscept	
ibility 1		
1		
	autism spectrum disorder d	
ue to AUTS2 deficiency		
1		
	autism spectrum disorder - epilep	
sy - arthrogryposis syndrome		

1		
		auriculocondylar
syndrome 4		
1		auriculocondylar
syndrome 3		
1		auriculocondylar
syndrome 2		
1		auriculocondylar
syndrome 1		
1		aural atresia,
congenital		
1		auditory neuropathy-opt
ic atrophy syndrome		
1		auditory neuropathy, au
tosomal dominant 3		
1		auditory neuropathy, au
tosomal dominant 2		
1		amyotrophic lateral sclerosis-pa
rkinsonism-dementia complex		
1		amyotrophic lateral sclerosi
s, susceptibility to, 25		
1		amyotrophic lateral sclerosi
s, susceptibility to, 24		
1		XYLT1-congenital disord
er of glycosylation		
1		abetalipopro
teinemia		
1		abdominal obesity-met
abolic syndrome 4		
1		abdominal obesity-met
abolic syndrome 3		
1		Zimmermann-Laban
d syndrome 3		
1		Zimmermann-Laban
d syndrome 2		
1		Zimmermann-Laban
d syndrome 1		
1		Zaki synd

rome	
1	
	ZTTK synd
rome	
1	
	Yuksel-Vogel-Bau
er syndrome	
1	
	Yoon-Bellen neurodevel
opmental syndrome	
1	
	Xq25 microduplica
tion syndrome	
1	
	XFE progeroid
syndrome	
1	
	amyotrophic lateral
sclerosis type 9	
1	
	X-linked spondyloepimet
aphyseal dysplasia	
1	
	X-linked spasticity-intellectual
disability-epilepsy syndrome	
1	
	X-linked sideroblastic
anemia with ataxia	
1	
	X-linked siderobla
stic anemia 1	
1	
	X-linked severe conge
nital neutropenia	
1	
	X-linked scapulooperoneal
muscular dystrophy	
1	
	X-linked reti
noschisis	
1	
	X-linked reticulate pi
gmentary disorder	
1	
	X-linked recessive
ocular albinism	
1	
	X-linked parkinsonism-s
pasticity syndrome	
1	
	X-linked osteoporosi
s with fractures	
1	
	ablepharon macrost
omia syndrome	
1	

	absence of fingerprints-con
genital milia syndrome	
1	
	acatala
sia	
1	
	aceruloplas
minemia	
1	
	acrofacial dysostosi
s Rodriguez type	
1	
	acrofacial dysostosis
Cincinnati type	
1	
	acro dysostosis 2 with or wit
hout hormone resistance	
1	
	acrodermatitis e
nteropathica	
1	
	acrocapitofemora
l dysplasia	
1	
	acral self-healing
collodion baby	
1	
	acral dystrophic epid
ermolysis bullosa	
1	
	acne inversa, f
amilial, 3	
1	
	acne inversa, f
amilial, 2	
1	
	acne inversa, f
amilial, 1	
1	
	acid sphingomyelin
ase deficiency	
1	
	achromatop
sia 7	
1	
	achromatop
sia 4	
1	
	achromatop
sia 3	
1	
	achromatop
sia 2	
1	
	achondrop
lasia	

1		
		achondrogenesi
s type II		
1		achondrogenesi
s type IB		
1		achondrogenesi
s type IA		
1		acheirop
ody		
1		acetyl-coa carboxyl
ase deficiency		
1		acetyl-CoA acetyltransf
erase-2 deficiency		
1		acetazolamide-respo
nsive myotonia		
1		X-linked non progressiv
e cerebellar ataxia		
1		X-linked myotubu
lar myopathy		
1		X-linked myopathy with po
stural muscle atrophy		
1		X-linked female restricted facial dysmorphism-short
stature-choanal atresia-intellectual disability		
1		X-linked erythropoiet
ic protoporphyria		
1		X-linked dystonia
-parkinsonism		
1		X-linked dyserythropoetic anemia with
abnormal platelets and neutropenia		
1		X-linked dominant hypop
hosphatemic rickets		
1		X-linked dominant chondrodyspla
sia, Chassaing-Lacombe type		
1		X-linked distal spinal mu
scular atrophy type 3		
1		X-linked congenital
hemolytic anemia		
1		X-linked congenital gener

alized hypertrichosis	
1	
dystrophy 3	X-linked cone-rod
1	
tic paraplegia type 1	X-linked complicated spas
1	
s callosum dysgenesis	X-linked complicated corpora
1	
intellectual disability-short stature syndrome	X-linked colobomatous microphthalmia-microcephaly-
1	
lasia punctata 2	X-linked chondrodysp
1	
lasia punctata 1	X-linked chondrodysp
1	
enoleukodystrophy	X-linked cerebral adre
1	
with late-onset testicular enlargement	X-linked central congenital hypothyroidism
1	
plasia congenita	X-linked adrenal hypo
1	
BBB syndrome	X-linked Opitz G/
1	
cterial diseases due to CYBB deficiency	X-linked Mendelian susceptibility to mycobac
1	
nlos syndrome	X-linked Ehlers-Da
1	
t syndrome	X-linked Alpor
1	
i syndrome	Woodhouse-Sakat
1	
sease	Wolman di
1	
ernal auditory canal-facial dysmorphism syndrome	X-linked external auditory canal atresia-dilated int
1	
omic neuropathy with hearing loss	X-linked hereditary sensory and auton
1	
excessive autophagy	X-linked myopathy with
1	

	X-linked hydrocephalus with stenosis of the aqueduct of Sylvius	
1		
	X-linked mixed hearing loss with perilymphatic gusher	
1		
	X-linked microcephaly-growth retardation-prognathism-cryptorchidism syndrome	
1		
	X-linked lymphoproliferative disease due to XIAP deficiency	
1		
	X-linked lymphoproliferative disease due to SH2D1A deficiency	
1		
	X-linked lissencephaly with abnormal genitalia	
1		
	X-linked intellectual disability-short stature-overweight syndrome	
1		
	X-linked intellectual disability-psoriasis-macroorchidism syndrome	
1		
	X-linked intellectual disability-limb spasticity-retinal dystrophy-diabetes insipidus syndrome	
1		
	X-linked intellectual disability-hypotonia-movement disorder syndrome	
1		
	X-linked intellectual disability-global development delay-facial dysmorphism-sacral caudal remnant syndrome	
1		
	X-linked intellectual disability-cerebellar hypoplasia-spondylo-epiphyseal dysplasia syndrome	
1		
	X-linked intellectual disability-cerebellar hypoplasia syndrome	
1		
	X-linked intellectual disability-cardiomegaly-congestive heart failure syndrome	
1		
	X-linked intellectual disability, van Esch type	
1		
	X-linked intellectual disability, Sutherland-Haan type	
1		
	X-linked intellectual disability, Stocco dos Santos type	
1		
	X-linked intellectual disability, Porteous type	
1		
	X-linked intellectual disability, Golabi-Ito-hall type	

1		
		X-linked intellectual disa
bility, Cantagrel type		
1		
		X-linked intellectual dis
ability, Cabezas type		
1		
		X-linked intellectual disability with
isolated growth hormone deficiency		
1		
		X-linked intellectual disabili
ty due to GRIA3 anomalies		
1		
		X-linked immunodeficiency with magnesium defect,
Epstein-Barr virus infection and neoplasia		
1		
		acrokeratosis ve
rruciformis		
1		
		acromegaloid facial ap
pearance syndrome		
1		
		acromelic frontona
sal dysostosis		
1		
		amelogenesis imperfecta, hy
omaturation type, IIA6		
1		
		amelogenesis imper
fecta type 3B		
1		
		amelogenesis imper
fecta type 2A1		
1		
		amelogenesis imper
fecta type 1H		
1		
		amelogenesis imper
fecta type 1G		
1		
		amelogenesis imper
fecta type 1F		
1		
		amelogenesis imper
fecta type 1E		
1		
		amelogenesis imper
fecta type 1C		
1		
		amelogenesis imper
fecta type 1B		
1		
		amelogenesis imper
fecta type 1A		
1		
		amelogenesis imperfecta hy

1	pomaturation type 2A5	
1		amelogenesis imperfecta hy
1	pomaturation type 2A4	
1		amelogenesis imperfecta hy
1	pomaturation type 2A3	
1		amelogenesis imperfecta hy
1	pomaturation type 2A2	
1		alveolar capillary dysplasia with m
1	isalignment of pulmonary veins	
1		alternating hemiplegi
1	a of childhood 2	
1		alternating hemiplegi
1	a of childhood 1	
1		alpha-methylacyl-CoA r
1	acemase deficiency	
1		alpha-manno
1	sidosis	
1		alpha-actin
1	opathy	
1		alpha-N-acetylgalactosamini
1	dase deficiency type 3	
1		alpha-N-acetylgalactosamini
1	dase deficiency type 2	
1		alpha-N-acetylgalactosamini
1	dase deficiency type 1	
1		alpha-N-acetylgalactosa
1	minidase deficiency	
1		amelogenesis imper
1	fecta, IIa 1K	
1		amelogenesis imper
1	fecta, type 3C	
1		alpha thalassemia-X-linked intel
1	lectual disability syndrome	
1		aminoacylase 1
1	deficiency	
1		amyotrophic lateral
1	sclerosis type 8	
1		

		amyotrophic lateral
sclerosis type 6		
1		
		amyotrophic lateral
sclerosis type 5		
1		
		amyotrophic lateral
sclerosis type 4		
1		
		amyotrophic lateral s
clerosis type 23		
1		
		amyotrophic lateral s
clerosis type 22		
1		
		amyotrophic lateral s
clerosis type 21		
1		
		amyotrophic lateral s
clerosis type 20		
1		
		amyotrophic lateral scler
osis type 2, juvenile		
1		
		amyotrophic lateral s
clerosis type 19		
1		
		amyotrophic lateral s
clerosis type 18		
1		
		amyotrophic lateral s
clerosis type 17		
1		
		amyotrophic lateral s
clerosis type 16		
1		
		amyotrophic lateral s
clerosis type 15		
1		
		amyotrophic lateral s
clerosis type 12		
1		
		amyotrophic lateral s
clerosis type 11		
1		
		amyotrophic lateral s
clerosis type 10		
1		
		amyotrophic lateral scl
erosis 27, juvenile		
1		
		amyotrophic lateral sclerosis 26 with
or without frontotemporal dementia		
1		
		amyloidosis, primary loc
alized cutaneous, 3		

1		
		amyloidosis, primary loc
alized cutaneous, 2		
1		amyloidosis, primary loc
alized cutaneous, 1		
1		amyloidosis, heredi
tary systemic 1		alpha-2-plasmin inhi
1		
		alpha 1-antitryps
bitor deficiency		
1		acromesomelic
in deficiency		adult-onset citrul
1		
		adult polyglucosan
dysplasia		
1		adult hypopho
linemia type I		adrenomyelone
1		
		adrenoleukod
body disease		
1		adrenocortical carci
sphatasia		adenylosuccinate ly
1		
		adenosine kinase
uropathy		
1		adenine phosphoribosyltr
ystrophy		acyl-CoA dehydrogena
1		
		acyl-CoA binding domain contai
noma, hereditary		
1		acute promyelocy
ase deficiency		acute neonatal citru
1		
deficiency		
1		
ansferase deficiency		
1		
se 9 deficiency		
1		
ning protein 5 deficiency		
1		
tic leukemia		
1		

1	linemia type I	
1		acute lymphoblastic leukemia
1		acute intermittent porphyria
1		acute infantile liver failure-cerebellar ataxia-peripheral sensory motor neuropathy syndrome
1		acute infantile liver failure due to synthesis defect of mtDNA-encoded proteins
1		action myoclonus-renal failure syndrome
1		acroosteolysis-keloid-like lesions-premature aging syndrome
1		acroosteolysis disorder
1		acromesomelic dysplasia
1		acromesomelic dysplasia
1		acromesomelic dysplasia 2
1		acromesomelic dysplasia
1		adult-onset autosomal dominant distal myelininating leukodystrophy
1		adult-onset distal myopathy due to VCP mutation
1		alopecia-intellectual disability syndrome 4
1		adult-onset proximal spinal muscular atrophy, autosomal dominant
1		alopecia-intellectual disability syndrome 1
1		alopecia universalis congenita
1		alkylglycerone-phosphate synthase deficiency

	alkapton
uria	
1	
	alkaline ceramidas
e 3 deficiency	
1	
	aldosterone-producing adenoma with seiz
ures and neurological abnormalities	
1	
	albini
sm	
1	
	alanine glyoxylate aminot
ransferase deficiency	
1	
	alacrima, achalasia, and intell
ectual disability syndrome	
1	
	agnathia-otoceph
aly complex	
1	
	agenesis of the corpus callosum
with peripheral neuropathy	
1	
	agenesis of corpus callosum, cardia
c, ocular, and genital syndrome	
1	
	age related macular
degeneration 13	
1	
	agammaglobulinemia 9, a
utosomal recessive	
1	
	agammaglobulinemia 8,
autosomal dominant	
1	
	agammaglobulinemia 7, a
utosomal recessive	
1	
	agammaglobulinemia 6, a
utosomal recessive	
1	
	agammaglobulinemia 5,
autosomal dominant	
1	
	agammaglobulinemia 4, a
utosomal recessive	
1	
	agammaglobulinemia 3, a
utosomal recessive	
1	
	agammaglobulinemia 2, a
utosomal recessive	
1	
	agammaglobulinemia 10,
autosomal dominant	

1		
		advanced sleep pha
se syndrome 1		
1		
		Mendelian susceptibility to mycobacterial dise
ases due to complete IFNgammaR1 deficiency		
1		
		Meier-Gorlin s
ndrome 8		
1		
		heparin cofactor
2 deficiency		
1		
		Charcot-Marie-Tooth dis
ease axonal type 2C		
1		
		Charcot-Marie-Tooth dis
ease axonal type 2V		
1		
		Charcot-Marie-Tooth dis
ease axonal type 2U		
1		
		Charcot-Marie-Tooth dis
ease axonal type 2S		
1		
		Charcot-Marie-Tooth dis
ease axonal type 2Q		
1		
		Charcot-Marie-Tooth dis
ease axonal type 2P		
1		
		Charcot-Marie-Tooth dis
ease axonal type 2O		
1		
		Charcot-Marie-Tooth dis
ease axonal type 2N		
1		
		Charcot-Marie-Tooth dis
ease axonal type 2L		
1		
		Charcot-Marie-Tooth dis
ease axonal type 2K		
1		
		Charcot-Marie-Tooth dis
ease axonal type 2F		
1		
		Charcot-Marie-Tooth dise
ase axonal type 2CC		
1		
		Charcot-Marie-Tooth diseas
e X-linked recessive 5		
1		
		Charcot-Marie-Tooth
disease type 2Y		
1		
		Charcot-Marie-Tooth diseas

e X-linked recessive 4	
1	
e X-linked dominant 6	Charcot-Marie-Tooth diseases
1	
e X-linked dominant 1	Charcot-Marie-Tooth diseases
1	
rome	Char synd
1	
deficiency	Cernunnos-XLF
1	
ellar ataxia	Cayman type cereb
1	
drome	Catifa syn
1	
syndrome	Catel-Manzke
1	
x, type 1	Carney comple
1	
udocamptodactyly syndrome	Carney complex - trismus - pse
1	
mplex	Carney co
1	
ease axonal type 2X	Charcot-Marie-Tooth dis
1	
ease axonal type 2Z	Charcot-Marie-Tooth dis
1	
dominant intermediate B	Charcot-Marie-Tooth disease
1	
dominant intermediate C	Charcot-Marie-Tooth disease
1	
disease type 2R	Charcot-Marie-Tooth
1	
disease type 2J	Charcot-Marie-Tooth
1	
disease type 2I	Charcot-Marie-Tooth
1	
disease type 2E	Charcot-Marie-Tooth
1	

		Charcot-Marie-Tooth
disease type 2D		
1		
		Charcot-Marie-Tooth
disease type 2B5		
1		
		Charcot-Marie-Tooth
disease type 2B1		
1		
		Charcot-Marie-Tooth
disease type 2B		
1		
		Charcot-Marie-Tooth
disease type 2A2		
1		
		Charcot-Marie-Tooth
disease type 2A1		
1		
		Charcot-Marie-Tooth
disease type 1F		
1		
		Charcot-Marie-Tooth
disease type 1E		
1		
		Charcot-Marie-Tooth
disease type 1D		
1		
		Charcot-Marie-Tooth
disease type 1C		
1		
		Charcot-Marie-Tooth
disease type 1B		
1		
		Charcot-Marie-Tooth
disease type 1A		
1		
		Charcot-Marie-Tooth disease
recessive intermediate D		
1		
		Charcot-Marie-Tooth disease
recessive intermediate C		
1		
		Charcot-Marie-Tooth disease
recessive intermediate B		
1		
		Charcot-Marie-Tooth disease
recessive intermediate A		
1		
		Charcot-Marie-Tooth disease
dominant intermediate F		
1		
		Charcot-Marie-Tooth disease
dominant intermediate E		
1		
		Charcot-Marie-Tooth disease
dominant intermediate D		

1		
		Carey-Fineman-Zit
er syndrome 1		
1		Carey-Fineman-Zi
ter syndrome		
1		Canavan di
sease		
1		CERKL-related r
etinopathy		
1		CEDNIK syn
drome		CEBALID sy
1		
ndrome		CDKL5 dis
1		
order		CDH1-related diffuse gastric and l
1		
obular breast cancer syndrome		
1		CCDC115-
CDG		
1		CBL-related
disorder		
1		CARASIL sy
ndrome		
1		CAPN5-related vitr
eoetinopathy		
1		CAMOS syn
drome		
1		CAMK2D-related neurodevelopmental dis
order and dilated cardiomyopathy		
1		C1 inhibitor d
eficiency		
1		C syndr
ome		Buschke-Ollendor
1		
ff syndrome		
1		Burkitt ly
mphoma		
1		Buratti-Harel

syndrome	
1	
	Bryant-Li-Bhoj neurodeve
lopmental syndrome 2	
1	
	Bryant-Li-Bhoj neurodeve
lopmental syndrome 1	
1	
	Bruton-type agamm
aglobulinemia	
1	
	Brunner sy
ndrome	
1	
	Brunet-Wagner neurodeve
lopmental syndrome	
1	
	Brugada syn
drome 9	
1	
	Brugada syn
drome 8	
1	
	Brugada syn
drome 7	
1	
	CEP290-related
ciliopathy	
1	
	CHEK2-related cancer
predisposition	
1	
	Camurati-Engelm
ann disease	
1	
	CHILD syn
drome	
1	
	Camptosynpolydact
lyly, complex	
1	
	Caffey di
sease	
1	
	CYP1B1-related glaucoma with or with
out anterior segment dysgenesis	
1	
	CTR9-related neurodeve
lopmental disorder	
1	
	CTNNA1-related diffuse gastric and
lobular breast cancer syndrome	
1	
	CPOX-related heredita
ry coproporphyrria	
1	

	COL4A1 or COL4A2-related cere
bral small vessel disease	
1	
	COG8-congenital disorde
r of glycosylation	
1	
	COG7-congenital disorde
r of glycosylation	
1	
	COG6-congenital disorde
r of glycosylation	
1	
	COG5-congenital disorde
r of glycosylation	
1	
	COG4-congenital disorde
r of glycosylation	
1	
	COG1-congenital disorde
r of glycosylation	
1	
	CODAS syn
drome	
1	
	CNGB3-related r
etinopathy	
1	
	CNGB1-related r
etinopathy	
1	
	CNGA3-related r
etinopathy	
1	
	CNGA1-related r
etinopathy	
1	
	CK syndr
ome	
1	
	CINCA syn
drome	
1	
	CIDEC-related familial p
artial lipodystrophy	
1	
	CHRNA1-associated hypo-akinesia
disorder of prenatal onset	
1	
	CHIME syn
drome	
1	
	Charcot-Marie-Tooth
disease type 2T	
1	
	Charcot-Marie-Tooth
disease type 4	

1		
		Brugada syn
drome 5		
1		
		D,L-2-hydroxyglut
aric aciduria		
1		
		DNA ligase IV
deficiency		
1		
		DKC1-related
disorder		
1		
		DK1-congenital disorde
r of glycosylation		
1		
		DICER1-related tumor
predisposition		
1		
		DICER1 syn
drome		
1		
		DIAPH1-related sensorineural hearing
loss-thrombocytopenia syndrome		
1		
		DEGCAGS sy
ndrome		
1		
		DDX41-related hematologic malign
ancy predisposition syndrome		
1		
		DDOST-congenital disord
er of glycosylation		
1		
		D-glyceric a
ciduria		
1		
		D-2-hydroxyglutar
ic aciduria 1		
1		
		Cushing syndrome due to macron
odular adrenal hyperplasia		
1		
		Charcot-Marie-Tooth
disease type 4A		
1		
		Curry-Jones
syndrome		
1		
		Currarino
triad		
1		
		Crouzon syndrome-acanthos
is nigricans syndrome		
1		
		Crigler-Najjar sy

ndrome type 2	1	
ndrome type 1	1	Crigler-Najjar sy
ndrome 7	1	Cowden synd
ndrome 6	1	Cowden synd
ndrome 5	1	Cowden synd
ndrome 4	1	Cowden synd
ndrome 1	1	Cowden synd
e syndrome 6	1	Cornelia de Lang
ciency	1	DOCK2 defi
er of glycosylation	1	DPAGT1-congenital disord
r of glycosylation	1	DPM3-congenital disorde
l disability syndrome	1	DYRK1A-related intellectua
n anemia 18	1	Diamond-Blackfa
n anemia 17	1	Diamond-Blackfa
n anemia 16	1	Diamond-Blackfa
mandibulofacial dysostosis	1	Diamond-Blackfan anemia 15 with
mandibulofacial dysostosis	1	Diamond-Blackfan anemia 14 with
n anemia 13	1	Diamond-Blackfa
n anemia 12	1	Diamond-Blackfa

		Diamond-Blackfa
n anemia 11		
1		
		Diamond-Blackfa
n anemia 10		
1		
		Diamond-Blackfa
n anemia 1		
1		
		DiGeorge sy
ndrome		
1		
		Desbuquois dy
splasia 2		
1		
		Desbuquois dy
splasia 1		
1		
		Denys-Drash
syndrome		
1		
		Dentici-Novelli neurodev
elopmental syndrome		
1		
		Dent disease
type 2		
1		
		Dent disease
type 1		
1		
		Delpire-McNeil
l syndrome		
1		
		DeSanto-Shinawi syndrome du
e to WAC point mutation		
1		
		DeSanto-Shinaw
i syndrome		
1		
		Darier di
sease		
1		
		Danon dis
ease		
1		
		Dandy-Walker
syndrome		
1		
		Cornelia de Lang
e syndrome 5		
1		
		Cornelia de Lang
e syndrome 4		
1		
		Cornelia de Lang
e syndrome 3		

1		
		Chitayat sy
ndrome		
1		
		Chediak-Higash
i syndrome		
1		
		Charcot-Marie-tooth disea
se, axonal, type 2JJ		
1		
		Charcot-Marie-tooth disea
se, axonal, type 2DD		
1		
		Charcot-Marie-Tooth disease,
demyelinating, type 1J		
1		
		Charcot-Marie-Tooth disease,
demyelinating, type 1G		
1		
		Charcot-Marie-Tooth diseas
e, demyelinating, IIA 1I		
1		
		Charcot-Marie-Tooth diseas
e, demyelinating, IIA 1H		
1		
		Charcot-Marie-Tooth disea
se, axonal, type 2FF		
1		
		Charcot-Marie-Tooth disea
se, axonal, type 2EE		
1		
		Charcot-Marie-Tooth disease, axonal,
autosomal recessive, type 2a2b;		
1		
		Charcot-Marie-Tooth disea
se, axonal, Type 2HH		
1		
		Charcot-Marie-Tooth dise
ase, axonal, IIa 2II		
1		
		Charcot-Marie-Tooth
disease type 4K		
1		
		Charcot-Marie-Tooth
disease type 4J		
1		
		Charcot-Marie-Tooth
disease type 4H		
1		
		Charcot-Marie-Tooth
disease type 4G		
1		
		Charcot-Marie-Tooth
disease type 4F		
1		
		Charcot-Marie-Tooth

disease type 4E	1	
disease type 4D	1	Charcot-Marie-Tooth
disease type 4C	1	Charcot-Marie-Tooth
disease type 4B3	1	Charcot-Marie-Tooth
disease type 4B2	1	Charcot-Marie-Tooth
disease type 4B1	1	Charcot-Marie-Tooth
developmental syndrome	1	Chilton-Okur-Chung neurod
don syndrome	1	Chopra-Amiel-Gor
e syndrome 2	1	Cornelia de Lang
syndrome	1	Christianson
e syndrome 1	1	Cornelia de Lang
drome	1	Cooks syn
Congenital adrenal insufficiency with 46, XY sex reversal OR 46,XY disorder of sex development-adrenal insufficiency due to CYP11A1 deficiency	1	
congenital myopathy	1	Compton-North conge
syndrome 2	1	Cole-Carpenter
syndrome 1	1	Cole-Carpenter
syndrome	1	Cohen-Gibson
drome	1	Cohen syn
syndrome 8	1	Coffin-Siris s

	Coffin-Siris s
ndrome 7	
1	
	Coffin-Siris s
ndrome 6	
1	
	Coffin-Siris s
ndrome 5	
1	
	Coffin-Siris s
ndrome 12	
1	
	Coffin-Siris s
ndrome 11	
1	
	Coffin-Siris s
ndrome 10	
1	
	Coffin-Siris s
ndrome 1	
1	
	Coffin-Lowry
syndrome	
1	
	Cockayne syndr
ome type 1	
1	
	Cockayne spectrum with or without ce
rebrooculofacioskeletal syndrome	
1	
	Clouston sy
ndrome	
1	
	Clark-Baraitse
r syndrome	
1	
	Chuvash poly
cythemia	
1	
	Chudley-McCullou
gh syndrome	
1	
	Brugada syn
drome 6	
1	
	Brugada syn
drome 4	
1	
	Meier-Gorlin s
ndrome 7	
1	
	ALG12-congenital disord
er of glycosylation	
1	
	APC-related attenuated famili
al adenomatous polyposis	

1		
		ANE synd
rome		
1		
		AMED syndrom
e, digenic		
1		
		ALys amylo
idosis		
1		
		ALS2-related motor
neuron disease		
1		
		ALG9-congenital disorde
r of glycosylation		
1		
		ALG9-associated autosomal domina
nt polycystic kidney disease		
1		
		ALG8-congenital disorde
r of glycosylation		
1		
		ALG6-congenital disorder
of glycosylation 1C		
1		
		ALG3-congenital disorde
r of glycosylation		
1		
		ALG2-congenital disorde
r of glycosylation		
1		
		ALG11-congenital disord
er of glycosylation		
1		
		Alagille syndrome due to a
NOTCH2 point mutation		
1		
		ALG1-congenital disorde
r of glycosylation		
1		
		ALDH18A1-related de
Barsy syndrome		
1		
		AKT2-related familial pa
rtial lipodystrophy		
1		
		AIPL1-related r
etinopathy		
1		
		AICA-ribos
iduria		
1		
		AHDC1-related intellectual disability - obstruct
ive sleep apnea - mild dysmorphism syndrome		
1		
		AGAT defic

iciency		
1		
		AFib amylo
idosis		
1		
		ADan amylo
idosis		
1		
		ADULT syn
drome		
1		
		ADNP-related multiple congenital anomalies - intel
lectual disability - autism spectrum disorder		
1		
		ASAH1-related sph
ingolipidosis		
1		
		ATF6-related r
etinopathy		
1		
		ATP1A3-associated neur
ological disorder		
1		
		ATP6AP2-relate
d disorder		
1		
		Alagille sy
ndrome		
1		
		Al-Raqad sy
ndrome		
1		
		Al Kaissi s
yndrome		
1		
		Aicardi-Goutiere
s syndrome 9		
1		
		Aicardi-Goutiere
s syndrome 8		
1		
		Aicardi-Goutiere
s syndrome 7		
1		
		Aicardi-Goutiere
s syndrome 6		
1		
		Aicardi-Goutiere
s syndrome 5		
1		
		Aicardi-Goutiere
s syndrome 4		
1		
		Aicardi-Goutiere
s syndrome 3		
1		

	Aicardi-Goutiere
s syndrome 2	
1	Aicardi-Goutiere
s syndrome 1	Adams-Oliver s
1	
	Adams-Oliver s
yndrome 6	
1	Adams-Oliver s
yndrome 5	Adams-Oliver s
1	
	Adams-Oliver s
yndrome 4	
1	Adams-Oliver s
yndrome 3	Adams-Oliver s
1	
	Adams-Oliver s
yndrome 2	
1	Adams-Oliver s
yndrome 1	Acrodysostosis 1 with or wit
1	
	Abruzzo-Erickso
hout hormone resistance	
1	Aarskog-Scott synd
n syndrome	ATTRV122I amy
1	
	ATR-X-related
rome, X-linked	
1	ADAM9-related r
loidosis	ACys amylo
1	
	ACTL6A-related
syndrome	
1	46,XY disorder of s
etinopathy	46,XX sex re
1	
	46,XX ovarian dysgenesis-s
idosis	
1	
BAFopathy	
1	
ex development	
1	
versal 3	
1	
hort stature syndrome	

1		
		46,XX disorder of s
ex development		
1		
		3MC syndr
ome 3		
1		
		3MC syndr
ome 2		
1		
		3MC syndr
ome 1		
1		
		3M syndro
me 3		
1		
		3M syndro
me 2		
1		
		3M syndro
me 1		
1		
		3-methylglutaconic ac
iduria, type VIIB		
1		
		3-methylglutaconic aciduria with deafness,
encephalopathy, and Leigh-like syndrome		
1		
		3-methylglutaconic
aciduria type 9		
1		
		3-methylglutaconic
aciduria type 8		
1		
		3-methylglutaconic
aciduria type 5		
1		
		3-methylglutaconic
aciduria type 1		
1		
		3-methylcrotonyl-CoA carb
oxylase 2 deficiency		
1		
		3-methylcrotonyl-CoA carb
oxylase 1 deficiency		
1		
		3-hydroxyisobutyryl-CoA
hydrolase deficiency		
1		
		3-hydroxyisobuty
ric aciduria		
1		
		3-hydroxyacyl-CoA dehyd
rogenase deficiency		
1		
		3-hydroxy-3-methylglutaryl-

CoA synthase deficiency	
1	
utaric aciduria	3-hydroxy-3-methylgl
1	
on syndrome	2q37 microdeleti
1	
versal 4	46,XX sex re
1	
a-hydroxysteroid dehydrogenase 3 deficiency	46,XY disorder of sex development due to 17-bet
1	
r adrenal hyperplasia 2	ACTH-independent macronodula
1	
to 5-alpha-reductase 2 deficiency	46,XY disorder of sex development due
1	
ic thrombocytopenia	ACTB-associated syndrom
1	
drome	ACCES syn
1	
idosis	ABri amylo
1	
loidosis	ABetaL34V amy
1	
loidosis	ABetaA21G amy
1	
s, dutch type	ABeta amyloidosi
1	
Italian type	ABeta amyloidosis,
1	
s, Iowa type	ABeta amyloidosi
1	
s, Arctic type	ABeta amyloidosi
1	
rome	ABCD synd
1	
etinopathy	ABCA4-related r
1	
oidosis	AApoAI amyl
1	

	8q24.3 microdelet
ion syndrome	
1	5-oxoprolinase
deficiency	46,xx sex re
1	
	46,XY sex re
versal 5	
1	46,XY sex re
versal 9	46,XY sex re
1	
	46,XY sex re
versal 6	
1	46,XY sex re
versal 5	46,XY sex re
1	
	46,XY sex re
versal 3	
1	46,XY sex re
versal 2	46,XY sex rev
1	
	46,XY sex re
ersal 11	
1	46,XY gonadal dysgenesis-motor an
d sensory neuropathy syndrome	Alagille syndrome due to
1	
	Aland island e
a JAG1 point mutation	
1	Brugada syn
ye disease	Bartter disea
1	
	Beemer-Langer
drome 3	
1	Beckwith-Wiedemann syndrom
se type 1	Beckwith-Wiedemann syndrome
1	
	Becker muscular
syndrome	
1	
e due to NSD1 mutation	
1	
due to CDKN1C mutation	
1	
dystrophy	

1		
		Beck-Fahrner
syndrome		
1		
		Beare-Stevenson cutis
gyrata syndrome		
1		
		Basilicata-Akht
ar syndrome		
1		
		Bartter disea
se type 5		
1		
		Bartter diseas
e type 4A		
1		
		Bartter disea
se type 3		
1		
		Bartter disea
se type 2		
1		
		Bartsocas-Papas
syndrome 2		
1		
		Alazami-Yuan
syndrome		
1		
		Bartsocas-Papas
syndrome 1		
1		
		Barth syn
drome		
1		
		Bart-Pumphrey
syndrome		
1		
		Bardet-Biedl s
yndrome 9		
1		
		Bardet-Biedl s
yndrome 8		
1		
		Bardet-Biedl s
yndrome 7		
1		
		Bardet-Biedl s
yndrome 6		
1		
		Bardet-Biedl s
yndrome 5		
1		
		Bardet-Biedl s
yndrome 4		
1		
		Bardet-Biedl s

ndrome 3		
1		
		Bardet-Biedl s
ndrome 22		
1		
		Behr synd
rome		
1		
		Bernard-Soulier syndrome, typ
e A2, autosomal dominant		
1		
		Bethlem myo
pathy 2		
1		
		Bietti crystalline corn
eo retinal dystrophy		
1		
		Brugada syn
drome 2		
1		
		Bruck synd
rome 2		
1		
		Bruck synd
rome 1		
1		
		Brown-Vialetto-van L
aere syndrome 2		
1		
		Brown-Vialetto-van L
aere syndrome 1		
1		
		Brooke-Spiegle
r syndrome		
1		
		Brody myo
pathy		
1		
		Bowen-Conradi
syndrome		
1		
		Boudin-Mortier
syndrome		
1		
		Bothnia retinal
dystrophy		
1		
		Bosley-Salih-Alor
ainy syndrome		
1		
		Bosch-Boonstra-Schaaf op
tic atrophy syndrome		
1		
		Borjeson-Forssman-L
ehmann syndrome		
1		

|
splasia
1 |
|
syndrome
1 |
|
drome
1 |
|
rome
1 |
|
yndrome
1 |
|
syndrome 2
1 |
|
syndrome 1
1 |
|
syndrome
1 |
|
yndrome
1 |
|
deficiency
1 |
|
yndrome 20
1 |
|
yndrome 2
1 |
|
yndrome 19
1 |
|
ndrome type 1
1 |
|
rome 2
1 |
|
rome 1
1 |
|
ndrome
1 |
|
rome
1 |
|
ndrome

Boomerang dy
|
Bohring-Opitz
|
Bloom syn
|
Blau synd
|
Bjornstad s
|
Birt-Hogg-Dube
|
Birt-Hogg-Dube
|
Birt-Hogg-Dube
|
Birk-Barel s
|
Birbeck granule
|
Bardet-Biedl s
|
Bardet-Biedl s
|
Bardet-Biedl s
|
Axenfeld-Rieger sy
|
Atelis synd
|
Atelis synd
|
Asperger sy
|
Arts synd
|
Armfield sy
|

1		
		Apert syn
drome		
1		
		Antley-Bixler syndrome without genital an
omalies or disordered steroidogenesis		
1		
		Antley-Bixler syndrome with genital anom
alies and disordered steroidogenesis		
1		
		Angelman sy
ndrome		
1		
		Angel-shaped phalango-e
piphyseal dysplasia		
1		
		Amish lethal mi
crocephaly		
1		
		Alzheimer di
sease 9		
1		
		Alzheimer di
sease 4		
1		
		Alzheimer di
sease 3		
1		
		Alzheimer di
sease 2		
1		
		Alzahrani-Kuwaha
ra syndrome		
1		
		Alstrom sy
ndrome		
1		
		Alport syndrome-intellectual disability-mid
face hypoplasia-elliptocytosis syndrome		
1		
		Alport syndrome 3b, au
tosomal recessive		
1		
		Allan-Herndon-Dud
ley syndrome		
1		
		Alkuraya-Kucinsk
as syndrome		
1		
		Alexander disea
se type II		
1		
		Alexander d
isease		
1		
		Au-Kline sy

ndrome	
1	
ndrome type 3	Axenfeld-Rieger sy
1	
yndrome 18	Bardet-Biedl s
1	
yndrome	Ayme-Gripp s
1	
yndrome 17	Bardet-Biedl s
1	
yndrome 16	Bardet-Biedl s
1	
yndrome 15	Bardet-Biedl s
1	
yndrome 14	Bardet-Biedl s
1	
yndrome 13	Bardet-Biedl s
1	
yndrome 12	Bardet-Biedl s
1	
yndrome 11	Bardet-Biedl s
1	
yndrome 10	Bardet-Biedl s
1	
yndrome	Barber-Say s
1	
syndrome	Baralle-Macken
1	
syndrome 2	Baraitser-winter
1	
syndrome 1	Baraitser-Winter
1	
lcaba syndrome	Bannayan-Riley-Ruva
1	
s syndrome	Bamforth-Lazaru
1	
syndrome	Baller-Gerold
1	

1		Bailey-Bloch conge
nital myopathy		
1		
		BRESEK syn
drome		
1		
		BNAR synd
rome		
1		
		BENTA dis
ease		
1		
		BDV synd
rome		
1		
		BAP1-related tumor pred
isposition syndrome		
1		
		B4GALT1-congenital disor
der of glycosylation		
1		
		B-cell immunodeficiency, distal limb ano
malies, and urogenital malformations		
1		
		Diamond-Blackfa
n anemia 19		
1		
		Diamond-Blackfa
n anemia 20		
1		
		Diamond-Blackfa
n anemia 21		
1		
		Kindler sy
ndrome		
1		
		Knobloch syn
drome 1		
1		
		Knobloch sy
ndrome		
1		
		Kniest dys
plasia		
1		
		Klippel-Feil syndrome 3,
autosomal dominant		
1		
		Klippel-Feil syndrome 2,
autosomal recessive		
1		
		Klippel-Feil syndrome 1,
autosomal dominant		
1		
		Klippel-Feil anomaly-myopathy-f
acial dysmorphism syndrome		

1		
		Kleefstra sy
ndrome 2		
1		
		Kleefstra sy
ndrome 1		
1		
		Kleefstra s
yndrome		
1		
		King-Denboroug
h syndrome		
1		
		Kilquist sy
ndrome		
1		
		Leber congenital
amaurosis 10		
1		
		Khan-Khan-Katsan
is syndrome		
1		
		Keutel syn
drome		
1		
		Kennedy di
sease		
1		
		Keipert sy
ndrome		
1		
		Kearns-Sayre
syndrome		
1		
		Kaya-Barakat-Mas
son syndrome		
1		
		Karayol-Borroto-Haghshenas ne
urodevelopmental syndrome		
1		
		Kaposi sarcoma, sus
ceptibility to		
1		
		Kabuki synd
rome 2		
1		
		Kabuki synd
rome 1		
1		
		KINSSHIP sy
ndrome		
1		
		Knobloch syn
drome 2		
1		
		Koolen-de Vrie

s syndrome		
1		
ue to a point mutation		Koolen-de Vries syndrome d
1		
ndrome		Kostmann sy
1		
syndrome		Laurence-Moon
1		
e, B3GAT3 type		Larsen-like syndrom
1		
drome		Laron syn
1		
c dysplasia		Langer mesomeli
1		
r syndrome		Landau-Kleffne
1		
syndrome		Lamb-Shaffer
1		
wannomatosis		LZTR1-related sch
1		
ous progeria syndrome		LMNA-related cardiocutane
1		
rtial lipodystrophy		LIPE-related familial pa
1		
drome 3		LEOPARD syn
1		
drome 2		LEOPARD syn
1		
drome 1		LEOPARD syn
1		
etinopathy		LCA5-related r
1		
set nephrotic syndrome		LAMB2-related infantile-on
1		
ystemic syndrome		LAMA5-related multis
1		
ular dystrophy		LAMA2-related musc
1		

ome		L1 syndr
1		
ficiency		L-ferritin de
1		
ric aciduria		L-2-hydroxygluta
1		
syndrome		Kury-Isidor
1		
syndrome		Kufor-Rakeb
1		
aposisin A deficiency		Krabbe disease due to s
1		
sease		Krabbe di
1		
d disorder		KCNH1 associate
1		
sorder with or without seizures		KCND2-related neurodevelopmental di
1		
rome		KBG synd
1		
rome 17		Joubert synd
1		
rome 15		Joubert synd
1		
rome 14		Joubert synd
1		
rome 13		Joubert synd
1		
rome 10		Joubert synd
1		
drome 1		Joubert syn
1		
rd syndrome		Johanson-Blizza
1		
elsen syndrome 2		Jervell and Lange-Ni
1		
elsen syndrome 1		Jervell and Lange-Ni

1		
		Jeffries-Lakhani neurode
developmental syndrome		
1		
		Jawad syn
drome		
1		
		Jalili syn
drome		
1		
		Jaberi-Elahi
syndrome		
1		
		Imagawa-Matsumo
to syndrome		
1		
		IgA nephropathy, susc
eptibility to, 3		
1		
		IVIC synd
rome		
1		
		ITPKB defi
ciency		
1		
		IRIDA syn
drome		
1		
		IRF6-related
condition		
1		
		IL21-related infantile infl
ammatory bowel disease		
1		
		IKBKG-related immunodeficiency with
or without ectodermal dysplasia		
1		
		IFT140-related reces
sive ciliopathy		
1		
		IFAP syndr
ome 2		
1		
		IFAP syndrome 1, with or wi
thout BRESHECK syndrome		
1		
		Joubert synd
rome 16		
1		
		Joubert synd
rome 18		
1		
		KAT6B-related multiple conge
nital anomalies syndrome		
1		
		Joubert syn

drome 2	
1	
	Joubert syn
drome 9	
1	
	Joubert syn
drome 8	
1	
	Joubert syn
drome 7	
1	
	Joubert syn
drome 6	
1	
	Joubert syn
drome 5	
1	
	Joubert synd
rome 39	
1	
	Joubert synd
rome 37	
1	
	Joubert synd
rome 36	
1	
	Joubert synd
rome 35	
1	
	Joubert synd
rome 33	
1	
	Joubert synd
rome 32	
1	
	Joubert synd
rome 31	
1	
	Joubert synd
rome 30	
1	
	Joubert syn
drome 3	
1	
	Joubert synd
rome 28	
1	
	Joubert synd
rome 27	
1	
	Joubert synd
rome 26	
1	
	Joubert synd
rome 25	
1	

	Joubert synd
rome 24	
1	
	Joubert synd
rome 23	
1	
	Joubert synd
rome 22	
1	
	Joubert synd
rome 21	
1	
	Joubert synd
rome 20	
1	
	Leber congenital
amaurosis 1	
1	
	Leber congenital
amaurosis 11	
1	
	Diamond-Blackfa
n anemia 3	
1	
	MPI-congenital disorde
r of glycosylation	
1	
	Majeed syn
drome	
1	
	Maffucci sy
ndrome	
1	
	Machado-Joseph di
sease type 3	
1	
	Machado-Joseph di
sease type 2	
1	
	Machado-Joseph di
sease type 1	
1	
	Machado-Josep
h disease	
1	
	MYPN-related
myopathy	
1	
	MYH7-related skel
etal myopathy	
1	
	MYH10-related neurodevelopmental dis
order with congenital anomalies	
1	
	MSH3-related attenuated famil
ial adenomatous polyposis	

1		
		MRCS synd
rome		
1		
		MPDU1-congenital disord
er of glycosylation		
1		
		Leber congenital
amaurosis 12		
1		
		MORM synd
rome		
1		
		MOGS-congenital disorde
r of glycosylation		
1		
		MMEP synd
rome		
1		
		MME-related autosomal dominant Char
cot Marie Tooth disease type 2		
1		
		MIRAGE syn
drome		
1		
		MHC class I de
ficiency 1		
1		
		MGAT2-congenital disord
er of glycosylation		
1		
		MERTK-related r
etinopathy		
1		
		MEND synd
rome		
1		
		MEHMO syn
drome		
1		
		MEGF8-related Carp
enter syndrome		
1		
		Malan overgrowt
h syndrome		
1		
		Marbach-Rustad prog
eroid syndrome		
1		
		Marbach-Schaaf neurodev
elopmental syndrome		
1		
		Marden-Walker
syndrome		
1		
		Meier-Gorlin s

yndrome 6	1		
			Meier-Gorlin s
yndrome 5	1		
			Meier-Gorlin s
yndrome 4	1		
			Meier-Gorlin s
yndrome 3	1		
			Meier-Gorlin s
yndrome 2	1		
			Meier-Gorlin s
yndrome 1	1		
			Meester-Loeys
syndrome	1		
			Meckel syndrom
e, type 9	1		
			Meckel syndrom
e, type 5	1		
			Meckel syndrom
e, type 3	1		
			Meckel syndrom
e, type 10	1		
			Meckel syndrom
e, type 1	1		
			Meckel synd
rome 13	1		
			McLeod neuroacanthoc
ytosis syndrome	1		
			McKusick-Kaufma
n syndrome	1		
			McCune-Albrigh
t syndrome	1		
			Mayer-Rokitansky-Kuste
r-Hauser syndrome	1		
			May-Hegglin
anomaly	1		
			Martsolf syn
drome 1	1		

	Martinez-Frias
syndrome	
1	
	Marshall-Smith
syndrome	
1	
	Marshall sy
ndrome	
1	
	Marfan syn
drome	
1	
	MEGF10-related
myopathy	
1	
	MED12-related intellectua
l disability syndrome	
1	
	MECOM-associate
d syndrome	
1	
	Li-Campeau s
yndrome	
1	
	Leydig cell hypop
lasia, type 1	
1	
	Lessel-Kreienka
mp syndrome	
1	
	Lesch-Nyhan
syndrome	
1	
	Leri-Weill dysch
ondrosteosis	
1	
	Lenz-Majewski hypero
stotic dwarfism	
1	
	Legius syn
drome	
1	
	Legg-Calve-Pert
hes disease	
1	
	Leber hereditary optic neurop
athy, autosomal recessive	
1	
	Leber congenital amaurosis w
ith early-onset deafness	
1	
	Leber congenital
amaurosis 9	
1	
	Leber congenital
amaurosis 8	

1		
		Leber congenital
amaurosis 7		
1		Leber congenital
amaurosis 6		
1		Leber congenital
amaurosis 5		
1		Leber congenital
amaurosis 4		
1		Leber congenital
amaurosis 3		
1		Leber congenital
amaurosis 2		
1		Leber congenital
amaurosis 19		
1		Leber congenital
amaurosis 17		
1		Leber congenital
amaurosis 16		
1		Leber congenital
amaurosis 15		
1		Leber congenital
amaurosis 14		
1		Leber congenital
amaurosis 13		
1		Lhermitte-Ducl
os disease		
1		Li-Fraumeni s
yndrome 1		
1		MASA synd
rome		
1		Li-Fraumeni s
yndrome 2		
1		MAN2B2 defi
ciency		
1		MAN1B1-congenital disord
er of glycosylation		
1		Lynch synd

rome 8	1	Lynch synd
rome 5	1	Lynch synd
rome 4	1	Lynch synd
rome 2	1	Luscan-Lumish
syndrome	1	Luo-Schoch-Yamam
oto syndrome	1	Lowry-Wood s
yndrome	1	Lopes-Maciel-Rod
an syndrome	1	Long-Olsen-Distelm
aier syndrome	1	Loeys-Dietz s
yndrome 6	1	Loeys-Dietz s
yndrome 4	1	Loeys-Dietz s
yndrome 2	1	Loeys-Dietz s
yndrome 1	1	Lisch epithelial co
rneal dystrophy	1	Liddle synd
rome 3	1	Liddle synd
rome 2	1	Liddle synd
rome 1	1	Lichtenstein-Kno
rr syndrome	1	Liberfarb s
yndrome	1	

		Liang-Wang s
syndrome		
1		
		Li-Ghorbani-Weisz-Hu
bushman syndrome		
1		
		IDH3B-related r
retinopathy		
1		
		Hurler-Scheie
syndrome		
1		
		Hurler syn
drome		
1		
		Fabry dis
ease		
1		
		Fanconi anemia comple
mentation group L		
1		
		Fanconi anemia comple
mentation group J		
1		
		Fanconi anemia comple
mentation group I		
1		
		Fanconi anemia comple
mentation group G		
1		
		Fanconi anemia comple
mentation group F		
1		
		Fanconi anemia comple
mentation group E		
1		
		Fanconi anemia complem
entation group D2		
1		
		Fanconi anemia complem
entation group D1		
1		
		Fanconi anemia comple
mentation group C		
1		
		Fanconi anemia comple
mentation group B		
1		
		Fanconi anemia comple
mentation group A		
1		
		FRAAXE intellectua
l disability		
1		
		Huppke-Brendel
syndrome		

1		
		FOXG1 dis
order		
1		
		FNIP1-associate
d syndrome		
1		
		FLVCR1-related retinopathy
with or without ataxia		
1		
		FG syndro
me 4		
1		
		FG syndro
me 1		
1		
		FBLN1-related developmental delay-central ne
rvous system anomaly-syndactyly syndrome		
1		
		FASTKD2-related infantile mitoc
hondrial encephalomyopathy		
1		
		FADD-related immu
nodeficiency		
1		
		Emery-Dreifuss muscular dystro
phy 7, autosomal dominant		
1		
		Emery-Dreifuss muscular dystro
phy 5, autosomal dominant		
1		
		Emery-Dreifuss muscular dystro
phy 4, autosomal dominant		
1		
		Fanconi anemia comple
mentation group N		
1		
		Fanconi anemia comple
mentation group O		
1		
		Fanconi anemia comple
mentation group P		
1		
		Fanconi anemia comple
mentation group Q		
1		
		Frank-Ter Haar
syndrome		
1		
		Fowler syn
drome		
1		
		Floating-Harbo
r syndrome		
1		
		Fliedner-Zweie

r syndrome		
1		
		Finnish type a
myloidosis		
1		
		Fine-Lubinsky
syndrome		
1		
		Filippi sy
ndrome		
1		
		Ferguson-Bonni neurodev
elopmental syndrome		
1		
		Feingold syndr
ome type 2		
1		
		Feingold syndr
ome type 1		
1		
		Faundes-Banka
syndrome		
1		
		Farber lipogran
ulomatosis		
1		
		Fanconi renotubul
ar syndrome 5		
1		
		Fanconi renotubular syndrome 4 with ma
turity-onset diabetes of the young		
1		
		Fanconi renotubul
ar syndrome 3		
1		
		Fanconi renotubul
ar syndrome 2		
1		
		Fanconi renotubul
ar syndrome 1		
1		
		Fanconi anemia, comple
mentation group W		
1		
		Fanconi anemia, comple
mentation group S		
1		
		Fanconi anemia comple
mentation group V		
1		
		Fanconi anemia comple
mentation group U		
1		
		Fanconi anemia comple
mentation group T		
1		

|
 mentation group R
 1 |
 |
 phy 3, autosomal recessive
 1 |
 |
 phy 2, autosomal dominant
 1 |
 |
 syndrome
 1 |
 |
 drome
 1 |
 |
 ausen disease
 1 |
 |
 velopmental syndrome
 1 |
 |
 r dystrophy
 1 |
 |
 uscular dystrophy
 1 |
 |
 syndrome
 1 |
 |
 y syndrome
 1 |
 |
 with or without deafness
 1 |
 |
 syndrome 2
 1 |
 |
 disease 4
 1 |
 |
 disease 2
 1 |
 |
 disease 1
 1 |
 |
 in disease
 1 |
 |
 ndrome
 1 |
 |
 syndrome

Fanconi anemia comple
 |
 Emery-Dreifuss muscular dystro
 |
 Emery-Dreifuss muscular dystro
 |
 Elsayh-Waters
 |
 EDICT syn
 |
 Dyggve-Melchior-Cl
 |
 Dworschak-Punetha neurode
 |
 Duchenne muscula
 |
 Duchenne and Becker m
 |
 Dubin-Johnson
 |
 Duane-radial ra
 |
 Duane retraction syndrome 3
 |
 Duane retraction
 |
 Dowling-Degos
 |
 Dowling-Degos
 |
 Dowling-Degos
 |
 Dorfman-Chanar
 |
 Donohue sy
 |
 Donnai-Barrow
 |

1		
		Diets-Jongmans
syndrome		
1		
		Dias-Logan s
yndrome		
1		
		Diamond-Blackfan
anemia-like		
1		
		Diamond-Blackfa
n anemia 9		
1		
		Diamond-Blackfa
n anemia 8		
1		
		Diamond-Blackfa
n anemia 7		
1		
		Diamond-Blackfa
n anemia 6		
1		
		Diamond-Blackfa
n anemia 5		
1		
		Diamond-Blackfa
n anemia 4		
1		
		EAST synd
rome		
1		
		EEC synd
rome		
1		
		El Hayek-Chahrour neurode
velopmental disorder		
1		
		EEM synd
rome		
1		
		Eiken syn
drome		
1		
		Ehlers-Danlos syndrom
e, vascular type		
1		
		Ehlers-Danlos syndrome, spo
ndylodysplastic type, 2		
1		
		Ehlers-Danlos syndrome, spo
ndylodysplastic type, 1		
1		
		Ehlers-Danlos syndrome, sp
ondylodysplastic type		
1		
		Ehlers-Danlos syndrome, spond

glocheiroidysplastic type	1		Ehlers–Danlos syndrome,
periodontal type 2	1		Ehlers–Danlos syndrome, mus
culocontractural type 2	1		Ehlers–Danlos syndrome, mus
culocontractural type 1	1		Ehlers–Danlos syndrome, k
kyphoscoliotic type, 2	1		Ehlers–Danlos syndrome, k
kyphoscoliotic type 1	1		Ehlers–Danlos syndrome,
dermatosparaxis type	1		Ehlers–Danlos syndrom
e, classic-like, 3	1		Ehlers–Danlos syndrom
e, classic-like, 2	1		Ehlers–Danlos syndrom
e, classic type, 2	1		Ehlers–Danlos syndrome, c
cardiac valvular type	1		Ehlers–Danlos syndrome, a
orthostatic type, 2	1		Ehlers–Danlos syndrome due t
alpha-1(I) procollagen propeptidase deficiency	1		Ebstein an
bleeding disorder	1		East Texas bleed
retinopathy	1		EYS-related re
malformation spectrum	1		EPHB4-associated vascular
limb-only type	1		ENDOVE syndrome, l
Marfan syndrome 2	1		Fraser synd

	Fraser synd
rome 3	
1	
	Frasier sy
ndrome	
1	
	Hennekam lymphangiectasia-
lymphedema syndrome 1	
1	
	Heimler syn
drome 2	
1	
	Heimler syn
drome 1	
1	
	Hartsfield-Bixler-D
emyer syndrome	
1	
	Harel-Yoon s
yndrome	
1	
	Hao-Fountain syndrome d
ue to USP7 mutation	
1	
	Hao-Fountain
syndrome	
1	
	Hallermann-Strei
ff syndrome	
1	
	Haim-Munk s
yndrome	
1	
	Hailey-Hailey
disease	
1	
	HTRA1-related autosomal dominant
cerebral small vessel disease	
1	
	HSD10 mitochondr
ial disease	
1	
	HSD10 disease, n
eonatal type	
1	
	HSD10 disease, in
fantile type	
1	
	HELIX syn
drome	
1	
	HELIOS defi
ciency	
1	
	HAND2 related congeni
tal heart defect	

1		
		H syndr
ome		
1		
		Guttmacher s
ndrome		
1		
		Griscelli syndr
ome type 2		
1		
		Griscelli syndr
ome type 1		
1		
		Greig cephalopolysyn
dactyly syndrome		
1		
		Greenberg dy
splasia		
1		
		Gordon syn
drome		
1		
		Hengel-Marooofian-S
chols syndrome		
1		
		Hennekam lymphangiectasia-
lymphedema syndrome 2		
1		
		Goldberg-Shprint
zen syndrome		
1		
		Hermansky-Pudla
k syndrome		
1		
		Huntington dise
ase-like 2		
1		
		Huntington dise
ase-like 1		
1		
		Hoxha-Aliu s
ndrome		
1		
		Houge-Janssens
syndrome 3		
1		
		Houge-Janssens
syndrome 2		
1		
		Holt-Oram s
ndrome		
1		
		Hogue-Janssens
syndrome 1		
1		
		Hirschsprung disease, s

usceptibility to, 4	
1	
	Hirschsprung disease, s
usceptibility to, 2	
1	
	Hirschsprung disease, s
usceptibility to, 1	
1	
	Hiatt-Neu-Cooper neurode
velopmental syndrome	
1	
	Heyn-Sproul-Jack
son syndrome	
1	
	Hermansky-Pudlak
syndrome 9	
1	
	Hermansky-Pudlak
syndrome 8	
1	
	Hermansky-Pudlak
syndrome 7	
1	
	Hermansky-Pudlak
syndrome 6	
1	
	Hermansky-Pudlak
syndrome 5	
1	
	Hermansky-Pudlak
syndrome 4	
1	
	Hermansky-Pudlak
syndrome 3	
1	
	Hermansky-Pudlak
syndrome 2	
1	
	Hermansky-Pudlak
syndrome 11	
1	
	Hermansky-Pudlak
syndrome 10	
1	
	Hermansky-Pudlak
syndrome 1	
1	
	Goldmann-Favre
syndrome	
1	
	Glanzmann throm
basthenia 2	
1	
	Friedreich
ataxia	
1	

		GTP cyclohydrolase
I deficiency		
1		
		GRM6-related r
etinopathy		
1		
		GRACILE sy
ndrome		
1		
		GPR179-related
retinopathy		
1		
		GNPTG-mucoli
pidosis		
1		
		GNPTAB-mucol
ipidosis		
1		
		GNE myop
athy		
1		
		GNAT2-related r
etinopathy		
1		
		GM3 synthase d
eficiency		
1		
		GM1 gangliosido
sis type 3		
1		
		GM1 gangliosido
sis type 2		
1		
		GM1 gangliosido
sis type 1		
1		
		GM1 ganglio
sidosis		
1		
		GLUT1 deficienc
y syndrome		
1		
		GCGR-related hype
rglucagonemia		
1		
		GATA6-related congenital heart disease with or wit
hout pancreatic agenesis or neonatal diabetes		
1		
		GATA2 deficiency with susc
eptibility to MDS/AML		
1		
		GATA1-Related X-Li
nked Cytopenia		
1		
		GAP0 synd
rome		

1		
		GABA aminotransamin
ase deficiency		
1		
		G6PD defic
iency		
1		
		Fuhrmann sy
ndrome		
1		
		Fryns syn
drome		
1		
		Friedreich a
taxia 1		
1		
		GRN-related frontotemporal lobar deg
eneration with Tdp43 inclusions		
1		
		GTP cyclohydrolase I deficiency
with hyperphenylalaninemia		
1		
		Glanzmann throm
basthenia 1		
1		
		GUCY2D-related domin
ant retinopathy		
1		
		Gillessen-Kaesbach-Ni
shimura syndrome		
1		
		Gilbert sy
ndrome		
1		
		Gerstmann-Straussler-S
cheinker syndrome		
1		
		Gaucher disease-ophthalmoplegia-cardi
ovascular calcification syndrome		
1		
		Gaucher diseas
e type III		
1		
		Gaucher diseas
e type II		
1		
		Gaucher disea
se type I		
1		
		Gaucher disease pe
rinatal lethal		
1		
		Gaucher disease due to s
aposin C deficiency		
1		
		Gaucher di

sease		
1		
		Garg-Mishra proge
roid syndrome		
1		
		Gamstorp-Wohlf
rt syndrome		
1		
		Galloway-Mowat
syndrome 8		
1		
		Galloway-Mowat
syndrome 7		
1		
		Galloway-Mowat
syndrome 6		
1		
		Galloway-Mowat
syndrome 5		
1		
		Galloway-Mowat
syndrome 4		
1		
		Galloway-Mowat
syndrome 3		
1		
		Galloway-Mowat syndr
ome 2, X-linked		
1		
		Galloway-Mowat
syndrome 10		
1		
		Galloway-Mowat
syndrome 1		
1		
		Gabriele de Vri
es syndrome		
1		
		GUCY2D-related reces
sive retinopathy		
1		
		autosomal dominant keratitis-icht
hyosis-hearing loss syndrome		
1		
		autosomal dominant limb-girdle
muscular dystrophy type 1C		
1		
		autosomal dominant limb-girdle musc
ular dystrophy type 1D (DNAJB6)		
1		
		dilated cardiom
yopathy 1L		
1		
		dilated cardiom
yopathy 1Z		
1		

	dilated cardiomyopathy
yopathy 1Y	
1	dilated cardiomyopathy
yopathy 1X	dilated cardiomyopathy
1	
	dilated cardiomyopathy
yopathy 1W	
1	dilated cardiomyopathy
yopathy 1U	dilated cardiomyopathy
1	
	dilated cardiomyopathy
yopathy 1S	
1	dilated cardiomyopathy
yopathy 1R	dilated cardiomyopathy
1	
	dilated cardiomyopathy
yopathy 1P	
1	dilated cardiomyopathy
yopathy 1O	dilated cardiomyopathy
1	
	dilated cardiomyopathy
yopathy 1NN	
1	dilated cardiomyopathy
yopathy 1M	dilated cardiomyopathy
1	
	dilated cardiomyopathy
yopathy 1KK	
1	dilated cardiomyopathy
	dilated cardiomyopathy
autosomal recessive 2	
1	dilated cardiomyopathy
yopathy 1JJ	dilated cardiomyopathy
1	
	dilated cardiomyopathy
yopathy 1J	
1	dilated cardiomyopathy
yopathy 1II	dilated cardiomyopathy
1	
	dilated cardiomyopathy
yopathy 1I	
1	dilated cardiomyopathy
yopathy 1HH	dilated cardiomyopathy
1	
	dilated cardiomyopathy
yopathy 1GG	
1	dilated cardiomyopathy
yopathy 1G	dilated cardiomyopathy

1		
		dilated cardiomyopathy
yopathy 1FF		
1		dilated cardiomyopathy
yopathy 1E		
1		dilated cardiomyopathy
yopathy 1DD		
1		dilated cardiomyopathy
yopathy 1D		
1		dilated cardiomyopathy
yopathy 2A		
1		dilated cardiomyopathy
yopathy 2B		
1		dilated cardiomyopathy
yopathy 3B		
1		dilated cardiomyopathy
otrophic hypogonadism syndrome		dilated cardiomyopathy-hypogonadism
1		
autosomal dominant 6		dyskeratosis congenita,
1		
autosomal dominant 3		dyskeratosis congenita,
1		
autosomal dominant 2		dyskeratosis congenita,
1		
autosomal dominant 1		dyskeratosis congenita,
1		
nita, X-linked		dyskeratosis congenita
1		
alis hereditaria 3		dyschromatosis universalis
1		
alis hereditaria 1		dyschromatosis universalis
1		
salis hereditaria		dyschromatosis universalis
1		
rica hereditaria		dyschromatosis symmetric
1		
piapterin reductase deficiency		dopa-responsive dystonia due to sepiapterin reductase deficiency
1		
		dominant beta-thalassemia

halassemia	
1	
	distal myopathy,
Welander type	
1	
	distal myopathy,
Tateyama type	
1	
	distal myopathy with v
ocal cord weakness	
1	
	distal myopathy with posterior leg
and anterior hand involvement	
1	
	distal myopathy with an
terior tibial onset	
1	
	distal arthrogryp
osis type 5D	
1	
	disorder of protein
N-glycosylation	
1	
	disorder of iron metabo
lism and transport	
1	
	disorder of gly
cosylation	
1	
	disorder of fatty acid and
ketone body metabolism	
1	
	disabling pansclerotic m
orpha of childhood	
1	
	dimethylglycine dehydr
ogenase deficiency	
1	
	dilated cardiom
yopathy 1CC	
1	
	dilated cardiom
yopathy 1BB	
1	
	dilated cardiom
yopathy 1AA	
1	
	diabetes mellitus, per
manent neonatal 4	
1	
	diabetes mellitus, per
manent neonatal 2	
1	
	diabetes insipidus, nep
hrogenic, autosomal	
1	

	diabetes insipidus, nep	
hrogenic, X-linked		
1		
	developmental malformations-d	
deafness-dystonia syndrome		
1		
	developmental dyspla	
sia of the hip 3		
1		
	developmental dyspl	
asia of the hip		
1		
	developmental delay, language impai	
ment, and ocular abnormalities		
1		
	developmental delay, intellectual disabil	
ity, obesity, and dysmorphic features		
1		
	developmental delay, impaired speech, and behavi	
oral abnormalities, with or without seizures		
1		
	developmental delay, impaired speec	
h, and behavioral abnormalities		
1		
	developmental delay, impaired growth, dys	
morphic facies, and axonal neuropathy		
1		
	developmental delay, hypotonia, musculoskelet	
al defects, and behavioral abnormalities		
1		
	developmental delay, hypotoni	
a, and impaired language		
1		
	developmental delay, dysmorphic	
facies, and brain anomalies		
1		
	developmental delay, behavioral abnormali	
ties, and neuropsychiatric disorders		
1		
	developmental delay with variable ne	
urologic and brain abnormalities		
1		
	developmental delay with variable intellectu	
l impairment and behavioral abnormalities		
1		
	developmental delay with variable intellec	
tual disability and dysmorphic facies		
1		
	developmental delay with short stature, dysmo	
rphic facial features, and sparse hair 1		
1		
	developmental delay with or without intellectu	
al impairment or behavioral abnormalities		
1		
	developmental delay with or without	
t dysmorphic facies and autism		

1		
		developmental delay with hypotonia, m
yopathy, and brain abnormalities		
1		
		developmental delay with dysmorphi
c facies and dental anomalies		
1		
		diabetes mellitus, per
manent neonatal 3		
1		
		diabetes mellitus, tra
nsient neonatal, 1		
1		
		dilated cardiom
yopathy 1A		
1		
		diabetes mellitus, tra
nsient neonatal, 2		
1		
		dihydropyrim
idinuria		
1		
		dihydropyrimidine dehyd
rogenase deficiency		
1		
		dihydropteridine redu
ctase deficiency		
1		
		diffuse palmoplantar keratode
rma with painful fissures		
1		
		diffuse nonepidermolytic pa
lmoplantar keratoderma		
1		
		diffuse lymphatic
malformation		
1		
		diffuse cerebral and cerebellar atrophy – intractab
le seizures – progressive microcephaly syndrome		
1		
		diencephalic–mesencephalic jun
ction dysplasia syndrome 2		
1		
		diencephalic–mesencephalic jun
ction dysplasia syndrome 1		
1		
		diencephalic–mesencephali
c junction dysplasia		
1		
		dicarboxylic am
inoaciduria		
1		
		diazoxide-resistant focal hyperins
ulinism due to SUR1 deficiency		
1		
		diazoxide-resistant focal hyperinsu

linism due to Kir6.2 deficiency	1	
1		
		diastrophic dysplasia
1		
		diarrhea-vomiting due to
trehalase deficiency	1	
1		
		diarrhea
a 9	1	
1		
		diarrhea
13	1	
1		
		diarrhea 10, protein-losing enteropathy type
1		
		diaphyseal medullary stenosis
-bone malignancy syndrome	1	
1		
		diaphragmatic hernia 4, with
cardiovascular defects	1	
1		
		diaphragmatic
hernia 3	1	
1		
		diaphanospondyl
odysostosis	1	
1		
		diabetes mellitus, transient neonatal, 3
1		
		dyskeratosis congenita, autosomal recessive 1
1		
		dyskeratosis congenita, autosomal recessive 3
1		
		developmental delay and seizures with
or without movement abnormalities	1	
1		
		encephalopathy, acute, infection-induced, susceptibility to, 9
1		
		epidermodysplasia verruciformis, susceptibility to, 1
1		
		epidermal disease
1		
		eosinophil peroxidase deficiency
1		
		endosteal sclerosis-cerebellar hypoplasia syndrome
1		

	endocrine-cerebro-osteo
dysplasia syndrome	
1	
	endocrine syste
m disorder	
1	
	encephalopathy-hypertrophic cardiomyopa
thy-renal tubular disease syndrome	
1	
	encephalopathy, progressive, with
amyotrophy and optic atrophy	
1	
	encephalopathy, progressive, early-on
set, with episodic rhabdomyolysis	
1	
	encephalopathy, neonatal severe, with lac
tic acidosis and brain abnormalities	
1	
	encephalopathy, lethal, due to defective
mitochondrial peroxisomal fission 1	
1	
	encephalopathy, acute, infection-
induced, susceptibility to, 4	
1	
	dyskeratosis congenita, a
utosomal recessive 5	
1	
	encephalopathy, acute, infection-induced
(herpes-specific), susceptibility to, 8	
1	
	encephalopathy, acute,
infection-induced	
1	
	encephalopathy due to h
ydroxykynureninuria	
1	
	encephalopathy due to defective mitoch
ondrial and peroxisomal fission 2	
1	
	encephalopathy due to beta-mercapt
olactate-cysteine disulfiduria	
1	
	encephalopathy due to
GLUT1 deficiency	
1	
	encephalocraniocutan
eous lipomatosis	
1	
	encephalitis/encephalopathy, mild, wit
h reversible myelin vacuolization	
1	
	encephalitis, acute, infection-in
duced, susceptibility to, 12	
1	
	encephalitis, acute, infection (vira
l)-induced, susceptibility to, 11	

1		
		elliptocyt
osis 3		
1		
		epidermodysplasia verruciform
is, susceptibility to, 2		
1		
		epidermodysplasia verruciform
is, susceptibility to, 3		
1		
		epidermodysplasia verruciform
is, susceptibility to, 4		
1		
		epidermodysplasia verruciform
is, susceptibility to, 5		
1		
		epilepsy, childhood absenc
e, susceptibility to, 5		
1		
		epilepsy, X-linked 2, with or without impaired int
ellectual development and dysmorphic features		
1		
		epilepsy, X-linked 1, with variable learni
ng disabilities and behavior disorders		
1		
		epilepsy with myocloni
c atonic seizures		
1		
		epilepsy of infancy with mi
grating focal seizures		
1		
		epidermolytic palmoplan
tar keratoderma, 1		
1		
		epidermolytic hyperkeratosis
2B, autosomal recessive		
1		
		epidermolytic hyperkeratosis
2A, autosomal dominant		
1		
		epidermolysis bullosa, junctional 7, with inter
stitial lung disease and nephrotic syndrome		
1		
		epidermolysis bullosa, junction
al 6, with pyloric atresia		
1		
		epidermolysis bullosa, junc
tional 4, intermediate		
1		
		epidermolysis bullosa with congenital localiz
ed absence of skin and deformity of nails		
1		
		epidermolysis bullosa simpl
ex with nail dystrophy		
1		
		epidermolysis bullosa simplex du

e to plakophilin deficiency	
1	
	epidermolysis bullosa simplex 7,
with nephropathy and deafness	
1	
	epidermolysis bullosa simplex 6, genera
lized, with scarring and hair loss	
1	
	epidermolysis bullosa simplex 5
B, with muscular dystrophy	
1	
	epidermolysis bullosa si
mplex 5A, Ogna type	
1	
	epidermolysis bullosa simplex 4, localized or ge
neralized intermediate, autosomal recessive	
1	
	epidermolysis bullosa simplex 3, localized or gen
eralized intermediate, with BP230 deficiency	
1	
	epidermolysis bullosa simplex 2d, generalized,
intermediate or severe, autosomal recessive	
1	
	epidermolysis bullosa simplex 2E, wi
th migratory circinate erythema	
1	
	epidermolysis bullosa simplex 2
B, generalized intermediate	
1	
	elliptocyt
osis 2	
1	
	elliptocyt
osis 1	
1	
	ectrodactyly, ectodermal dysplasia,
and cleft lip-palate syndrome 3	
1	
	early-onset Lafora
body disease	
1	
	dystrophic epidermolysis
bullosa, nails only	
1	
	dystrophic epidermolysis
bullosa pruriginosa	
1	
	dystonia, early-onset, an
d/or spastic paraplegia	
1	
	dystonia, childhood-onset, with optic atr
ophy and basal ganglia abnormalities	
1	
	dystoni
a 9	
1	

	dystoni
a 5	
1	
	dystonia 37, early-onset,
with striatal lesions	
1	
	dystonia
33	
1	
	dystonia
30	
1	
	dystonia 28, chi
ldhood-onset	
1	
	dystonia
27	
1	
	dystonia
25	
1	
	dystonia
24	
1	
	dystonia
23	
1	
	dystonia
16	
1	
	dystonia
12	
1	
	dysspondyloencho
ndromatosis	
1	
	dysostosis multiple
x, Ain-Naz type	
1	
	dysosto
sis	
1	
	dyskinesia, limb and orofa
cial, infantile-onset	
1	
	dyskinesia with orofacial invol
vement, autosomal dominant	
1	
	dyskeratosis congenita, a
utosomal recessive 8	
1	
	dyskeratosis congenita, a
utosomal recessive 6	
1	
	early infantile epilep
tic encephalopathy	

1		
		early-onset Parkin
son disease 20		
1		ectopia lentis
et pupillae		
1		early-onset calcifying leukoencep
halopathy-skeletal dysplasia		
1		ectopia lentis 2, isolate
d, autosomal recessive		
1		ectopia lentis 1, isolate
d, autosomal dominant		
1		ectodermal dysplasia-sy
ndactyly syndrome 1		
1		ectodermal dysplasia-s
yndactyly syndrome		
1		ectodermal dysplasia with facial dysmorphism
and acral, ocular, and brain anomalies		
1		ectodermal dysplasia and
immunodeficiency 2		
1		ectodermal dysplasia and
immunodeficiency 1		
1		ectodermal dysplasia
WNT10A related		
1		ectodermal dysplasia
9, hair/nail type		
1		ectodermal dysplasia
4, hair/nail type		
1		ectodermal dysplasia 15, h
ypohidrotic/hair type		
1		ectodermal dysplasia 14, hair/tooth t
ype with or without hypohidrosis		
1		ectodermal dysplasia 1
3, hair/tooth type		
1		ectodermal dysplasia 12, hypohi
drotic/hair/tooth/nail type		
1		ectodermal dysplasia 11B, hypohidrotic/h
air/tooth type, autosomal recessive		
1		ectodermal dysplasia 11A, hypohidrotic/

hair/tooth type, autosomal dominant	
1	
	ectodermal dysplasia 10B, hypohidrotic/h
air/tooth type, autosomal recessive	
1	
	ectodermal dysplasia 10A, hypohidrotic/
hair/nail type, autosomal dominant	
1	
	early-onset progressive neurodegeneration
-blindness-ataxia-spasticity syndrome	
1	
	early-onset progressive encephalopathy-hearing
loss-pons hypoplasia-brain atrophy syndrome	
1	
	early-onset parkinsonism-intell
ectual disability syndrome	
1	
	early-onset myopathy with
fatal cardiomyopathy	
1	
	early-onset epileptic encephalopathy and intel
lectual disability due to GRIN2A mutation	
1	
	developmental delay with autism spect
rum disorder and gait instability	
1	
	developmental and speech dela
y due to SOX5 deficiency	
1	
	autosomal dominant limb-girdle
muscular dystrophy type 1F	
1	
	cutis l
axa	
1	
	cystathion
inuria	
1	
	cyclic hemat
opoiesis	
1	
	cyanosis, transi
ent neonatal	
1	
	cutis laxa, autosomal
recessive, type 2E	
1	
	cutis laxa, autosomal
recessive, type 1d	
1	
	cutis laxa, autosomal
recessive, type 1B	
1	
	cutis laxa, autosomal
recessive, type 1A	
1	

	cutis laxa, autoso
mal dominant 3	
1	
	cutis laxa, autoso
mal dominant 2	
1	
	cutis laxa, autoso
mal dominant 1	
1	
	cutis laxa with severe pulmonary, gast
rointestinal and urinary anomalies	
1	
	cutaneous mas
tocytosis	
1	
	desmoid t
umor	
1	
	cryptosporidiosis-chronic cholan
gitis-liver disease syndrome	
1	
	cryptogenic multifocal ulcer
ous stenosing enteritis	
1	
	cryopyrin-associated
periodic syndrome	
1	
	cryohydroc
ytosis	
1	
	creatine transport
er deficiency	
1	
	craniotubular dysplas
ia, Ikegawa type	
1	
	craniosynostosis-anal anomali
es-porokeratosis syndrome	
1	
	craniosynostosis and
dental anomalies	
1	
	craniosynos
tosis 7	
1	
	craniosynos
tosis 6	
1	
	craniosynos
tosis 4	
1	
	cystic kidney
disease	
1	
	cystino
sis	

1		
		cystinuria
type A		
1		
		cystinuria
type B		
1		
		dermatopathia pigmen
tosa reticularis		
1		
		depressive d
isorder		
1		
		dentinogenesis imp
erfecta type 3		
1		
		dentinogenesis imp
erfecta type 2		
1		
		dentinogenesis
imperfecta		
1		
		dentin dysplas
ia type II		
1		
		dentatorubral-pallid
oluysian atrophy		
1		
		dehydrated hereditary stomatocytosis with or with
out pseudohyperkalemia and/or perinatal edema		
1		
		dehydrated hereditary
stomatocytosis 2		
1		
		deficiency of adenos
ine deaminase 2		
1		
		deficiency in anterior pituitary function
- variable immunodeficiency syndrome		
1		
		deep dermato
phytosis		
1		
		deafness-lymphedema-l
eukemia syndrome		
1		
		deafness-encephaloneuropathy-ob
esity-valvulopathy syndrome		
1		
		deafness, cataract, impaired intellectu
al development, and polyneuropathy		
1		
		deafness, autosomal dominant 39, wi
th dentinogenesis imperfecta 1		
1		
		deafness, Y-

linked 2	
1	
	deafness with labyrinthine aplasi
a, microtia, and microdontia	
1	
	deafness dyston
ia syndrome	
1	
	deaf blind hypopigmentation
syndrome, Yemenite type	
1	
	d-bifunctional prot
ein deficiency	
1	
	d-2-hydroxyglutar
ic aciduria 2	
1	
	cytosolic phospholipase-A2 alpha defic
iciency associated bleeding disorder	
1	
	craniosynos
tosis 2	
1	
	craniometaphyseal dysplasi
a, autosomal recessive	
1	
	craniometaphyseal dysplasi
a, autosomal dominant	
1	
	cortical dysplasia, complex, with
other brain malformations 11	
1	
	corpus callosum agenesis-intellectual disa
bility-coloboma-micrognathia syndrome	
1	
	corpus callosum agenesis-abn
ormal genitalia syndrome	
1	
	coronary arter
y disorder	
1	
	coronary artery disease,
autosomal dominant 2	
1	
	corneal intraepithelial dyskeratosis-palmoplantar
hyperkeratosis-laryngeal dyskeratosis syndrome	
1	
	corneal dystrophy-percept
ive deafness syndrome	
1	
	corneal dystrophy, punctiform an
d polychromatic pre-descemet	
1	
	corneal dystroph
y, Meesmann, 2	
1	

	corneal dystrophy, Fuc
hs endothelial, 8	
1	
	corneal dystrophy, Fuc
hs endothelial, 6	
1	
	corneal dystrophy, Fuc
hs endothelial, 4	
1	
	corneal dystrophy, Fuc
hs endothelial, 3	
1	
	corneal dystrophy, Fuc
hs endothelial, 1	
1	
	cornea pl
ana 2	
1	
	cornea p
lana	
1	
	coralliform
cataract	
1	
	contractures, pterygia, and variabl
e skeletal fusions syndrome 1B	
1	
	contractures, pterygia, and spondyl
ocarpotarsal fusion syndrome 1A	
1	
	continuous spikes and
waves during sleep	
1	
	constitutional megaloblastic anemia
with severe neurologic disease	
1	
	congenital thrombotic thr
ombocytopenic purpura	
1	
	congenital sucrase-iso
maltase deficiency	
1	
	congenital stromal c
orneal dystrophy	
1	
	cortical dysplasia, complex, with
other brain malformations 10	
1	
	cortical dysplasia, complex, with
other brain malformations 12	
1	
	craniolenticulosut
ural dysplasia	
1	
	cortical dysplasia, complex, with
other brain malformations 9	

1		
		craniofrontonas
al syndrome		
1		
		craniofacial-deafnes
s-hand syndrome		
1		
		craniofacial mi
rosomia 2		
1		
		craniofacial dysplasia-short stature-ectodermal
anomalies-intellectual disability syndrome		
1		
		craniofacial dysplasia -
osteopenia syndrome		
1		
		craniofacial dysmorphism, skeletal anomalies, and
impaired intellectual development syndrome 2		
1		
		craniofacial dysmorphism, skeletal anomalies,
and impaired intellectual development 1		
1		
		craniofacial anomalies and anterio
r segment dysgenesis syndrome		
1		
		cranioectodermal
dysplasia 4		
1		
		cranioectodermal
dysplasia 3		
1		
		cranioectodermal
dysplasia 2		
1		
		cranioectodermal
dysplasia 1		
1		
		craniodiaphyseal dysplasi
a, autosomal dominant		
1		
		craniodiaphyseal
l dysplasia		
1		
		cranio-osteoar
thropathy		
1		
		coxopodopatella
r syndrome		
1		
		coumarin res
istance		
1		
		cortisone reductas
e deficiency 2		
1		
		cortisone reductas

e deficiency 1	
1	
ase type 2 deficiency	corticosterone methyloxid
1	
ase type 1 deficiency	corticosterone methyloxid
1	
globulin deficiency	corticosteroid-binding
1	
l epilepsy syndrome	cortical dysplasia-foca
1	
llory body-like inclusions	desmin-related myopathy with Ma
1	
losis	desmostero
1	
ic encephalopathy, 90	developmental and epilept
1	
ic encephalopathy, 53	developmental and epilept
1	
ic encephalopathy, 64	developmental and epilept
1	
ic encephalopathy, 63	developmental and epilept
1	
ic encephalopathy, 62	developmental and epilept
1	
ic encephalopathy, 61	developmental and epilept
1	
ic encephalopathy, 60	developmental and epilept
1	
ic encephalopathy, 59	developmental and epilept
1	
ic encephalopathy, 58	developmental and epilept
1	
ic encephalopathy, 57	developmental and epilept
1	
ic encephalopathy, 56	developmental and epilept
1	
ic encephalopathy, 55	developmental and epilept
1	

1		
		developmental and epilept
ic encephalopathy, 88		
1		
		developmental and epilept
ic encephalopathy, 87		
1		
		developmental and epilept
ic encephalopathy, 86		
1		
		developmental and epileptic encephalopathy, 8
5, with or without midline brain defects		
1		
		developmental and epilept
ic encephalopathy, 84		
1		
		developmental and epilept
ic encephalopathy, 83		
1		
		developmental and epilept
ic encephalopathy, 82		
1		
		developmental and epilept
ic encephalopathy, 81		
1		
		developmental and epilept
ic encephalopathy, 80		
1		
		developmental and epilept
ic encephalopathy, 8		
1		
		developmental and epilept
ic encephalopathy, 79		
1		
		developmental and epilept
ic encephalopathy, 78		
1		
		developmental and epilept
ic encephalopathy, 77		
1		
		developmental and epilept
ic encephalopathy, 76		
1		
		developmental and epilept
ic encephalopathy, 75		
1		
		developmental and epilept
ic encephalopathy, 74		
1		
		developmental and epilept
ic encephalopathy, 73		
1		
		developmental and epilept
ic encephalopathy, 72		
1		
		developmental and epilept

ic encephalopathy, 71	
1	
	developmental and epilept
ic encephalopathy, 70	
1	
	developmental and epilept
ic encephalopathy, 7	
1	
	developmental and epilept
ic encephalopathy, 6A	
1	
	developmental and epilept
ic encephalopathy, 41	
1	
	developmental and epilept
ic encephalopathy, 40	
1	
	developmental and epilept
ic encephalopathy, 4	
1	
	developmental and epilept
ic encephalopathy, 13	
1	
	developmental and epilept
ic encephalopathy, 11	
1	
	developmental and epilept
ic encephalopathy 99	
1	
	developmental and epilept
ic encephalopathy 98	
1	
	developmental and epilept
ic encephalopathy 97	
1	
	developmental and epilept
ic encephalopathy 96	
1	
	developmental and epilept
ic encephalopathy 94	
1	
	developmental and epilept
ic encephalopathy 92	
1	
	developmental and epilept
ic encephalopathy 91	
1	
	developmental and epilept
ic encephalopathy 89	
1	
	developmental and epilept
ic encephalopathy 116	
1	
	developmental and epilept
ic encephalopathy 114	
1	

		developmental and epilept
ic encephalopathy 113		
1		
		developmental and epilept
ic encephalopathy 112		
1		
		developmental and epilept
ic encephalopathy 110		
1		
		developmental and epilept
ic encephalopathy 109		
1		
		developmental and epilept
ic encephalopathy 108		
1		
		developmental and epilept
ic encephalopathy 107		
1		
		developmental and epilept
ic encephalopathy 106		
1		
	developmental and epileptic encephal	
opathy 105 with hypopituitarism		
1		
		developmental and epilept
ic encephalopathy 104		
1		
		developmental and epilept
ic encephalopathy 103		
1		
		developmental and epilept
ic encephalopathy 102		
1		
		developmental and epilept
ic encephalopathy 101		
1		
		developmental and epilept
ic encephalopathy, 12		
1		
		developmental and epilept
ic encephalopathy, 14		
1		
		developmental and epilept
ic encephalopathy, 39		
1		
		developmental and epilept
ic encephalopathy, 15		
1		
		developmental and epilept
ic encephalopathy, 38		
1		
		developmental and epilept
ic encephalopathy, 37		
1		
		developmental and epilept
ic encephalopathy, 36		

ic encephalopathy, 18	
1	
	developmental and epilept
ic encephalopathy, 17	
1	
	epilepsy, childhood absenc
e, susceptibility to, 6	
1	
	epilepsy, early-onset, 3, with or
without developmental delay	
1	
	epilepsy, early-onset, v
itamin B6-dependent	
1	
	glucocorticoid d
eficiency 4	
1	
	glycine N-methyltrans
ferase deficiency	
1	
	glyceronephosphate 0-acylt
ransferase deficiency	
1	
	glutathione synthetase defici
ency with 5-oxoprolinuria	
1	
	glutaryl-CoA dehydrog
enase deficiency	
1	
	glutaric acide
mia type 3	
1	
	glutaminase d
eficiency	
1	
	glutamate pyruvate trans
aminase 2 deficiency	
1	
	glucose-galactose
malabsorption	
1	
	glucocorticoid-remedia
ble aldosteronism	
1	
	glucocorticoid
resistance	
1	
	glucocorticoid d
eficiency 5	
1	
	glucocorticoid d
eficiency 2	
1	
	glycosylphosphatidylinositol
biosynthesis defect 16	
1	

	glucocorticoid d
eficiency 1	
1	
	glomavenous ma
lformation	
1	
	glomerulopathy with fib
ronectin deposits 2	
1	
	glomerulocystic kidney disease with
hyperuricemia and isosthenuria	
1	
	global developmental delay-visual anomalies-progressi
ve cerebellar atrophy-truncal hypotonia syndrome	
1	
	global developmental delay, progressi
ve ataxia, and elevated glutamine	
1	
	global developmental delay, absent or hypoplas
tic corpus callosum, and dysmorphic facies	
1	
	global developmental delay with spee
ch and behavioral abnormalities	
1	
	global developmental delay - lung cysts
- overgrowth - Wilms tumor syndrome	
1	
	glioma suscept
ibility 9	
1	
	glioma suscept
ibility 2	
1	
	glycine enceph
alopathy 1	
1	
	glycine enceph
alopathy 2	
1	
	glycogen storage
disease II	
1	
	glycogen storage
disease III	
1	
	glycogen storage disorder due to hepa
tic glycogen synthase deficiency	
1	
	glycogen storage disease type
1 due to SLC37A4 mutation	
1	
	glycogen storage disease due to pho
sphoglycerate mutase deficiency	
1	
	glycogen storage disease due to phos
phoglycerate kinase 1 deficiency	

1		
		glycogen storage disease due to m
uscle beta-enolase deficiency		
1		
		glycogen storage disease due to muscle an
d heart glycogen synthase deficiency		
1		
		glycogen storage disease due to lactate
dehydrogenase M-subunit deficiency		
1		
		glycogen storage disease due to lactate
dehydrogenase H-subunit deficiency		
1		
		glycogen storage disease due to glyc
ogen branching enzyme deficiency		
1		
		glycogen storage disease due to glucos
e-6-phosphatase deficiency type IA		
1		
		glycogen storage disease due
to aldolase A deficiency		
1		
		glycogen storage disease due to acid
maltase deficiency, late-onset		
1		
		glycogen storage disease due to acid m
altase deficiency, infantile onset		
1		
		glycogen storage disease d
ue to GLUT2 deficiency		
1		
		glycogen storage
disease XV		
1		
		glycogen storage
disease VII		
1		
		glycogen storage
disease VI		
1		
		glycogen storag
e disease V		
1		
		glycogen storage
disease Ib		
1		
		glycogen storage
disease IXd		
1		
		glycogen storage
disease IXc		
1		
		glycogen storage
disease IXb		
1		
		glycogen storage

disease IXa1	
1	
	glioma suscept
ibility 1	
1	
	glaucoma, normal tensio
n, susceptibility to	
1	
	glaucoma secondary to spherophakia/
ectopia lentis and megalocornea	
1	
	gelatinous drop-like
corneal dystrophy	
1	
	gaze palsy, familial horizontal,
with progressive scoliosis 1	
1	
	gastrointestinal defects and i
mmunodeficiency syndrome 1	
1	
	gastric neuroendoc
rine neoplasm	
1	
	gastric car
cinoma	
1	
	gastric c
ancer	
1	
	gastric adenocarcinoma and proxi
mal polyposis of the stomach	
1	
	gamma-glutamylcysteine s
ynthetase deficiency	
1	
	gamma-glutamyl transpe
ptidase deficiency	
1	
	gallbladder d
isease 1	
1	
	galactosial
idosis	
1	
	galactose
mia 4	
1	
	galactos
emia	
1	
	galactose epimera
se deficiency	
1	
	galactokinase
deficiency	
1	

	fumaric ac
iduria	
1	
	fucosyltransferase
6 deficiency	
1	
	fucosido
sis	
1	
	fructose-1,6-bisphosp
hatase deficiency	
1	
	frontotemporal dementia and/or am
yotrophic lateral sclerosis 8	
1	
	frontotemporal dementia and/or am
yotrophic lateral sclerosis 7	
1	
	frontotemporal dementia and/or am
yotrophic lateral sclerosis 6	
1	
	frontotemporal dementia and/or am
yotrophic lateral sclerosis 5	
1	
	frontotemporal dementia and/or am
yotrophic lateral sclerosis 4	
1	
	gaze palsy, familial horizontal,
with progressive scoliosis, 2	
1	
	geleophysic dy
splasia 1	
1	
	glaucoma
3A	
1	
	geleophysic dy
splasia 3	
1	
	glaucoma 3, primary
congenital, E	
1	
	glaucoma 3, primary
congenital, D	
1	
	glaucoma 1, ope
n angle, 0	
1	
	glaucoma 1, ope
n angle, G	
1	
	glaucoma 1, ope
n angle, F	
1	
	glaucoma 1, ope
n angle, A	

1		
		gingival fibromatosis-hyp
ertrichosis syndrome		
1		
		giant axonal ne
uropathy 2		
1		
		giant axonal ne
uropathy 1		
1		
		ghosal hematodiaphy
seal dysplasia		
1		
		genitourinary and/or brain
malformation syndrome		
1		
		genitopatellar
syndrome		
1		
		genetic hyperferritinemia
without iron overload		
1		
		genetic cerebral smal
l vessel disease		
1		
		generalized lip
odystrophy		
1		
		generalized epilepsy-paroxys
mal dyskinesia syndrome		
1		
		generalized epilepsy with febr
ile seizures plus, type 9		
1		
		generalized epilepsy with febr
ile seizures plus, type 7		
1		
		generalized epilepsy with febr
ile seizures plus, type 2		
1		
		generalized epilepsy with febr
ile seizures plus, type 10		
1		
		generalized epilepsy with febr
ile seizures plus, type 1		
1		
		generalized
epilepsy		
1		
		generalized dominant dystroph
ic epidermolysis bullosa		
1		
		glycosylphosphatidylinositol
biosynthesis defect 15		
1		
		glycosylphosphatidylinositol

biosynthesis defect 17	
1	
	epilepsy, early-onset, with or
without developmental delay	
1	
	hearing loss, autosom
al recessive 114	
1	
	heart and brain malfo
rmation syndrome	
1	
	hearing loss, autoso
mal recessive 99	
1	
	hearing loss, autoso
mal recessive 94	
1	
	hearing loss, autoso
mal recessive 57	
1	
	hearing loss, autosom
al recessive 125	
1	
	hearing loss, autosom
al recessive 121	
1	
	hearing loss, autosom
al recessive 120	
1	
	hearing loss, autosom
al recessive 119	
1	
	hearing loss, autosom
al recessive 117	
1	
	hearing loss, autosom
al recessive 116	
1	
	hearing loss, autosom
al recessive 115	
1	
	hearing loss, autosom
al recessive 113	
1	
	glycosylphosphatidylinositol
biosynthesis defect 18	
1	
	hearing loss, autosom
al recessive 112	
1	
	hearing loss, autosom
al recessive 111	
1	
	hearing loss, autosom
al recessive 110	
1	

	hearing loss, autosom
al recessive 109	
1	
	hearing loss, autosom
al recessive 108	
1	
	hearing loss, autosom
al recessive 107	
1	
	hearing loss, autosom
al recessive 106	
1	
	hearing loss, autosom
al recessive 100	
1	
	hearing loss, autoso
mal dominant 90	
1	
	hearing loss, autoso
mal dominant 89	
1	
	hearing loss, autoso
mal dominant 88	
1	
	heart defect - tongue hamartoma
- polysyndactyly syndrome	
1	
	heart-hand syndrome,
Slovenian type	
1	
	helicoid peripapillary chor
ioretinal degeneration	
1	
	heme oxygenase 1
deficiency	
1	
	hennekam lymphangiectasia-
lymphedema syndrome 3	
1	
	hemorrhagic disease due to alpha-1-
antitrypsin Pittsburgh mutation	
1	
	hemophil
ia B	
1	
	hemophil
ia A	
1	
	hemophagocytic lymphohisti
ocytosis, familial, 6	
1	
	hemolytic uremic syndrome, atypical,
8, with rhizomelic short stature	
1	
	hemolytic anemia due to pyrimidin
e 5' nucleotidase deficiency	

1		
		hemolytic anemia due to glutat
hione reductase deficiency		
1		
		hemolytic anemia due to glucopho
sphate isomerase deficiency		
1		
		hemolytic anemia due to diphosph
oglycerate mutase deficiency		
1		
		hemolytic anemia due to aden
ylate kinase deficiency		
1		
		hemoglobinopathy
Toms River		
1		
		hemoglobin, high alt
itude adaptation		
1		
		hemoglobin E-beta-tha
lassemia syndrome		
1		
		hemoglobin E
disease		
1		
		hemoglobin C-beta-tha
lassemia syndrome		
1		
		hemoglobin C
disease		
1		
		hemochromatos
is type 4		
1		
		hemochromatos
is type 3		
1		
		hemochromatosi
s type 2B		
1		
		hemochromatosi
s type 2A		
1		
		hemochromatos
is type 1		
1		
		hemiplegic migraine-developmental and
epileptic encephalopathy spectrum		
1		
		hearing loss, autoso
mal dominant 87		
1		
		hearing loss, autoso
mal dominant 85		
1		
		hearing loss, autoso

mal dominant 84	
1	
	growth hormone insensitivity with immune
dysregulation 1, autosomal recessive	
1	
	growth hormone insensitivity syndr
ome with immune dysregulation	
1	
	growth hormone insens
itivity syndrome	
1	
	growth delay due to insulin-like g
rowth factor type 1 deficiency	
1	
	growth delay due to insulin-like
growth factor I resistance	
1	
	growth and developmental delay-hypotonia-vis
ion impairment-lactic acidosis syndrome	
1	
	gray platelet
syndrome	
1	
	granulomatous disease, chronic, autosomal r
cessive, cytochrome b-positive, type 3	
1	
	granulomatous disease, chronic, autosomal r
cessive, cytochrome b-positive, type 2	
1	
	granulomatous disease, chronic, autosomal r
cessive, cytochrome b-positive, type 1	
1	
	granulomatous disease, chronic, autosom
al recessive, cytochrome b-negative	
1	
	granulomatous disease, chroni
c, autosomal recessive, 5	
1	
	granulomatous disease,
chronic, X-linked	
1	
	granulocytopenia with immu
noglobulin abnormality	
1	
	granular corneal dy
strophy type II	
1	
	granular corneal dy
strophy type I	
1	
	grange syn
drome	
1	
	gonadal dysgenesis, dysmorphic facie
s, retinal dystrophy, and myopathy	
1	

	gonadal dys
genesis	
1	
	goiter, multinodular 1, with or wit
hout Sertoli-Leydig cell tumors	
1	
	gnb5-related intellectual disabili
ty-cardiac arrhythmia syndrome	
1	
	gnathodiaphyseal
l dysplasia	
1	
	glycosylphosphatidylinositol
biosynthesis defect 25	
1	
	glycosylphosphatidylinositol
biosynthesis defect 21	
1	
	growth hormone insensitivity syndrome with im
mune dysregulation 2, autosomal dominant	
1	
	growth hormone secreting
pituitary adenoma 1	
1	
	hearing loss, autoso
mal dominant 82	
1	
	growth retardation, intellectual development
al disorder, hypotonia, and hepatopathy	
1	
	hearing loss, autoso
mal dominant 80	
1	
	hearing loss, autoso
mal dominant 79	
1	
	hearing loss, autoso
mal dominant 78	
1	
	hearing loss, autoso
mal dominant 77	
1	
	hearing loss, autoso
mal dominant 76	
1	
	hearing loss, autoso
mal dominant 75	
1	
	hearing loss, autoso
mal dominant 74	
1	
	hearing loss, autoso
mal dominant 73	
1	
	hearing loss, autoso
mal dominant 72	

1		
		hearing loss, autoso
mal dominant 71		
1		hearing loss, autoso
mal dominant 37		
1		hearing loss,
X-linked 6		hearing loss,
1		
		hearing loss,
X-linked 4		
1		hearing loss,
X-linked 1		
1		hearing impairment and in
fertile male syndrome		
1		head and nec
k cancer		hawkinsin
1		
uria		hatipoglu immunodefi
1		
ciency syndrome		harderopor
1		
phyria		hand-foot-genit
1		
al syndrome		hamel cerebro-palato-
1		
cardiac syndrome		guanidinoacetate methyltr
1		
ansferase deficiency		growth retardation-mild developmental
1		
delay-chronic hepatitis syndrome		frontotemporal dementia and/or am
1		
yotrophic lateral sclerosis 3		frontotemporal dementia and/or am
1		
yotrophic lateral sclerosis 2		frontotemporal dementia and/or am
1		
yotrophic lateral sclerosis 1		facioscapulohumeral muscula
1		
r dystrophy 3, digenic		familial adenomato
1		

us polyposis 2	1		
			familial adenomato
us polyposis 1	1		
			familial acute necrotiz
ing encephalopathy	1		
			familial Mediterranean fev
er, autosomal dominant	1		
			familial Mediter
ranean fever	1		
			familial Alzheimer-li
ke prion disease	1		
			factor XIII, b subuni
t, deficiency of	1		
			factor XIII, A subuni
t, deficiency of	1		
			factor VII de
ficiency	1		
			factor V and factor VIII, comb
ined deficiency of, type 1	1		
			factor 5 and Factor VIII, co
mbined deficiency of, 2	1		
			facial paresis, heredi
tary congenital, 3	1		
			frontotemporal dementia and/or a
myotrophic lateral sclerosis	1		
			facial palsy, congenital, with ptosi
s and velopharyngeal dysfunction	1		
			facial dysmorphism-lens dislocation-anterior segment a
bnormalities-spontaneous filtering blebs syndrome	1		
			facial dysmorphism-immunodeficiency
-livedo-short stature syndrome	1		
			facial dysmorphism, hypertrichosis, epilepsy, intellectua
l/developmental delay, and gingival overgrowth syndrome	1		
			exudative vitreor
etinopathy 7	1		
			exudative vitreor
etinopathy 6	1		

	exudative vitreor
etinopathy 5	
1	
	exudative vitreor
etinopathy 4	
1	
	exudative vitreoretino
pathy 2, X-linked	
1	
	exudative vitreor
etinopathy 1	
1	
	extraoral halitosis due to meth
anethiol oxidase deficiency	
1	
	familial adenomato
us polyposis 3	
1	
	familial adenomato
us polyposis 4	
1	
	familial amyloid
neuropathy	
1	
	familial amyotrophic
lateral sclerosis	
1	
	familial hemiple
gic migraine	
1	
	familial gestational
hyperthyroidism	
1	
	familial generalize
d lentiginosis	
1	
	familial gastric type 1
neuroendocrine tumor	
1	
	familial expansil
e osteolysis	
1	
	familial episodic pain syndrome with pr
edominantly upper body involvement	
1	
	familial episodic pain syndrome with pr
edominantly lower limb involvement	
1	
	familial encephalopathy with ne
uroserpin inclusion bodies	
1	
	familial dyskinesia an
d facial myokymia	
1	
	familial digital arthro
pathy-brachydactyly	

1		
		familial cylin
dromatosis		
1		
		familial cutaneous telangiectasia and oroph
aryngeal predisposition cancer syndrome		
1		
		familial congenital nasolac
rimal duct obstruction		
1		
		familial cold autoinfla
mmatory syndrome 4		
1		
		familial cold autoinfla
mmatory syndrome 3		
1		
		familial cold autoinfla
mmatory syndrome 2		
1		
		familial cold autoinfla
mmatory syndrome 1		
1		
		familial cold autoinfl
ammatory syndrome		
1		
		familial cavitary op
tic disk anomaly		
1		
		familial benign f
lecked retina		
1		
		familial atypical multiple
mole melanoma syndrome		
1		
		familial atri
al myxoma		
1		
		familial apolipoprotei
n C-II deficiency		
1		
		exostoses, multi
ple, type 2		
1		
		exostoses, multi
ple, type 1		
1		
		exercise-induced h
yperinsulinism		
1		
		episodic atax
ia type 1		
1		
		epiphyseal dysplasi
a, multiple, 3		
1		
		epiphyseal dysplasi

a, multiple, 2	
1	
	epilepsy, progressiv
e myoclonic, 1B	
1	
	epilepsy, progressiv
e myoclonic, 12	
1	
	epilepsy, progressiv
e myoclonic, 11	
1	
	epilepsy, idiopathic generaliz
zed, susceptibility to, 9	
1	
	epilepsy, idiopathic generaliz
zed, susceptibility to, 8	
1	
	epilepsy, idiopathic generaliz
ed, susceptibility to, 15	
1	
	epilepsy, idiopathic generaliz
ed, susceptibility to, 14	
1	
	epilepsy, idiopathic generaliz
ed, susceptibility to, 13	
1	
	epilepsy, idiopathic generaliz
ed, susceptibility to, 12	
1	
	epilepsy, idiopathic generaliz
ed, susceptibility to, 11	
1	
	epilepsy, idiopathic generaliz
ed, susceptibility to, 10	
1	
	epilepsy, familial t
emporal lobe, 1	
1	
	epilepsy, familial focal,
with variable foci 4	
1	
	epilepsy, familial focal,
with variable foci 3	
1	
	epilepsy, familial focal,
with variable foci 2	
1	
	epilepsy, familial focal,
with variable foci 1	
1	
	epilepsy, familial ad
ult myoclonic, 7	
1	
	epilepsy, familial ad
ult myoclonic, 6	
1	

	epilepsy, familial ad
ult myoclonic, 5	
1	epilepsy, familial ad
ult myoclonic, 2	epilepsy, familial ad
1	
	epilepsy, familial ad
ult myoclonic, 1	
1	epiphyseal dysplasi
a, multiple, 6	episodic atax
1	
	exercise intolerance, ri
ia type 2	
1	episodic atax
boflavin-responsive	even-plus s
1	
	ethylmalonic enc
ia type 5	
1	estrogen resista
yndrome	essential
1	
	essential fru
cephalopathy	
1	erythrokeratodermia variab
nce syndrome	erythrokeratodermia variab
1	
	erythrokeratodermia variab
tremor	
1	erythrokeratodermia variab
ctosuria	erythrokeratodermia variab
1	
	erythrokeratodermia variab
ilis et progressiva 7	
1	erythrocytosis,
ilis et progressiva 6	
1	
ilis et progressiva 4	
1	
ilis et progressiva 3	
1	
ilis et progressiva 2	
1	
ilis et progressiva 1	
1	
familial, 6	

1		
		erythrocytosis,
familial, 5		
1		erythrocytosis,
familial, 4		
1		erythrocytosis,
familial, 3		
1		epsilon-trimethyllysine h
ydroxylase deficiency		
1		epithelial-stromal
TGFBI dystrophy		
1		epithelial recurrent
erosion dystrophy		
1		epithelial basement m
embrane dystrophy		
1		episodic pain syndro
me, familial, 2		
1		episodic kinesigeni
c dyskinesia 3		
1		episodic ataxi
a, type 9		
1		episodic atax
ia type 6		
1		familial hemophagocytic l
ymphohistiocytosis 2		
1		familial hemophagocytic l
ymphohistiocytosis 3		
1		familial hemophagocytic l
ymphohistiocytosis 4		
1		focal segmental glom
erulosclerosis 1		
1		focal facial dermal d
ysplasia type IV		
1		focal facial dermal d
ysplasia type III		
1		focal epilepsy-intellectual disabilit
y-cerebro-cerebellar malformation		
1		focal dermal h

ypoplasia	
1	
dystrophy	fleck corneal
1	
isease	fish eye d
1	
and cerebral angiomatosis	fibrosis, neurodegeneration,
1	
uscles, congenital, 5	fibrosis of extraocular m
1	
scles, congenital, 3b	fibrosis of extraocular mu
1	
A, with or without extraocular involvement	fibrosis of extraocular muscles, congenital, 3
1	
uscles, congenital, 2	fibrosis of extraocular m
1	
merulopathy	fibronectin glo
1	
ingival, 5	fibromatosis, g
1	
ingival, 1	fibromatosis, g
1	
cans progressiva	fibrodysplasia ossifi
1	
enesis 1	fibrochondrog
1	
tinal hemorrhage syndrome	fetal akinesia-cerebral and re
1	
cephaly, polymicrogyria, and dysmorphic facies	fetal akinesia, respiratory insufficiency, microc
1	
ation sequence 4	fetal akinesia deform
1	
ation sequence 3	fetal akinesia deform
1	
drogenase deficiency	2-methylbutyryl-CoA dehy
1	
neous syndrome	ferro-cerebro-cuta
1	

	female reproductive
system disorder	
1	
	focal palmo-plantar keratoder
ma with joint keratoses	
1	
	focal segmental glom
erulosclerosis 2	
1	
	female infe
rtility	
1	
	focal segmental glomeruloscler
osis 3, susceptibility to	
1	
	frontorh
iny	
1	
	frontonasal dysplasia with alo
pecia and genital anomaly	
1	
	frontonasal dysplasia - severe microphthal
mia - severe facial clefting syndrome	
1	
	frontonasal d
ysplasia	
1	
	frontometaphyseal
dysplasia 2	
1	
	frontometaphyseal
dysplasia 1	
1	
	frontal lobe
epilepsy	
1	
	fried syn
drome	
1	
	free sialic acid storage d
isease, infantile form	
1	
	free sialic acid s
torage disease	
1	
	fragile X-associated tre
mor/ataxia syndrome	
1	
	fragile X s
yndrome	
1	
	foveal hypoplasia-preseni
le cataract syndrome	
1	
	foveal hypoplasia - optic nerve decussation def
ect - anterior segment dysgenesis syndrome	

1		
		foveal hypo
plasia		
1		
		formiminoglutam
ic aciduria		
1		
		focal segmental glomerulosclerosis
and neurodevelopmental syndrome		
1		
		focal segmental glom
erulosclerosis 9		
1		
		focal segmental glom
erulosclerosis 8		
1		
		focal segmental glom
erulosclerosis 7		
1		
		focal segmental glom
erulosclerosis 6		
1		
		focal segmental glom
erulosclerosis 5		
1		
		focal segmental glomeruloscler
osis 4, susceptibility to		
1		
		female infertility due to
oocyte meiotic arrest		
1		
		febrile seizures,
familial, 8		
1		
		familial hemophagocytic l
ymphohistiocytosis 5		
1		
		familial multiple tr
ichoepithelioma		
1		
		familial mesial temporal lobe ep
ilepsy with febrile seizures		
1		
		familial male-limited
precocious puberty		
1		
		familial lipoprotein
lipase deficiency		
1		
		familial juvenile hyperuric
emic nephropathy type 2		
1		
		familial juvenile hyperuric
emic nephropathy type 1		
1		
		familial isolated

trichomegaly	
1	
	familial isolated pi
pituitary adenoma	
1	
	familial isolated hypoparathyroidism du
due to agenesis of parathyroid gland	
1	
	familial isolated defic
iciency of vitamin E	
1	
	familial isolated arrhythmogenic
right ventricular dysplasia	
1	
	familial hypocalciuri
idiopathic hypercalcemia 3	
1	
	familial hypocalciuri
idiopathic hypercalcemia 2	
1	
	familial hypocalciuri
idiopathic hypercalcemia 1	
1	
	familial hypobetali
hyperproteinemia 2	
1	
	familial hypobetali
hyperproteinemia 1	
1	
	familial hypoal
hyperaldosteronism	
1	
	familial hypertr
tryptophanemia	
1	
	familial hyperthyroidism due to
mutations in TSH receptor	
1	
	familial hyperreninemic hy
hyperaldosteronism type 2	
1	
	familial hyperpr
hyperlactinemia	
1	
	familial hyper
hyperinsulinism	
1	
	familial hyperaldost
hyperaldosteronism type III	
1	
	familial hyperaldost
hyperaldosteronism type II	
1	
	familial mitral v
mitral valve prolapse	
1	

	familial nonmedullary
thyroid carcinoma	
1	
	febrile seizures,
familial, 4	
1	
	familial ovaria
n carcinoma	
1	
	febrile seizures,
familial, 11	
1	
	fatty acyl-CoA reducta
se 1 upregulation	
1	
	fatty acyl-CoA reduct
ase 1 deficiency	
1	
	fatal post-viral neurode
generative disorder	
1	
	fatal mitochondrial disease due to combined
oxidative phosphorylation defect type 3	
1	
	fatal infantile hypertonic
myofibrillar myopathy	
1	
	fatal familial
insomnia	
1	
	familial viscer
al myopathy	
1	
	familial temporal l
obe epilepsy 8	
1	
	familial temporal l
obe epilepsy 7	
1	
	familial temporal l
obe epilepsy 5	
1	
	familial steroid-resistant nephrotic sy
ndrome with sensorineural deafness	
1	
	familial spontaneou
s pneumothorax	
1	
	familial scaphocephaly synd
rome, McGillivray type	
1	
	familial retinal arter
ial macroaneurysm	
1	
	familial renal
glucosuria	

1		
		familial pseudoh
yperkalemia		
1		
		familial progressive retinal dystrophy-iris
coloboma-congenital cataract syndrome		
1		
		familial progressive
hyperpigmentation		
1		
		familial progressive hyper
- and hypopigmentation		
1		
		familial porphyria
cutanea tarda		
1		
		familial pityriasis
rubra pilaris		
1		
		familial partial lipodyst
rophy, Dunnigan type		
1		
		congenital stationary night bli
ndness autosomal dominant 3		
1		
		congenital stationary night bli
ndness autosomal dominant 2		
1		
		congenital stationary night bli
ndness autosomal dominant 1		
1		
		bipolar di
sorder		
1		
		blue color b
lindness		
1		
		blistering, acantholytic, of
oral and laryngeal mucosa		
1		
		blepharophimosis, ptosis, and e
picanthus inversus syndrome		
1		
		blepharophimosis - intellectual di
sability syndrome, SBBYS type		
1		
		blepharophimosis - intellectual d
isability syndrome, MKB type		
1		
		blepharocheilodont
ic syndrome 2		
1		
		blepharocheilodont
ic syndrome 1		
1		
		bleeding disorder, p

latelet-type, 24	1		
latelet-type, 22	1		bleeding disorder, p
latelet-type, 21	1		bleeding disorder, p
boxane synthesis deficiency	1		bleeding diathesis due to throm
eficiency	1		biotinidase d
id syndrome	1		brain-lung-thyro
l ganglia disease	1		biotin-responsive basa
and skeletal syndrome	1		biliary, renal, neurologic,
ion, primary, 2	1		bile acid malabsorpt
ion, primary, 1	1		bile acid malabsorpt
tion defect 1	1		bile acid conjuga
yltransferase deficiency	1		bile acid CoA:amino acid N-ac
-occipital polymicrogyria	1		bilateral parasagittal parieto
-cleft palate syndrome	1		bilateral microtia-deafness
d polymicrogyria	1		bilateral generalize
al polymicrogyria	1		bilateral frontopariet
ase deficiency	1		beta-ureidopropion
tive trait locus 19	1		body mass index quantita
ative trait locus 9	1		body mass index quantit

	body skin hyperlaxity due to vitamin K-de
pendent coagulation factor deficiency	
1	
	bone developme
nt disease	
1	
	brain small vessel disease 1 with
or without ocular anomalies	
1	
	brain malformations with or wit
hout urinary tract defects	
1	
	brain dopamine-serotonin ves
icular transport disease	
1	
	brain abnormalities, neurodegener
ation, and dysosteosclerosis	
1	
	braddock-carey
syndrome 2	
1	
	brachyolmia-amelogenesis
imperfecta syndrome	
1	
	brachydactyly-synda
ctyly syndrome	
1	
	brachydactyly-arterial h
ypertension syndrome	
1	
	brachydactyly
type E2	
1	
	brachydactyl
y type C	
1	
	brachydactyly
type B2	
1	
	brachydactyly
type B1	
1	
	brachydactyly
type A1D	
1	
	brachydactyly
type A1C	
1	
	brachydac
tyly	
1	
	brachycephaly, trichomegaly,
and developmental delay	
1	
	bone mineral density quant
itative trait locus 1	

1		
		bone marrow failu
re syndrome 6		
1		bone marrow failu
re syndrome 5		
1		bone marrow failu
re syndrome 4		
1		bone marrow failu
re syndrome 3		
1		bone marrow failure and dia
betes mellitus syndrome		
1		bone fragility with contractures,
arterial rupture, and deafness		
1		beta-thalassemia-X-linked th
rombocytopenia syndrome		
1		beta-thalasse
mia major		beta-thalassemia
1		
intermedia		axial spondylometaph
1		
yseal dysplasia		autosomal semi-dominant severe
1		
lipodystrophic laminopathy		autosomal recessive spondylometaphy
1		
seal dysplasia, Megarbane type		autosomal recessive spin
1		
ocerebellar ataxia 7		autosomal recessive spino
1		
cerebellar ataxia 20		autosomal recessive spin
1		
ocerebellar ataxia 2		autosomal recessive spino
1		
cerebellar ataxia 18		autosomal recessive spino
1		
cerebellar ataxia 17		autosomal recessive spino
1		
cerebellar ataxia 16		autosomal recessive spino
1		

cerebellar ataxia 15	1	autosomal recessive spino
cerebellar ataxia 14	1	autosomal recessive spino
cerebellar ataxia 13	1	autosomal recessive spino
cerebellar ataxia 12	1	autosomal recessive spino
cerebellar ataxia 11	1	autosomal recessive spino
cerebellar ataxia 10	1	autosomal recessive spino
ic paraplegia type 78	1	autosomal recessive spast
ic paraplegia type 76	1	autosomal recessive spast
ic paraplegia type 71	1	autosomal recessive spast
ic paraplegia type 70	1	autosomal recessive spast
ic paraplegia type 69	1	autosomal recessive spast
ic paraplegia type 67	1	autosomal recessive spast
ic paraplegia type 66	1	autosomal recessive spast
ic paraplegia type 60	1	autosomal recessive spast
ic paraplegia type 59	1	autosomal recessive spast
oral head, primary, 1	1	avascular necrosis of fem
nd sensory neuropathy	1	axonal hereditary motor a
a HBB/LCRB	1	beta-thalassemi
opathy	1	axonal neur

1	beta-mannos
idosis	
1	
	beta-ketothiolas
e deficiency	
1	
	beta thala
ssemia	
1	
	bent bone dysplas
ia syndrome 1	
1	
	benign recurrent intrahepa
tic cholestasis type 2	
1	
	benign paroxysmal tort
icollis of infancy	
1	
	benign familial mesial te
mporal lobe epilepsy	
1	
	benign Samaritan con
genital myopathy	
1	
	bathing suit i
chthyosis	
1	
	basal lamina
r drusen	
1	
	basal ganglia calcification, idiop
athic, 9, autosomal recessive	
1	
	basal ganglia calcification, idiop
athic, 8, autosomal recessive	
1	
	basal ganglia calcification, idiop
athic, 7, autosomal recessive	
1	
	basal ganglia calcifica
tion, idiopathic, 6	
1	
	basal ganglia calcifica
tion, idiopathic, 5	
1	
	basal ganglia calcifica
tion, idiopathic, 4	
1	
	basal ganglia calcification, idiop
athic, 10, autosomal recessive	
1	
	basal ganglia calcifica
tion, idiopathic, 1	
1	
	basal cell nevus
syndrome 2	

1		
		basal cell nevus
syndrome 1		
1		
		bardet-biedl s
syndrome 21		
1		
		band heterotopi
a of brain		
1		
		azoospermia, obstructive,
with nephrolithiasis		
1		
		brain small vess
el disease 3		
1		
		branched-chain keto acid dehyd
rogenase kinase deficiency		
1		
		cataract 5 mult
iple types		
1		
		carnitine palmitoyl trans
ferase 1A deficiency		
1		
		cataract 12 mul
tiple types		
1		
		cataract 11 mul
tiple types		
1		
		cataract 10 mul
tiple types		
1		
		cataract 1 mult
iple types		
1		
		cartilage-hair
hypoplasia		
1		
		carpal tunnel
syndrome		
1		
		carnitine-acylcarnitine t
ranslocase deficiency		
1		
		carnitine palmitoyltrans
ferase II deficiency		
1		
		carnitine palmitoyl transferase II d
efficiency, severe infantile form		
1		
		carnitine palmitoyl transferase
II deficiency, neonatal form		
1		
		carnitine palmitoyl transferase I

I deficiency, myopathic form	
1	
	cardiospondylocarpo
facial syndrome	
1	
	branchial arch abnormalities, choanal atresia, athe
lia, hearing loss, and hypothyroidism syndrome	
1	
	cardiomyopathy-hypotonia-la
ctic acidosis syndrome	
1	
	cardiomyopathy, famili
al restrictive, 6	
1	
	cardiomyopathy, famili
al restrictive, 3	
1	
	cardiomyopathy, famili
al restrictive, 1	
1	
	cardiomyopathy, familial hypertrophi
c, 29, with polyglucosan bodies	
1	
	cardiomyopathy, familia
l hypertrophic, 28	
1	
	cardiomyopathy, famili
al hypertrophic 27	
1	
	cardiomyopathy, dilated, with wooly hai
r, keratoderma, and tooth agenesis	
1	
	cardiomyopathy,
dilated, 2j	
1	
	cardiomyopathy,
dilated, 2c	
1	
	cardiomyopathy,
dilated, 2I	
1	
	cataract 13 with ad
ult I phenotype	
1	
	cataract 14 mul
tiple types	
1	
	cataract 15 mul
tiple types	
1	
	cataract 16 mul
tiple types	
1	
	cataract
48	
1	

|
 enile-onset
 1 |
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 45
 1 |
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 44
 1 |
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 43
 1 |
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 42
 1 |
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 41
 1 |
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 iple types
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 tiple types
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 38
 1 |
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 36
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 33
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 tiple types
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 30
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 iple types
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cataract 46 juv
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 cataract
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 cataract
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 cataract 4 mult
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 cataract 39 mul
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 cataract
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 cataract
 |
 cataract 31 mul
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 cataract
 |
 cataract 3 mult
 |
 cataract
 |
 cataract 22 mul
 |
 cataract 21 mul
 |
 cataract 20 mul
 |
 cataract 2, mul
 |

1		
		cataract 19 mul
multiple types		
1		
		cataract
18		
1		
		cataract 17 mul
multiple types		
1		
		cardiomyopathy,
dilated, 2G		
1		
		cardiomyopathy,
dilated, 2F		
1		
		cardiomyopathy,
dilated, 2E		
1		
		candidiasis, f
amilial, 9		
1		
		candidiasis, f
amilial, 6		
1		
		camptodactyly-tall stature-scol
iosis-hearing loss syndrome		
1		
		camptodactyly-arthropathy-coxa
vara-pericarditis syndrome		
1		
		camptodactyly
of fingers		
1		
		campomelic d
ysplasia		
1		
		calvarial doughnut lesions-
bone fragility syndrome		
1		
		c11orf73-related autosomal recessive
hypomyelinating leukodystrophy		
1		
		butyrylcholinester
ase deficiency		
1		
		bronchiectasis with or without
elevated sweat chloride 3		
1		
		bronchiectasis with or without
elevated sweat chloride 2		
1		
		bronchiectasis with or without
elevated sweat chloride 1		
1		
		brittle cornea

syndrome 2	
1	
	brittle cornea
syndrome 1	
1	
	breasts and/or nipples, apla
sia or hypoplasia of, 2	
1	
	breast-ovarian cancer, famili
al, susceptibility to, 4	
1	
	breast-ovarian cancer, famili
al, susceptibility to, 3	
1	
	breast-ovarian cancer, famili
al, susceptibility to, 2	
1	
	breast-ovarian cancer, famili
al, susceptibility to, 1	
1	
	branchiootorenal
syndrome 2	
1	
	branchiootorenal
syndrome 1	
1	
	branchiootic s
syndrome 3	
1	
	branchiootic s
syndrome 1	
1	
	branchiooculofac
ial syndrome	
1	
	candidiasis, f
amilial, 8	
1	
	capillary malformation-arter
iovenous malformation 1	
1	
	cardiomyopathy,
dilated, 2D	
1	
	capillary malformation-arter
iovenous malformation 2	
1	
	cardiomyopathy,
dilated, 100	
1	
	cardiofacioneurodevel
opmental syndrome	
1	
	cardiofaciocutaneo
us syndrome 4	
1	

	cardiofaciocutaneo
us syndrome 3	
1	
	cardiofaciocutaneo
us syndrome 2	
1	
	cardiofaciocutaneo
us syndrome 1	
1	
	cardioencephalomyopathy, fatal infantile, d
ue to cytochrome c oxidase deficiency 4	
1	
	cardioencephalomyopathy, fatal infantile, d
ue to cytochrome c oxidase deficiency 3	
1	
	cardioencephalomyopathy, fatal infantile, d
ue to cytochrome c oxidase deficiency 2	
1	
	cardioencephalomyopathy, fatal infantile, d
ue to cytochrome c oxidase deficiency 1	
1	
	cardioacrofacial
dysplasia 2	
1	
	cardioacrofacial
dysplasia 1	
1	
	cardiac-urogenit
al syndrome	
1	
	cardiac, facial, and digital anoma
lies with developmental delay	
1	
	cardiac valvular dys
plasia, X-linked	
1	
	cardiac valvular
dysplasia 2	
1	
	cardiac valvular defe
ct, developmental	
1	
	cardiac rhyth
m disease	
1	
	cardiac malformation, cleft lip/palate,
microcephaly, and digital anomalies	
1	
	cardiac arrhythmia, a
nkyrin-B-related	
1	
	cardiac anomalies - developmental del
ay - facial dysmorphism syndrome	
1	
	carboxypeptidase
N deficiency	

1		
		carbamoyl phosphate synthetase I deficiency disease
1		
		autosomal recessive severe congenital neutropenia due to JAGN1 deficiency
1		
		autosomal recessive severe congenital neutropenia due to G6PC3 deficiency
1		
		autosomal recessive severe congenital neutropenia due to CXCR2 deficiency
1		
		autosomal dominant vitiligo
1		
		autosomal recessive amelia
1		
		autosomal recessive agammaglobulinemia 1
1		
		autosomal recessive Parkinson disease 14
1		
		autosomal recessive Mendelian susceptibility to mycobacterial diseases due to partial IFNGammaR2 deficiency
1		
		autosomal recessive Mendelian susceptibility to mycobacterial diseases due to partial IFNGammaR1 deficiency
1		
		autosomal recessive Mendelian susceptibility to mycobacterial diseases due to complete IFNGammaR2 deficiency
1		
		autosomal recessive Kenny-Caffey syndrome
1		
		autosomal recessive Emery-Dreifuss muscular dystrophy
1		
		autosomal erythropoietic protoporphyria
1		
		autosomal dominant tight woolly hair
1		
		autosomal dominant vitreoretinopathy
1		
		autosomal dominant striatal neurodegeneration type 1
1		
		autosomal recessive severe congenital neutropenia due to CSF3R deficiency
1		
		autosomal dominant spondyloepiphyseal dysplasia

ylcostal dysostosis	
1	
rve conduction velocity	autosomal dominant slowed ne
1	
eroblastic anemia	autosomal dominant sid
1	
ensory ataxia 1	autosomal dominant s
1	
enous retinal detachment	autosomal dominant rhegmatog
1	
poaldosteronism type 1	autosomal dominant pseudohy
1	
t prognathism	autosomal dominan
1	
-upperback hypertrichosis syndrome	autosomal dominant preaxial polydactyly
1	
al pterygium syndrome	autosomal dominant poplite
1	
toderma and congenital alopecia	autosomal dominant palmoplantar kera
1	
clerosis, Worth type	autosomal dominant osteos
1	
due to PEX10 deficiency	autosomal recessive ataxia
1	
to ubiquinone deficiency	autosomal recessive ataxia due
1	
axia, Beauce type	autosomal recessive at
1	
h disease due to copper metabolism defect	autosomal recessive axonal charcot-marie-toot
1	
pinal muscular atrophy 2	autosomal recessive distal s
1	
pinal muscular atrophy 1	autosomal recessive distal s
1	
utis laxa type 2D	autosomal recessive c
1	
utis laxa type 2C	autosomal recessive c
1	

	autosomal recessive c
utis laxa type 2B	
1	
	autosomal recessive c
utis laxa type 2A	
1	
	autosomal recessive cong
enital ichthyosis 9	
1	
	autosomal recessive cong
enital ichthyosis 8	
1	
	autosomal recessive cong
enital ichthyosis 6	
1	
	autosomal recessive cong
enital ichthyosis 5	
1	
	autosomal recessive cong
enital ichthyosis 4B	
1	
	autosomal recessive cong
enital ichthyosis 4A	
1	
	autosomal recessive cong
enital ichthyosis 3	
1	
	autosomal recessive cong
enital ichthyosis 2	
1	
	autosomal recessive cong
enital ichthyosis 10	
1	
	autosomal recessive cong
enital ichthyosis 1	
1	
	autosomal recessive complex s
pastic paraplegia type 9B	
1	
	autosomal recessive
cerebral atrophy	
1	
	autosomal recessive cerebellar atax
ia-blindness-deafness syndrome	
1	
	autosomal recessive cerebellar ata
xia with late-onset spasticity	
1	
	autosomal recessive c
erebellar ataxia	
1	
	autosomal recessiv
e brachyolmia	
1	
	autosomal recessive
bestrophinopathy	

1		
		autosomal dominant
osteopetrosis 2		
1		
		autosomal dominant
osteopetrosis 1		
1		
		autosomal dominant optic
atrophy plus syndrome		
1		
		autosomal dominant nonsynd
romic hearing loss 27		
1		
		autosomal dominant nonsynd
romic hearing loss 23		
1		
		autosomal dominant nonsynd
romic hearing loss 22		
1		
		autosomal dominant nonsynd
romic hearing loss 21		
1		
		autosomal dominant nonsynd
romic hearing loss 20		
1		
		autosomal dominant nonsynd
romic hearing loss 17		
1		
		autosomal dominant nonsynd
romic hearing loss 15		
1		
		autosomal dominant nonsynd
romic hearing loss 13		
1		
		autosomal dominant nonsynd
romic hearing loss 12		
1		
		autosomal dominant nonsynd
romic hearing loss 11		
1		
		autosomal dominant nonsynd
romic hearing loss 10		
1		
		autosomal dominant nonsyn
dromic hearing loss 1		
1		
		autosomal dominant nocturnal
frontal lobe epilepsy 5		
1		
		autosomal dominant nocturnal
frontal lobe epilepsy 4		
1		
		autosomal dominant nocturnal
frontal lobe epilepsy 3		
1		
		autosomal dominant nocturnal

frontal lobe epilepsy 1	
1	
	autosomal dominant neovascular in
inflammatory vitreoretinopathy	
1	
	autosomal dominant myopia-midfacial retrusion-sensori
neural hearing loss-rhizomelic dysplasia syndrome	
1	
	autosomal dominant mitochondrial myo
pathy with exercise intolerance	
1	
	autosomal dominant mendelian susceptibility to mycobact
erial diseases due to partial IFNgammaR2 deficiency	
1	
	autosomal dominant mendelian susceptibility to mycobact
erial diseases due to partial IFNgammaR1 deficiency	
1	
	autosomal dominant medullary cystic kidney
disease with or without hyperuricemia	
1	
	autosomal dominant medullary cystic k
idney disease with hyperuricemia	
1	
	autosomal dominant limb-girdle
muscular dystrophy type 1G	
1	
	autosomal dominant nonsynd
romic hearing loss 25	
1	
	autosomal dominant nonsynd
romic hearing loss 28	
1	
	autosomal dominant
omodysplasia	
1	
	autosomal dominant nonsynd
romic hearing loss 2A	
1	
	autosomal dominant nonsyn
dromic hearing loss 9	
1	
	autosomal dominant nonsynd
romic hearing loss 70	
1	
	autosomal dominant nonsyn
dromic hearing loss 7	
1	
	autosomal dominant nonsynd
romic hearing loss 69	
1	
	autosomal dominant nonsynd
romic hearing loss 68	
1	
	autosomal dominant nonsynd
romic hearing loss 67	
1	

	autosomal dominant nonsynd
romic hearing loss 66	
1	autosomal dominant nonsynd
romic hearing loss 65	autosomal dominant nonsynd
1	
	autosomal dominant nonsynd
romic hearing loss 64	
1	autosomal dominant nonsyn
dromic hearing loss 6	autosomal dominant nonsynd
1	
	autosomal dominant nonsynd
romic hearing loss 58	
1	autosomal dominant nonsynd
romic hearing loss 56	autosomal dominant nonsynd
1	
	autosomal dominant nonsynd
romic hearing loss 50	
1	autosomal dominant nonsyn
dromic hearing loss 5	autosomal dominant nonsynd
1	
	autosomal dominant nonsynd
romic hearing loss 4B	
1	autosomal dominant nonsynd
romic hearing loss 4A	autosomal dominant nonsynd
1	
	autosomal dominant nonsynd
romic hearing loss 44	
1	autosomal dominant nonsynd
romic hearing loss 41	autosomal dominant nonsynd
1	
	autosomal dominant nonsynd
romic hearing loss 40	
1	autosomal dominant nonsynd
romic hearing loss 3B	autosomal dominant nonsynd
1	
	autosomal dominant nonsynd
romic hearing loss 3A	
1	autosomal dominant nonsynd
romic hearing loss 36	autosomal dominant nonsynd
1	
	autosomal dominant nonsynd
romic hearing loss 2B	
1	autosomal recessive early-on
set Parkinson disease 23	autosomal recessive early-on
1	
set Parkinson disease 6	

1		
		autosomal recessive early-on
set Parkinson disease 7		
1		
		autosomal recessive nonsyn
dromic hearing loss 77		
1		
		autosomal recessive nonsyn
dromic hearing loss 74		
1		
		autosomal recessive nonsyn
dromic hearing loss 70		
1		
		autosomal recessive nonsyn
dromic hearing loss 7		
1		
		autosomal recessive nonsyn
dromic hearing loss 68		
1		
		autosomal recessive nonsyn
dromic hearing loss 67		
1		
		autosomal recessive nonsyn
dromic hearing loss 66		
1		
		autosomal recessive nonsyn
dromic hearing loss 63		
1		
		autosomal recessive nonsyn
dromic hearing loss 61		
1		
		autosomal recessive nonsyn
dromic hearing loss 6		
1		
		autosomal recessive nonsyn
dromic hearing loss 59		
1		
		autosomal recessive nonsyn
dromic hearing loss 53		
1		
		autosomal recessive nonsyn
dromic hearing loss 49		
1		
		autosomal recessive nonsyn
dromic hearing loss 48		
1		
		autosomal recessive nonsyn
dromic hearing loss 44		
1		
		autosomal recessive nonsyn
dromic hearing loss 42		
1		
		autosomal recessive nonsyn
dromic hearing loss 39		
1		
		autosomal recessive nonsyn

dromic hearing loss 37		
1		
		autosomal recessive nonsyn
dromic hearing loss 36		
1		
		autosomal recessive nonsyn
dromic hearing loss 35		
1		
		autosomal recessive nonsyn
dromic hearing loss 32		
1		
		autosomal recessive nonsyn
dromic hearing loss 31		
1		
		autosomal recessive nonsyn
dromic hearing loss 30		
1		
		autosomal recessive nonsyn
dromic hearing loss 3		
1		
		autosomal recessive nonsyn
dromic hearing loss 76		
1		
		autosomal recessive nonsyn
dromic hearing loss 79		
1		
		autosomal recessive nonsyn
dromic hearing loss 28		
1		
		autosomal recessive nonsyn
dromic hearing loss 8		
1		
		autosomal recessive secondary polycyth
emia not associated with VHL gene		
1		
		autosomal recessive proximal
renal tubular acidosis		
1		
		autosomal recessive primary immunodeficiency with defe
ctive spontaneous natural killer cell cytotoxicity		
1		
		autosomal recessive palmoplantar ker
atoderma and congenital alopecia		
1		
		autosomal recessive
osteopetrosis 8		
1		
		autosomal recessive
osteopetrosis 7		
1		
		autosomal recessive
osteopetrosis 5		
1		
		autosomal recessive
osteopetrosis 4		
1		

	autosomal recessive
osteopetrosis 3	
1	
	autosomal recessive
osteopetrosis 2	
1	
	autosomal recessive
osteopetrosis 1	
1	
	autosomal recessive opti
c atrophy, OPA7 type	
1	
	autosomal recessiv
e omodysplasia	
1	
	autosomal recessive nonsyn
dromic hearing loss 98	
1	
	autosomal recessive nonsyn
dromic hearing loss 97	
1	
	autosomal recessive nonsyn
dromic hearing loss 93	
1	
	autosomal recessive nonsyn
dromic hearing loss 91	
1	
	autosomal recessive nonsyn
dromic hearing loss 9	
1	
	autosomal recessive nonsyn
dromic hearing loss 89	
1	
	autosomal recessive nonsyn
dromic hearing loss 88	
1	
	autosomal recessive nonsyn
dromic hearing loss 86	
1	
	autosomal recessive nonsynd
romic hearing loss 84B	
1	
	autosomal recessive nonsynd
romic hearing loss 84A	
1	
	autosomal recessive nonsyn
dromic hearing loss 29	
1	
	autosomal recessive nonsyn
dromic hearing loss 26	
1	
	autosomal recessive epid
ermolytic ichthyosis	
1	
	autosomal recessive limb-girdle
muscular dystrophy type 2T	

1		
		autosomal recessive limb-girdle
muscular dystrophy type 2Q		
1		autosomal recessive limb-girdle
muscular dystrophy type 2P		
1		autosomal recessive limb-girdle
muscular dystrophy type 2O		
1		autosomal recessive limb-girdle
muscular dystrophy type 2N		
1		autosomal recessive limb-girdle
muscular dystrophy type 2M		
1		autosomal recessive limb-girdle
muscular dystrophy type 2L		
1		autosomal recessive limb-girdle
muscular dystrophy type 2K		
1		autosomal recessive limb-girdle
muscular dystrophy type 2J		
1		autosomal recessive limb-girdle
muscular dystrophy type 2I		
1		autosomal recessive limb-girdle
muscular dystrophy type 2H		
1		autosomal recessive limb-girdle
muscular dystrophy type 2G		
1		autosomal recessive limb-girdle
muscular dystrophy type 2F		
1		autosomal recessive limb-girdle
muscular dystrophy type 2E		
1		autosomal recessive limb-girdle
muscular dystrophy type 2D		
1		autosomal recessive limb-girdle
muscular dystrophy type 2C		
1		autosomal recessive limb-girdle
muscular dystrophy type 2B		
1		autosomal recessive limb-girdle
muscular dystrophy type 2A		
1		
		autosomal recessive juveni
le Parkinson disease 2		
1		
		autosomal recessive hyperinsuli

nism due to SUR1 deficiency	1		
			autosomal recessive hyperinsulin
ism due to Kir6.2 deficiency	1		
			autosomal recessive hereditary demyelin
ating motor and sensory neuropathy	1		
			autosomal recessive famili
al Mediterranean fever	1		
			autosomal recessive ex
tra-oral halitosis	1		
			autosomal recessive limb-girdle
muscular dystrophy type 2R1	1		
			autosomal recessive limb-girdle
muscular dystrophy type 2U	1		
			autosomal recessive nonsyn
dromic hearing loss 25	1		
			autosomal recessive limb-girdle
muscular dystrophy type 2W	1		
			autosomal recessive nonsyn
dromic hearing loss 24	1		
			autosomal recessive nonsyn
dromic hearing loss 23	1		
			autosomal recessive nonsyn
dromic hearing loss 22	1		
			autosomal recessive nonsyn
dromic hearing loss 21	1		
			autosomal recessive nonsyn
dromic hearing loss 2	1		
			autosomal recessive nonsyn
dromic hearing loss 1B	1		
			autosomal recessive nonsynd
romic hearing loss 18B	1		
			autosomal recessive nonsynd
romic hearing loss 18A	1		
			autosomal recessive nonsyn
dromic hearing loss 16	1		
			autosomal recessive nonsyn
dromic hearing loss 15	1		

	autosomal recessive nonsynd
romic hearing loss 124	
1	
	autosomal recessive nonsynd
romic hearing loss 104	
1	
	autosomal recessive nonsynd
romic hearing loss 103	
1	
	autosomal recessive nonsynd
romic hearing loss 102	
1	
	autosomal recessive nonsynd
romic hearing loss 101	
1	
	autosomal recessive nonsy
ndromic deafness 105	
1	
	autosomal recessive
nail dysplasia	
1	
	autosomal recessive myogenic arthr
ogryposis multiplex congenita	
1	
	autosomal recessive mendelian susceptibility to mycobacteria
l diseases due to complete RORgamma receptor deficiency	
1	
	autosomal recessive lympho
proliferative disease	
1	
	autosomal recessive limb-girdle
muscular dystrophy type R18	
1	
	autosomal recessive limb-girdle
muscular dystrophy type 2Y	
1	
	autosomal recessive limb-girdle
muscular dystrophy type 2X	
1	
	cataract
49	
1	
	cataract 50 with or
without glaucoma	
1	
	congenital stationary
night blindness 2A	
1	
	congenital anomalies of kidney and urinary tract syndrome with or
without hearing loss, abnormal ears, or developmental delay	
1	
	congenital contractur
al arachnodactyly	
1	
	congenital cataracts-facial dysm
orphism-neuropathy syndrome	

1		
		congenital cataract-progressive muscular hypoton
ia-hearing loss-developmental delay syndrome		
1		
		congenital cataract-microcephaly-nevus flammeus si
mplex-severe intellectual disability syndrome		
1		
		congenital brain dysgenesis due to
glutamine synthetase deficiency		
1		
		congenital bile acid s
ynthesis defect 6		
1		
		congenital bile acid s
ynthesis defect 5		
1		
		congenital bile acid s
ynthesis defect 4		
1		
		congenital bile acid s
ynthesis defect 3		
1		
		congenital bile acid s
ynthesis defect 2		
1		
		congenital bile acid s
ynthesis defect 1		
1		
		congenital anomalies of kid
ney and urinary tract 3		
1		
		congenital dyserythropoiet
ic anemia type type 1B		
1		
		congenital anomalies of kid
ney and urinary tract 2		
1		
		congenital anomalies of kid
ney and urinary tract 1		
1		
		congenital ana
lbuminemia		
1		
		congenital amegakaryocyt
c thrombocytopenia 1		
1		
		congenital adrenal hyperplasia due to cyto
chrome P450 oxidoreductase deficiency		
1		
		congenital adrenal hyperplasia due to 3-beta
-hydroxysteroid dehydrogenase deficiency		
1		
		congenital adrenal hyperplasia due to
17-alpha-hydroxylase deficiency		
1		
		congenital adrenal hyperplasia due t

o 11-beta-hydroxylase deficiency	
1	
	cone-rod synaptic disorder syndro
me, congenital nonprogressive	
1	
	cone-rod dystrophy an
d hearing loss 2	
1	
	cone-rod dystrophy an
d hearing loss 1	
1	
	congenital contractures of the limbs and fa
ce, hypotonia, and developmental delay	
1	
	congenital cor
nea plana	
1	
	congenital deficiency i
n alpha-fetoprotein	
1	
	congenital d
iarrhea	
1	
	congenital dyserythropo
ietic anemia type 3	
1	
	congenital dyserythropo
ietic anemia type 2	
1	
	congenital disorder of gl
ycosylation, type iit	
1	
	congenital disorder of glycosylatio
n, type Iw, autosomal dominant	
1	
	congenital disorder of gl
ycosylation, type IIz	
1	
	congenital disorder of gl
ycosylation, type IIy	
1	
	congenital disorder of gl
ycosylation, type IIw	
1	
	congenital disorder of gl
ycosylation, type IIr	
1	
	congenital disorder of gl
ycosylation, type IIq	
1	
	congenital disorder of gly
cosylation, type IIbb	
1	
	congenital disorder of gly
cosylation, type IIaa	
1	

	congenital disorder of gl
ycosylation, type IAA	
1	
	congenital disorder of gl
ycosylation, type 2v	
1	
	congenital disorder of glycosylatio
n with defective fucosylation 2	
1	
	congenital disorder of glycosylatio
n with defective fucosylation 1	
1	
	congenital disorder of g
lycosylation type II	
1	
	congenital disorder of g
lycosylation type I	
1	
	congenital disorder of g
lycosylation type 1E	
1	
	congenital disorder
of glycosylation	
1	
	congenital disorder of
deglycosylation 2	
1	
	congenital disorder of
deglycosylation 1	
1	
	congenital di
arrhea 6	
1	
	congenital diarrhea 5 wit
h tufting enteropathy	
1	
	cone-rod dystrophy a
nd hearing loss	
1	
	cone-rod dys
trophy 9	
1	
	cone-rod dys
trophy 7	
1	
	complex cortical dysplasia with
other brain malformations 1	
1	
	complete androgen inse
nsitivity syndrome	
1	
	complement factor
b deficiency	
1	
	complement factor
I deficiency	

1		
		complement factor
H deficiency		
1		
		complement component
C1s deficiency		
1		
		complement componen
t 9 deficiency		
1		
		complement componen
t 7 deficiency		
1		
		complement componen
t 6 deficiency		
1		
		complement componen
t 5 deficiency		
1		
		complement componen
t 4b deficiency		
1		
		complement componen
t 4a deficiency		
1		
		complement componen
t 3 deficiency		
1		
		complement componen
t 2 deficiency		
1		
		combined oxidative phospho
rylation deficiency 58		
1		
		combined oxidative phospho
rylation deficiency 57		
1		
		combined oxidative phospho
rylation deficiency 56		
1		
		combined oxidative phospho
rylation deficiency 55		
1		
		combined oxidative phospho
rylation deficiency 54		
1		
		combined oxidative phospho
rylation deficiency 53		
1		
		combined oxidative phospho
rylation deficiency 52		
1		
		combined oxidative phospho
rylation deficiency 51		
1		
		combined oxidative phospho

rylation deficiency 50	1		
rylation deficiency 49	1		combined oxidative phospho
other brain malformations	1		complex cortical dysplasia with
other brain malformations 2	1		complex cortical dysplasia with
trophy 6	1		cone-rod dys
other brain malformations 3	1		complex cortical dysplasia with
trophy 5	1		cone-rod dys
trophy 3	1		cone-rod dys
rophy 24	1		cone-rod dyst
rophy 21	1		cone-rod dyst
rophy 20	1		cone-rod dyst
trophy 2	1		cone-rod dys
rophy 19	1		cone-rod dyst
rophy 18	1		cone-rod dyst
rophy 16	1		cone-rod dyst
rophy 15	1		cone-rod dyst
rophy 14	1		cone-rod dyst
rophy 13	1		cone-rod dyst
rophy 12	1		cone-rod dyst

	cone-rod dyst
rophy 11	
1	
	cone-rod dyst
rophy 10	
1	
	cone dystrophy with supe
rnormal rod response	
1	
	cone dyst
rophy 4	
1	
	cone dyst
rophy 3	
1	
	complex lethal osteo
chondrodysplasia	
1	
	complex cortical dysplasia with
other brain malformations 7	
1	
	complex cortical dysplasia with
other brain malformations 6	
1	
	complex cortical dysplasia with
other brain malformations 5	
1	
	complex cortical dysplasia with
other brain malformations 4	
1	
	congenital dyserythropo
ietic anemia type 4	
1	
	congenital enteropathy due to
enteropeptidase deficiency	
1	
	cataract 6 mult
iple types	
1	
	congenital myasthe
nic syndrome 6	
1	
	congenital myopathy 2a, typi
cal, autosomal dominant	
1	
	congenital my
opathy 25	
1	
	congenital my
opathy 23	
1	
	congenital myopath
y 22A, classic	
1	
	congenital myopathy 21 with e
arly respiratory failure	

1		
		congenital my
opathy 18		
1		congenital my
opathy 15		congenital my
1		
opathy 11		
1		
		congenital myasthe
nic syndrome 9		
1		congenital myasthe
nic syndrome 8		congenital myasthe
1		
nic syndrome 7		
1		congenital myasthe
nic syndrome 5		congenital myasthe
1		
V deficiency		congenital factor
1		
		congenital myasthen
ic syndrome 4C		
1		congenital myasthen
ic syndrome 4B		congenital myasthen
1		
ic syndrome 4A		
1		congenital myasthen
ic syndrome 3C		congenital myasthen
1		
ic syndrome 3B		congenital myasthen
1		
ic syndrome 3A		congenital myasthen
1		
ic syndrome 2C		congenital myasthen
1		
ic syndrome 2A		congenital myasthen
1		
ic syndrome 21		congenital myasthen
1		
ic syndrome 20		congenital myasthen
1		
		congenital myasthen

ic syndrome 19	
1	
nfantile, autosomal dominant	congenital myopathy 2c, severe i
1	
autosomal recessive	congenital myopathy 4B,
1	
storage, autosomal dominant	congenital myopathy 7A, myosin
1	
cess of thin filaments	congenital myopathy with ex
1	
night blindness 1H	congenital stationary
1	
night blindness 1G	congenital stationary
1	
night blindness 1F	congenital stationary
1	
night blindness 1E	congenital stationary
1	
night blindness 1D	congenital stationary
1	
night blindness 1C	congenital stationary
1	
night blindness 1B	congenital stationary
1	
night blindness 1A	congenital stationary
1	
	congenital sideroblastic anemia-B-cell immunodeficie
ncy-periodic fever-developmental delay syndrome	
1	
	congenital siali
dosis type 2	
1	
ome, autosomal recessive	congenital short bowel syndr
1	
sodium diarrhea 8	congenital secretory
1	
sodium diarrhea 3	congenital secretory
1	
hloride diarrhea 1	congenital secretory c
1	

		congenital prothrom	
bin deficiency			
1			
		congenital prim	
ary aphakia			
1			
		congenital plasminogen activator	
inhibitor type 1 deficiency			
1			
		congenital nongoitrous	
hypothyroidism 6			
1			
		congenital nervous	
system disorder			
1			
		congenital nephrotic syn	
drome, Finnish type			
1			
		congenital myopath	
y, Paradas type			
1			
		congenital myopathy with redu	
ced type 2 muscle fibers			
1			
		congenital myopathy with interna	
l nuclei and atypical cores			
1			
		congenital myasthen	
ic syndrome 18			
1			
		congenital myasthen	
ic syndrome 17			
1			
		congenital myasthen	
ic syndrome 16			
1			
		congenital high-molecular-wei	
ght kininogen deficiency			
1			
		congenital hereditary endot	
helial dystrophy type I			
1			
		congenital hereditary endothe	
lial dystrophy of cornea			
1			
		congenital heart	
malformation			
1			
		congenital heart defect	
s, multiple types, 9			
1			
		congenital heart defects, multiple typ	
es, 8, with or without heterotaxy			
1			
		congenital heart defect	
s, multiple types, 7			

1		
		congenital heart defect
s, multiple types, 6		
1		
		congenital heart defect
s, multiple types, 5		
1		
		congenital heart defect
s, multiple types, 4		
1		
		congenital heart defect
s, multiple types, 2		
1		
		congenital heart defects, dysmorphic facial featu
res, and intellectual developmental disorder		
1		
		congenital heart defects and ske
letal malformations syndrome		
1		
		congenital heart defects an
d ectodermal dysplasia		
1		
		congenital generalized l
ipodystrophy type 4		
1		
		congenital generalized l
ipodystrophy type 3		
1		
		congenital generalized l
ipodystrophy type 2		
1		
		congenital generalized l
ipodystrophy type 1		
1		
		congenital generalized hypercontrac
tile muscle stiffness syndrome		
1		
		congenital fibrosis of extr
aocular muscles type 1		
1		
		congenital factor
XII deficiency		
1		
		congenital factor
XI deficiency		
1		
		congenital factor
X deficiency		
1		
		congenital factor
VII deficiency		
1		
		congenital hereditary facial paralysi
s-variable hearing loss syndrome		
1		
		congenital hypothalamic

hamartoma syndrome	
1	
	congenital myasthen
ic syndrome 15	
1	
	congenital hypo
thyroidism	
1	
	congenital myasthen
ic syndrome 14	
1	
	congenital myasthen
ic syndrome 13	
1	
	congenital myasthen
ic syndrome 12	
1	
	congenital myasthen
ic syndrome 11	
1	
	congenital myasthen
ic syndrome 10	
1	
	congenital muscular dystrophy with intell
ectual disability and severe epilepsy	
1	
	congenital muscular dystrophy with cat
aracts and intellectual disability	
1	
	congenital muscular dystrophy due
to integrin alpha-7 deficiency	
1	
	congenital muscular dystrop
hy due to LMNA mutation	
1	
	congenital multicore myopathy wi
th external ophthalmoplegia	
1	
	congenital microcephaly – severe encephalopath
y – progressive cerebral atrophy syndrome	
1	
	congenital merosin-deficien
t muscular dystrophy 1A	
1	
	congenital membranous nephropathy due to maternal
anti-neutral endopeptidase alloimmunization	
1	
	congenital malabsorp
tive diarrhea 4	
1	
	congenital lipid adrenal hyperp
lasia due to STAR deficiency	
1	
	congenital lactic acidosis, Sa
guenay-Lac-Saint-Jean type	
1	

1		congenital lactas	
e deficiency			
1			
		congenital isolated adrenocorti	
cotrophic hormone deficiency			
1			
		congenital insensitivity to p	
ain-hypohidrosis syndrome			
1			
		congenital insensitivity to pain wit	
h severe intellectual disability			
1			
		congenital ichthyosis-intellectual disab	
ility-spastic quadriplegia syndrome			
1			
		congenital hypotrichosis with	
juvenile macular dystrophy			
1			
		congenital hypotonia, epilepsy, develop	
mental delay, and digital anomalies			
1			
		combined oxidative phospho	
rylation deficiency 48			
1			
		combined oxidative phospho	
rylation deficiency 47			
1			
		combined oxidative phospho	
rylation deficiency 46			
1			
		cholestasis, progressive fa	
milial intrahepatic, 10			
1			
		chondrodysplasia B	
lomstrand type			
1			
		chondrodys	
plasia			
1			
		chondrocalci	
nosis 2			
1			
		cholesteryl ester s	
torage disease			
1			
		cholestasis-pigmentary retinop	
athy-cleft palate syndrome			
1			
		cholestasis, progressive fa	
milial intrahepatic, 9			
1			
		cholestasis, progressive fa	
milial intrahepatic, 8			
1			
		cholestasis, progressive familial intrahep	
atic, 7, with or without hearing loss			

1		
		cholestasis, progressive fa
milial intrahepatic, 6		
1		cholestasis, progressive fa
milial intrahepatic, 5		
1		cholestasis, progressive fa
milial intrahepatic, 4		
1		cholestasis, intrahepat
ic, of pregnancy, 1		
1		combined oxidative phospho
rylation deficiency 45		
1		choanal atresia-athelia-hypothyroidism-de
layed puberty-short stature syndrome		
1		childhood-onset progressive contractures-limb-
girdle weakness-muscle dystrophy syndrome		
1		childhood-onset motor and cognitive regression sy
ndrome with extrapyramidal movement disorder		
1		childhood-onset epi
lepsy syndrome		
1		childhood-onset benign chorea
with striatal involvement		
1		childhood-onset autosomal recessive myo
pathy with external ophthalmoplegia		
1		childhood hypop
hosphatasia		
1		childhood encephalopathy due to thiam
ine pyrophosphokinase deficiency		
1		childhood apraxi
a of speech		
1		chilblain l
upus 2		
1		chilblain l
upus 1		
1		chondrodysplasia punctata
2, X-linked dominant		
1		chondrodysplasia with joint
dislocations, gPAPP type		
1		chondrodysplasia-pseudoher

maphroditism syndrome	
1	
rcoma	chondrosa
1	
a, primary, 39	ciliary dyskinesi
1	
a, primary, 38	ciliary dyskinesi
1	
a, primary, 37	ciliary dyskinesi
1	
mary, 36, X-linked	ciliary dyskinesia, pri
1	
tion disease	chylomicron reten
1	
urfactant metabolism deficiency	chronic respiratory distress with s
1	
ulmonary disease	chronic obstructive p
1	
ated with SLC02A1 gene	chronic enteropathy associ
1	
stinal dysrhythmia	chronic atrial and inte
1	
ication syndrome	chromosome Xq28 dupl
1	
elation syndrome	chromosome 6q24-q25 d
1	
etion syndrome	chromosome 5q12 del
1	
elation syndrome	chromosome 2q32-q33 d
1	
letion syndrome	chromosome 2p16.3 de
1	
elation syndrome	chromosome 1p32-p31 d
1	
letion syndrome	chromosome 15q24 de
1	
-obesity syndrome	choroideremia-deafness
1	

		choroider
emia		
1		
		choroidal dystrophy,
central areolar 2		
1		
		choroid plexus
carcinoma		
1		
		chorea-acanth
ocytosis		
1		
		chorea, childhood-onset, with
psychomotor retardation		
1		
		chordo
ma		
1		
		chilblain
lupus		
1		
		cherubi
sm		
1		
		channelopathy-associated congenital insen
sitivity to pain, autosomal recessive		
1		
		cerebellar dysfunction with variable cog
nitive and behavioral abnormalities		
1		
		cerebellar atrophy, developme
ntal delay, and seizures		
1		
		cerebellar atrophy with seizures an
d variable developmental delay		
1		
		cerebellar ataxia-areflexia-pes cavus-optic at
rophy-sensorineural hearing loss syndrome		
1		
		cerebellar ataxia, neuropathy, and
vestibular areflexia syndrome		
1		
		cerebellar ataxia, intellectual disabil
ity, and dysequilibrium syndrome 4		
1		
		cerebellar ataxia, intellectual disabil
ity, and dysequilibrium syndrome 3		
1		
		cerebellar ataxia, intellectual disabil
ity, and dysequilibrium syndrome 2		
1		
		cerebellar ataxia, intellectual disabil
ity, and dysequilibrium syndrome 1		
1		
		cerebellar ataxia, brain abnormalitie
s, and cardiac conduction defects		

1		
		cerebellar ataxia with neuropathy and bil
		ateral vestibular areflexia syndrome
1		
		central precocio
		us puberty 1
1		
		central hypoventilation syndrome, conge
		nital, 2, and autonomic dysfunction
1		
		central hypoventilation syndrome, congenital,
		1, with or without Hirschsprung disease
1		
		central core
		myopathy
1		
		caveolino
		pathy
1		
		caudal dupl
		ication
1		
		catecholaminergic polymorphic
		ventricular tachycardia 5
1		
		catecholaminergic polymorphic
		ventricular tachycardia 4
1		
		catecholaminergic polymorphic
		ventricular tachycardia 3
1		
		catecholaminergic polymorphic
		ventricular tachycardia 2
1		
		catecholaminergic polymorphic
		ventricular tachycardia 1
1		
		cataract-growth hormone deficiency-sensory neuropathy-se
		nsorineural hearing loss-skeletal dysplasia syndrome
1		
		cataract 9 mult
		iple types
1		
		cerebellar atrophy, visual impairmen
		t, and psychomotor retardation;
1		
		cerebellar dysfunction, impaired intellectual de
		velopment, and hypogonadotropic hypogonadism
1		
		ceroid lipofuscinosis, neu
		ronal, 6B (Kufs type)
1		
		cerebellar hypoplasia-intellectual disability-congenital mi
		crocephaly-dystonia-anemia-growth retardation syndrome
1		
		ceroid lipofuscinosi

s, neuronal, 6A	1		
1			
uronal, 4 (Kufs type)	1		ceroid lipofuscinosis, ne
1			
xanthomatosis	1		cerebrotendinous
1			
th calcifications and cysts 3	1		cerebroretinal microangiopathy wi
1			
th calcifications and cysts 2	1		cerebroretinal microangiopathy wi
1			
th calcifications and cysts 1	1		cerebroretinal microangiopathy wi
1			
letal syndrome 4	1		cerebrooculofacioske
1			
letal syndrome 3	1		cerebrooculofacioske
1			
letal syndrome 2	1		cerebrooculofacioske
1			
cic dysplasia	1		cerebrofaciothora
1			
ular syndrome	1		cerebrocostomandib
1			
c quadriplegic, 3	1		cerebral palsy, spasti
1			
c quadriplegic, 2	1		cerebral palsy, spasti
1			
c quadriplegic, 1	1		cerebral palsy, spasti
1			
deficiency	1		cerebral folate
1			
malformation 3	1		cerebral cavernous
1			
malformation 2	1		cerebral cavernous
1			
malformation 1	1		cerebral cavernous
1			
ortical infarcts and leukoencephalopathy, type 2	1		cerebral arteriopathy, autosomal dominant, with subc
1			

	cerebral arteriopathy, autosomal dominant, with subc
ortical infarcts and leukoencephalopathy, type 1	
1	
	cerebral amyloid angio
pathy, APP-related	
1	
	cerebellar-facial-d
ental syndrome	
1	
	cerebellar, ocular, craniofac
ial, and genital syndrome	
1	
	ciliary dyskinesi
a, primary, 40	
1	
	ciliary dyskinesi
a, primary, 41	
1	
	ciliary dyskinesi
a, primary, 42	
1	
	combined oxidative phosphor
ylation defect type 24	
1	
	combined oxidative phosphor
ylation defect type 21	
1	
	combined oxidative phospho
rylation defect type 2	
1	
	combined oxidative phosphor
ylation defect type 17	
1	
	combined oxidative phosphor
ylation defect type 15	
1	
	combined oxidative phosphor
ylation defect type 14	
1	
	combined oxidative phosphor
ylation defect type 13	
1	
	combined oxidative phosphor
ylation defect type 11	
1	
	combined osteogenesis imperfecta
and Ehlers-Danlos syndrome 2	
1	
	combined osteogenesis imperfecta
and Ehlers-Danlos syndrome 1	
1	
	combined malonic and met
hylmalonic acidemia	
1	
	combined immunodefic
iciency, X-linked	

1		
		combined immunodeficiency due
to partial RAG1 deficiency		
1		
		combined immunodeficiency d
ue to moesin deficiency		
1		
		combined immunodeficiency d
ue to ZAP70 deficiency		
1		
		combined immunodeficiency d
ue to STIM1 deficiency		
1		
		combined immunodeficiency due
to RELA haploinsufficiency		
1		
		combined immunodeficiency
due to OX40 deficiency		
1		
		combined immunodeficiency d
ue to ORAI1 deficiency		
1		
		combined immunodeficiency d
ue to MALT1 deficiency		
1		
		combined immunodeficiency
due to LRBA deficiency		
1		
		combined immunodeficiency d
ue to GINS1 deficiency		
1		
		combined immunodeficiency d
ue to DOCK8 deficiency		
1		
		combined immunodeficiency du
e to CD3gamma deficiency		
1		
		combined oxidative phosphor
ylation defect type 23		
1		
		combined oxidative phosphor
ylation defect type 25		
1		
		combined immuno
deficiency		
1		
		combined oxidative phosphor
ylation defect type 26		
1		
		combined oxidative phospho
rylation deficiency 44		
1		
		combined oxidative phospho
rylation deficiency 43		
1		
		combined oxidative phospho

rylation deficiency 42	
1	
	combined oxidative phospho
rylation deficiency 40	
1	
	combined oxidative phospho
rylation deficiency 39	
1	
	combined oxidative phospho
rylation deficiency 38	
1	
	combined oxidative phospho
rylation deficiency 36	
1	
	combined oxidative phospho
rylation deficiency 35	
1	
	combined oxidative phospho
rylation deficiency 34	
1	
	combined oxidative phospho
rylation deficiency 33	
1	
	combined oxidative phospho
rylation deficiency 32	
1	
	combined oxidative phospho
rylation deficiency 29	
1	
	combined oxidative phospho
rylation deficiency 28	
1	
	combined oxidative phospho
rylation deficiency 22	
1	
	combined oxidative phospho
rylation deficiency 19	
1	
	combined oxidative phosph
orylation deficiency	
1	
	combined oxidative phospho
rylation defect type 9	
1	
	combined oxidative phospho
rylation defect type 8	
1	
	combined oxidative phospho
rylation defect type 7	
1	
	combined oxidative phospho
rylation defect type 4	
1	
	combined oxidative phosphor
ylation defect type 30	
1	

	combined oxidative phosphor
ylation defect type 29	
1	
	combined oxidative phosphor
ylation defect type 27	
1	
	combined immunodeficiency and megaloblastic an
emia with or without hyperhomocysteinemia	
1	
	combined PSAP
deficiency	
1	
	ciliary dyskinesi
a, primary, 43	
1	
	cleft lip with or wit
hout cleft palate	
1	
	classic or attenuated famili
al adenomatous polyposis	
1	
	classic homoc
ystinuria	
1	
	classic gala
ctosemia	
1	
	classic dopamine transport
er deficiency syndrome	
1	
	classic congenital adrenal hyperplasia due to 21-h
ydroxylase deficiency, simple virilizing form	
1	
	classic congenital adrenal hyperplasia due to 2
1-hydroxylase deficiency, salt wasting form	
1	
	classic congenital adrenal hyperplasia
due to 21-hydroxylase deficiency	
1	
	classic Hodgki
n lymphoma	
1	
	class I glucose-6-phosphate
dehydrogenase deficiency	
1	
	citrullinemia, type
II, adult-onset	
1	
	citrullinemia
type II	
1	
	citrullinemi
a type I	
1	
	citrin defi
ciency	

1		
a - hypermanganesemia syndrome		cirrhosis - dystonia - polycythemi
1		
a, primary, 52		ciliary dyskinesi
1		
a, primary, 51		ciliary dyskinesi
1		
a, primary, 50		ciliary dyskinesi
1		
9, without situs inversus		ciliary dyskinesia, primary, 4
1		
8, without situs inversus		ciliary dyskinesia, primary, 4
1		
y, 47, and lissencephaly		ciliary dyskinesia, primar
1		
a, primary, 46		ciliary dyskinesi
1		
a, primary, 45		ciliary dyskinesi
1		
a, primary, 44		ciliary dyskinesi
1		
ketonuria		classic phenyl
1		
al dysplasia syndrome		cleft lip/palate-ectoderm
1		
ceptibility to, 12		colorectal cancer, sus
1		
ankyloglossia, X-linked		cleft palate with or without
1		
ceptibility to, 10		colorectal cancer, sus
1		
ceptibility to, 1		colorectal cancer, sus
1		
ry nonpolyposis, type 7		colorectal cancer, heredita
1		
ophy-chorioretinopathy syndrome		colobomatous optic disc-macular atr
1		
		colobomatous microphthalmia-rhi

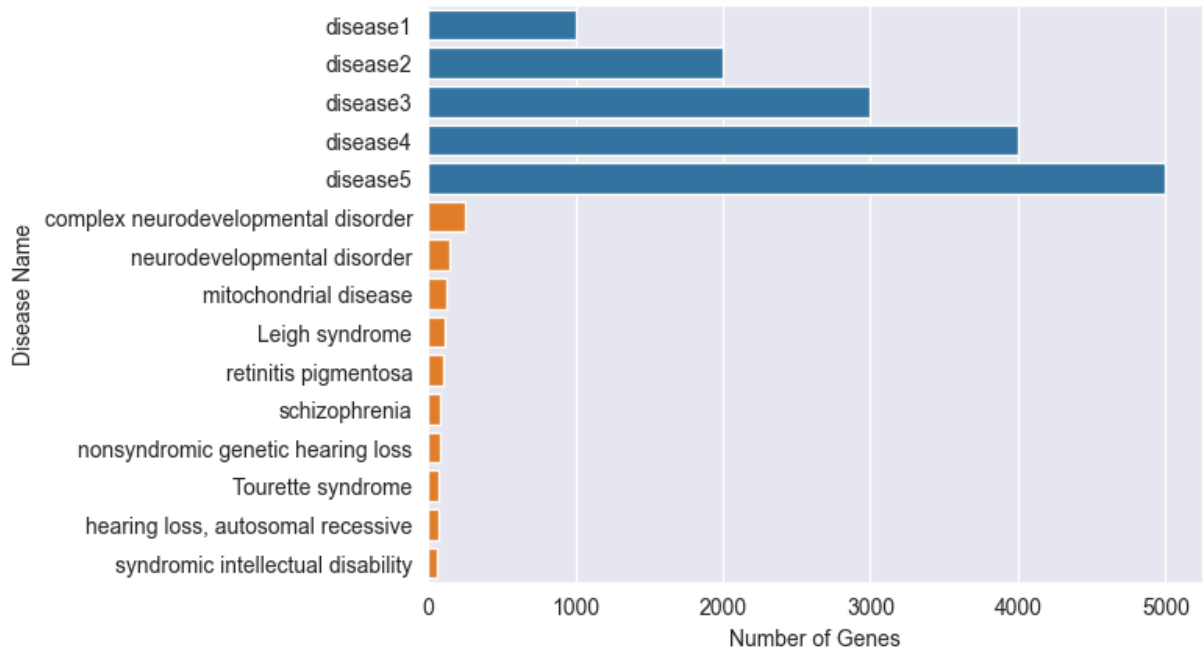
zomelic dysplasia syndrome	
1	
	colobomatous macrophthalmi
a-microcornea syndrome	
1	
	coloboma, osteopetrosis, microphthalmia,
macrocephaly, albinism, and deafness	
1	
	coloboma, ocular, aut
osomal recessive	
1	
	coloboma, ocular, au
tosomal dominant	
1	
	cold-induced sweat
ing syndrome 2	
1	
	cognitive impairment with or
without cerebellar ataxia	
1	
	cognitive impairment - coarse facies - heart defects - obesity - pul
monary involvement - short stature - skeletal dysplasia syndrome	
1	
	coenzyme q10 deficie
ncy, primary, 9	
1	
	coenzyme Q10 deficie
ncy, primary, 3	
1	
	coenzyme Q10 deficie
ncy, primary, 1	
1	
	coenzyme Q10 d
eficiency	
1	
	cocoon syn
drome	
1	
	cobblestone lissencephaly without
muscular or ocular involvement	
1	
	clubfo
ot	
1	
	cleidocranial d
ysplasia 2	
1	
	cleidocranial d
ysplasia 1	
1	
	cleft soft
palate	
1	
	cleft palate, proliferative retino
pathy, and developmental delay	
1	


```

|
yopathy
1
+-----+
|
zebra body m
|
+-----+

```

Out[7]: <Axes: xlabel='Number of Genes', ylabel='Disease Name'>



```

In [8]: # Step Three – Gene–Disease Confidence

'''When making decisions about gene disease relationships there are a comple
of assertions. These are stored in the 'classification_title' field of the or

#pseudocode 9 – group the data by classification_title (confidence categorie
'''###YOUR CODE HERE###'''
grouped = df.groupby('classification_title').agg({'gene_symbol': 'nunique'})

#pseudocode 10 – sort the data in descending order
'''###YOUR CODE HERE###'''
grouped = grouped.sort_values(by='gene_symbol', ascending=False)

#pseudocode 11 – look at the first few rows of the dataframe
'''###YOUR CODE HERE###'''
grouped.head()

#pseudocode 12 – create a table using the PrettyTable package
'''###YOUR CODE HERE###'''
table = PrettyTable()
table.field_names = ["Classification Title", "Number of Genes"]
for index, row in grouped.iterrows():
    table.add_row([row['classification_title'], row['gene_symbol']])
print(table)

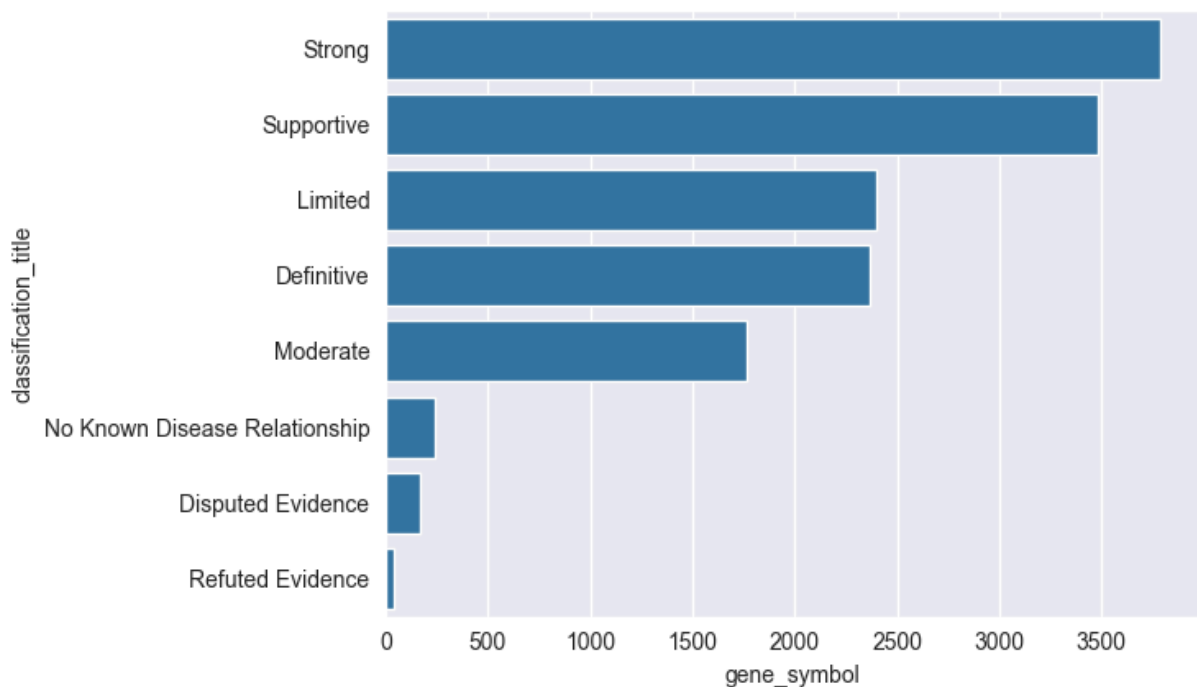
#pseudocode 13 – plot the classification_titles with the most genes
'''###YOUR CODE HERE###'''

```

```
top10 = grouped.head(10)
sns.barplot(x='gene_symbol', y='classification_title', data=top10)
```

Classification Title	Number of Genes
Strong	3790
Supportive	3479
Limited	2399
Definitive	2363
Moderate	1765
No Known Disease Relationship	239
Disputed Evidence	172
Refuted Evidence	36

Out[8]: <Axes: xlabel='gene_symbol', ylabel='classification_title'>



In [11]: *# Step Four – Modes of Inheritance*

```
'''Many of these diseases are Mendelian diseases. This means that they are c
GenCC captures this information in the 'moi_curie' field'''

#pseudocode 14 – group the data by moi_curie (modes of inheritance) and cour
'''###YOUR CODE HERE###'''
grouped = df.groupby('moi_curie').agg({'gene_symbol': 'nunique'}).reset_index
grouped = grouped.rename(columns={'moi_curie': 'Mode of Inheritance Curie',
print(grouped)

'''This is good but the values we have here are accession ids that aren't ve
We know that terms (classes) like 'HP:0000006' are actually from the Human F
We're going to use the BioPortal API to get infromation about these classes
```

	Mode of Inheritance	Curie	Number of Genes
0		HP:0000005	879
1		HP:0000006	2772
2		HP:0000007	3398
3		HP:0001417	277
4		HP:0001419	2
5		HP:0001427	43
6		HP:0001450	3
7		HP:0032113	118

Out[11]: "This is good but the values we have here are accession ids that aren't very informative\nWe know that terms (classes) like 'HP:0000006' are actually from the Human Phenotype Ontology (HPO)\nWe're going to use the BioPortal API to get information about these classes and then finish by merging these data with our table"

In [12]: '''first we need an API key from BioPortal. You can create an account on BioPortal at <https://bioportal.bioontology.org/>\nFor this example I will use my API key – note how I have it stored in a file

```
#load my API key for BioPortal
with open("../api_keys/bioportal.txt", "r") as file:
    apikey = file.read().strip()

'''Here is an example of how you can use the requests library to get information from the BioPortal API'''

#find details for an ontology term using the BioPortal API
import requests
import json

#set the URL for the BioPortal API – here we query with a fixed HP class id
url = 'http://data.bioontology.org/ontologies/HP/classes/HP:0000006?apikey=' + apikey

#send the request
response = requests.get(url)

#parse the JSON in the response
class_info = response.json()

#display the JSON (for interest!)
print(json.dumps(class_info, indent=2))

#display the keys in the JSON (for interest!)
print(class_info.keys())

#display the values for the keys
print(class_info['@id'], class_info['prefLabel'])
print(class_info['definition'][0])
```

```

{
  "prefLabel": "Autosomal dominant inheritance",
  "synonym": [
    "Autosomal dominant form",
    "Autosomal dominant type",
    "Autosomal dominant",
    "monoallelic_autosomal"
  ],
  "definition": [
    "A mode of inheritance that is observed for traits related to a gene encoded on one of the autosomes (i.e., the human chromosomes 1-22) in which a trait manifests in heterozygotes. In the context of medical genetics, an autosomal dominant disorder is caused when a single copy of the mutant allele is present. Males and females are affected equally, and can both transmit the disorder with a risk of 50% for each child of inheriting the mutant allele."
  ],
  "cui": [],
  "semanticType": [],
  "obsolete": false,
  "created": null,
  "modified": null,
  "memberOf": [],
  "inScheme": [],
  "@id": "http://purl.obolibrary.org/obo/HP_0000006",
  "@type": "http://www.w3.org/2002/07/owl#Class",
  "links": {
    "self": "https://data.bioontology.org/ontologies/HP/classes/http%3A%2F%2Fpurl.obolibrary.org%2Fobo%2FHP_0000006",
    "ontology": "https://data.bioontology.org/ontologies/HP",
    "children": "https://data.bioontology.org/ontologies/HP/classes/http%3A%2F%2Fpurl.obolibrary.org%2Fobo%2FHP_0000006/children",
    "parents": "https://data.bioontology.org/ontologies/HP/classes/http%3A%2F%2Fpurl.obolibrary.org%2Fobo%2FHP_0000006/parents",
    "descendants": "https://data.bioontology.org/ontologies/HP/classes/http%3A%2F%2Fpurl.obolibrary.org%2Fobo%2FHP_0000006/descendants",
    "ancestors": "https://data.bioontology.org/ontologies/HP/classes/http%3A%2F%2Fpurl.obolibrary.org%2Fobo%2FHP_0000006/ancestors",
    "instances": "https://data.bioontology.org/ontologies/HP/classes/http%3A%2F%2Fpurl.obolibrary.org%2Fobo%2FHP_0000006/instances",
    "tree": "https://data.bioontology.org/ontologies/HP/classes/http%3A%2F%2Fpurl.obolibrary.org%2Fobo%2FHP_0000006/tree",
    "notes": "https://data.bioontology.org/ontologies/HP/classes/http%3A%2F%2Fpurl.obolibrary.org%2Fobo%2FHP_0000006/notes",
    "mappings": "https://data.bioontology.org/ontologies/HP/classes/http%3A%2F%2Fpurl.obolibrary.org%2Fobo%2FHP_0000006/mappings",
    "ui": "http://bioportal.bioontology.org/ontologies/HP?p=classes&conceptid=http%3A%2F%2Fpurl.obolibrary.org%2Fobo%2FHP_0000006",
    "@context": {
      "self": "http://www.w3.org/2002/07/owl#Class",
      "ontology": "http://data.bioontology.org/metadata/Ontology",
      "children": "http://www.w3.org/2002/07/owl#Class",
      "parents": "http://www.w3.org/2002/07/owl#Class",
      "descendants": "http://www.w3.org/2002/07/owl#Class",
      "ancestors": "http://www.w3.org/2002/07/owl#Class",
      "instances": "http://data.bioontology.org/metadata/Instance",
      "tree": "http://www.w3.org/2002/07/owl#Class",

```

```

        "notes": "http://data.bioontology.org/metadata/Note",
        "mappings": "http://data.bioontology.org/metadata/Mapping",
        "ui": "http://www.w3.org/2002/07/owl#Class"
    }
},
"@context": {
    "@vocab": "http://data.bioontology.org/metadata/",
    "prefLabel": "http://www.w3.org/2004/02/skos/core#prefLabel",
    "synonym": "http://www.w3.org/2004/02/skos/core#altLabel",
    "definition": "http://www.w3.org/2004/02/skos/core#definition",
    "obsolete": "http://www.w3.org/2002/07/owl#deprecated",
    "semanticType": "http://bioportal.bioontology.org/ontologies/umls/hasST
Y",
    "cui": "http://bioportal.bioontology.org/ontologies/umls/cui",
    "inScheme": "http://www.w3.org/2004/02/skos/core#inScheme",
    "created": "http://purl.org/dc/terms/created",
    "modified": "http://purl.org/dc/terms/modified",
    "@language": "en"
}
}
dict_keys(['prefLabel', 'synonym', 'definition', 'cui', 'semanticType', 'obsolete', 'created', 'modified', 'memberOf', 'inScheme', '@id', '@type', 'links', '@context'])
http://purl.obolibrary.org/obo/HP_000006 Autosomal dominant inheritance
A mode of inheritance that is observed for traits related to a gene encoded on one of the autosomes (i.e., the human chromosomes 1-22) in which a trait manifests in heterozygotes. In the context of medical genetics, an autosomal dominant disorder is caused when a single copy of the mutant allele is present. Males and females are affected equally, and can both transmit the disorder with a risk of 50% for each child of inheriting the mutant allele.

```

```

In [13]: #pseudocode 15 - using the unique accession ids in the moi_curie field, get
'''###YOUR CODE HERE###'''
moi_classes = {}
for moi in grouped['Mode of Inheritance Curie'].unique():
    url = f'http://data.bioontology.org/ontologies/HP/classes/{moi}?apikey={
    response = requests.get(url)
    if response.status_code == 200:
        class_info = response.json()
        moi_classes[moi] = {
            'prefLabel': class_info.get('prefLabel', 'N/A'),
            'definition': class_info.get('definition', ['N/A'])[0]
        }
    else:
        moi_classes[moi] = {'prefLabel': 'N/A', 'definition': 'N/A'}

#pseudocode 16 - merge the data with the moi_curie field with the data from
# the mode of inheritance and the number of genes for each and the name of t
'''###YOUR CODE HERE###'''
grouped['Mode of Inheritance Name'] = grouped['Mode of Inheritance Curie'].n
print(grouped)

```

	Mode of Inheritance	Curie	Number of Genes	Mode of Inheritance Name
e				
0		HP:0000005	879	Mode of inheritance
e				
1		HP:0000006	2772	Autosomal dominant inheritance
e				
2		HP:0000007	3398	Autosomal recessive inheritance
e				
3		HP:0001417	277	X-linked inheritance
e				
4		HP:0001419	2	X-linked recessive inheritance
e				
5		HP:0001427	43	Mitochondrial inheritance
e				
6		HP:0001450	3	Y-linked inheritance
e				
7		HP:0032113	118	Semidominant inheritance
e				

Well done!

You've now practised FTP download, API use and also completed some data manipulation using Pandas, PrettyTable, and plotting with Seaborn